

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				2 *****
				3 *
				4 * Zvector E7 instruction tests for VRR-d encoded:
				5 *
				6 * E7A9 VMALH - Vector Multiply and Add Logical High
				7 * E7AA VMAL - Vector Multiply and Add Low
				8 * E7AB VMAH - Vector Multiply and Add High
				9 * E7AC VMALE - Vector Multiply and Add Logical Even
				10 * E7AD VMALO - Vector Multiply and Add Logical Odd
				11 * E7AE VMAE - Vector Multiply and Add Even
				12 * E7AF VMAO - Vector Multiply and Add Odd
				13 *
				14 * James Wekel March 2025
				15 * July 2025 - Vector-enhancements facility 3 update
				16 *****
				18 *****
				19 *
				20 * basic instruction tests
				21 *
				22 *****
				23 * This program tests proper functioning of the z/arch E7 VRR-d vector
				24 * multiply and add (logical high, low, high, logical even,
				25 * logical odd, even, and odd) instructions.
				26 * Exceptions are not tested.
				27 *
				28 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
				29 * obvious coding errors. None of the tests are thorough. They are
				30 * NOT designed to test all aspects of any of the instructions.
				31 *
				32 *****
				33 *
				34 * *Testcase zvector-e7-10-multiplyAdd
				35 * *
				36 * * Zvector E7 instruction tests for VRR-d encoded:
				37 * *
				38 * * E7A9 VMALH - Vector Multiply and Add Logical High
				39 * * E7AA VMAL - Vector Multiply and Add Low
				40 * * E7AB VMAH - Vector Multiply and Add High
				41 * * E7AC VMALE - Vector Multiply and Add Logical Even
				42 * * E7AD VMALO - Vector Multiply and Add Logical Odd
				43 * * E7AE VMAE - Vector Multiply and Add Even
				44 * * E7AF VMAO - Vector Multiply and Add Odd
				45 * *
				46 * * # -----
				47 * * # This tests only the basic function of the instruction.
				48 * * # Exceptions are NOT tested.
				49 * * # -----
				50 * *
				51 * mainsize 2
				52 * numcpu 1
				53 * sysclear
				54 * archlvl z/Arch
				55 * *
				56 * loadcore "\${testpath}/zvector-e7-10-multiplyAdd.core" 0x0

```

57 *
58 *   diag8cmd   enable   # (needed for messages to Hercules console)
59 *   runtest    10       # (2 secs if intrinsic used, 10 otherwise!)
60 *   diag8cmd   disable  # (reset back to default)
61 *
62 *   *Done
63 *
64 ****

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LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				66 *****
				67 * FCHECK Macro - Is a Facility Bit set?
				68 *
				69 * If the facility bit is NOT set, an message is issued and
				70 * the test is skipped.
				71 *
				72 * Fcheck uses R0, R1 and R2
				73 *
				74 * eg. FCHECK 134,'vector-packed-decimal'
				75 *****
				76 MACRO
				77 FCHECK &BITNO,&NOTSETMSG
				78 .* &BITNO : facility bit number to check
				79 .* &NOTSETMSG : 'facility name'
				80 LCLA &FBBYTE Facility bit in Byte
				81 LCLA &FBBIT Facility bit within Byte
				82
				83 LCLA &L(8)
				84 &L(1) SetA 128,64,32,16,8,4,2,1 bit positions within byte
				85
				86 &FBBYTE SETA &BITNO/8
				87 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)
				88 .* MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
				89
				90 B X&SYSNDX
				91 * Fcheck data area
				92 * skip messgae
				93 SKT&SYSNDX DC C' Skipping tests: '
				94 DC C&NOTSETMSG
				95 DC C' (bit &BITNO) is not installed.'
				96 SKL&SYSNDX EQU *-SKT&SYSNDX
				97 * facility bits
				98 DS FD gap
				99 FB&SYSNDX DS 4FD
				100 DS FD gap
				101 *
				102 X&SYSNDX EQU *
				103 LA R0,((X&SYSNDX-FB&SYSNDX)/8)-1
				104 STFLE FB&SYSNDX get facility bits
				105
				106 XGR R0,R0
				107 IC R0,FB&SYSNDX+&FBBYTE get fbit byte
				108 N R0,=F'&FBBIT' is bit set?
				109 BNZ XC&SYSNDX
				110 *
				111 * facility bit not set, issue message and exit
				112 *
				113 LA R0,SKL&SYSNDX message length
				114 LA R1,SKT&SYSNDX message address
				115 BAL R2,MSG
				116
				117 B EOJ
				118 XC&SYSNDX EQU *
				119 MEND

LOC	OBJECT CODE	ADDR1	ADDR2	STMT	
				121	*****
				122	* Low core PSWs
				123	*****
00000000		00000000	00008CDB	124	ZVE7TST START 0
		00000000		125	USING ZVE7TST,R0 Low core addressability
		00000140	00000000	126	
				127	SVOLDPSW EQU ZVE7TST+X'140' z/Arch Supervisor call old PSW
00000000		00000000	000001A0	129	ORG ZVE7TST+X'1A0' z/Architecure RESTART PSW
000001A0	00000001 80000000			130	DC X'0000000180000000'
000001A8	00000000 00000200			131	DC AD(BEGIN)
000001B0		000001B0	000001D0	133	ORG ZVE7TST+X'1D0' z/Architecure PROGRAM CHECK PSW
000001D0	00020001 80000000			134	DC X'0002000180000000'
000001D8	00000000 0000DEAD			135	DC AD(X'DEAD')
000001E0		000001E0	00000200	137	ORG ZVE7TST+X'200' Start of actual test program...
				139	*****
				140	* The actual "ZVE7TST" program itself...
				141	*****
				142	*
				143	* Architecture Mode: z/Arch
				144	* Register Usage:
				145	*
				146	* R0 (work)
				147	* R1-4 (work)
				148	* R5 Testing control table - current test base
				149	* R6-R7 (work)
				150	* R8 First base register
				151	* R9 Second base register
				152	* R10 Third base register
				153	* R11 E7TEST call return
				154	* R12 E7TESTS register
				155	* R13 (work)
				156	* R14 Subroutine call
				157	* R15 Secondary Subroutine call or work
				158	*
				159	*****
00000200		00000200		161	USING BEGIN,R8 FIRST Base Register
00000200		00001200		162	USING BEGIN+4096,R9 SECOND Base Register
00000200		00002200		163	USING BEGIN+8192,R10 THIRD Base Register
00000200	0580			165	BEGIN BALR R8,0 Initalize FIRST base register
00000202	0680			166	BCTR R8,0 Initalize FIRST base register
00000204	0680			167	BCTR R8,0 Initalize FIRST base register
00000206	4190 8800		00000800	169	LA R9,2048(,R8) Initalize SECOND base register
0000020A	4190 9800		00000800	170	LA R9,2048(,R9) Initalize SECOND base register
				171	

[illegible]

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				378 *****
				379 * Normal completion or Abnormal termination PSWs
				380 *****
00000510	00020001 80000000			382 EOJPSW DC 0D'0',X'0002000180000000',AD(0)
00000520	B2B2 8310		00000510	384 EOJ LPSWE EOJPSW Normal completion
00000528	00020001 80000000			386 FAILPSW DC 0D'0',X'0002000180000000',AD(X'BAD')
00000538	B2B2 8328		00000528	388 FAILTEST LPSWE FAILPSW Abnormal termination
				390 *****
				391 * Working Storage
				392 *****
0000053C	00000000			394 CTLR0 DS F CR0
00000540	00000000			395 DS F
00000544				397 LTORG , Literals pool
00000544	00000040			398 =F'64'
00000548	00000002			399 =F'2'
0000054C	00008A44			400 =A(E7TESTS)
00000550	00000001			401 =F'1'
00000554	0000			402 =H'0'
00000556	005F			403 =AL2(L'MSGMSG)
				404
				405 * some constants
				406
		00000400	00000001	407 K EQU 1024 One KB
		00001000	00000001	408 PAGE EQU (4*K) Size of one page
		00010000	00000001	409 K64 EQU (64*K) 64 KB
		00100000	00000001	410 MB EQU (K*K) 1 MB
				411
		AABBCCDD	00000001	412 REG2PATT EQU X'AABBCCDD' Polluted Register pattern
		000000DD	00000001	413 REG2LOW EQU X'DD' (last byte above)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				456 *****
				457 * E7TEST DSECT
				458 *****
				460 E7TEST DSECT ,
00000000	00000000			461 TSUB DC A(0) pointer to test
00000004	0000			462 TNUM DC H'00' Test Number
00000006	00			463 DC X'00'
00000007	00			464 M5 DC HL1'00' m4 used
				465
00000008	40404040	40404040		466 OPNAME DC CL8' ' E7 name
00000010	00000000			467 V2ADDR DC A(0) address of v2 source
00000014	00000000			468 V3ADDR DC A(0) address of v3 source
00000018	00000000			469 V4ADDR DC A(0) address of v4 source
0000001C	00000000			470 RELEN DC A(0) RESULT LENGTH
00000020	00000000			471 READDR DC A(0) result (expected) address
00000028	00000000	00000000		472 DS FD gap
00000030	00000000	00000000		473 V10OUTPUT DS XL16 V1 Output
00000040	00000000	00000000		474 DS FD gap
				475
				476 * test routine will be here (from VRR-d macro)
				477 *
				478 * followed by
				479 * EXPECTED RESULT
000010B4		00000000	00008CDB	481 ZVE7TST CSECT ,
				482 DS 0F
				484 *****
				485 * Macros to help build test tables
				486 *****
				488 *
				489 * macro to generate individual test
				490 *
				491 MACRO
				492 VRR_D &INST,&M5
				493 .* &INST - VRR-d instruction under test
				494 .* &m5 - m5 field
				495
				496 GBLA &TNUM
				497 &TNUM SETA &TNUM+1
				498
				499 DS 0FD
				500 USING *,R5 base for test data and test routine
				501
				502 T&TNUM DC A(X&TNUM) address of test routine
				503 DC H'&TNUM' test number
				504 DC X'00'
				505 DC HL1'&M5' m5
				506 DC CL8'&INST' instruction name

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				507	DC	A(RE&TNUM+16)	address of v2 source
				508	DC	A(RE&TNUM+32)	address of v3 source
				509	DC	A(RE&TNUM+48)	address of v4 source
				510	DC	A(16)	result length
				511	REA&TNUM	DC	A(RE&TNUM)
				512	DS	FD	gap
				513	V10&TNUM	DS	XL16
				514	DS	FD	gap
				515	.	*	
				516	*		
				517	X&TNUM	DS	0F
				518	LGF	R1,V2ADDR	load v2 source
				519	VL	v22,0(R1)	use v22 to test decoder
				520			
				521	LGF	R1,V3ADDR	load v3 source
				522	VL	v23,0(R1)	use v23 to test decoder
				523			
				524	LGF	R1,V4ADDR	load v4 source
				525	VL	v24,0(R1)	use v24 to test decoder
				526			
				527	&INST	V22,V22,V23,V24,&M5	test instruction (dest is a source)
				528	VST	V22,V10&TNUM	save v1 output
				529			
				530	BR	R11	return
				531			
				532	RE&TNUM	DC	0F
				533			xl16 expected result
				534	DROP	R5	
				535	MEND		
				537	*		
				538	*	macro to generate table of pointers to individual tests	
				539	*		
				540		MACRO	
				541		PTTABLE	
				542		GBLA	&TNUM
				543		LCLA	&CUR
				544	&CUR	SETA	1
				545	.	*	
				546	TTABLE	DS	0F
				547	.LOOP	ANOP	
				548	.	*	
				549		DC	A(T&CUR)
				550	.	*	
				551	&CUR	SETA	&CUR+1
				552	AIF		(&CUR LE &TNUM).LOOP
				553	*		
				554		DC	A(0)
				555		DC	A(0)
				556	.	*	
				557		MEND	
				558			

LOC	OBJECT CODE	ADDR1	ADDR2	STMT
				560 *****
				561 * E7 VRR-d tests
				562 *****
000010B8	00000000 00000000			563 PRINT DATA
				564 DS FD
				565 *
				566 * E7A9 VMALH - Vector Multiply and Add Logical High
				567 * E7AA VMAL - Vector Multiply and Add Low
				568 * E7AB VMAH - Vector Multiply and Add High
				569 * E7AC VMALE - Vector Multiply and Add Logical Even
				570 * E7AD VMALO - Vector Multiply and Add Logical Odd
				571 * E7AE VMAE - Vector Multiply and Add Even
				572 * E7AF VMAO - Vector Multiply and Add Odd
				573 *
				574 * VRR-d instruction, m5
				575 * followed by
				576 * 16 byte expected result (V1)
				577 * 16 byte V2 source
				578 * 16 byte V3 source
				579 * 16 byte V4 source
				580 *-----
				581 * VMALH - Vector Multiply and Add Logical High
				582 *-----
				583 * Byte
000010C0				584 VRR_D VMALH,0
000010C0		000010C0		585+ DS 0FD
000010C0	00001108			586+ USING *,R5
000010C4	0001			587+T1 DC A(X1)
000010C6	00			588+ DC H'1'
000010C7	00			589+ DC X'00'
000010C8	E5D4C1D3 C8404040			590+ DC HL1'0'
000010D0	0000114C			591+ DC CL8'VMALH'
000010D4	0000115C			592+ DC A(RE1+16)
000010D8	0000116C			593+ DC A(RE1+32)
000010DC	00000010			594+ DC A(RE1+48)
000010E0	0000113C			595+ DC A(16)
000010E8	00000000 00000000			596+REA1 DC A(RE1)
000010F0	00000000 00000000			597+ DS FD
000010F8	00000000 00000000			598+V101 DS XL16
00001100	00000000 00000000			599+ DS FD
				600+*
00001108				601+X1 DS 0F
00001108	E310 5010 0014	00000010		602+ LGF R1,V2ADDR
0000110E	E761 0000 0806	00000000		603+ VL v22,0(R1)
00001114	E310 5014 0014	00000014		604+ LGF R1,V3ADDR
0000111A	E771 0000 0806	00000000		605+ VL v23,0(R1)
00001120	E310 5018 0014	00000018		606+ LGF R1,V4ADDR
00001126	E781 0000 0806	00000000		607+ VL v24,0(R1)
0000112C	E766 7000 8FA9			608+ VMALH V22,V22,V23,V24,0
00001132	E760 5030 080E	000010F0		609+ VST V22,V101
00001138	07FB			610+ BR R11
0000113C				611+RE1 DC 0F
0000113C				612+ DROP R5
0000113C	FE000000 00000002			613 DC XL16'FE00000000000002 0000000C000000F4'
00001144	0000000C 000000F4			

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000114C	FF000000 00000019			614	DC	XL16'FF0000000000000019 000000380000000FA'	v2
00001154	00000038 000000FA						
0000115C	FF000000 00000019			615	DC	XL16'FF0000000000000019 000000380000000FA'	v3
00001164	00000038 000000FA						
0000116C	00000000 00000000			616	DC	XL16'0000000000000000 0000000000000000'	v4
00001174	00000000 00000000						
				617			
				618	VRR_D	VMALH,0	
00001180				619+	DS	0FD	
00001180		00001180		620+	USING	*,R5	base for test data and test routine
00001180	000011C8			621+T2	DC	A(X2)	address of test routine
00001184	0002			622+	DC	H'2'	test number
00001186	00			623+	DC	X'00'	
00001187	00			624+	DC	HL1'0'	m5
00001188	E5D4C1D3 C8404040			625+	DC	CL8'VMALH'	instruction name
00001190	0000120C			626+	DC	A(RE2+16)	address of v2 source
00001194	0000121C			627+	DC	A(RE2+32)	address of v3 source
00001198	0000122C			628+	DC	A(RE2+48)	address of v4 source
0000119C	00000010			629+	DC	A(16)	result length
000011A0	000011FC			630+REA2	DC	A(RE2)	result address
000011A8	00000000 00000000			631+	DS	FD	gap
000011B0	00000000 00000000			632+V102	DS	XL16	V1 output
000011B8	00000000 00000000						
000011C0	00000000 00000000			633+	DS	FD	gap
				634+*			
000011C8				635+X2	DS	0F	
000011C8	E310 5010 0014		00000010	636+	LGF	R1,V2ADDR	load v2 source
000011CE	E761 0000 0806		00000000	637+	VL	v22,0(R1)	use v22 to test decoder
000011D4	E310 5014 0014		00000014	638+	LGF	R1,V3ADDR	load v3 source
000011DA	E771 0000 0806		00000000	639+	VL	v23,0(R1)	use v23 to test decoder
000011E0	E310 5018 0014		00000018	640+	LGF	R1,V4ADDR	load v4 source
000011E6	E781 0000 0806		00000000	641+	VL	v24,0(R1)	use v24 to test decoder
000011EC	E766 7000 8FA9			642+	VMALH	V22,V22,V23,V24,0	test instruction (dest is a source)
000011F2	E760 5030 080E		000011B0	643+	VST	V22,V102	save v1 output
000011F8	07FB			644+	BR	R11	return
000011FC				645+RE2	DC	0F	xl16 expected result
000011FC				646+	DROP	R5	
000011FC	FE000001 00000006			647	DC	XL16'FE00000100000006 0000000C0000000F4'	result
00001204	0000000C 000000F4						
0000120C	FF0000FF 00000029			648	DC	XL16'FF0000FF00000029 000000380000000FA'	v2
00001214	00000038 000000FA						
0000121C	FF000001 00000029			649	DC	XL16'FF00000100000029 000000380000000FA'	v3
00001224	00000038 000000FA						
0000122C	00000001 0000002F			650	DC	XL16'000000010000002F 00000000000000002'	v4
00001234	00000000 00000002						
				651			
				652	VRR_D	VMALH,0	
00001240				653+	DS	0FD	
00001240		00001240		654+	USING	*,R5	base for test data and test routine
00001240	00001288			655+T3	DC	A(X3)	address of test routine
00001244	0003			656+	DC	H'3'	test number
00001246	00			657+	DC	X'00'	
00001247	00			658+	DC	HL1'0'	m5
00001248	E5D4C1D3 C8404040			659+	DC	CL8'VMALH'	instruction name
00001250	000012CC			660+	DC	A(RE3+16)	address of v2 source
00001254	000012DC			661+	DC	A(RE3+32)	address of v3 source

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00001258	000012EC			662+	DC	A(RE3+48)	address of v4 source
0000125C	00000010			663+	DC	A(16)	result length
00001260	000012BC			664+REA3	DC	A(RE3)	result address
00001268	00000000 00000000			665+	DS	FD	gap
00001270	00000000 00000000			666+V103	DS	XL16	V1 output
00001278	00000000 00000000						
00001280	00000000 00000000			667+	DS	FD	gap
				668+*			
00001288				669+X3	DS	0F	
00001288	E310 5010 0014		00000010	670+	LGF	R1,V2ADDR	load v2 source
0000128E	E761 0000 0806		00000000	671+	VL	v22,0(R1)	use v22 to test decoder
00001294	E310 5014 0014		00000014	672+	LGF	R1,V3ADDR	load v3 source
0000129A	E771 0000 0806		00000000	673+	VL	v23,0(R1)	use v23 to test decoder
000012A0	E310 5018 0014		00000018	674+	LGF	R1,V4ADDR	load v4 source
000012A6	E781 0000 0806		00000000	675+	VL	v24,0(R1)	use v24 to test decoder
000012AC	E766 7000 8FA9			676+	VMALH	V22,V22,V23,V24,0	test instruction (dest is a source)
000012B2	E760 5030 080E		00001270	677+	VST	V22,V103	save v1 output
000012B8	07FB			678+	BR	R11	return
000012BC				679+RE3	DC	0F	xl16 expected result
000012BC				680+	DROP	R5	
000012BC	FF000000 00000000			681	DC	XL16'FF00000000000000 0000000000000001'	result
000012C4	00000000 00000001						
000012CC	FF020304 05060708			682	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000012D4	090A0B0C 0D0E0F10						
000012DC	FF020304 05060708			683	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000012E4	090A0B0C 0D0E0F10						
000012EC	FF020304 05060708			684	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000012F4	090A0B0C 0D0E0F10						
				685			
00001300				686	VRR_D	VMALH,0	
00001300		00001300		687+	DS	0FD	
00001300	00001348			688+	USING	*,R5	base for test data and test routine
00001304	0004			689+T4	DC	A(X4)	address of test routine
00001306	00			690+	DC	H'4'	test number
00001307	00			691+	DC	X'00'	
00001308	E5D4C1D3 C8404040			692+	DC	HL1'0'	m5
00001310	0000138C			693+	DC	CL8'VMALH'	instruction name
00001314	0000139C			694+	DC	A(RE4+16)	address of v2 source
00001318	000013AC			695+	DC	A(RE4+32)	address of v3 source
0000131C	00000010			696+	DC	A(RE4+48)	address of v4 source
00001320	0000137C			697+	DC	A(16)	result length
00001328	00000000 00000000			698+REA4	DC	A(RE4)	result address
00001330	00000000 00000000			699+	DS	FD	gap
00001338	00000000 00000000			700+V104	DS	XL16	V1 output
00001340	00000000 00000000			701+	DS	FD	gap
				702+*			
00001348				703+X4	DS	0F	
00001348	E310 5010 0014		00000010	704+	LGF	R1,V2ADDR	load v2 source
0000134E	E761 0000 0806		00000000	705+	VL	v22,0(R1)	use v22 to test decoder
00001354	E310 5014 0014		00000014	706+	LGF	R1,V3ADDR	load v3 source
0000135A	E771 0000 0806		00000000	707+	VL	v23,0(R1)	use v23 to test decoder
00001360	E310 5018 0014		00000018	708+	LGF	R1,V4ADDR	load v4 source
00001366	E781 0000 0806		00000000	709+	VL	v24,0(R1)	use v24 to test decoder
0000136C	E766 7000 8FA9			710+	VMALH	V22,V22,V23,V24,0	test instruction (dest is a source)
00001372	E760 5030 080E		00001330	711+	VST	V22,V104	save v1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001378	07FB			712+	BR	R11	return
0000137C				713+RE4	DC	0F	xl16 expected result
0000137C				714+	DROP	R5	
0000137C	FF000000 00000000			715	DC	XL16'FF00000000000000 0000000000000000'	result
00001384	00000000 00000000						
0000138C	FF020304 05060708			716	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001394	090A0B0C 0D0E0F10						
0000139C	FF010102 02030304			717	DC	XL16'FF01010202030304 0405050606070708'	v3
000013A4	04050506 06070708						
000013AC	FF020304 05060708			718	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000013B4	090A0B0C 0D0E0F10						
				719			
000013C0				720	VRR_D	VMALH,0	
000013C0		000013C0		721+	DS	0FD	
000013C0	00001408			722+	USING	*,R5	base for test data and test routine
000013C4	0005			723+T5	DC	A(X5)	address of test routine
000013C6	00			724+	DC	H'5'	test number
000013C7	00			725+	DC	X'00'	
000013C8	E5D4C1D3 C8404040			726+	DC	HL1'0'	m5
000013D0	0000144C			727+	DC	CL8'VMALH'	instruction name
000013D4	0000145C			728+	DC	A(RE5+16)	address of v2 source
000013D8	0000146C			729+	DC	A(RE5+32)	address of v3 source
000013DC	00000010			730+	DC	A(RE5+48)	address of v4 source
000013E0	0000143C			731+	DC	A(16)	result length
000013E8	00000000 00000000			732+REA5	DC	A(RE5)	result address
000013F0	00000000 00000000			733+	DS	FD	gap
000013F8	00000000 00000000			734+V105	DS	XL16	V1 output
00001400	00000000 00000000						
				735+	DS	FD	gap
				736+*			
00001408				737+X5	DS	0F	
00001408	E310 5010 0014		00000010	738+	LGF	R1,V2ADDR	load v2 source
0000140E	E761 0000 0806		00000000	739+	VL	v22,0(R1)	use v22 to test decoder
00001414	E310 5014 0014		00000014	740+	LGF	R1,V3ADDR	load v3 source
0000141A	E771 0000 0806		00000000	741+	VL	v23,0(R1)	use v23 to test decoder
00001420	E310 5018 0014		00000018	742+	LGF	R1,V4ADDR	load v4 source
00001426	E781 0000 0806		00000000	743+	VL	v24,0(R1)	use v24 to test decoder
0000142C	E766 7000 8FA9			744+	VMALH	V22,V22,V23,V24,0	test instruction (dest is a source)
00001432	E760 5030 080E		000013F0	745+	VST	V22,V105	save v1 output
00001438	07FB			746+	BR	R11	return
0000143C				747+RE5	DC	0F	xl16 expected result
0000143C				748+	DROP	R5	
0000143C	FF000000 00000000			749	DC	XL16'FF00000000000000 0000000000000000'	result
00001444	00000000 00000000						
0000144C	FF020304 05060708			750	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001454	090A0B0C 0D0E0F10						
0000145C	FF000000 00000001			751	DC	XL16'FF000000000000001 0101010101010102'	v3
00001464	01010101 01010102						
0000146C	FF020304 05060708			752	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001474	090A0B0C 0D0E0F10						
				753			
				754 * Halfword			
				755	VRR_D	VMALH,1	
00001480				756+	DS	0FD	
00001480		00001480		757+	USING	*,R5	base for test data and test routine
00001480	000014C8			758+T6	DC	A(X6)	address of test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001484	0006			759+	DC	H'6'	test number
00001486	00			760+	DC	X'00'	
00001487	01			761+	DC	HL1'1'	m5
00001488	E5D4C1D3 C8404040			762+	DC	CL8'VMALH'	instruction name
00001490	0000150C			763+	DC	A(RE6+16)	address of v2 source
00001494	0000151C			764+	DC	A(RE6+32)	address of v3 source
00001498	0000152C			765+	DC	A(RE6+48)	address of v4 source
0000149C	00000010			766+	DC	A(16)	result length
000014A0	000014FC			767+REA6	DC	A(RE6)	result address
000014A8	00000000 00000000			768+	DS	FD	gap
000014B0	00000000 00000000			769+V106	DS	XL16	V1 output
000014B8	00000000 00000000						
000014C0	00000000 00000000			770+	DS	FD	gap
				771+*			
000014C8				772+X6	DS	0F	
000014C8	E310 5010 0014		00000010	773+	LGF	R1,V2ADDR	load v2 source
000014CE	E761 0000 0806		00000000	774+	VL	v22,0(R1)	use v22 to test decoder
000014D4	E310 5014 0014		00000014	775+	LGF	R1,V3ADDR	load v3 source
000014DA	E771 0000 0806		00000000	776+	VL	v23,0(R1)	use v23 to test decoder
000014E0	E310 5018 0014		00000018	777+	LGF	R1,V4ADDR	load v4 source
000014E6	E781 0000 0806		00000000	778+	VL	v24,0(R1)	use v24 to test decoder
000014EC	E766 7100 8FA9			779+	VMALH	V22,V22,V23,V24,1	test instruction (dest is a source)
000014F2	E760 5030 080E		000014B0	780+	VST	V22,V106	save v1 output
000014F8	07FB			781+	BR	R11	return
000014FC				782+RE6	DC	0F	xl16 expected result
000014FC				783+	DROP	R5	
000014FC	FE010000 00000000			784	DC	XL16'FE01000000000000 0000000000000000'	result
00001504	00000000 00000000						
0000150C	FF000000 00000019			785	DC	XL16'FF00000000000019 00000038000000FA'	v2
00001514	00000038 000000FA						
0000151C	FF000000 00000019			786	DC	XL16'FF00000000000019 00000038000000FA'	v3
00001524	00000038 000000FA						
0000152C	00000000 00000000			787	DC	XL16'0000000000000000 0000000000000000'	v4
00001534	00000000 00000000						
				788			
00001540				789	VRR_D	VMALH,1	
00001540		00001540		790+	DS	0FD	
00001540	00001588			791+	USING	*,R5	base for test data and test routine
00001544	0007			792+T7	DC	A(X7)	address of test routine
00001546	00			793+	DC	H'7'	test number
00001547	01			794+	DC	X'00'	
00001548	E5D4C1D3 C8404040			795+	DC	HL1'1'	m5
00001550	000015CC			796+	DC	CL8'VMALH'	instruction name
00001554	000015DC			797+	DC	A(RE7+16)	address of v2 source
00001558	000015EC			798+	DC	A(RE7+32)	address of v3 source
0000155C	00000010			799+	DC	A(RE7+48)	address of v4 source
00001560	000015BC			800+	DC	A(16)	result length
00001568	00000000 00000000			801+REA7	DC	A(RE7)	result address
00001570	00000000 00000000			802+	DS	FD	gap
00001578	00000000 00000000			803+V107	DS	XL16	V1 output
00001580	00000000 00000000			804+	DS	FD	gap
				805+*			
00001588				806+X7	DS	0F	
00001588	E310 5010 0014		00000010	807+	LGF	R1,V2ADDR	load v2 source
0000158E	E761 0000 0806		00000000	808+	VL	v22,0(R1)	use v22 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001594	E310 5014 0014		00000014	809+	LGF	R1,V3ADDR	load v3 source
0000159A	E771 0000 0806		00000000	810+	VL	v23,0(R1)	use v23 to test decoder
000015A0	E310 5018 0014		00000018	811+	LGF	R1,V4ADDR	load v4 source
000015A6	E781 0000 0806		00000000	812+	VL	v24,0(R1)	use v24 to test decoder
000015AC	E766 7100 8FA9			813+	VMALH	V22,V22,V23,V24,1	test instruction (dest is a source)
000015B2	E760 5030 080E		00001570	814+	VST	V22,V107	save v1 output
000015B8	07FB			815+	BR	R11	return
000015BC				816+RE7	DC	0F	xl16 expected result
000015BC				817+	DROP	R5	
000015BC	FE010000 00000000			818	DC	XL16'FE01000000000000 0000000000000000'	result
000015C4	00000000 00000000						
000015CC	FF0000FF 00000029			819	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
000015D4	00000038 000000FA						
000015DC	FF000001 00000029			820	DC	XL16'FF00000100000029 00000038000000FA'	v3
000015E4	00000038 000000FA						
000015EC	00000001 0000002F			821	DC	XL16'000000010000002F 0000000000000002'	v4
000015F4	00000000 00000002						
				822			
				823	VRR_D	VMALH,1	
00001600				824+	DS	0FD	
00001600		00001600		825+	USING	*,R5	base for test data and test routine
00001600	00001648			826+T8	DC	A(X8)	address of test routine
00001604	0008			827+	DC	H'8'	test number
00001606	00			828+	DC	X'00'	
00001607	01			829+	DC	HL1'1'	m5
00001608	E5D4C1D3 C8404040			830+	DC	CL8'VMALH'	instruction name
00001610	0000168C			831+	DC	A(RE8+16)	address of v2 source
00001614	0000169C			832+	DC	A(RE8+32)	address of v3 source
00001618	000016AC			833+	DC	A(RE8+48)	address of v4 source
0000161C	00000010			834+	DC	A(16)	result length
00001620	0000167C			835+REA8	DC	A(RE8)	result address
00001628	00000000 00000000			836+	DS	FD	gap
00001630	00000000 00000000			837+V108	DS	XL16	V1 output
00001638	00000000 00000000						
00001640	00000000 00000000			838+	DS	FD	gap
				839+*			
00001648				840+X8	DS	0F	
00001648	E310 5010 0014		00000010	841+	LGF	R1,V2ADDR	load v2 source
0000164E	E761 0000 0806		00000000	842+	VL	v22,0(R1)	use v22 to test decoder
00001654	E310 5014 0014		00000014	843+	LGF	R1,V3ADDR	load v3 source
0000165A	E771 0000 0806		00000000	844+	VL	v23,0(R1)	use v23 to test decoder
00001660	E310 5018 0014		00000018	845+	LGF	R1,V4ADDR	load v4 source
00001666	E781 0000 0806		00000000	846+	VL	v24,0(R1)	use v24 to test decoder
0000166C	E766 7100 8FA9			847+	VMALH	V22,V22,V23,V24,1	test instruction (dest is a source)
00001672	E760 5030 080E		00001630	848+	VST	V22,V108	save v1 output
00001678	07FB			849+	BR	R11	return
0000167C				850+RE8	DC	0F	xl16 expected result
0000167C				851+	DROP	R5	
0000167C	FE050009 00190031			852	DC	XL16'FE05000900190031 0051007A00AA00E2'	result
00001684	0051007A 00AA00E2						
0000168C	FF020304 05060708			853	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001694	090A0B0C 0D0E0F10						
0000169C	FF020304 05060708			854	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000016A4	090A0B0C 0D0E0F10						
000016AC	FF020304 05060708			855	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000016B4	090A0B0C 0D0E0F10						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				856		
				857	VRR_D VMALH,1	
000016C0				858+	DS 0FD	
000016C0		000016C0		859+	USING *,R5	base for test data and test routine
000016C0	00001708			860+T9	DC A(X9)	address of test routine
000016C4	0009			861+	DC H'9'	test number
000016C6	00			862+	DC X'00'	
000016C7	01			863+	DC HL1'1'	m5
000016C8	E5D4C1D3 C8404040			864+	DC CL8'VMALH'	instruction name
000016D0	0000174C			865+	DC A(RE9+16)	address of v2 source
000016D4	0000175C			866+	DC A(RE9+32)	address of v3 source
000016D8	0000176C			867+	DC A(RE9+48)	address of v4 source
000016DC	00000010			868+	DC A(16)	result length
000016E0	0000173C			869+REA9	DC A(RE9)	result address
000016E8	00000000 00000000			870+	DS FD	gap
000016F0	00000000 00000000			871+V109	DS XL16	V1 output
000016F8	00000000 00000000					
00001700	00000000 00000000			872+	DS FD	gap
				873+*		
00001708				874+X9	DS 0F	
00001708	E310 5010 0014		00000010	875+	LGF R1,V2ADDR	load v2 source
0000170E	E761 0000 0806		00000000	876+	VL v22,0(R1)	use v22 to test decoder
00001714	E310 5014 0014		00000014	877+	LGF R1,V3ADDR	load v3 source
0000171A	E771 0000 0806		00000000	878+	VL v23,0(R1)	use v23 to test decoder
00001720	E310 5018 0014		00000018	879+	LGF R1,V4ADDR	load v4 source
00001726	E781 0000 0806		00000000	880+	VL v24,0(R1)	use v24 to test decoder
0000172C	E766 7100 8FA9			881+	VMALH V22,V22,V23,V24,1	test instruction (dest is a source)
00001732	E760 5030 080E		000016F0	882+	VST V22,V109	save v1 output
00001738	07FB			883+	BR R11	return
0000173C				884+RE9	DC 0F	xl16 expected result
0000173C				885+	DROP R5	
0000173C	FE040003 000A0015			886	DC XL16'FE040003000A0015 00240037004E0069'	result
00001744	00240037 004E0069					
0000174C	FF020304 05060708			887	DC XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001754	090A0B0C 0D0E0F10					
0000175C	FF010102 02030304			888	DC XL16'FF01010202030304 0405050606070708'	v3
00001764	04050506 06070708					
0000176C	FF020304 05060708			889	DC XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001774	090A0B0C 0D0E0F10					
				890		
				891	VRR_D VMALH,1	
00001780				892+	DS 0FD	
00001780		00001780		893+	USING *,R5	base for test data and test routine
00001780	000017C8			894+T10	DC A(X10)	address of test routine
00001784	000A			895+	DC H'10'	test number
00001786	00			896+	DC X'00'	
00001787	01			897+	DC HL1'1'	m5
00001788	E5D4C1D3 C8404040			898+	DC CL8'VMALH'	instruction name
00001790	0000180C			899+	DC A(RE10+16)	address of v2 source
00001794	0000181C			900+	DC A(RE10+32)	address of v3 source
00001798	0000182C			901+	DC A(RE10+48)	address of v4 source
0000179C	00000010			902+	DC A(16)	result length
000017A0	000017FC			903+REA10	DC A(RE10)	result address
000017A8	00000000 00000000			904+	DS FD	gap
000017B0	00000000 00000000			905+V1010	DS XL16	V1 output
000017B8	00000000 00000000					

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000017C0	00000000 00000000			906+ 907+*	DS	FD	gap
000017C8				908+X10	DS	0F	
000017C8	E310 5010 0014		00000010	909+	LGF	R1,V2ADDR	load v2 source
000017CE	E761 0000 0806		00000000	910+	VL	v22,0(R1)	use v22 to test decoder
000017D4	E310 5014 0014		00000014	911+	LGF	R1,V3ADDR	load v3 source
000017DA	E771 0000 0806		00000000	912+	VL	v23,0(R1)	use v23 to test decoder
000017E0	E310 5018 0014		00000018	913+	LGF	R1,V4ADDR	load v4 source
000017E6	E781 0000 0806		00000000	914+	VL	v24,0(R1)	use v24 to test decoder
000017EC	E766 7100 8FA9			915+	VMALH	V22,V22,V23,V24,1	test instruction (dest is a source)
000017F2	E760 5030 080E		000017B0	916+	VST	V22,V1010	save v1 output
000017F8	07FB			917+	BR	R11	return
000017FC				918+RE10	DC	0F	xl16 expected result
000017FC				919+	DROP	R5	
000017FC	FE030000 00000000			920	DC	XL16'FE03000000000000 0009000B000D000F'	result
00001804	0009000B 000D000F						
0000180C	FF020304 05060708			921	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001814	090A0B0C 0D0E0F10						
0000181C	FF000000 00000001			922	DC	XL16'FF00000000000001 0101010101010102'	v3
00001824	01010101 01010102						
0000182C	FF020304 05060708			923	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001834	090A0B0C 0D0E0F10						
				924			
				925 * Word			
				926	VRR_D	VMALH,2	
00001840				927+	DS	0FD	
00001840		00001840		928+	USING	*,R5	base for test data and test routine
00001840	00001888			929+T11	DC	A(X11)	address of test routine
00001844	000B			930+	DC	H'11'	test number
00001846	00			931+	DC	X'00'	
00001847	02			932+	DC	HL1'2'	m5
00001848	E5D4C1D3 C8404040			933+	DC	CL8'VMALH'	instruction name
00001850	000018CC			934+	DC	A(RE11+16)	address of v2 source
00001854	000018DC			935+	DC	A(RE11+32)	address of v3 source
00001858	000018EC			936+	DC	A(RE11+48)	address of v4 source
0000185C	00000010			937+	DC	A(16)	result length
00001860	000018BC			938+REA11	DC	A(RE11)	result address
00001868	00000000 00000000			939+	DS	FD	gap
00001870	00000000 00000000			940+V1011	DS	XL16	V1 output
00001878	00000000 00000000						
00001880	00000000 00000000			941+	DS	FD	gap
				942+*			
00001888				943+X11	DS	0F	
00001888	E310 5010 0014		00000010	944+	LGF	R1,V2ADDR	load v2 source
0000188E	E761 0000 0806		00000000	945+	VL	v22,0(R1)	use v22 to test decoder
00001894	E310 5014 0014		00000014	946+	LGF	R1,V3ADDR	load v3 source
0000189A	E771 0000 0806		00000000	947+	VL	v23,0(R1)	use v23 to test decoder
000018A0	E310 5018 0014		00000018	948+	LGF	R1,V4ADDR	load v4 source
000018A6	E781 0000 0806		00000000	949+	VL	v24,0(R1)	use v24 to test decoder
000018AC	E766 7200 8FA9			950+	VMALH	V22,V22,V23,V24,2	test instruction (dest is a source)
000018B2	E760 5030 080E		00001870	951+	VST	V22,V1011	save v1 output
000018B8	07FB			952+	BR	R11	return
000018BC				953+RE11	DC	0F	xl16 expected result
000018BC				954+	DROP	R5	
000018BC	FE010000 00000000			955	DC	XL16'FE01000000000000 0000000000000000'	result
000018C4	00000000 00000000						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000018CC	FF000000 00000019			956	DC	XL16'FF0000000000000019 000000380000000FA'	v2
000018D4	00000038 000000FA						
000018DC	FF000000 00000019			957	DC	XL16'FF0000000000000019 000000380000000FA'	v3
000018E4	00000038 000000FA						
000018EC	00000000 00000000			958	DC	XL16'0000000000000000 0000000000000000'	v4
000018F4	00000000 00000000						
				959			
				960	VRR_D	VMALH, 2	
00001900				961+	DS	0FD	
00001900		00001900		962+	USING	*, R5	base for test data and test routine
00001900	00001948			963+T12	DC	A(X12)	address of test routine
00001904	000C			964+	DC	H'12'	test number
00001906	00			965+	DC	X'00'	
00001907	02			966+	DC	HL1'2'	m5
00001908	E5D4C1D3 C8404040			967+	DC	CL8'VMALH'	instruction name
00001910	0000198C			968+	DC	A(RE12+16)	address of v2 source
00001914	0000199C			969+	DC	A(RE12+32)	address of v3 source
00001918	000019AC			970+	DC	A(RE12+48)	address of v4 source
0000191C	00000010			971+	DC	A(16)	result length
00001920	0000197C			972+REA12	DC	A(RE12)	result address
00001928	00000000 00000000			973+	DS	FD	gap
00001930	00000000 00000000			974+V1012	DS	XL16	V1 output
00001938	00000000 00000000						
00001940	00000000 00000000			975+	DS	FD	gap
				976+*			
00001948				977+X12	DS	0F	
00001948	E310 5010 0014		00000010	978+	LGF	R1, V2ADDR	load v2 source
0000194E	E761 0000 0806		00000000	979+	VL	v22, 0(R1)	use v22 to test decoder
00001954	E310 5014 0014		00000014	980+	LGF	R1, V3ADDR	load v3 source
0000195A	E771 0000 0806		00000000	981+	VL	v23, 0(R1)	use v23 to test decoder
00001960	E310 5018 0014		00000018	982+	LGF	R1, V4ADDR	load v4 source
00001966	E781 0000 0806		00000000	983+	VL	v24, 0(R1)	use v24 to test decoder
0000196C	E766 7200 8FA9			984+	VMALH	V22, V22, V23, V24, 2	test instruction (dest is a source)
00001972	E760 5030 080E		00001930	985+	VST	V22, V1012	save v1 output
00001978	07FB			986+	BR	R11	return
0000197C				987+RE12	DC	0F	xl16 expected result
0000197C				988+	DROP	R5	
0000197C	FE0100FF 00000000			989	DC	XL16'FE0100FF00000000 0000000000000000'	result
00001984	00000000 00000000						
0000198C	FF0000FF 00000029			990	DC	XL16'FF0000FF00000029 000000380000000FA'	v2
00001994	00000038 000000FA						
0000199C	FF000001 00000029			991	DC	XL16'FF00000100000029 000000380000000FA'	v3
000019A4	00000038 000000FA						
000019AC	00000001 0000002F			992	DC	XL16'000000010000002F 00000000000000002'	v4
000019B4	00000000 00000002						
				993			
				994	VRR_D	VMALH, 2	
000019C0				995+	DS	0FD	
000019C0		000019C0		996+	USING	*, R5	base for test data and test routine
000019C0	00001A08			997+T13	DC	A(X13)	address of test routine
000019C4	000D			998+	DC	H'13'	test number
000019C6	00			999+	DC	X'00'	
000019C7	02			1000+	DC	HL1'2'	m5
000019C8	E5D4C1D3 C8404040			1001+	DC	CL8'VMALH'	instruction name
000019D0	00001A4C			1002+	DC	A(RE13+16)	address of v2 source
000019D4	00001A5C			1003+	DC	A(RE13+32)	address of v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000019D8	00001A6C			1004+	DC	A(RE13+48)	address of v4 source
000019DC	00000010			1005+	DC	A(16)	result length
000019E0	00001A3C			1006+REA13	DC	A(RE13)	result address
000019E8	00000000 00000000			1007+	DS	FD	gap
000019F0	00000000 00000000			1008+V1013	DS	XL16	V1 output
000019F8	00000000 00000000						
00001A00	00000000 00000000			1009+	DS	FD	gap
				1010+*			
00001A08				1011+X13	DS	0F	
00001A08	E310 5010 0014		00000010	1012+	LGF	R1,V2ADDR	load v2 source
00001A0E	E761 0000 0806		00000000	1013+	VL	v22,0(R1)	use v22 to test decoder
00001A14	E310 5014 0014		00000014	1014+	LGF	R1,V3ADDR	load v3 source
00001A1A	E771 0000 0806		00000000	1015+	VL	v23,0(R1)	use v23 to test decoder
00001A20	E310 5018 0014		00000018	1016+	LGF	R1,V4ADDR	load v4 source
00001A26	E781 0000 0806		00000000	1017+	VL	v24,0(R1)	use v24 to test decoder
00001A2C	E766 7200 8FA9			1018+	VMALH	V22,V22,V23,V24,2	test instruction (dest is a source)
00001A32	E760 5030 080E		000019F0	1019+	VST	V22,V1013	save v1 output
00001A38	07FB			1020+	BR	R11	return
00001A3C				1021+RE13	DC	0F	xl16 expected result
00001A3C				1022+	DROP	R5	
00001A3C	FE050207 00193C6A			1023	DC	XL16'FE05020700193C6A 0051B52B00AA6E4D'	result
00001A44	0051B52B 00AA6E4D						
00001A4C	FF020304 05060708			1024	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001A54	090A0B0C 0D0E0F10						
00001A5C	FF020304 05060708			1025	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00001A64	090A0B0C 0D0E0F10						
00001A6C	FF020304 05060708			1026	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001A74	090A0B0C 0D0E0F10						
				1027			
00001A80				1028	VRR_D	VMALH,2	
00001A80		00001A80		1029+	DS	0FD	
00001A80	00001AC8			1030+	USING	*,R5	base for test data and test routine
00001A84	000E			1031+T14	DC	A(X14)	address of test routine
00001A86	00			1032+	DC	H'14'	test number
00001A87	02			1033+	DC	X'00'	
00001A88	E5D4C1D3 C8404040			1034+	DC	HL1'2'	m5
00001A88				1035+	DC	CL8'VMALH'	instruction name
00001A90	00001B0C			1036+	DC	A(RE14+16)	address of v2 source
00001A94	00001B1C			1037+	DC	A(RE14+32)	address of v3 source
00001A98	00001B2C			1038+	DC	A(RE14+48)	address of v4 source
00001A9C	00000010			1039+	DC	A(16)	result length
00001AA0	00001AFC			1040+REA14	DC	A(RE14)	result address
00001AA8	00000000 00000000			1041+	DS	FD	gap
00001AB0	00000000 00000000			1042+V1014	DS	XL16	V1 output
00001AB8	00000000 00000000						
00001AC0	00000000 00000000			1043+	DS	FD	gap
				1044+*			
00001AC8				1045+X14	DS	0F	
00001AC8	E310 5010 0014		00000010	1046+	LGF	R1,V2ADDR	load v2 source
00001ACE	E761 0000 0806		00000000	1047+	VL	v22,0(R1)	use v22 to test decoder
00001AD4	E310 5014 0014		00000014	1048+	LGF	R1,V3ADDR	load v3 source
00001ADA	E771 0000 0806		00000000	1049+	VL	v23,0(R1)	use v23 to test decoder
00001AE0	E310 5018 0014		00000018	1050+	LGF	R1,V4ADDR	load v4 source
00001AE6	E781 0000 0806		00000000	1051+	VL	v24,0(R1)	use v24 to test decoder
00001AEC	E766 7200 8FA9			1052+	VMALH	V22,V22,V23,V24,2	test instruction (dest is a source)
00001AF2	E760 5030 080E		00001AB0	1053+	VST	V22,V1014	save v1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001AF8	07FB			1054+	BR	R11	return
00001AFC				1055+RE14	DC	0F	xl16 expected result
00001AFC				1056+	DROP	R5	
00001AFC	FE040104 000A1B2F			1057	DC	XL16'FE040104000A1B2F 0024558B004EB018'	result
00001B04	0024558B 004EB018						
00001B0C	FF020304 05060708			1058	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001B14	090A0B0C 0D0E0F10						
00001B1C	FF010102 02030304			1059	DC	XL16'FF01010202030304 0405050606070708'	v3
00001B24	04050506 06070708						
00001B2C	FF020304 05060708			1060	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001B34	090A0B0C 0D0E0F10						
				1061			
00001B40				1062	VRR_D	VMALH, 2	
00001B40		00001B40		1063+	DS	0FD	
00001B40	00001B88			1064+	USING	*, R5	base for test data and test routine
00001B44	000F			1065+T15	DC	A(X15)	address of test routine
00001B46	00			1066+	DC	H'15'	test number
00001B47	02			1067+	DC	X'00'	
00001B48	E5D4C1D3 C8404040			1068+	DC	HL1'2'	m5
00001B50	00001BCC			1069+	DC	CL8'VMALH'	instruction name
00001B54	00001BDC			1070+	DC	A(RE15+16)	address of v2 source
00001B58	00001BEC			1071+	DC	A(RE15+32)	address of v3 source
00001B5C	00000010			1072+	DC	A(RE15+48)	address of v4 source
00001B60	00001BBC			1073+	DC	A(16)	result length
00001B68	00000000 00000000			1074+REA15	DC	A(RE15)	result address
00001B70	00000000 00000000			1075+	DS	FD	gap
00001B78	00000000 00000000			1076+V1015	DS	XL16	V1 output
00001B80	00000000 00000000						
				1077+	DS	FD	gap
00001B88				1078+*			
00001B88	E310 5010 0014		00000010	1079+X15	DS	0F	
00001B8E	E761 0000 0806		00000000	1080+	LGF	R1, V2ADDR	load v2 source
00001B94	E310 5014 0014		00000014	1081+	VL	v22, 0(R1)	use v22 to test decoder
00001B9A	E771 0000 0806		00000000	1082+	LGF	R1, V3ADDR	load v3 source
00001BA0	E310 5018 0014		00000018	1083+	VL	v23, 0(R1)	use v23 to test decoder
00001BA6	E781 0000 0806		00000000	1084+	LGF	R1, V4ADDR	load v4 source
00001BAC	E766 7200 8FA9		00000000	1085+	VL	v24, 0(R1)	use v24 to test decoder
00001BB2	E760 5030 080E		00001B70	1086+	VMALH	V22, V22, V23, V24, 2	test instruction (dest is a source)
00001BB8	07FB			1087+	VST	V22, V1015	save v1 output
00001BBC				1088+	BR	R11	return
00001BBC				1089+RE15	DC	0F	xl16 expected result
00001BBC				1090+	DROP	R5	
00001BBC	FE030101 00000000			1091	DC	XL16'FE03010100000000 0009131E000D1B2A'	result
00001BC4	0009131E 000D1B2A						
00001BCC	FF020304 05060708			1092	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00001BD4	090A0B0C 0D0E0F10						
00001BDC	FF000000 00000001			1093	DC	XL16'FF00000000000001 0101010101010102'	v3
00001BE4	01010101 01010102						
00001BEC	FF020304 05060708			1094	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00001BF4	090A0B0C 0D0E0F10						
				1095			
				1096 * Doubleword			
				1097	VRR_D	VMALH, 3	
00001C00				1098+	DS	0FD	
00001C00		00001C00		1099+	USING	*, R5	base for test data and test routine
00001C00	00001C48			1100+T16	DC	A(X16)	address of test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00001C04	0010			1101+	DC	H'16'	test number
00001C06	00			1102+	DC	X'00'	
00001C07	03			1103+	DC	HL1'3'	m5
00001C08	E5D4C1D3 C8404040			1104+	DC	CL8'VMALH'	instruction name
00001C10	00001C8C			1105+	DC	A(RE16+16)	address of v2 source
00001C14	00001C9C			1106+	DC	A(RE16+32)	address of v3 source
00001C18	00001CAC			1107+	DC	A(RE16+48)	address of v4 source
00001C1C	00000010			1108+	DC	A(16)	result length
00001C20	00001C7C			1109+REA16	DC	A(RE16)	result address
00001C28	00000000 00000000			1110+	DS	FD	gap
00001C30	00000000 00000000			1111+V1016	DS	XL16	V1 output
00001C38	00000000 00000000						
00001C40	00000000 00000000			1112+	DS	FD	gap
				1113+*			
00001C48				1114+X16	DS	0F	
00001C48	E310 5010 0014		00000010	1115+	LGF	R1,V2ADDR	load v2 source
00001C4E	E761 0000 0806		00000000	1116+	VL	v22,0(R1)	use v22 to test decoder
00001C54	E310 5014 0014		00000014	1117+	LGF	R1,V3ADDR	load v3 source
00001C5A	E771 0000 0806		00000000	1118+	VL	v23,0(R1)	use v23 to test decoder
00001C60	E310 5018 0014		00000018	1119+	LGF	R1,V4ADDR	load v4 source
00001C66	E781 0000 0806		00000000	1120+	VL	v24,0(R1)	use v24 to test decoder
00001C6C	E766 7300 8FA9			1121+	VMALH	V22,V22,V23,V24,3	test instruction (dest is a source)
00001C72	E760 5030 080E		00001C30	1122+	VST	V22,V1016	save v1 output
00001C78	07FB			1123+	BR	R11	return
00001C7C				1124+RE16	DC	0F	xl16 expected result
00001C7C				1125+	DROP	R5	
00001C7C	FFFFFFFFE 00032000			1126	DC	XL16'FFFFFFFFE00032000 00000000000000C77'	result
00001C84	00000000 00000C77						
00001C8C	FFFFFFFF 00019000			1127	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00001C94	00000038 EEEEEEEFA						
00001C9C	FFFFFFFF 00019000			1128	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
00001CA4	00000038 0EEEEEEFA						
00001CAC	00000000 00000000			1129	DC	XL16'0000000000000000 0000000000000000'	v4
00001CB4	00000000 00000000						
				1130			
00001CC0				1131	VRR_D	VMALH,3	
00001CC0		00001CC0		1132+	DS	0FD	
00001CC0	00001D08			1133+	USING	*,R5	base for test data and test routine
00001CC4	0011			1134+T17	DC	A(X17)	address of test routine
00001CC6	00			1135+	DC	H'17'	test number
00001CC7	03			1136+	DC	X'00'	
00001CC8	E5D4C1D3 C8404040			1137+	DC	HL1'3'	m5
00001CD0	00001D4C			1138+	DC	CL8'VMALH'	instruction name
00001CD4	00001D5C			1139+	DC	A(RE17+16)	address of v2 source
00001CD8	00001D6C			1140+	DC	A(RE17+32)	address of v3 source
00001CDC	00000010			1141+	DC	A(RE17+48)	address of v4 source
00001CE0	00001D3C			1142+	DC	A(16)	result length
00001CE8	00000000 00000000			1143+REA17	DC	A(RE17)	result address
00001CF0	00000000 00000000			1144+	DS	FD	gap
00001CF8	00000000 00000000			1145+V1017	DS	XL16	V1 output
00001D00	00000000 00000000						
				1146+	DS	FD	gap
				1147+*			
00001D08				1148+X17	DS	0F	
00001D08	E310 5010 0014		00000010	1149+	LGF	R1,V2ADDR	load v2 source
00001D0E	E761 0000 0806		00000000	1150+	VL	v22,0(R1)	use v22 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00001D14	E310 5014 0014		00000014	1151+	LGF	R1,V3ADDR	load v3 source	
00001D1A	E771 0000 0806		00000000	1152+	VL	v23,0(R1)	use v23 to test decoder	
00001D20	E310 5018 0014		00000018	1153+	LGF	R1,V4ADDR	load v4 source	
00001D26	E781 0000 0806		00000000	1154+	VL	v24,0(R1)	use v24 to test decoder	
00001D2C	E766 7300 8FA9			1155+	VMALH	V22,V22,V23,V24,3	test instruction (dest is a source)	
00001D32	E760 5030 080E		00001CF0	1156+	VST	V22,V1017	save v1 output	
00001D38	07FB			1157+	BR	R11	return	
00001D3C				1158+RE17	DC	0F	xl16 expected result	
00001D3C				1159+	DROP	R5		
00001D3C	01010308 111F3397			1160	DC	XL16'01010308111F3397 0051B52F8692B4F6'	result	
00001D44	0051B52F 8692B4F6							
00001D4C	FF020304 05060750			1161	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
00001D54	090A0B0C 0D0E0F7F							
00001D5C	01020304 05060750			1162	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3	
00001D64	090A0B78 0D0E0F7F							
00001D6C	FF000000 00000000			1163	DC	XL16'FF00000000000000 2000000000000000'	v4	
00001D74	20000000 00000000							
00001D80				1164				
00001D80		00001D80		1165	VRR_D	VMALH,3		
00001D80	00001DC8			1166+	DS	0FD		
00001D84	0012			1167+	USING	*,R5	base for test data and test routine	
00001D86	00			1168+T18	DC	A(X18)	address of test routine	
00001D87	03			1169+	DC	H'18'	test number	
00001D88	E5D4C1D3 C8404040			1170+	DC	X'00'		
00001D90	00001E0C			1171+	DC	HL1'3'	m5	
00001D94	00001E1C			1172+	DC	CL8'VMALH'	instruction name	
00001D98	00001E2C			1173+	DC	A(RE18+16)	address of v2 source	
00001D9C	00000010			1174+	DC	A(RE18+32)	address of v3 source	
00001DA0	00001DFC			1175+	DC	A(RE18+48)	address of v4 source	
00001DA8	00000000 00000000			1176+	DC	A(16)	result length	
00001DB0	00000000 00000000			1177+REA18	DC	A(RE18)	result address	
00001DB8	00000000 00000000			1178+	DS	FD	gap	
00001DC0	00000000 00000000			1179+V1018	DS	XL16	V1 output	
00001DC8				1180+	DS	FD	gap	
00001DC8				1181+*				
00001DC8	E310 5010 0014		00000010	1182+X18	DS	0F		
00001DCE	E761 0000 0806		00000000	1183+	LGF	R1,V2ADDR	load v2 source	
00001DD4	E310 5014 0014		00000014	1184+	VL	v22,0(R1)	use v22 to test decoder	
00001DDA	E771 0000 0806		00000000	1185+	LGF	R1,V3ADDR	load v3 source	
00001DE0	E310 5018 0014		00000018	1186+	VL	v23,0(R1)	use v23 to test decoder	
00001DE6	E781 0000 0806		00000000	1187+	LGF	R1,V4ADDR	load v4 source	
00001DEC	E766 7300 8FA9			1188+	VL	v24,0(R1)	use v24 to test decoder	
00001DF2	E760 5030 080E		00001DB0	1189+	VMALH	V22,V22,V23,V24,3	test instruction (dest is a source)	
00001DF8	07FB			1190+	VST	V22,V1018	save v1 output	
00001DFC				1191+	BR	R11	return	
00001DFC				1192+RE18	DC	0F	xl16 expected result	
00001DFC				1193+	DROP	R5		
00001DFC	00010003 050C1344			1194	DC	XL16'00010003050C1344 0024558DB838C863'	result	
00001E04	0024558D B838C863							
00001E0C	FF020304 05060750			1195	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
00001E14	090A0B0C 0D0E0F7F							
00001E1C	00010102 02030328			1196	DC	XL16'0001010202030328 0405053C0607073F'	v3	
00001E24	0405053C 0607073F							
00001E2C	7FFFFFFFF FFFFFFFF			1197	DC	XL16'7FFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF'	v4	
00001E34	FFFFFFFF FFFFFFFF							

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				1198		
				1199	VRR_D VMALH, 3	
00001E40				1200+	DS 0FD	
00001E40		00001E40		1201+	USING *, R5	base for test data and test routine
00001E40	00001E88			1202+T19	DC A(X19)	address of test routine
00001E44	0013			1203+	DC H'19'	test number
00001E46	00			1204+	DC X'00'	
00001E47	03			1205+	DC HL1'3'	m5
00001E48	E5D4C1D3 C8404040			1206+	DC CL8'VMALH'	instruction name
00001E50	00001ECC			1207+	DC A(RE19+16)	address of v2 source
00001E54	00001EDC			1208+	DC A(RE19+32)	address of v3 source
00001E58	00001EEC			1209+	DC A(RE19+48)	address of v4 source
00001E5C	00000010			1210+	DC A(16)	result length
00001E60	00001EBC			1211+REA19	DC A(RE19)	result address
00001E68	00000000 00000000			1212+	DS FD	gap
00001E70	00000000 00000000			1213+V1019	DS XL16	V1 output
00001E78	00000000 00000000					
00001E80	00000000 00000000			1214+	DS FD	gap
				1215+*		
00001E88				1216+X19	DS 0F	
00001E88	E310 5010 0014		00000010	1217+	LGF R1, V2ADDR	load v2 source
00001E8E	E761 0000 0806		00000000	1218+	VL v22, 0(R1)	use v22 to test decoder
00001E94	E310 5014 0014		00000014	1219+	LGF R1, V3ADDR	load v3 source
00001E9A	E771 0000 0806		00000000	1220+	VL v23, 0(R1)	use v23 to test decoder
00001EA0	E310 5018 0014		00000018	1221+	LGF R1, V4ADDR	load v4 source
00001EA6	E781 0000 0806		00000000	1222+	VL v24, 0(R1)	use v24 to test decoder
00001EAC	E766 7300 8FA9			1223+	VMALH V22, V22, V23, V24, 3	test instruction (dest is a source)
00001EB2	E760 5030 080E		00001E70	1224+	VST V22, V1019	save v1 output
00001EB8	07FB			1225+	BR R11	return
00001EBC				1226+RE19	DC 0F	xl16 expected result
00001EBC				1227+	DROP R5	
00001EBC	00000000 0000000A			1228	DC XL16'000000000000000A 0009131EA8C3DFFE'	result
00001EC4	0009131E A8C3DFFE					
00001ECC	FF020304 05060750			1229	DC XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00001ED4	090A0B0C 0D0E0F7F					
00001EDC	00000000 0000000A			1230	DC XL16'000000000000000A 0101010F0101010F'	v3
00001EE4	0101010F 0101010F					
00001EEC	FFFFFFFF FFFFFFFF			1231	DC XL16'FFFFFFFFFFFFFFFF 7FFFFFFFFFFFFFFFFF'	v4
00001EF4	7FFFFFFFF FFFFFFFF					
				1232		
				1233 * Quadword		
				1234	VRR_D VMALH, 4	
00001F00				1235+	DS 0FD	
00001F00		00001F00		1236+	USING *, R5	base for test data and test routine
00001F00	00001F48			1237+T20	DC A(X20)	address of test routine
00001F04	0014			1238+	DC H'20'	test number
00001F06	00			1239+	DC X'00'	
00001F07	04			1240+	DC HL1'4'	m5
00001F08	E5D4C1D3 C8404040			1241+	DC CL8'VMALH'	instruction name
00001F10	00001F8C			1242+	DC A(RE20+16)	address of v2 source
00001F14	00001F9C			1243+	DC A(RE20+32)	address of v3 source
00001F18	00001FAC			1244+	DC A(RE20+48)	address of v4 source
00001F1C	00000010			1245+	DC A(16)	result length
00001F20	00001F7C			1246+REA20	DC A(RE20)	result address
00001F28	00000000 00000000			1247+	DS FD	gap
00001F30	00000000 00000000			1248+V1020	DS XL16	V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00001F38	00000000 00000000								
00001F40	00000000 00000000			1249+	DS	FD	gap		
				1250+*					
00001F48				1251+X20	DS	0F			
00001F48	E310 5010 0014		00000010	1252+	LGF	R1,V2ADDR	load v2 source		
00001F4E	E761 0000 0806		00000000	1253+	VL	v22,0(R1)	use v22 to test decoder		
00001F54	E310 5014 0014		00000014	1254+	LGF	R1,V3ADDR	load v3 source		
00001F5A	E771 0000 0806		00000000	1255+	VL	v23,0(R1)	use v23 to test decoder		
00001F60	E310 5018 0014		00000018	1256+	LGF	R1,V4ADDR	load v4 source		
00001F66	E781 0000 0806		00000000	1257+	VL	v24,0(R1)	use v24 to test decoder		
00001F6C	E766 7400 8FA9			1258+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00001F72	E760 5030 080E		00001F30	1259+	VST	V22,V1020	save v1 output		
00001F78	07FB			1260+	BR	R11	return		
00001F7C				1261+RE20	DC	0F	xl16 expected result		
00001F7C				1262+	DROP	R5			
00001F7C	E1000000 00000000			1263	DC	XL16'E1000000 00000000 00000000 00000000'	result		
00001F84	00000000 00000000								
00001F8C	F0000000 00000000			1264	DC	XL16'F0000000 00000000 00000000 00000000'	v2		
00001F94	00000000 00000000								
00001F9C	F0000000 00000000			1265	DC	XL16'F0000000 00000000 00000000 00000000'	v3		
00001FA4	00000000 00000000								
00001FAC	FF000000 00000000			1266	DC	XL16'FF000000 00000000 20000000 00000000'	v4		
00001FB4	20000000 00000000								
				1267					
				1268	VRR_D	VMALH,4			
00001FC0				1269+	DS	0FD			
00001FC0		00001FC0		1270+	USING	*,R5	base for test data and test routine		
00001FC0	00002008			1271+T21	DC	A(X21)	address of test routine		
00001FC4	0015			1272+	DC	H'21'	test number		
00001FC6	00			1273+	DC	X'00'			
00001FC7	04			1274+	DC	HL1'4'	m5		
00001FC8	E5D4C1D3 C8404040			1275+	DC	CL8'VMALH'	instruction name		
00001FD0	0000204C			1276+	DC	A(RE21+16)	address of v2 source		
00001FD4	0000205C			1277+	DC	A(RE21+32)	address of v3 source		
00001FD8	0000206C			1278+	DC	A(RE21+48)	address of v4 source		
00001FDC	00000010			1279+	DC	A(16)	result length		
00001FE0	0000203C			1280+REA21	DC	A(RE21)	result address		
00001FE8	00000000 00000000			1281+	DS	FD	gap		
00001FF0	00000000 00000000			1282+V1021	DS	XL16	V1 output		
00001FF8	00000000 00000000								
00002000	00000000 00000000			1283+	DS	FD	gap		
				1284+*					
00002008				1285+X21	DS	0F			
00002008	E310 5010 0014		00000010	1286+	LGF	R1,V2ADDR	load v2 source		
0000200E	E761 0000 0806		00000000	1287+	VL	v22,0(R1)	use v22 to test decoder		
00002014	E310 5014 0014		00000014	1288+	LGF	R1,V3ADDR	load v3 source		
0000201A	E771 0000 0806		00000000	1289+	VL	v23,0(R1)	use v23 to test decoder		
00002020	E310 5018 0014		00000018	1290+	LGF	R1,V4ADDR	load v4 source		
00002026	E781 0000 0806		00000000	1291+	VL	v24,0(R1)	use v24 to test decoder		
0000202C	E766 7400 8FA9			1292+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00002032	E760 5030 080E		00001FF0	1293+	VST	V22,V1021	save v1 output		
00002038	07FB			1294+	BR	R11	return		
0000203C				1295+RE21	DC	0F	xl16 expected result		
0000203C				1296+	DROP	R5			
0000203C	E101C200 E1000000			1297	DC	XL16'E101C200 E1000000 00000000 00000000'	result		
00002044	00000000 00000000								

LOC	OBJECT CODE	ADDR1	ADDR2	STMT						
0000204C	F000F000	00000000		1298	DC	XL16'	F000F000	00000000	00000000	00000000' v2
00002054	00000000	00000000								
0000205C	F000F000	00000000		1299	DC	XL16'	F000F000	00000000	00000000	00000000' v3
00002064	00000000	00000000								
0000206C	FF000000	00000000		1300	DC	XL16'	FF000000	00000000	20000000	00000000' v4
00002074	20000000	00000000								
				1301						
				1302	VRR_D	VMALH, 4				
00002080				1303+	DS	0FD				
00002080		00002080		1304+	USING	*, R5				base for test data and test routine
00002080	000020C8			1305+T22	DC	A(X22)				address of test routine
00002084	0016			1306+	DC	H'22'				test number
00002086	00			1307+	DC	X'00'				
00002087	04			1308+	DC	HL1'4'				m5
00002088	E5D4C1D3	C8404040		1309+	DC	CL8'VMALH'				instruction name
00002090	0000210C			1310+	DC	A(RE22+16)				address of v2 source
00002094	0000211C			1311+	DC	A(RE22+32)				address of v3 source
00002098	0000212C			1312+	DC	A(RE22+48)				address of v4 source
0000209C	00000010			1313+	DC	A(16)				result length
000020A0	000020FC			1314+REA22	DC	A(RE22)				result address
000020A8	00000000	00000000		1315+	DS	FD				gap
000020B0	00000000	00000000		1316+V1022	DS	XL16				V1 output
000020B8	00000000	00000000								
000020C0	00000000	00000000		1317+	DS	FD				gap
				1318+*						
000020C8				1319+X22	DS	0F				
000020C8	E310 5010 0014		00000010	1320+	LGF	R1, V2ADDR				load v2 source
000020CE	E761 0000 0806		00000000	1321+	VL	v22, 0(R1)				use v22 to test decoder
000020D4	E310 5014 0014		00000014	1322+	LGF	R1, V3ADDR				load v3 source
000020DA	E771 0000 0806		00000000	1323+	VL	v23, 0(R1)				use v23 to test decoder
000020E0	E310 5018 0014		00000018	1324+	LGF	R1, V4ADDR				load v4 source
000020E6	E781 0000 0806		00000000	1325+	VL	v24, 0(R1)				use v24 to test decoder
000020EC	E766 7400 8FA9			1326+	VMALH	V22, V22, V23, V24, 4				test instruction (dest is a source)
000020F2	E760 5030 080E		000020B0	1327+	VST	V22, V1022				save v1 output
000020F8	07FB			1328+	BR	R11				return
000020FC				1329+RE22	DC	0F				xl16 expected result
000020FC				1330+	DROP	R5				
000020FC	E101C201	C200E100		1331	DC	XL16'	E101C201	C200E100	00000000	00000000' result
00002104	00000000	00000000								
0000210C	F000F000	F0000000		1332	DC	XL16'	F000F000	F0000000	00000000	00000000' v2
00002114	00000000	00000000								
0000211C	F000F000	00000000		1333	DC	XL16'	F000F000	00000000	00000000	00000000' v3
00002124	00000000	00000000								
0000212C	FF000000	00000000		1334	DC	XL16'	FF000000	00000000	20000000	00000000' v4
00002134	20000000	00000000								
				1335						
				1336	VRR_D	VMALH, 4				
00002140				1337+	DS	0FD				
00002140		00002140		1338+	USING	*, R5				base for test data and test routine
00002140	00002188			1339+T23	DC	A(X23)				address of test routine
00002144	0017			1340+	DC	H'23'				test number
00002146	00			1341+	DC	X'00'				
00002147	04			1342+	DC	HL1'4'				m5
00002148	E5D4C1D3	C8404040		1343+	DC	CL8'VMALH'				instruction name
00002150	000021CC			1344+	DC	A(RE23+16)				address of v2 source
00002154	000021DC			1345+	DC	A(RE23+32)				address of v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00002158	000021EC			1346+	DC	A(RE23+48)	address of v4 source		
0000215C	00000010			1347+	DC	A(16)	result length		
00002160	000021BC			1348+REA23	DC	A(RE23)	result address		
00002168	00000000 00000000			1349+	DS	FD	gap		
00002170	00000000 00000000			1350+V1023	DS	XL16	V1 output		
00002178	00000000 00000000								
00002180	00000000 00000000			1351+	DS	FD	gap		
				1352+*					
00002188				1353+X23	DS	0F			
00002188	E310 5010 0014		00000010	1354+	LGF	R1,V2ADDR	load v2 source		
0000218E	E761 0000 0806		00000000	1355+	VL	v22,0(R1)	use v22 to test decoder		
00002194	E310 5014 0014		00000014	1356+	LGF	R1,V3ADDR	load v3 source		
0000219A	E771 0000 0806		00000000	1357+	VL	v23,0(R1)	use v23 to test decoder		
000021A0	E310 5018 0014		00000018	1358+	LGF	R1,V4ADDR	load v4 source		
000021A6	E781 0000 0806		00000000	1359+	VL	v24,0(R1)	use v24 to test decoder		
000021AC	E766 7400 8FA9			1360+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)		
000021B2	E760 5030 080E		00002170	1361+	VST	V22,V1023	save v1 output		
000021B8	07FB			1362+	BR	R11	return		
000021BC				1363+RE23	DC	0F	xl16 expected result		
000021BC				1364+	DROP	R5			
000021BC	E101C201 C200E100			1365	DC	XL16'E101C201 C200E100 E100E100 E1000000'	result		
000021C4	E100E100 E1000000								
000021CC	F000F000 F0000000			1366	DC	XL16'F000F000 F0000000 00000000 00000000'	v2		
000021D4	00000000 00000000								
000021DC	F000F000 00000000			1367	DC	XL16'F000F000 00000000 F0000000 00000000'	v3		
000021E4	F0000000 00000000								
000021EC	FF000000 00000000			1368	DC	XL16'FF000000 00000000 20000000 00000000'	v4		
000021F4	20000000 00000000								
				1369					
00002200				1370	VRR_D	VMALH,4			
00002200		00002200		1371+	DS	0FD			
00002200	00002248			1372+	USING	*,R5	base for test data and test routine		
00002204	0018			1373+T24	DC	A(X24)	address of test routine		
00002206	00			1374+	DC	H'24'	test number		
00002207	04			1375+	DC	X'00'			
00002208	E5D4C1D3 C8404040			1376+	DC	HL1'4'	m5		
00002210	0000228C			1377+	DC	CL8'VMALH'	instruction name		
00002214	0000229C			1378+	DC	A(RE24+16)	address of v2 source		
00002218	000022AC			1379+	DC	A(RE24+32)	address of v3 source		
0000221C	00000010			1380+	DC	A(RE24+48)	address of v4 source		
00002220	0000227C			1381+	DC	A(16)	result length		
00002220	0000227C			1382+REA24	DC	A(RE24)	result address		
00002228	00000000 00000000			1383+	DS	FD	gap		
00002230	00000000 00000000			1384+V1024	DS	XL16	V1 output		
00002238	00000000 00000000								
00002240	00000000 00000000			1385+	DS	FD	gap		
				1386+*					
00002248				1387+X24	DS	0F			
00002248	E310 5010 0014		00000010	1388+	LGF	R1,V2ADDR	load v2 source		
0000224E	E761 0000 0806		00000000	1389+	VL	v22,0(R1)	use v22 to test decoder		
00002254	E310 5014 0014		00000014	1390+	LGF	R1,V3ADDR	load v3 source		
0000225A	E771 0000 0806		00000000	1391+	VL	v23,0(R1)	use v23 to test decoder		
00002260	E310 5018 0014		00000018	1392+	LGF	R1,V4ADDR	load v4 source		
00002266	E781 0000 0806		00000000	1393+	VL	v24,0(R1)	use v24 to test decoder		
0000226C	E766 7400 8FA9			1394+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00002272	E760 A030 080E		00002230	1395+	VST	V22,V1024	save v1 output		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00002278	07FB			1396+	BR	R11	return		
0000227C				1397+RE24	DC	0F	xl16 expected result		
0000227C				1398+	DROP	R5			
0000227C	00000001 00000001			1399	DC	XL16'00000001 00000001 00000001 00000001'	result		
00002284	00000001 00000001								
0000228C	FFFFFFFF FFFFFFFF			1400	DC	XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'	v2		
00002294	FFFFFFFF FFFFFFFF								
0000229C	00000001 00000001			1401	DC	XL16'00000001 00000001 00000001 00000001'	v3		
000022A4	00000001 00000001								
000022AC	FF000000 00000000			1402	DC	XL16'FF000000 00000000 20000000 00000000'	v4		
000022B4	20000000 00000000								
				1403					
000022C0				1404	VRR_D	VMALH,4			
000022C0		000022C0		1405+	DS	0FD			
000022C0	00002308			1406+	USING	*,R5	base for test data and test routine		
000022C4	0019			1407+T25	DC	A(X25)	address of test routine		
000022C6	00			1408+	DC	H'25'	test number		
000022C7	04			1409+	DC	X'00'			
000022C8	E5D4C1D3 C8404040			1410+	DC	HL1'4'	m5		
000022D0	0000234C			1411+	DC	CL8'VMALH'	instruction name		
000022D4	0000235C			1412+	DC	A(RE25+16)	address of v2 source		
000022D8	0000236C			1413+	DC	A(RE25+32)	address of v3 source		
000022DC	00000010			1414+	DC	A(RE25+48)	address of v4 source		
000022E0	0000233C			1415+	DC	A(16)	result length		
000022E8	00000000 00000000			1416+REA25	DC	A(RE25)	result address		
000022F0	00000000 00000000			1417+	DS	FD	gap		
000022F8	00000000 00000000			1418+V1025	DS	XL16	V1 output		
00002300	00000000 00000000								
				1419+	DS	FD	gap		
00002308				1420+*					
00002308	E310 5010 0014		00000010	1421+X25	DS	0F			
0000230E	E761 0000 0806		00000000	1422+	LGF	R1,V2ADDR	load v2 source		
00002314	E310 5014 0014		00000014	1423+	VL	v22,0(R1)	use v22 to test decoder		
0000231A	E771 0000 0806		00000000	1424+	LGF	R1,V3ADDR	load v3 source		
00002320	E310 5018 0014		00000018	1425+	VL	v23,0(R1)	use v23 to test decoder		
00002326	E781 0000 0806		00000000	1426+	LGF	R1,V4ADDR	load v4 source		
0000232C	E766 7400 8FA9		00000000	1427+	VL	v24,0(R1)	use v24 to test decoder		
00002332	E760 5030 080E		000022F0	1428+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00002338	07FB			1429+	VST	V22,V1025	save v1 output		
0000233C				1430+	BR	R11	return		
0000233C				1431+RE25	DC	0F	xl16 expected result		
0000233C				1432+	DROP	R5			
0000233C	FFFFFFFFE 00032000			1433	DC	XL16'FFFFFFFFE00032000 FFFCE0736EDDDD83'	result		
00002344	FFFCE073 6EDDDD83								
0000234C	FFFFFFFF 00019000			1434	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2		
00002354	00000038 EEEEEEEFA								
0000235C	FFFFFFFF 00019000			1435	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3		
00002364	00000038 0EEEEEEFA								
0000236C	00000000 00000000			1436	DC	XL16'0000000000000000 0000000000000000'	v4		
00002374	00000000 00000000								
				1437					
00002380				1438	VRR_D	VMALH,4			
00002380		00002380		1439+	DS	0FD			
00002380	000023C8			1440+	USING	*,R5	base for test data and test routine		
00002384	001A			1441+T26	DC	A(X26)	address of test routine		
				1442+	DC	H'26'	test number		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002386	00			1443+	DC	X'00'	
00002387	04			1444+	DC	HL1'4'	m5
00002388	E5D4C1D3 C8404040			1445+	DC	CL8'VMALH'	instruction name
00002390	0000240C			1446+	DC	A(RE26+16)	address of v2 source
00002394	0000241C			1447+	DC	A(RE26+32)	address of v3 source
00002398	0000242C			1448+	DC	A(RE26+48)	address of v4 source
0000239C	00000010			1449+	DC	A(16)	result length
000023A0	000023FC			1450+REA26	DC	A(RE26)	result address
000023A8	00000000 00000000			1451+	DS	FD	gap
000023B0	00000000 00000000			1452+V1026	DS	XL16	V1 output
000023B8	00000000 00000000						
000023C0	00000000 00000000			1453+	DS	FD	gap
				1454+*			
000023C8				1455+X26	DS	0F	
000023C8	E310 5010 0014		00000010	1456+	LGF	R1,V2ADDR	load v2 source
000023CE	E761 0000 0806		00000000	1457+	VL	v22,0(R1)	use v22 to test decoder
000023D4	E310 5014 0014		00000014	1458+	LGF	R1,V3ADDR	load v3 source
000023DA	E771 0000 0806		00000000	1459+	VL	v23,0(R1)	use v23 to test decoder
000023E0	E310 5018 0014		00000018	1460+	LGF	R1,V4ADDR	load v4 source
000023E6	E781 0000 0806		00000000	1461+	VL	v24,0(R1)	use v24 to test decoder
000023EC	E766 7400 8FA9			1462+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)
000023F2	E760 5030 080E		000023B0	1463+	VST	V22,V1026	save v1 output
000023F8	07FB			1464+	BR	R11	return
000023FC				1465+RE26	DC	0F	xl16 expected result
000023FC				1466+	DROP	R5	
000023FC	01010308 111F3396			1467	DC	XL16'01010308111F3396 72BF86C3CAFA7484'	result
00002404	72BF86C3 CAFA7484						
0000240C	FF020304 05060750			1468	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00002414	090A0B0C 0D0E0F7F						
0000241C	01020304 05060750			1469	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3
00002424	090A0B78 0D0E0F7F						
0000242C	FF000000 00000000			1470	DC	XL16'FF00000000000000 2000000000000000'	v4
00002434	20000000 00000000						
				1471			
				1472	VRR_D	VMALH,4	
00002440				1473+	DS	0FD	
00002440		00002440		1474+	USING	*,R5	base for test data and test routine
00002440	00002488			1475+T27	DC	A(X27)	address of test routine
00002444	001B			1476+	DC	H'27'	test number
00002446	00			1477+	DC	X'00'	
00002447	04			1478+	DC	HL1'4'	m5
00002448	E5D4C1D3 C8404040			1479+	DC	CL8'VMALH'	instruction name
00002450	000024CC			1480+	DC	A(RE27+16)	address of v2 source
00002454	000024DC			1481+	DC	A(RE27+32)	address of v3 source
00002458	000024EC			1482+	DC	A(RE27+48)	address of v4 source
0000245C	00000010			1483+	DC	A(16)	result length
00002460	000024BC			1484+REA27	DC	A(RE27)	result address
00002468	00000000 00000000			1485+	DS	FD	gap
00002470	00000000 00000000			1486+V1027	DS	XL16	V1 output
00002478	00000000 00000000						
00002480	00000000 00000000			1487+	DS	FD	gap
				1488+*			
00002488				1489+X27	DS	0F	
00002488	E310 5010 0014		00000010	1490+	LGF	R1,V2ADDR	load v2 source
0000248E	E761 0000 0806		00000000	1491+	VL	v22,0(R1)	use v22 to test decoder
00002494	E310 5014 0014		00000014	1492+	LGF	R1,V3ADDR	load v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
0000249A	E771 0000 0806		00000000	1493+	VL	v23,0(R1)	use v23 to test decoder	
000024A0	E310 5018 0014		00000018	1494+	LGF	R1,V4ADDR	load v4 source	
000024A6	E781 0000 0806		00000000	1495+	VL	v24,0(R1)	use v24 to test decoder	
000024AC	E766 7400 8FA9			1496+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)	
000024B2	E760 5030 080E		00002470	1497+	VST	V22,V1027	save v1 output	
000024B8	07FB			1498+	BR	R11	return	
000024BC				1499+RE27	DC	0F	xl16 expected result	
000024BC				1500+	DROP	R5		
000024BC	00010003 050C1344			1501	DC	XL16'00010003050C1344 0AD40FD0ABE579A2'	result	
000024C4	0AD40FD0 ABE579A2							
000024CC	FF020304 05060750			1502	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
000024D4	090A0B0C 0D0E0F7F							
000024DC	00010102 02030328			1503	DC	XL16'0001010202030328 0405053C0607073F'	v3	
000024E4	0405053C 0607073F							
000024EC	7FFFFFFF FFFFFFFF			1504	DC	XL16'7FFFFFFF7FFFFFFF FFFFFFFF7FFFFFFF'	v4	
000024F4	FFFFFFF FFFFFFFF							
				1505				
00002500				1506	VRR_D	VMALH,4		
00002500		00002500		1507+	DS	0FD		
00002500	00002548			1508+	USING	*,R5	base for test data and test routine	
00002504	001C			1509+T28	DC	A(X28)	address of test routine	
00002506	00			1510+	DC	H'28'	test number	
00002507	04			1511+	DC	X'00'		
00002508	E5D4C1D3 C8404040			1512+	DC	HL1'4'	m5	
00002510	0000258C			1513+	DC	CL8'VMALH'	instruction name	
00002514	0000259C			1514+	DC	A(RE28+16)	address of v2 source	
00002518	000025AC			1515+	DC	A(RE28+32)	address of v3 source	
0000251C	00000010			1516+	DC	A(RE28+48)	address of v4 source	
00002520	0000257C			1517+	DC	A(16)	result length	
00002528	00000000 00000000			1518+REA28	DC	A(RE28)	result address	
00002530	00000000 00000000			1519+	DS	FD	gap	
00002538	00000000 00000000			1520+V1028	DS	XL16	V1 output	
00002540	00000000 00000000							
				1521+	DS	FD	gap	
00002548				1522+*				
00002548	E310 5010 0014		00000010	1523+X28	DS	0F		
0000254E	E761 0000 0806		00000000	1524+	LGF	R1,V2ADDR	load v2 source	
00002554	E310 5014 0014		00000014	1525+	VL	v22,0(R1)	use v22 to test decoder	
0000255A	E771 0000 0806		00000000	1526+	LGF	R1,V3ADDR	load v3 source	
00002560	E310 5018 0014		00000018	1527+	VL	v23,0(R1)	use v23 to test decoder	
00002566	E781 0000 0806		00000000	1528+	LGF	R1,V4ADDR	load v4 source	
0000256C	E766 7400 8FA9			1529+	VL	v24,0(R1)	use v24 to test decoder	
00002572	E760 5030 080E		00002530	1530+	VMALH	V22,V22,V23,V24,4	test instruction (dest is a source)	
00002578	07FB			1531+	VST	V22,V1028	save v1 output	
0000257C				1532+	BR	R11	return	
0000257C				1533+RE28	DC	0F	xl16 expected result	
0000257C				1534+	DROP	R5		
0000257C	00000000 00000009			1535	DC	XL16'0000000000000009 F714203B2D668782'	result	
00002584	F714203B 2D668782							
0000258C	FF020304 05060750			1536	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
00002594	090A0B0C 0D0E0F7F							
0000259C	00000000 0000000A			1537	DC	XL16'000000000000000A 0101010F0101010F'	v3	
000025A4	0101010F 0101010F							
000025AC	FFFFFFF FFFFFFFF			1538	DC	XL16'FFFFFFF7FFFFFFF 7FFFFFFF7FFFFFFF'	v4	
000025B4	7FFFFFFF FFFFFFFF							
				1539				

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
000025C0				1540	VRR_D VMALH,4	
000025C0				1541+	DS 0FD	
000025C0		000025C0		1542+	USING *,R5	base for test data and test routine
000025C0	00002608			1543+T29	DC A(X29)	address of test routine
000025C4	001D			1544+	DC H'29'	test number
000025C6	00			1545+	DC X'00'	
000025C7	04			1546+	DC HL1'4'	m5
000025C8	E5D4C1D3 C8404040			1547+	DC CL8'VMALH'	instruction name
000025D0	0000264C			1548+	DC A(RE29+16)	address of v2 source
000025D4	0000265C			1549+	DC A(RE29+32)	address of v3 source
000025D8	0000266C			1550+	DC A(RE29+48)	address of v4 source
000025DC	00000010			1551+	DC A(16)	result length
000025E0	0000263C			1552+REA29	DC A(RE29)	result address
000025E8	00000000 00000000			1553+	DS FD	gap
000025F0	00000000 00000000			1554+V1029	DS XL16	V1 output
000025F8	00000000 00000000					
00002600	00000000 00000000			1555+	DS FD	gap
				1556+*		
00002608				1557+X29	DS 0F	
00002608	E310 5010 0014		00000010	1558+	LGF R1,V2ADDR	load v2 source
0000260E	E761 0000 0806		00000000	1559+	VL v22,0(R1)	use v22 to test decoder
00002614	E310 5014 0014		00000014	1560+	LGF R1,V3ADDR	load v3 source
0000261A	E771 0000 0806		00000000	1561+	VL v23,0(R1)	use v23 to test decoder
00002620	E310 5018 0014		00000018	1562+	LGF R1,V4ADDR	load v4 source
00002626	E781 0000 0806		00000000	1563+	VL v24,0(R1)	use v24 to test decoder
0000262C	E766 7400 8FA9			1564+	VMALH V22,V22,V23,V24,4	test instruction (dest is a source)
00002632	E760 5030 080E		000025F0	1565+	VST V22,V1029	save v1 output
00002638	07FB			1566+	BR R11	return
0000263C				1567+RE29	DC 0F	xl16 expected result
0000263C				1568+	DROP R5	
0000263C	4665B5B7 6BD3F0C3			1569	DC XL16'4665B5B7 6BD3F0C3 4C8D873A A8D2B95E'	result
00002644	4C8D873A A8D2B95E					
0000264C	50515253 54555657			1570	DC XL16'50515253 54555657 58595A5B 5C5D5E5F'	v2
00002654	58595A5B 5C5D5E5F					
0000265C	E0616263 64656667			1571	DC XL16'E0616263 64656667 68696A6B 6C6D6E6F'	v3
00002664	68696A6B 6C6D6E6F					
0000266C	FFFFFFFF FFFFFFFF			1572	DC XL16'FFFFFFFF FFFFFFFF 7FFFFFFFF FFFFFFFF'	v4
00002674	7FFFFFFFF FFFFFFFF					
				1573		
				1574 *	-----	
				1575 * VMAL	- Vector Multiply and Add Low	
				1576 *	-----	
				1577 * Byte		
				1578	VRR_D VMAL,0	
00002680				1579+	DS 0FD	
00002680		00002680		1580+	USING *,R5	base for test data and test routine
00002680	000026C8			1581+T30	DC A(X30)	address of test routine
00002684	001E			1582+	DC H'30'	test number
00002686	00			1583+	DC X'00'	
00002687	00			1584+	DC HL1'0'	m5
00002688	E5D4C1D3 40404040			1585+	DC CL8'VMAL'	instruction name
00002690	0000270C			1586+	DC A(RE30+16)	address of v2 source
00002694	0000271C			1587+	DC A(RE30+32)	address of v3 source
00002698	0000272C			1588+	DC A(RE30+48)	address of v4 source
0000269C	00000010			1589+	DC A(16)	result length
000026A0	000026FC			1590+REA30	DC A(RE30)	result address

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
000026A8	00000000	00000000				1591+	DS	FD	gap
000026B0	00000000	00000000				1592+V1030	DS	XL16	V1 output
000026B8	00000000	00000000							
000026C0	00000000	00000000				1593+	DS	FD	gap
						1594+*			
000026C8						1595+X30	DS	0F	
000026C8	E310	5010	0014		00000010	1596+	LGF	R1,V2ADDR	load v2 source
000026CE	E761	0000	0806		00000000	1597+	VL	v22,0(R1)	use v22 to test decoder
000026D4	E310	5014	0014		00000014	1598+	LGF	R1,V3ADDR	load v3 source
000026DA	E771	0000	0806		00000000	1599+	VL	v23,0(R1)	use v23 to test decoder
000026E0	E310	5018	0014		00000018	1600+	LGF	R1,V4ADDR	load v4 source
000026E6	E781	0000	0806		00000000	1601+	VL	v24,0(R1)	use v24 to test decoder
000026EC	E766	7000	8FAA			1602+	VMAL	V22,V22,V23,V24,0	test instruction (dest is a source)
000026F2	E760	5030	080E		000026B0	1603+	VST	V22,V1030	save v1 output
000026F8	07FB					1604+	BR	R11	return
000026FC						1605+RE30	DC	0F	xl16 expected result
000026FC						1606+	DROP	R5	
000026FC	01000000	00000071				1607	DC	XL16'010000000000000071 00000040000000024'	result
00002704	00000040	00000024							
0000270C	FF000000	00000019				1608	DC	XL16'FF0000000000000019 000000380000000FA'	v2
00002714	00000038	000000FA							
0000271C	FF000000	00000019				1609	DC	XL16'FF0000000000000019 000000380000000FA'	v3
00002724	00000038	000000FA							
0000272C	00000000	00000000				1610	DC	XL16'000000000000000000 00000000000000000'	v4
00002734	00000000	00000000							
						1611			
						1612	VRR_D	VMAL,0	
00002740						1613+	DS	0FD	
00002740				00002740		1614+	USING	*,R5	base for test data and test routine
00002740	00002788					1615+T31	DC	A(X31)	address of test routine
00002744	001F					1616+	DC	H'31'	test number
00002746	00					1617+	DC	X'00'	
00002747	00					1618+	DC	HL1'0'	m5
00002748	E5D4C1D3	40404040				1619+	DC	CL8'VMAL'	instruction name
00002750	000027CC					1620+	DC	A(RE31+16)	address of v2 source
00002754	000027DC					1621+	DC	A(RE31+32)	address of v3 source
00002758	000027EC					1622+	DC	A(RE31+48)	address of v4 source
0000275C	00000010					1623+	DC	A(16)	result length
00002760	000027BC					1624+REA31	DC	A(RE31)	result address
00002768	00000000	00000000				1625+	DS	FD	gap
00002770	00000000	00000000				1626+V1031	DS	XL16	V1 output
00002778	00000000	00000000							
00002780	00000000	00000000				1627+	DS	FD	gap
						1628+*			
00002788						1629+X31	DS	0F	
00002788	E310	5010	0014		00000010	1630+	LGF	R1,V2ADDR	load v2 source
0000278E	E761	0000	0806		00000000	1631+	VL	v22,0(R1)	use v22 to test decoder
00002794	E310	5014	0014		00000014	1632+	LGF	R1,V3ADDR	load v3 source
0000279A	E771	0000	0806		00000000	1633+	VL	v23,0(R1)	use v23 to test decoder
000027A0	E310	5018	0014		00000018	1634+	LGF	R1,V4ADDR	load v4 source
000027A6	E781	0000	0806		00000000	1635+	VL	v24,0(R1)	use v24 to test decoder
000027AC	E766	7000	8FAA			1636+	VMAL	V22,V22,V23,V24,0	test instruction (dest is a source)
000027B2	E760	5030	080E		00002770	1637+	VST	V22,V1031	save v1 output
000027B8	07FB					1638+	BR	R11	return
000027BC						1639+RE31	DC	0F	xl16 expected result
000027BC						1640+	DROP	R5	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000027BC	01000000 000000C0			1641	DC	XL16'01000000000000C0 0000004300000026'	result
000027C4	00000043 00000026						
000027CC	FF0000FF 00000029			1642	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
000027D4	00000038 000000FA						
000027DC	FF000001 00000029			1643	DC	XL16'FF00000100000029 00000038000000FA'	v3
000027E4	00000038 000000FA						
000027EC	00000001 0000002F			1644	DC	XL16'000000010000002F 0000000300000002'	v4
000027F4	00000003 00000002						
				1645			
00002800				1646	VRR_D	VMAL,0	
00002800		00002800		1647+	DS	0FD	
00002800	00002848			1648+	USING	*,R5	base for test data and test routine
00002804	0020			1649+T32	DC	A(X32)	address of test routine
00002806	00			1650+	DC	H'32'	test number
00002807	00			1651+	DC	X'00'	
00002808	E5D4C1D3 40404040			1652+	DC	HL1'0'	m5
00002810	0000288C			1653+	DC	CL8'VMAL'	instruction name
00002814	0000289C			1654+	DC	A(RE32+16)	address of v2 source
00002818	000028AC			1655+	DC	A(RE32+32)	address of v3 source
0000281C	00000010			1656+	DC	A(RE32+48)	address of v4 source
00002820	0000287C			1657+	DC	A(16)	result length
00002828	00000000 00000000			1658+REA32	DC	A(RE32)	result address
00002830	00000000 00000000			1659+	DS	FD	gap
00002838	00000000 00000000			1660+V1032	DS	XL16	V1 output
00002840	00000000 00000000						
				1661+	DS	FD	gap
				1662+*			
00002848				1663+X32	DS	0F	
00002848	E310 5010 0014		00000010	1664+	LGF	R1,V2ADDR	load v2 source
0000284E	E761 0000 0806		00000000	1665+	VL	v22,0(R1)	use v22 to test decoder
00002854	E310 5014 0014		00000014	1666+	LGF	R1,V3ADDR	load v3 source
0000285A	E771 0000 0806		00000000	1667+	VL	v23,0(R1)	use v23 to test decoder
00002860	E310 5018 0014		00000018	1668+	LGF	R1,V4ADDR	load v4 source
00002866	E781 0000 0806		00000000	1669+	VL	v24,0(R1)	use v24 to test decoder
0000286C	E766 7000 8FAA			1670+	VMAL	V22,V22,V23,V24,0	test instruction (dest is a source)
00002872	E760 5030 080E		00002830	1671+	VST	V22,V1032	save v1 output
00002878	07FB			1672+	BR	R11	return
0000287C				1673+RE32	DC	0F	xl16 expected result
0000287C				1674+	DROP	R5	
0000287C	00060C14 1E2A3848			1675	DC	XL16'00060C141E2A3848 5A6E849CB6D2F010'	result
00002884	5A6E849C B6D2F010						
0000288C	FF020304 05060708			1676	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002894	090A0B0C 0D0E0F10						
0000289C	FF020304 05060708			1677	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000028A4	090A0B0C 0D0E0F10						
000028AC	FF020304 05060708			1678	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000028B4	090A0B0C 0D0E0F10						
				1679			
000028C0				1680	VRR_D	VMAL,0	
000028C0		000028C0		1681+	DS	0FD	
000028C0	00002908			1682+	USING	*,R5	base for test data and test routine
000028C4	0021			1683+T33	DC	A(X33)	address of test routine
000028C6	00			1684+	DC	H'33'	test number
000028C7	00			1685+	DC	X'00'	
000028C8	E5D4C1D3 40404040			1686+	DC	HL1'0'	m5
				1687+	DC	CL8'VMAL'	instruction name

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000028D0	0000294C			1688+	DC	A(RE33+16)	address of v2 source
000028D4	0000295C			1689+	DC	A(RE33+32)	address of v3 source
000028D8	0000296C			1690+	DC	A(RE33+48)	address of v4 source
000028DC	00000010			1691+	DC	A(16)	result length
000028E0	0000293C			1692+REA33	DC	A(RE33)	result address
000028E8	00000000 00000000			1693+	DS	FD	gap
000028F0	00000000 00000000			1694+V1033	DS	XL16	V1 output
000028F8	00000000 00000000						
00002900	00000000 00000000			1695+	DS	FD	gap
				1696+*			
00002908				1697+X33	DS	0F	
00002908	E310 5010 0014		00000010	1698+	LGF	R1,V2ADDR	load v2 source
0000290E	E761 0000 0806		00000000	1699+	VL	v22,0(R1)	use v22 to test decoder
00002914	E310 5014 0014		00000014	1700+	LGF	R1,V3ADDR	load v3 source
0000291A	E771 0000 0806		00000000	1701+	VL	v23,0(R1)	use v23 to test decoder
00002920	E310 5018 0014		00000018	1702+	LGF	R1,V4ADDR	load v4 source
00002926	E781 0000 0806		00000000	1703+	VL	v24,0(R1)	use v24 to test decoder
0000292C	E766 7000 8FAA			1704+	VMAL	V22,V22,V23,V24,0	test instruction (dest is a source)
00002932	E760 5030 080E		000028F0	1705+	VST	V22,V1033	save v1 output
00002938	07FB			1706+	BR	R11	return
0000293C				1707+RE33	DC	0F	xl16 expected result
0000293C				1708+	DROP	R5	
0000293C	0004060C 0F181C28			1709	DC	XL16'0004060C0F181C28 2D3C42545B707890'	result
00002944	2D3C4254 5B707890						
0000294C	FF020304 05060708			1710	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002954	090A0B0C 0D0E0F10						
0000295C	FF010102 02030304			1711	DC	XL16'FF01010202030304 0405050606070708'	v3
00002964	04050506 06070708						
0000296C	FF020304 05060708			1712	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00002974	090A0B0C 0D0E0F10						
				1713			
				1714	VRR_D	VMAL,0	
00002980				1715+	DS	0FD	
00002980		00002980		1716+	USING	*,R5	base for test data and test routine
00002980	000029C8			1717+T34	DC	A(X34)	address of test routine
00002984	0022			1718+	DC	H'34'	test number
00002986	00			1719+	DC	X'00'	
00002987	00			1720+	DC	HL1'0'	m5
00002988	E5D4C1D3 40404040			1721+	DC	CL8'VMAL'	instruction name
00002990	00002A0C			1722+	DC	A(RE34+16)	address of v2 source
00002994	00002A1C			1723+	DC	A(RE34+32)	address of v3 source
00002998	00002A2C			1724+	DC	A(RE34+48)	address of v4 source
0000299C	00000010			1725+	DC	A(16)	result length
000029A0	000029FC			1726+REA34	DC	A(RE34)	result address
000029A8	00000000 00000000			1727+	DS	FD	gap
000029B0	00000000 00000000			1728+V1034	DS	XL16	V1 output
000029B8	00000000 00000000						
000029C0	00000000 00000000			1729+	DS	FD	gap
				1730+*			
000029C8				1731+X34	DS	0F	
000029C8	E310 5010 0014		00000010	1732+	LGF	R1,V2ADDR	load v2 source
000029CE	E761 0000 0806		00000000	1733+	VL	v22,0(R1)	use v22 to test decoder
000029D4	E310 5014 0014		00000014	1734+	LGF	R1,V3ADDR	load v3 source
000029DA	E771 0000 0806		00000000	1735+	VL	v23,0(R1)	use v23 to test decoder
000029E0	E310 5018 0014		00000018	1736+	LGF	R1,V4ADDR	load v4 source
000029E6	E781 0000 0806		00000000	1737+	VL	v24,0(R1)	use v24 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000029EC	E766 7000 8FAA			1738+	VMAL	V22,V22,V23,V24,0	test instruction (dest is a source)
000029F2	E760 5030 080E		000029B0	1739+	VST	V22,V1034	save v1 output
000029F8	07FB			1740+	BR	R11	return
000029FC				1741+RE34	DC	0F	xl16 expected result
000029FC				1742+	DROP	R5	
000029FC	00020304 05060710			1743	DC	XL16'0002030405060710 121416181A1C1E30'	result
00002A04	12141618 1A1C1E30						
00002A0C	FF020304 05060708			1744	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002A14	090A0B0C 0D0E0F10						
00002A1C	FF000000 00000001			1745	DC	XL16'FF00000000000001 0101010101010102'	v3
00002A24	01010101 01010102						
00002A2C	FF020304 05060708			1746	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00002A34	090A0B0C 0D0E0F10						
				1747			
				1748 * Halfword			
				1749	VRR_D	VMAL,1	
00002A40				1750+	DS	0FD	
00002A40		00002A40		1751+	USING	*,R5	base for test data and test routine
00002A40	00002A88			1752+T35	DC	A(X35)	address of test routine
00002A44	0023			1753+	DC	H'35'	test number
00002A46	00			1754+	DC	X'00'	
00002A47	01			1755+	DC	HL1'1'	m5
00002A48	E5D4C1D3 40404040			1756+	DC	CL8'VMAL'	instruction name
00002A50	00002ACC			1757+	DC	A(RE35+16)	address of v2 source
00002A54	00002ADC			1758+	DC	A(RE35+32)	address of v3 source
00002A58	00002AEC			1759+	DC	A(RE35+48)	address of v4 source
00002A5C	00000010			1760+	DC	A(16)	result length
00002A60	00002ABC			1761+REA35	DC	A(RE35)	result address
00002A68	00000000 00000000			1762+	DS	FD	gap
00002A70	00000000 00000000			1763+V1035	DS	XL16	V1 output
00002A78	00000000 00000000						
00002A80	00000000 00000000			1764+	DS	FD	gap
				1765+*			
00002A88				1766+X35	DS	0F	
00002A88	E310 5010 0014		00000010	1767+	LGF	R1,V2ADDR	load v2 source
00002A8E	E761 0000 0806		00000000	1768+	VL	v22,0(R1)	use v22 to test decoder
00002A94	E310 5014 0014		00000014	1769+	LGF	R1,V3ADDR	load v3 source
00002A9A	E771 0000 0806		00000000	1770+	VL	v23,0(R1)	use v23 to test decoder
00002AA0	E310 5018 0014		00000018	1771+	LGF	R1,V4ADDR	load v4 source
00002AA6	E781 0000 0806		00000000	1772+	VL	v24,0(R1)	use v24 to test decoder
00002AAC	E766 7100 8FAA			1773+	VMAL	V22,V22,V23,V24,1	test instruction (dest is a source)
00002AB2	E760 5030 080E		00002A70	1774+	VST	V22,V1035	save v1 output
00002AB8	07FB			1775+	BR	R11	return
00002ABC				1776+RE35	DC	0F	xl16 expected result
00002ABC				1777+	DROP	R5	
00002ABC	00000000 00000271			1778	DC	XL16'0000000000000271 00000C400000F424'	result
00002AC4	00000C40 0000F424						
00002ACC	FF000000 00000019			1779	DC	XL16'FF00000000000019 00000038000000FA'	v2
00002AD4	00000038 000000FA						
00002ADC	FF000000 00000019			1780	DC	XL16'FF00000000000019 00000038000000FA'	v3
00002AE4	00000038 000000FA						
00002AEC	00000000 00000000			1781	DC	XL16'0000000000000000 0000000000000000'	v4
00002AF4	00000000 00000000						
				1782			
				1783	VRR_D	VMAL,1	
00002B00				1784+	DS	0FD	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00002B00		00002B00		1785+	USING *,R5	base for test data and test routine
00002B00	00002B48			1786+T36	DC A(X36)	address of test routine
00002B04	0024			1787+	DC H'36'	test number
00002B06	00			1788+	DC X'00'	
00002B07	01			1789+	DC HL1'1'	m5
00002B08	E5D4C1D3 40404040			1790+	DC CL8'VMAL'	instruction name
00002B10	00002B8C			1791+	DC A(RE36+16)	address of v2 source
00002B14	00002B9C			1792+	DC A(RE36+32)	address of v3 source
00002B18	00002BAC			1793+	DC A(RE36+48)	address of v4 source
00002B1C	00000010			1794+	DC A(16)	result length
00002B20	00002B7C			1795+REA36	DC A(RE36)	result address
00002B28	00000000 00000000			1796+	DS FD	gap
00002B30	00000000 00000000			1797+V1036	DS XL16	V1 output
00002B38	00000000 00000000					
00002B40	00000000 00000000			1798+	DS FD	gap
				1799+*		
00002B48				1800+X36	DS 0F	
00002B48	E310 5010 0014		00000010	1801+	LGF R1,V2ADDR	load v2 source
00002B4E	E761 0000 0806		00000000	1802+	VL v22,0(R1)	use v22 to test decoder
00002B54	E310 5014 0014		00000014	1803+	LGF R1,V3ADDR	load v3 source
00002B5A	E771 0000 0806		00000000	1804+	VL v23,0(R1)	use v23 to test decoder
00002B60	E310 5018 0014		00000018	1805+	LGF R1,V4ADDR	load v4 source
00002B66	E781 0000 0806		00000000	1806+	VL v24,0(R1)	use v24 to test decoder
00002B6C	E766 7100 8FAA			1807+	VMAL V22,V22,V23,V24,1	test instruction (dest is a source)
00002B72	E760 5030 080E		00002B30	1808+	VST V22,V1036	save v1 output
00002B78	07FB			1809+	BR R11	return
00002B7C				1810+RE36	DC 0F	xl16 expected result
00002B7C				1811+	DROP R5	
00002B7C	00000100 000006C0			1812	DC XL16'000001000000006C0 00000C430000F426'	result
00002B84	00000C43 0000F426					
00002B8C	FF0000FF 00000029			1813	DC XL16'FF0000FF00000029 00000038000000FA'	v2
00002B94	00000038 000000FA					
00002B9C	FF000001 00000029			1814	DC XL16'FF00000100000029 00000038000000FA'	v3
00002BA4	00000038 000000FA					
00002BAC	00000001 0000002F			1815	DC XL16'000000010000002F 0000000300000002'	v4
00002BB4	00000003 00000002					
				1816		
				1817	VRR_D VMAL,1	
00002BC0				1818+	DS 0FD	
00002BC0		00002BC0		1819+	USING *,R5	base for test data and test routine
00002BC0	00002C08			1820+T37	DC A(X37)	address of test routine
00002BC4	0025			1821+	DC H'37'	test number
00002BC6	00			1822+	DC X'00'	
00002BC7	01			1823+	DC HL1'1'	m5
00002BC8	E5D4C1D3 40404040			1824+	DC CL8'VMAL'	instruction name
00002BD0	00002C4C			1825+	DC A(RE37+16)	address of v2 source
00002BD4	00002C5C			1826+	DC A(RE37+32)	address of v3 source
00002BD8	00002C6C			1827+	DC A(RE37+48)	address of v4 source
00002BDC	00000010			1828+	DC A(16)	result length
00002BE0	00002C3C			1829+REA37	DC A(RE37)	result address
00002BE8	00000000 00000000			1830+	DS FD	gap
00002BF0	00000000 00000000			1831+V1037	DS XL16	V1 output
00002BF8	00000000 00000000					
00002C00	00000000 00000000			1832+	DS FD	gap
				1833+*		
00002C08				1834+X37	DS 0F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002C08	E310 5010 0014		00000010	1835+	LGF	R1,V2ADDR	load v2 source
00002C0E	E761 0000 0806		00000000	1836+	VL	v22,0(R1)	use v22 to test decoder
00002C14	E310 5014 0014		00000014	1837+	LGF	R1,V3ADDR	load v3 source
00002C1A	E771 0000 0806		00000000	1838+	VL	v23,0(R1)	use v23 to test decoder
00002C20	E310 5018 0014		00000018	1839+	LGF	R1,V4ADDR	load v4 source
00002C26	E781 0000 0806		00000000	1840+	VL	v24,0(R1)	use v24 to test decoder
00002C2C	E766 7100 8FAA			1841+	VMAL	V22,V22,V23,V24,1	test instruction (dest is a source)
00002C32	E760 5030 080E		00002BF0	1842+	VST	V22,V1037	save v1 output
00002C38	07FB			1843+	BR	R11	return
00002C3C				1844+RE37	DC	0F	xl16 expected result
00002C3C				1845+	DROP	R5	
00002C3C	FB061B14 412A7748			1846	DC	XL16'FB061B14412A7748 BD6E139C79D2F010'	result
00002C44	BD6E139C 79D2F010						
00002C4C	FF020304 05060708			1847	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002C54	090A0B0C 0D0E0F10						
00002C5C	FF020304 05060708			1848	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00002C64	090A0B0C 0D0E0F10						
00002C6C	FF020304 05060708			1849	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00002C74	090A0B0C 0D0E0F10						
				1850			
00002C80				1851	VRR_D	VMAL,1	
00002C80		00002C80		1852+	DS	0FD	
00002C80	00002CC8			1853+	USING	*,R5	base for test data and test routine
00002C84	0026			1854+T38	DC	A(X38)	address of test routine
00002C86	00			1855+	DC	H'38'	test number
00002C87	01			1856+	DC	X'00'	
00002C88	E5D4C1D3 40404040			1857+	DC	HL1'1'	m5
00002C90	00002D0C			1858+	DC	CL8'VMAL'	instruction name
00002C94	00002D1C			1859+	DC	A(RE38+16)	address of v2 source
00002C98	00002D2C			1860+	DC	A(RE38+32)	address of v3 source
00002C9C	00000010			1861+	DC	A(RE38+48)	address of v4 source
00002CA0	00002CFC			1862+	DC	A(16)	result length
00002CA8	00000000 00000000			1863+REA38	DC	A(RE38)	result address
00002CB0	00000000 00000000			1864+	DS	FD	gap
00002CB8	00000000 00000000			1865+V1038	DS	XL16	V1 output
00002CC0	00000000 00000000			1866+	DS	FD	gap
				1867+*			
00002CC8				1868+X38	DS	0F	
00002CC8	E310 5010 0014		00000010	1869+	LGF	R1,V2ADDR	load v2 source
00002CCE	E761 0000 0806		00000000	1870+	VL	v22,0(R1)	use v22 to test decoder
00002CD4	E310 5014 0014		00000014	1871+	LGF	R1,V3ADDR	load v3 source
00002CDA	E771 0000 0806		00000000	1872+	VL	v23,0(R1)	use v23 to test decoder
00002CE0	E310 5018 0014		00000018	1873+	LGF	R1,V4ADDR	load v4 source
00002CE6	E781 0000 0806		00000000	1874+	VL	v24,0(R1)	use v24 to test decoder
00002CEC	E766 7100 8FAA			1875+	VMAL	V22,V22,V23,V24,1	test instruction (dest is a source)
00002CF2	E760 5030 080E		00002CB0	1876+	VST	V22,V1038	save v1 output
00002CF8	07FB			1877+	BR	R11	return
00002CFC				1878+RE38	DC	0F	xl16 expected result
00002CFC				1879+	DROP	R5	
00002CFC	FC040D0C 20183B28			1880	DC	XL16'FC040D0C20183B28 5E3C8954BC70F790'	result
00002D04	5E3C8954 BC70F790						
00002D0C	FF020304 05060708			1881	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002D14	090A0B0C 0D0E0F10						
00002D1C	FF010102 02030304			1882	DC	XL16'FF01010202030304 0405050606070708'	v3
00002D24	04050506 06070708						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002D2C	FF020304 05060708			1883	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00002D34	090A0B0C 0D0E0F10						
				1884			
				1885	VRR_D	VMAL,1	
00002D40				1886+	DS	0FD	
00002D40		00002D40		1887+	USING	*,R5	base for test data and test routine
00002D40	00002D88			1888+T39	DC	A(X39)	address of test routine
00002D44	0027			1889+	DC	H'39'	test number
00002D46	00			1890+	DC	X'00'	
00002D47	01			1891+	DC	HL1'1'	m5
00002D48	E5D4C1D3 40404040			1892+	DC	CL8'VMAL'	instruction name
00002D50	00002DCC			1893+	DC	A(RE39+16)	address of v2 source
00002D54	00002DDC			1894+	DC	A(RE39+32)	address of v3 source
00002D58	00002DEC			1895+	DC	A(RE39+48)	address of v4 source
00002D5C	00000010			1896+	DC	A(16)	result length
00002D60	00002DBC			1897+REA39	DC	A(RE39)	result address
00002D68	00000000 00000000			1898+	DS	FD	gap
00002D70	00000000 00000000			1899+V1039	DS	XL16	V1 output
00002D78	00000000 00000000						
00002D80	00000000 00000000			1900+	DS	FD	gap
				1901+*			
00002D88				1902+X39	DS	0F	
00002D88	E310 5010 0014		00000010	1903+	LGF	R1,V2ADDR	load v2 source
00002D8E	E761 0000 0806		00000000	1904+	VL	v22,0(R1)	use v22 to test decoder
00002D94	E310 5014 0014		00000014	1905+	LGF	R1,V3ADDR	load v3 source
00002D9A	E771 0000 0806		00000000	1906+	VL	v23,0(R1)	use v23 to test decoder
00002DA0	E310 5018 0014		00000018	1907+	LGF	R1,V4ADDR	load v4 source
00002DA6	E781 0000 0806		00000000	1908+	VL	v24,0(R1)	use v24 to test decoder
00002DAC	E766 7100 8FAA			1909+	VMAL	V22,V22,V23,V24,1	test instruction (dest is a source)
00002DB2	E760 5030 080E		00002D70	1910+	VST	V22,V1039	save v1 output
00002DB8	07FB			1911+	BR	R11	return
00002DBC				1912+RE39	DC	0F	xl16 expected result
00002DBC				1913+	DROP	R5	
00002DBC	FD020304 05060E10			1914	DC	XL16'FD02030405060E10 1C142218281C3D30'	result
00002DC4	1C142218 281C3D30						
00002DCC	FF020304 05060708			1915	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00002DD4	090A0B0C 0D0E0F10						
00002DDC	FF000000 00000001			1916	DC	XL16'FF0000000000000001 0101010101010102'	v3
00002DE4	01010101 01010102						
00002DEC	FF020304 05060708			1917	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00002DF4	090A0B0C 0D0E0F10						
				1918			
				1919 * Word			
				1920	VRR_D	VMAL,2	
00002E00				1921+	DS	0FD	
00002E00		00002E00		1922+	USING	*,R5	base for test data and test routine
00002E00	00002E48			1923+T40	DC	A(X40)	address of test routine
00002E04	0028			1924+	DC	H'40'	test number
00002E06	00			1925+	DC	X'00'	
00002E07	02			1926+	DC	HL1'2'	m5
00002E08	E5D4C1D3 40404040			1927+	DC	CL8'VMAL'	instruction name
00002E10	00002E8C			1928+	DC	A(RE40+16)	address of v2 source
00002E14	00002E9C			1929+	DC	A(RE40+32)	address of v3 source
00002E18	00002EAC			1930+	DC	A(RE40+48)	address of v4 source
00002E1C	00000010			1931+	DC	A(16)	result length
00002E20	00002E7C			1932+REA40	DC	A(RE40)	result address

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
00002E28	00000000	00000000				1933+	DS	FD	gap
00002E30	00000000	00000000				1934+V1040	DS	XL16	V1 output
00002E38	00000000	00000000							
00002E40	00000000	00000000				1935+	DS	FD	gap
						1936+*			
00002E48						1937+X40	DS	0F	
00002E48	E310	5010	0014		00000010	1938+	LGF	R1,V2ADDR	load v2 source
00002E4E	E761	0000	0806		00000000	1939+	VL	v22,0(R1)	use v22 to test decoder
00002E54	E310	5014	0014		00000014	1940+	LGF	R1,V3ADDR	load v3 source
00002E5A	E771	0000	0806		00000000	1941+	VL	v23,0(R1)	use v23 to test decoder
00002E60	E310	5018	0014		00000018	1942+	LGF	R1,V4ADDR	load v4 source
00002E66	E781	0000	0806		00000000	1943+	VL	v24,0(R1)	use v24 to test decoder
00002E6C	E766	7200	8FAA			1944+	VMAL	V22,V22,V23,V24,2	test instruction (dest is a source)
00002E72	E760	5030	080E		00002E30	1945+	VST	V22,V1040	save v1 output
00002E78	07FB					1946+	BR	R11	return
00002E7C						1947+RE40	DC	0F	xl16 expected result
00002E7C						1948+	DROP	R5	
00002E7C	00000000	00000271				1949	DC	XL16'00000000000000271 00000C400000F424'	result
00002E84	00000C40	0000F424							
00002E8C	FF000000	00000019				1950	DC	XL16'FF00000000000019 00000038000000FA'	v2
00002E94	00000038	000000FA							
00002E9C	FF000000	00000019				1951	DC	XL16'FF00000000000019 00000038000000FA'	v3
00002EA4	00000038	000000FA							
00002EAC	00000000	00000000				1952	DC	XL16'0000000000000000 0000000000000000'	v4
00002EB4	00000000	00000000							
						1953			
						1954	VRR_D	VMAL,2	
00002EC0						1955+	DS	0FD	
00002EC0				00002EC0		1956+	USING	*,R5	base for test data and test routine
00002EC0	00002F08					1957+T41	DC	A(X41)	address of test routine
00002EC4	0029					1958+	DC	H'41'	test number
00002EC6	00					1959+	DC	X'00'	
00002EC7	02					1960+	DC	HL1'2'	m5
00002EC8	E5D4C1D3	40404040				1961+	DC	CL8'VMAL'	instruction name
00002ED0	00002F4C					1962+	DC	A(RE41+16)	address of v2 source
00002ED4	00002F5C					1963+	DC	A(RE41+32)	address of v3 source
00002ED8	00002F6C					1964+	DC	A(RE41+48)	address of v4 source
00002EDC	00000010					1965+	DC	A(16)	result length
00002EE0	00002F3C					1966+REA41	DC	A(RE41)	result address
00002EE8	00000000	00000000				1967+	DS	FD	gap
00002EF0	00000000	00000000				1968+V1041	DS	XL16	V1 output
00002EF8	00000000	00000000							
00002F00	00000000	00000000				1969+	DS	FD	gap
						1970+*			
00002F08						1971+X41	DS	0F	
00002F08	E310	5010	0014		00000010	1972+	LGF	R1,V2ADDR	load v2 source
00002F0E	E761	0000	0806		00000000	1973+	VL	v22,0(R1)	use v22 to test decoder
00002F14	E310	5014	0014		00000014	1974+	LGF	R1,V3ADDR	load v3 source
00002F1A	E771	0000	0806		00000000	1975+	VL	v23,0(R1)	use v23 to test decoder
00002F20	E310	5018	0014		00000018	1976+	LGF	R1,V4ADDR	load v4 source
00002F26	E781	0000	0806		00000000	1977+	VL	v24,0(R1)	use v24 to test decoder
00002F2C	E766	7200	8FAA			1978+	VMAL	V22,V22,V23,V24,2	test instruction (dest is a source)
00002F32	E760	5030	080E		00002EF0	1979+	VST	V22,V1041	save v1 output
00002F38	07FB					1980+	BR	R11	return
00002F3C						1981+RE41	DC	0F	xl16 expected result
00002F3C						1982+	DROP	R5	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00002F3C	00000100 000006C0			1983	DC	XL16'000001000000006C0 00000C430000F426'	result
00002F44	00000C43 0000F426						
00002F4C	FF0000FF 00000029			1984	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00002F54	00000038 000000FA						
00002F5C	FF000001 00000029			1985	DC	XL16'FF00000100000029 00000038000000FA'	v3
00002F64	00000038 000000FA						
00002F6C	00000001 0000002F			1986	DC	XL16'000000010000002F 0000000300000002'	v4
00002F74	00000003 00000002						
				1987			
				1988	VRR_D	VMAL, 2	
00002F80				1989+	DS	0FD	
00002F80		00002F80		1990+	USING	*, R5	base for test data and test routine
00002F80	00002FC8			1991+T42	DC	A(X42)	address of test routine
00002F84	002A			1992+	DC	H'42'	test number
00002F86	00			1993+	DC	X'00'	
00002F87	02			1994+	DC	HL1'2'	m5
00002F88	E5D4C1D3 40404040			1995+	DC	CL8'VMAL'	instruction name
00002F90	0000300C			1996+	DC	A(RE42+16)	address of v2 source
00002F94	0000301C			1997+	DC	A(RE42+32)	address of v3 source
00002F98	0000302C			1998+	DC	A(RE42+48)	address of v4 source
00002F9C	00000010			1999+	DC	A(16)	result length
00002FA0	00002FFC			2000+REA42	DC	A(RE42)	result address
00002FA8	00000000 00000000			2001+	DS	FD	gap
00002FB0	00000000 00000000			2002+V1042	DS	XL16	V1 output
00002FB8	00000000 00000000						
00002FC0	00000000 00000000			2003+	DS	FD	gap
				2004+*			
00002FC8				2005+X42	DS	0F	
00002FC8	E310 5010 0014		00000010	2006+	LGF	R1, V2ADDR	load v2 source
00002FCE	E761 0000 0806		00000000	2007+	VL	v22, 0(R1)	use v22 to test decoder
00002FD4	E310 5014 0014		00000014	2008+	LGF	R1, V3ADDR	load v3 source
00002FDA	E771 0000 0806		00000000	2009+	VL	v23, 0(R1)	use v23 to test decoder
00002FE0	E310 5018 0014		00000018	2010+	LGF	R1, V4ADDR	load v4 source
00002FE6	E781 0000 0806		00000000	2011+	VL	v24, 0(R1)	use v24 to test decoder
00002FEC	E766 7200 8FAA			2012+	VMAL	V22, V22, V23, V24, 2	test instruction (dest is a source)
00002FF2	E760 5030 080E		00002FB0	2013+	VST	V22, V1042	save v1 output
00002FF8	07FB			2014+	BR	R11	return
00002FFC				2015+RE42	DC	0F	xl16 expected result
00002FFC				2016+	DROP	R5	
00002FFC	031B1B14 A9977748			2017	DC	XL16'031B1B14A9977748 BE74139C53B0F010'	result
00003004	BE74139C 53B0F010						
0000300C	FF020304 05060708			2018	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00003014	090A0B0C 0D0E0F10						
0000301C	FF020304 05060708			2019	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00003024	090A0B0C 0D0E0F10						
0000302C	FF020304 05060708			2020	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00003034	090A0B0C 0D0E0F10						
				2021			
				2022	VRR_D	VMAL, 2	
00003040				2023+	DS	0FD	
00003040		00003040		2024+	USING	*, R5	base for test data and test routine
00003040	00003088			2025+T43	DC	A(X43)	address of test routine
00003044	002B			2026+	DC	H'43'	test number
00003046	00			2027+	DC	X'00'	
00003047	02			2028+	DC	HL1'2'	m5
00003048	E5D4C1D3 40404040			2029+	DC	CL8'VMAL'	instruction name

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003050	000030CC			2030+	DC	A(RE43+16)	address of v2 source
00003054	000030DC			2031+	DC	A(RE43+32)	address of v3 source
00003058	000030EC			2032+	DC	A(RE43+48)	address of v4 source
0000305C	00000010			2033+	DC	A(16)	result length
00003060	000030BC			2034+REA43	DC	A(RE43)	result address
00003068	00000000 00000000			2035+	DS	FD	gap
00003070	00000000 00000000			2036+V1043	DS	XL16	V1 output
00003078	00000000 00000000						
00003080	00000000 00000000			2037+	DS	FD	gap
				2038+*			
00003088				2039+X43	DS	0F	
00003088	E310 5010 0014		00000010	2040+	LGF	R1,V2ADDR	load v2 source
0000308E	E761 0000 0806		00000000	2041+	VL	v22,0(R1)	use v22 to test decoder
00003094	E310 5014 0014		00000014	2042+	LGF	R1,V3ADDR	load v3 source
0000309A	E771 0000 0806		00000000	2043+	VL	v23,0(R1)	use v23 to test decoder
000030A0	E310 5018 0014		00000018	2044+	LGF	R1,V4ADDR	load v4 source
000030A6	E781 0000 0806		00000000	2045+	VL	v24,0(R1)	use v24 to test decoder
000030AC	E766 7200 8FAA			2046+	VMAL	V22,V22,V23,V24,2	test instruction (dest is a source)
000030B2	E760 5030 080E		00003070	2047+	VST	V22,V1043	save v1 output
000030B8	07FB			2048+	BR	R11	return
000030BC				2049+RE43	DC	0F	xl16 expected result
000030BC				2050+	DROP	R5	
000030BC	FE0D0D0C 504B3B28			2051	DC	XL16'FE0D0D0C504B3B28 D8B98954A157F790'	result
000030C4	D8B98954 A157F790						
000030CC	FF020304 05060708			2052	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000030D4	090A0B0C 0D0E0F10						
000030DC	FF010102 02030304			2053	DC	XL16'FF01010202030304 0405050606070708'	v3
000030E4	04050506 06070708						
000030EC	FF020304 05060708			2054	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000030F4	090A0B0C 0D0E0F10						
				2055			
00003100				2056	VRR_D	VMAL,2	
00003100		00003100		2057+	DS	0FD	
00003100	00003148			2058+	USING	*,R5	base for test data and test routine
00003104	002C			2059+T44	DC	A(X44)	address of test routine
00003106	00			2060+	DC	H'44'	test number
00003106	00			2061+	DC	X'00'	
00003107	02			2062+	DC	HL1'2'	m5
00003108	E5D4C1D3 40404040			2063+	DC	CL8'VMAL'	instruction name
00003110	0000318C			2064+	DC	A(RE44+16)	address of v2 source
00003114	0000319C			2065+	DC	A(RE44+32)	address of v3 source
00003118	000031AC			2066+	DC	A(RE44+48)	address of v4 source
0000311C	00000010			2067+	DC	A(16)	result length
00003120	0000317C			2068+REA44	DC	A(RE44)	result address
00003128	00000000 00000000			2069+	DS	FD	gap
00003130	00000000 00000000			2070+V1044	DS	XL16	V1 output
00003138	00000000 00000000						
00003140	00000000 00000000			2071+	DS	FD	gap
				2072+*			
00003148				2073+X44	DS	0F	
00003148	E310 5010 0014		00000010	2074+	LGF	R1,V2ADDR	load v2 source
0000314E	E761 0000 0806		00000000	2075+	VL	v22,0(R1)	use v22 to test decoder
00003154	E310 5014 0014		00000014	2076+	LGF	R1,V3ADDR	load v3 source
0000315A	E771 0000 0806		00000000	2077+	VL	v23,0(R1)	use v23 to test decoder
00003160	E310 5018 0014		00000018	2078+	LGF	R1,V4ADDR	load v4 source
00003166	E781 0000 0806		00000000	2079+	VL	v24,0(R1)	use v24 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000316C	E766 7200 8FAA			2080+	VMAL	V22,V22,V23,V24,2	test instruction (dest is a source)
00003172	E760 5030 080E		00003130	2081+	VST	V22,V1044	save v1 output
00003178	07FB			2082+	BR	R11	return
0000317C				2083+RE44	DC	0F	xl16 expected result
0000317C				2084+	DROP	R5	
0000317C	FB020304 0A0C0E10			2085	DC	XL16'FB0203040A0C0E10 332B221854493D30'	result
00003184	332B2218 54493D30						
0000318C	FF020304 05060708			2086	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00003194	090A0B0C 0D0E0F10						
0000319C	FF000000 00000001			2087	DC	XL16'FF00000000000001 0101010101010102'	v3
000031A4	01010101 01010102						
000031AC	FF020304 05060708			2088	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000031B4	090A0B0C 0D0E0F10						
				2089			
				2090 * Doubleword			
000031C0				2091	VRR_D	VMAL,3	
000031C0		000031C0		2092+	DS	0FD	
000031C0	00003208			2093+	USING	*,R5	base for test data and test routine
000031C4	002D			2094+T45	DC	A(X45)	address of test routine
000031C6	00			2095+	DC	H'45'	test number
000031C7	03			2096+	DC	X'00'	
000031C8	E5D4C1D3 40404040			2097+	DC	HL1'3'	m5
000031D0	0000324C			2098+	DC	CL8'VMAL'	instruction name
000031D4	0000325C			2099+	DC	A(RE45+16)	address of v2 source
000031D8	0000326C			2100+	DC	A(RE45+32)	address of v3 source
000031DC	00000010			2101+	DC	A(RE45+48)	address of v4 source
000031E0	00000010			2102+	DC	A(16)	result length
000031E8	0000323C			2103+REA45	DC	A(RE45)	result address
000031E8	00000000 00000000			2104+	DS	FD	gap
000031F0	00000000 00000000			2105+V1045	DS	XL16	V1 output
000031F8	00000000 00000000						
00003200	00000000 00000000			2106+	DS	FD	gap
				2107+*			
00003208				2108+X45	DS	0F	
00003208	E310 5010 0014		00000010	2109+	LGF	R1,V2ADDR	load v2 source
0000320E	E761 0000 0806		00000000	2110+	VL	v22,0(R1)	use v22 to test decoder
00003214	E310 5014 0014		00000014	2111+	LGF	R1,V3ADDR	load v3 source
0000321A	E771 0000 0806		00000000	2112+	VL	v23,0(R1)	use v23 to test decoder
00003220	E310 5018 0014		00000018	2113+	LGF	R1,V4ADDR	load v4 source
00003226	E781 0000 0806		00000000	2114+	VL	v24,0(R1)	use v24 to test decoder
0000322C	E766 7300 8FAA			2115+	VMAL	V22,V22,V23,V24,3	test instruction (dest is a source)
00003232	E760 5030 080E		000031F0	2116+	VST	V22,V1045	save v1 output
00003238	07FB			2117+	BR	R11	return
0000323C				2118+RE45	DC	0F	xl16 expected result
0000323C				2119+	DROP	R5	
0000323C	FFFCE002 71000000			2120	DC	XL16'FFFCE00271000000 96789F9F4FEDCC24'	result
00003244	96789F9F 4FEDCC24						
0000324C	FFFFFFFF 00019000			2121	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00003254	00000038 EEEEEEEFA						
0000325C	FFFFFFFF 00019000			2122	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
00003264	00000038 0EEEEEEFA						
0000326C	00000000 00000000			2123	DC	XL16'0000000000000000 0000000000000000'	v4
00003274	00000000 00000000						
				2124			
				2125	VRR_D	VMAL,3	
00003280				2126+	DS	0FD	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00003280		00003280		2127+	USING *,R5	base for test data and test routine
00003280	000032C8			2128+T46	DC A(X46)	address of test routine
00003284	002E			2129+	DC H'46'	test number
00003286	00			2130+	DC X'00'	
00003287	03			2131+	DC HL1'3'	m5
00003288	E5D4C1D3 40404040			2132+	DC CL8'VMAL'	instruction name
00003290	0000330C			2133+	DC A(RE46+16)	address of v2 source
00003294	0000331C			2134+	DC A(RE46+32)	address of v3 source
00003298	0000332C			2135+	DC A(RE46+48)	address of v4 source
0000329C	00000010			2136+	DC A(16)	result length
000032A0	000032FC			2137+REA46	DC A(RE46)	result address
000032A8	00000000 00000000			2138+	DS FD	gap
000032B0	00000000 00000000			2139+V1046	DS XL16	V1 output
000032B8	00000000 00000000					
000032C0	00000000 00000000			2140+	DS FD	gap
				2141+*		
000032C8				2142+X46	DS 0F	
000032C8	E310 5010 0014		00000010	2143+	LGF R1,V2ADDR	load v2 source
000032CE	E761 0000 0806		00000000	2144+	VL v22,0(R1)	use v22 to test decoder
000032D4	E310 5014 0014		00000014	2145+	LGF R1,V3ADDR	load v3 source
000032DA	E771 0000 0806		00000000	2146+	VL v23,0(R1)	use v23 to test decoder
000032E0	E310 5018 0014		00000018	2147+	LGF R1,V4ADDR	load v4 source
000032E6	E781 0000 0806		00000000	2148+	VL v24,0(R1)	use v24 to test decoder
000032EC	E766 7300 8FAA			2149+	VMAL V22,V22,V23,V24,3	test instruction (dest is a source)
000032F2	E760 5030 080E		000032B0	2150+	VST V22,V1046	save v1 output
000032F8	07FB			2151+	BR R11	return
000032FC				2152+RE46	DC 0F	xl16 expected result
000032FC				2153+	DROP R5	
000032FC	69B556ED 77F57901			2154	DC XL16'69B556ED77F57901 152B55D498D42102'	result
00003304	152B55D4 98D42102					
0000330C	FF020304 05060750			2155	DC XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00003314	090A0B0C 0D0E0F7F					
0000331C	01020304 05060750			2156	DC XL16'0102030405060750 090A0B780D0E0F7F'	v3
00003324	090A0B78 0D0E0F7F					
0000332C	00000000 00000001			2157	DC XL16'0000000000000001 0000000000000001'	v4
00003334	00000000 00000001					
				2158		
				2159	VRR_D VMAL,3	
00003340				2160+	DS 0FD	
00003340		00003340		2161+	USING *,R5	base for test data and test routine
00003340	00003388			2162+T47	DC A(X47)	address of test routine
00003344	002F			2163+	DC H'47'	test number
00003346	00			2164+	DC X'00'	
00003347	03			2165+	DC HL1'3'	m5
00003348	E5D4C1D3 40404040			2166+	DC CL8'VMAL'	instruction name
00003350	000033CC			2167+	DC A(RE47+16)	address of v2 source
00003354	000033DC			2168+	DC A(RE47+32)	address of v3 source
00003358	000033EC			2169+	DC A(RE47+48)	address of v4 source
0000335C	00000010			2170+	DC A(16)	result length
00003360	000033BC			2171+REA47	DC A(RE47)	result address
00003368	00000000 00000000			2172+	DS FD	gap
00003370	00000000 00000000			2173+V1047	DS XL16	V1 output
00003378	00000000 00000000					
00003380	00000000 00000000			2174+	DS FD	gap
				2175+*		
00003388				2176+X47	DS 0F	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00003388	E310 5010 0014		00000010	2177+	LGF	R1,V2ADDR	load v2 source	
0000338E	E761 0000 0806		00000000	2178+	VL	v22,0(R1)	use v22 to test decoder	
00003394	E310 5014 0014		00000014	2179+	LGF	R1,V3ADDR	load v3 source	
0000339A	E771 0000 0806		00000000	2180+	VL	v23,0(R1)	use v23 to test decoder	
000033A0	E310 5018 0014		00000018	2181+	LGF	R1,V4ADDR	load v4 source	
000033A6	E781 0000 0806		00000000	2182+	VL	v24,0(R1)	use v24 to test decoder	
000033AC	E766 7300 8FAA			2183+	VMAL	V22,V22,V23,V24,3	test instruction (dest is a source)	
000033B2	E760 5030 080E		00003370	2184+	VST	V22,V1047	save v1 output	
000033B8	07FB			2185+	BR	R11	return	
000033BC				2186+RE47	DC	0F	xl16 expected result	
000033BC				2187+	DROP	R5		
000033BC	26D2FE70 90F71480			2188	DC	XL16'26D2FE7090F71480 B47CD8D5FF5B4940'	result	
000033C4	B47CD8D5 FF5B4940							
000033CC	FF020304 05060750			2189	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
000033D4	090A0B0C 0D0E0F7F							
000033DC	00010102 02030328			2190	DC	XL16'0001010202030328 0405053C0607073F'	v3	
000033E4	0405053C 0607073F							
000033EC	20000000 00000000			2191	DC	XL16'2000000000000000 FFFFFFFF'	v4	
000033F4	FFFFFFFF FFFFFFFF							
				2192				
				2193	VRR_D	VMAL,3		
00003400				2194+	DS	0FD		
00003400		00003400		2195+	USING	*,R5	base for test data and test routine	
00003400	00003448			2196+T48	DC	A(X48)	address of test routine	
00003404	0030			2197+	DC	H'48'	test number	
00003406	00			2198+	DC	X'00'		
00003407	03			2199+	DC	HL1'3'	m5	
00003408	E5D4C1D3 40404040			2200+	DC	CL8'VMAL'	instruction name	
00003410	0000348C			2201+	DC	A(RE48+16)	address of v2 source	
00003414	0000349C			2202+	DC	A(RE48+32)	address of v3 source	
00003418	000034AC			2203+	DC	A(RE48+48)	address of v4 source	
0000341C	00000010			2204+	DC	A(16)	result length	
00003420	0000347C			2205+REA48	DC	A(RE48)	result address	
00003428	00000000 00000000			2206+	DS	FD	gap	
00003430	00000000 00000000			2207+V1048	DS	XL16	V1 output	
00003438	00000000 00000000							
00003440	00000000 00000000			2208+	DS	FD	gap	
				2209+*				
00003448				2210+X48	DS	0F		
00003448	E310 5010 0014		00000010	2211+	LGF	R1,V2ADDR	load v2 source	
0000344E	E761 0000 0806		00000000	2212+	VL	v22,0(R1)	use v22 to test decoder	
00003454	E310 5014 0014		00000014	2213+	LGF	R1,V3ADDR	load v3 source	
0000345A	E771 0000 0806		00000000	2214+	VL	v23,0(R1)	use v23 to test decoder	
00003460	E310 5018 0014		00000018	2215+	LGF	R1,V4ADDR	load v4 source	
00003466	E781 0000 0806		00000000	2216+	VL	v24,0(R1)	use v24 to test decoder	
0000346C	E766 7300 8FAA			2217+	VMAL	V22,V22,V23,V24,3	test instruction (dest is a source)	
00003472	E760 5030 080E		00003430	2218+	VST	V22,V1048	save v1 output	
00003478	07FB			2219+	BR	R11	return	
0000347C				2220+RE48	DC	0F	xl16 expected result	
0000347C				2221+	DROP	R5		
0000347C	F6142E28 323C4921			2222	DC	XL16'F6142E28323C4921 191C345060616772'	result	
00003484	191C3450 60616772							
0000348C	FF020304 05060750			2223	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
00003494	090A0B0C 0D0E0F7F							
0000349C	00000000 0000000A			2224	DC	XL16'000000000000000A 0101010F0101010F'	v3	
000034A4	0101010F 0101010F							

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000034AC	00001000 00000001			2225	DC	XL16'00001000000000001 1000000000000001'	v4
000034B4	10000000 00000001						
				2226			
				2227	* Quadword		
				2228	VRR_D	VMAL, 4	
000034C0				2229+	DS	0FD	
000034C0		000034C0		2230+	USING	*, R5	base for test data and test routine
000034C0	00003508			2231+T49	DC	A(X49)	address of test routine
000034C4	0031			2232+	DC	H'49'	test number
000034C6	00			2233+	DC	X'00'	
000034C7	04			2234+	DC	HL1'4'	m5
000034C8	E5D4C1D3 40404040			2235+	DC	CL8'VMAL'	instruction name
000034D0	0000354C			2236+	DC	A(RE49+16)	address of v2 source
000034D4	0000355C			2237+	DC	A(RE49+32)	address of v3 source
000034D8	0000356C			2238+	DC	A(RE49+48)	address of v4 source
000034DC	00000010			2239+	DC	A(16)	result length
000034E0	0000353C			2240+REA49	DC	A(RE49)	result address
000034E8	00000000 00000000			2241+	DS	FD	gap
000034F0	00000000 00000000			2242+V1049	DS	XL16	V1 output
000034F8	00000000 00000000						
00003500	00000000 00000000			2243+	DS	FD	gap
				2244+*			
00003508				2245+X49	DS	0F	
00003508	E310 5010 0014		00000010	2246+	LGF	R1, V2ADDR	load v2 source
0000350E	E761 0000 0806		00000000	2247+	VL	v22, 0(R1)	use v22 to test decoder
00003514	E310 5014 0014		00000014	2248+	LGF	R1, V3ADDR	load v3 source
0000351A	E771 0000 0806		00000000	2249+	VL	v23, 0(R1)	use v23 to test decoder
00003520	E310 5018 0014		00000018	2250+	LGF	R1, V4ADDR	load v4 source
00003526	E781 0000 0806		00000000	2251+	VL	v24, 0(R1)	use v24 to test decoder
0000352C	E766 7400 8FAA			2252+	VMAL	V22, V22, V23, V24, 4	test instruction (dest is a source)
00003532	E760 5030 080E		000034F0	2253+	VST	V22, V1049	save v1 output
00003538	07FB			2254+	BR	R11	return
0000353C				2255+RE49	DC	0F	xl16 expected result
0000353C				2256+	DROP	R5	
0000353C	02D2AEB6 AACD4C77			2257	DC	XL16'02D2AEB6AACD4C77 96789F9F4FEDCC24'	result
00003544	96789F9F 4FEDCC24						
0000354C	FFFFFFFF 00019000			2258	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00003554	00000038 EEEEEEEFA						
0000355C	FFFFFFFF 00019000			2259	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
00003564	00000038 0EEEEEEFA						
0000356C	00000000 00000000			2260	DC	XL16'0000000000000000 0000000000000000'	v4
00003574	00000000 00000000						
				2261			
				2262	VRR_D	VMAL, 4	
00003580				2263+	DS	0FD	
00003580		00003580		2264+	USING	*, R5	base for test data and test routine
00003580	000035C8			2265+T50	DC	A(X50)	address of test routine
00003584	0032			2266+	DC	H'50'	test number
00003586	00			2267+	DC	X'00'	
00003587	04			2268+	DC	HL1'4'	m5
00003588	E5D4C1D3 40404040			2269+	DC	CL8'VMAL'	instruction name
00003590	0000360C			2270+	DC	A(RE50+16)	address of v2 source
00003594	0000361C			2271+	DC	A(RE50+32)	address of v3 source
00003598	0000362C			2272+	DC	A(RE50+48)	address of v4 source
0000359C	00000010			2273+	DC	A(16)	result length
000035A0	000035FC			2274+REA50	DC	A(RE50)	result address

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
000035A8	00000000	00000000				2275+	DS	FD	gap
000035B0	00000000	00000000				2276+V1050	DS	XL16	V1 output
000035B8	00000000	00000000							
000035C0	00000000	00000000				2277+	DS	FD	gap
						2278+*			
000035C8						2279+X50	DS	0F	
000035C8	E310	5010	0014		00000010	2280+	LGF	R1,V2ADDR	load v2 source
000035CE	E761	0000	0806		00000000	2281+	VL	v22,0(R1)	use v22 to test decoder
000035D4	E310	5014	0014		00000014	2282+	LGF	R1,V3ADDR	load v3 source
000035DA	E771	0000	0806		00000000	2283+	VL	v23,0(R1)	use v23 to test decoder
000035E0	E310	5018	0014		00000018	2284+	LGF	R1,V4ADDR	load v4 source
000035E6	E781	0000	0806		00000000	2285+	VL	v24,0(R1)	use v24 to test decoder
000035EC	E766	7400	8FAA			2286+	VMAL	V22,V22,V23,V24,4	test instruction (dest is a source)
000035F2	E760	5030	080E		000035B0	2287+	VST	V22,V1050	save v1 output
000035F8	07FB					2288+	BR	R11	return
000035FC						2289+RE50	DC	0F	xl16 expected result
000035FC						2290+	DROP	R5	
000035FC	599AFA6A	24295657				2291	DC	XL16'599AFA6A24295657 252B55D498D42102'	result
00003604	252B55D4	98D42102							
0000360C	FF020304	05060750				2292	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00003614	090A0B0C	0D0E0F7F							
0000361C	01020304	05060750				2293	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3
00003624	090A0B78	0D0E0F7F							
0000362C	10000000	00000001				2294	DC	XL16'1000000000000001 1000000000000001'	v4
00003634	10000000	00000001							
						2295			
						2296	VRR_D	VMAL,4	
00003640						2297+	DS	0FD	
00003640				00003640		2298+	USING	*,R5	base for test data and test routine
00003640	00003688					2299+T51	DC	A(X51)	address of test routine
00003644	0033					2300+	DC	H'51'	test number
00003646	00					2301+	DC	X'00'	
00003647	04					2302+	DC	HL1'4'	m5
00003648	E5D4C1D3	40404040				2303+	DC	CL8'VMAL'	instruction name
00003650	000036CC					2304+	DC	A(RE51+16)	address of v2 source
00003654	000036DC					2305+	DC	A(RE51+32)	address of v3 source
00003658	000036EC					2306+	DC	A(RE51+48)	address of v4 source
0000365C	00000010					2307+	DC	A(16)	result length
00003660	000036BC					2308+REA51	DC	A(RE51)	result address
00003668	00000000	00000000				2309+	DS	FD	gap
00003670	00000000	00000000				2310+V1051	DS	XL16	V1 output
00003678	00000000	00000000							
00003680	00000000	00000000				2311+	DS	FD	gap
						2312+*			
00003688						2313+X51	DS	0F	
00003688	E310	5010	0014		00000010	2314+	LGF	R1,V2ADDR	load v2 source
0000368E	E761	0000	0806		00000000	2315+	VL	v22,0(R1)	use v22 to test decoder
00003694	E310	5014	0014		00000014	2316+	LGF	R1,V3ADDR	load v3 source
0000369A	E771	0000	0806		00000000	2317+	VL	v23,0(R1)	use v23 to test decoder
000036A0	E310	5018	0014		00000018	2318+	LGF	R1,V4ADDR	load v4 source
000036A6	E781	0000	0806		00000000	2319+	VL	v24,0(R1)	use v24 to test decoder
000036AC	E766	7400	8FAA			2320+	VMAL	V22,V22,V23,V24,4	test instruction (dest is a source)
000036B2	E760	5030	080E		00003670	2321+	VST	V22,V1051	save v1 output
000036B8	07FB					2322+	BR	R11	return
000036BC						2323+RE51	DC	0F	xl16 expected result
000036BC						2324+	DROP	R5	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000036BC	A5AC7D14 92F5ADEA			2325	DC	XL16 'A5AC7D1492F5ADEA B47CD8D5FF5B4940'	result
000036C4	B47CD8D5 FF5B4940						
000036CC	FF020304 05060750			2326	DC	XL16 'FF02030405060750 090A0B0C0D0E0F7F'	v2
000036D4	090A0B0C 0D0E0F7F						
000036DC	00010102 02030328			2327	DC	XL16 '0001010202030328 0405053C0607073F'	v3
000036E4	0405053C 0607073F						
000036EC	FFFFFFFF FFFFFFFF			2328	DC	XL16 'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF'	v4
000036F4	FFFFFFFF FFFFFFFF						
				2329			
00003700				2330	VRR_D	VMAL, 4	
00003700		00003700		2331+	DS	0FD	
00003700				2332+	USING	*, R5	base for test data and test routine
00003700	00003748			2333+T52	DC	A(X52)	address of test routine
00003704	0034			2334+	DC	H'52'	test number
00003706	00			2335+	DC	X'00'	
00003707	04			2336+	DC	HL1'4'	m5
00003708	E5D4C1D3 40404040			2337+	DC	CL8'VMAL'	instruction name
00003710	0000378C			2338+	DC	A(RE52+16)	address of v2 source
00003714	0000379C			2339+	DC	A(RE52+32)	address of v3 source
00003718	000037AC			2340+	DC	A(RE52+48)	address of v4 source
0000371C	00000010			2341+	DC	A(16)	result length
00003720	0000377C			2342+REA52	DC	A(RE52)	result address
00003728	00000000 00000000			2343+	DS	FD	gap
00003730	00000000 00000000			2344+V1052	DS	XL16	V1 output
00003738	00000000 00000000						
00003740	00000000 00000000			2345+	DS	FD	gap
				2346+*			
00003748				2347+X52	DS	0F	
00003748	E310 5010 0014		00000010	2348+	LGF	R1, V2ADDR	load v2 source
0000374E	E761 0000 0806		00000000	2349+	VL	v22, 0(R1)	use v22 to test decoder
00003754	E310 5014 0014		00000014	2350+	LGF	R1, V3ADDR	load v3 source
0000375A	E771 0000 0806		00000000	2351+	VL	v23, 0(R1)	use v23 to test decoder
00003760	E310 5018 0014		00000018	2352+	LGF	R1, V4ADDR	load v4 source
00003766	E781 0000 0806		00000000	2353+	VL	v24, 0(R1)	use v24 to test decoder
0000376C	E766 7400 8FAA			2354+	VMAL	V22, V22, V23, V24, 4	test instruction (dest is a source)
00003772	E760 5030 080E		00003730	2355+	VST	V22, V1052	save v1 output
00003778	07FB			2356+	BR	R11	return
0000377C				2357+RE52	DC	0F	xl16 expected result
0000377C				2358+	DROP	R5	
0000377C	FD497B95 D40238A4			2359	DC	XL16 'FD497B95D40238A4 091C34506061676E'	result
00003784	091C3450 6061676E						
0000378C	FF020304 05060750			2360	DC	XL16 'FF02030405060750 090A0B0C0D0E0F7F'	v2
00003794	090A0B0C 0D0E0F7F						
0000379C	00000000 0000000A			2361	DC	XL16 '0000000000000000A 0101010F0101010F'	v3
000037A4	0101010F 0101010F						
000037AC	FFFFFFFF FFFFFFFF			2362	DC	XL16 'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFD'	v4
000037B4	FFFFFFFF FFFFFFFD						
				2363			
				2364	*-----		
				2365	* VMAH - Vector Multiply and Add High		
				2366	*-----		
				2367	* Byte		
				2368	VRR_D	VMAH, 0	
000037C0				2369+	DS	0FD	
000037C0		000037C0		2370+	USING	*, R5	base for test data and test routine
000037C0	00003808			2371+T53	DC	A(X53)	address of test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000037C4	0035			2372+	DC	H'53'	test number
000037C6	00			2373+	DC	X'00'	
000037C7	00			2374+	DC	HL1'0'	m5
000037C8	E5D4C1C8 40404040			2375+	DC	CL8'VMAH'	instruction name
000037D0	0000384C			2376+	DC	A(RE53+16)	address of v2 source
000037D4	0000385C			2377+	DC	A(RE53+32)	address of v3 source
000037D8	0000386C			2378+	DC	A(RE53+48)	address of v4 source
000037DC	00000010			2379+	DC	A(16)	result length
000037E0	0000383C			2380+REA53	DC	A(RE53)	result address
000037E8	00000000 00000000			2381+	DS	FD	gap
000037F0	00000000 00000000			2382+V1053	DS	XL16	V1 output
000037F8	00000000 00000000						
00003800	00000000 00000000			2383+	DS	FD	gap
				2384+*			
00003808				2385+X53	DS	0F	
00003808	E310 5010 0014		00000010	2386+	LGF	R1,V2ADDR	load v2 source
0000380E	E761 0000 0806		00000000	2387+	VL	v22,0(R1)	use v22 to test decoder
00003814	E310 5014 0014		00000014	2388+	LGF	R1,V3ADDR	load v3 source
0000381A	E771 0000 0806		00000000	2389+	VL	v23,0(R1)	use v23 to test decoder
00003820	E310 5018 0014		00000018	2390+	LGF	R1,V4ADDR	load v4 source
00003826	E781 0000 0806		00000000	2391+	VL	v24,0(R1)	use v24 to test decoder
0000382C	E766 7000 8FAB			2392+	VMAH	V22,V22,V23,V24,0	test instruction (dest is a source)
00003832	E760 5030 080E		000037F0	2393+	VST	V22,V1053	save v1 output
00003838	07FB			2394+	BR	R11	return
0000383C				2395+RE53	DC	0F	xl16 expected result
0000383C				2396+	DROP	R5	
0000383C	00000000 00000002			2397	DC	XL16'0000000000000002 0000000C00000000'	result
00003844	0000000C 00000000						
0000384C	FF000000 00000019			2398	DC	XL16'FF00000000000019 00000038000000FA'	v2
00003854	00000038 000000FA						
0000385C	FF000000 00000019			2399	DC	XL16'FF00000000000019 00000038000000FA'	v3
00003864	00000038 000000FA						
0000386C	00000000 00000000			2400	DC	XL16'0000000000000000 0000000000000000'	v4
00003874	00000000 00000000						
				2401			
00003880				2402	VRR_D	VMAH,0	
00003880		00003880		2403+	DS	0FD	
00003880	000038C8			2404+	USING	*,R5	base for test data and test routine
00003884	0036			2405+T54	DC	A(X54)	address of test routine
00003886	00			2406+	DC	H'54'	test number
00003887	00			2407+	DC	X'00'	
00003887	00			2408+	DC	HL1'0'	m5
00003888	E5D4C1C8 40404040			2409+	DC	CL8'VMAH'	instruction name
00003890	0000390C			2410+	DC	A(RE54+16)	address of v2 source
00003894	0000391C			2411+	DC	A(RE54+32)	address of v3 source
00003898	0000392C			2412+	DC	A(RE54+48)	address of v4 source
0000389C	00000010			2413+	DC	A(16)	result length
000038A0	000038FC			2414+REA54	DC	A(RE54)	result address
000038A8	00000000 00000000			2415+	DS	FD	gap
000038B0	00000000 00000000			2416+V1054	DS	XL16	V1 output
000038B8	00000000 00000000						
000038C0	00000000 00000000			2417+	DS	FD	gap
				2418+*			
000038C8				2419+X54	DS	0F	
000038C8	E310 5010 0014		00000010	2420+	LGF	R1,V2ADDR	load v2 source
000038CE	E761 0000 0806		00000000	2421+	VL	v22,0(R1)	use v22 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000038D4	E310 5014 0014		00000014	2422+	LGF	R1,V3ADDR	load v3 source
000038DA	E771 0000 0806		00000000	2423+	VL	v23,0(R1)	use v23 to test decoder
000038E0	E310 5018 0014		00000018	2424+	LGF	R1,V4ADDR	load v4 source
000038E6	E781 0000 0806		00000000	2425+	VL	v24,0(R1)	use v24 to test decoder
000038EC	E766 7000 8FAB			2426+	VMAH	V22,V22,V23,V24,0	test instruction (dest is a source)
000038F2	E760 5030 080E		000038B0	2427+	VST	V22,V1054	save v1 output
000038F8	07FB			2428+	BR	R11	return
000038FC				2429+RE54	DC	0F	xl16 expected result
000038FC				2430+	DROP	R5	
000038FC	00000000 00000006			2431	DC	XL16'0000000000000006 0000000C00000000'	result
00003904	0000000C 00000000						
0000390C	FF0000FF 00000029			2432	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00003914	00000038 000000FA						
0000391C	FF000001 00000029			2433	DC	XL16'FF00000100000029 00000038000000FA'	v3
00003924	00000038 000000FA						
0000392C	00000001 0000002F			2434	DC	XL16'000000010000002F 0000000300000002'	v4
00003934	00000003 00000002						
00003940				2435			
00003940				2436	VRR_D	VMAH,0	
00003940		00003940		2437+	DS	0FD	
00003940	00003988			2438+	USING	*,R5	base for test data and test routine
00003944	0037			2439+T55	DC	A(X55)	address of test routine
00003946	00			2440+	DC	H'55'	test number
00003947	00			2441+	DC	X'00'	
00003948	E5D4C1C8 40404040			2442+	DC	HL1'0'	m5
00003950	000039CC			2443+	DC	CL8'VMAH'	instruction name
00003954	000039DC			2444+	DC	A(RE55+16)	address of v2 source
00003958	000039EC			2445+	DC	A(RE55+32)	address of v3 source
0000395C	00000010			2446+	DC	A(RE55+48)	address of v4 source
00003960	000039BC			2447+	DC	A(16)	result length
00003968	00000000 00000000			2448+REA55	DC	A(RE55)	result address
00003970	00000000 00000000			2449+	DS	FD	gap
00003978	00000000 00000000			2450+V1055	DS	XL16	V1 output
00003980	00000000 00000000						
00003988				2451+	DS	FD	gap
00003988				2452+*			
00003988	E310 5010 0014		00000010	2453+X55	DS	0F	
0000398E	E761 0000 0806		00000000	2454+	LGF	R1,V2ADDR	load v2 source
00003994	E310 5014 0014		00000014	2455+	VL	v22,0(R1)	use v22 to test decoder
0000399A	E771 0000 0806		00000000	2456+	LGF	R1,V3ADDR	load v3 source
000039A0	E310 5018 0014		00000018	2457+	VL	v23,0(R1)	use v23 to test decoder
000039A6	E781 0000 0806		00000000	2458+	LGF	R1,V4ADDR	load v4 source
000039AC	E766 7000 8FAB			2459+	VL	v24,0(R1)	use v24 to test decoder
000039B2	E760 5030 080E			2460+	VMAH	V22,V22,V23,V24,0	test instruction (dest is a source)
000039B8	07FB		00003970	2461+	VST	V22,V1055	save v1 output
000039BC				2462+	BR	R11	return
000039BC				2463+RE55	DC	0F	xl16 expected result
000039BC				2464+	DROP	R5	
000039BC	FF000000 00000000			2465	DC	XL16'FF00000000000000 00000000000000FF'	result
000039C4	00000000 000000FF						
000039CC	FF020304 05060708			2466	DC	XL16'FF02030405060708 090A0B0C0D0E0FF0'	v2
000039D4	090A0B0C 0D0E0FF0						
000039DC	01020304 05060708			2467	DC	XL16'0102030405060708 090A0B0C0D0E0F10'	v3
000039E4	090A0B0C 0D0E0F10						
000039EC	FF020304 05060708			2468	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000039F4	090A0B0C 0D0E0F10						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				2469		
				2470	VRR_D	VMAH,0
00003A00				2471+	DS	0FD
00003A00		00003A00		2472+	USING	*,R5
00003A00	00003A48			2473+T56	DC	A(X56)
00003A04	0038			2474+	DC	H'56'
00003A06	00			2475+	DC	X'00'
00003A07	00			2476+	DC	HL1'0'
00003A08	E5D4C1C8	40404040		2477+	DC	CL8'VMAH'
00003A10	00003A8C			2478+	DC	A(RE56+16)
00003A14	00003A9C			2479+	DC	A(RE56+32)
00003A18	00003AAC			2480+	DC	A(RE56+48)
00003A1C	00000010			2481+	DC	A(16)
00003A20	00003A7C			2482+REA56	DC	A(RE56)
00003A28	00000000	00000000		2483+	DS	FD
00003A30	00000000	00000000		2484+V1056	DS	XL16
00003A38	00000000	00000000				
00003A40	00000000	00000000		2485+	DS	FD
				2486+*		gap
00003A48				2487+X56	DS	0F
00003A48	E310	5010	0014	2488+	LGF	R1,V2ADDR
00003A4E	E761	0000	0806	2489+	VL	v22,0(R1)
00003A54	E310	5014	0014	2490+	LGF	R1,V3ADDR
00003A5A	E771	0000	0806	2491+	VL	v23,0(R1)
00003A60	E310	5018	0014	2492+	LGF	R1,V4ADDR
00003A66	E781	0000	0806	2493+	VL	v24,0(R1)
00003A6C	E766	7000	8FAB	2494+	VMAH	V22,V22,V23,V24,0
00003A72	E760	5030	080E	2495+	VST	V22,V1056
00003A78	07FB			2496+	BR	R11
00003A7C				2497+RE56	DC	0F
00003A7C				2498+	DROP	R5
00003A7C	FF000000	00000000		2499	DC	XL16'FF00000000000000 00000000000000FF'
00003A84	00000000	000000FF				result
00003A8C	FF020304	05060708		2500	DC	XL16'FF02030405060708 090A0B0C0D0E0FF0'
00003A94	090A0B0C	0D0E0FF0				v2
00003A9C	00010102	02030304		2501	DC	XL16'0001010202030304 0405050606070708'
00003AA4	04050506	06070708				v3
00003AAC	FF020304	05060708		2502	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'
00003AB4	090A0B0C	0D0E0F10				v4
				2503		
				2504	VRR_D	VMAH,0
00003AC0				2505+	DS	0FD
00003AC0		00003AC0		2506+	USING	*,R5
00003AC0	00003B08			2507+T57	DC	A(X57)
00003AC4	0039			2508+	DC	H'57'
00003AC6	00			2509+	DC	X'00'
00003AC7	00			2510+	DC	HL1'0'
00003AC8	E5D4C1C8	40404040		2511+	DC	CL8'VMAH'
00003AD0	00003B4C			2512+	DC	A(RE57+16)
00003AD4	00003B5C			2513+	DC	A(RE57+32)
00003AD8	00003B6C			2514+	DC	A(RE57+48)
00003ADC	00000010			2515+	DC	A(16)
00003AE0	00003B3C			2516+REA57	DC	A(RE57)
00003AE8	00000000	00000000		2517+	DS	FD
00003AF0	00000000	00000000		2518+V1057	DS	XL16
00003AF8	00000000	00000000				V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003B00	00000000 00000000			2519+ 2520+*	DS	FD	gap
00003B08				2521+X57	DS	0F	
00003B08	E310 5010 0014		00000010	2522+	LGF	R1,V2ADDR	load v2 source
00003B0E	E761 0000 0806		00000000	2523+	VL	v22,0(R1)	use v22 to test decoder
00003B14	E310 5014 0014		00000014	2524+	LGF	R1,V3ADDR	load v3 source
00003B1A	E771 0000 0806		00000000	2525+	VL	v23,0(R1)	use v23 to test decoder
00003B20	E310 5018 0014		00000018	2526+	LGF	R1,V4ADDR	load v4 source
00003B26	E781 0000 0806		00000000	2527+	VL	v24,0(R1)	use v24 to test decoder
00003B2C	E766 7000 8FAB			2528+	VMAH	V22,V22,V23,V24,0	test instruction (dest is a source)
00003B32	E760 5030 080E		00003AF0	2529+	VST	V22,V1057	save v1 output
00003B38	07FB			2530+	BR	R11	return
00003B3C				2531+RE57	DC	0F	xl16 expected result
00003B3C				2532+	DROP	R5	
00003B3C	FF000000 00000000			2533	DC	XL16'FF00000000000000 00000000000000FF'	result
00003B44	00000000 000000FF						
00003B4C	FF020304 05060708			2534	DC	XL16'FF02030405060708 090A0B0C0D0E0FF0'	v2
00003B54	090A0B0C 0D0E0FF0						
00003B5C	00000000 00000001			2535	DC	XL16'0000000000000001 0101010101010102'	v3
00003B64	01010101 01010102						
00003B6C	FF020304 05060708			2536	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00003B74	090A0B0C 0D0E0F10						
				2537			
				2538 * Halfword			
00003B80				2539	VRR_D	VMAH,1	
00003B80		00003B80		2540+	DS	0FD	
00003B80	00003BC8			2541+	USING	*,R5	base for test data and test routine
00003B84	003A			2542+T58	DC	A(X58)	address of test routine
00003B86	00			2543+	DC	H'58'	test number
00003B87	01			2544+	DC	X'00'	
00003B88	E5D4C1C8 40404040			2545+	DC	HL1'1'	m5
00003B90	00003C0C			2546+	DC	CL8'VMAH'	instruction name
00003B94	00003C1C			2547+	DC	A(RE58+16)	address of v2 source
00003B98	00003C2C			2548+	DC	A(RE58+32)	address of v3 source
00003B9C	00000010			2549+	DC	A(RE58+48)	address of v4 source
00003BA0	00003BFC			2550+	DC	A(16)	result length
00003BA8	00000000 00000000			2551+REA58	DC	A(RE58)	result address
00003BB0	00000000 00000000			2552+	DS	FD	gap
00003BB8	00000000 00000000			2553+V1058	DS	XL16	V1 output
00003BC0	00000000 00000000						
				2554+	DS	FD	gap
				2555+*			
00003BC8				2556+X58	DS	0F	
00003BC8	E310 5010 0014		00000010	2557+	LGF	R1,V2ADDR	load v2 source
00003BCE	E761 0000 0806		00000000	2558+	VL	v22,0(R1)	use v22 to test decoder
00003BD4	E310 5014 0014		00000014	2559+	LGF	R1,V3ADDR	load v3 source
00003BDA	E771 0000 0806		00000000	2560+	VL	v23,0(R1)	use v23 to test decoder
00003BE0	E310 5018 0014		00000018	2561+	LGF	R1,V4ADDR	load v4 source
00003BE6	E781 0000 0806		00000000	2562+	VL	v24,0(R1)	use v24 to test decoder
00003BEC	E766 7100 8FAB			2563+	VMAH	V22,V22,V23,V24,1	test instruction (dest is a source)
00003BF2	E760 5030 080E		00003BB0	2564+	VST	V22,V1058	save v1 output
00003BF8	07FB			2565+	BR	R11	return
00003BFC				2566+RE58	DC	0F	xl16 expected result
00003BFC				2567+	DROP	R5	
00003BFC	00010000 00000000			2568	DC	XL16'0001000000000000 0000000000000000'	result
00003C04	00000000 00000000						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003C0C	FF000000 00000019			2569	DC	XL16'FF0000000000000019 000000380000000FA'	v2
00003C14	00000038 000000FA						
00003C1C	FF000000 00000019			2570	DC	XL16'FF0000000000000019 000000380000000FA'	v3
00003C24	00000038 000000FA						
00003C2C	00000000 00000000			2571	DC	XL16'0000000000000000 0000000000000000'	v4
00003C34	00000000 00000000						
				2572			
				2573	VRR_D	VMAH,1	
00003C40				2574+	DS	0FD	
00003C40		00003C40		2575+	USING	*,R5	base for test data and test routine
00003C40	00003C88			2576+T59	DC	A(X59)	address of test routine
00003C44	003B			2577+	DC	H'59'	test number
00003C46	00			2578+	DC	X'00'	
00003C47	01			2579+	DC	HL1'1'	m5
00003C48	E5D4C1C8 40404040			2580+	DC	CL8'VMAH'	instruction name
00003C50	00003CCC			2581+	DC	A(RE59+16)	address of v2 source
00003C54	00003CDC			2582+	DC	A(RE59+32)	address of v3 source
00003C58	00003CEC			2583+	DC	A(RE59+48)	address of v4 source
00003C5C	00000010			2584+	DC	A(16)	result length
00003C60	00003CBC			2585+REA59	DC	A(RE59)	result address
00003C68	00000000 00000000			2586+	DS	FD	gap
00003C70	00000000 00000000			2587+V1059	DS	XL16	V1 output
00003C78	00000000 00000000						
00003C80	00000000 00000000			2588+	DS	FD	gap
				2589+*			
00003C88				2590+X59	DS	0F	
00003C88	E310 5010 0014		00000010	2591+	LGF	R1,V2ADDR	load v2 source
00003C8E	E761 0000 0806		00000000	2592+	VL	v22,0(R1)	use v22 to test decoder
00003C94	E310 5014 0014		00000014	2593+	LGF	R1,V3ADDR	load v3 source
00003C9A	E771 0000 0806		00000000	2594+	VL	v23,0(R1)	use v23 to test decoder
00003CA0	E310 5018 0014		00000018	2595+	LGF	R1,V4ADDR	load v4 source
00003CA6	E781 0000 0806		00000000	2596+	VL	v24,0(R1)	use v24 to test decoder
00003CAC	E766 7100 8FAB			2597+	VMAH	V22,V22,V23,V24,1	test instruction (dest is a source)
00003CB2	E760 5030 080E		00003C70	2598+	VST	V22,V1059	save v1 output
00003CB8	07FB			2599+	BR	R11	return
00003CBC				2600+RE59	DC	0F	xl16 expected result
00003CBC				2601+	DROP	R5	
00003CBC	00010000 00000000			2602	DC	XL16'0001000000000000 0000000000000000'	result
00003CC4	00000000 00000000						
00003CCC	FF0000FF 00000029			2603	DC	XL16'FF0000FF00000029 000000380000000FA'	v2
00003CD4	00000038 000000FA						
00003CDC	FF000001 00000029			2604	DC	XL16'FF00000100000029 000000380000000FA'	v3
00003CE4	00000038 000000FA						
00003CEC	00000001 0000002F			2605	DC	XL16'000000010000002F 00000003000000002'	v4
00003CF4	00000003 00000002						
				2606			
				2607	VRR_D	VMAH,1	
00003D00				2608+	DS	0FD	
00003D00		00003D00		2609+	USING	*,R5	base for test data and test routine
00003D00	00003D48			2610+T60	DC	A(X60)	address of test routine
00003D04	003C			2611+	DC	H'60'	test number
00003D06	00			2612+	DC	X'00'	
00003D07	01			2613+	DC	HL1'1'	m5
00003D08	E5D4C1C8 40404040			2614+	DC	CL8'VMAH'	instruction name
00003D10	00003D8C			2615+	DC	A(RE60+16)	address of v2 source
00003D14	00003D9C			2616+	DC	A(RE60+32)	address of v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003D18	00003DAC			2617+	DC	A(RE60+48)	address of v4 source
00003D1C	00000010			2618+	DC	A(16)	result length
00003D20	00003D7C			2619+REA60	DC	A(RE60)	result address
00003D28	00000000 00000000			2620+	DS	FD	gap
00003D30	00000000 00000000			2621+V1060	DS	XL16	V1 output
00003D38	00000000 00000000						
00003D40	00000000 00000000			2622+	DS	FD	gap
				2623+*			
00003D48				2624+X60	DS	0F	
00003D48	E310 5010 0014		00000010	2625+	LGF	R1,V2ADDR	load v2 source
00003D4E	E761 0000 0806		00000000	2626+	VL	v22,0(R1)	use v22 to test decoder
00003D54	E310 5014 0014		00000014	2627+	LGF	R1,V3ADDR	load v3 source
00003D5A	E771 0000 0806		00000000	2628+	VL	v23,0(R1)	use v23 to test decoder
00003D60	E310 5018 0014		00000018	2629+	LGF	R1,V4ADDR	load v4 source
00003D66	E781 0000 0806		00000000	2630+	VL	v24,0(R1)	use v24 to test decoder
00003D6C	E766 7100 8FAB			2631+	VMAH	V22,V22,V23,V24,1	test instruction (dest is a source)
00003D72	E760 5030 080E		00003D30	2632+	VST	V22,V1060	save v1 output
00003D78	07FB			2633+	BR	R11	return
00003D7C				2634+RE60	DC	0F	xl16 expected result
00003D7C				2635+	DROP	R5	
00003D7C	FFFE0009 00190031			2636	DC	XL16'FFFE000900190031 0051007AFF57FF1F'	result
00003D84	0051007A FF57FF1F						
00003D8C	FF020304 05060708			2637	DC	XL16'FF02030405060708 090A0B0CF30EF110'	v2
00003D94	090A0B0C F30EF110						
00003D9C	01020304 05060708			2638	DC	XL16'0102030405060708 090A0B0C0D0E0F10'	v3
00003DA4	090A0B0C 0D0E0F10						
00003DAC	FF020304 05060708			2639	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00003DB4	090A0B0C 0D0E0F10						
				2640			
00003DC0				2641	VRR_D	VMAH,1	
00003DC0		00003DC0		2642+	DS	0FD	
00003DC0	00003E08			2643+	USING	*,R5	base for test data and test routine
00003DC4	003D			2644+T61	DC	A(X61)	address of test routine
00003DC6	00			2645+	DC	H'61'	test number
00003DC7	01			2646+	DC	X'00'	
00003DC8	E5D4C1C8 40404040			2647+	DC	HL1'1'	m5
00003DD0	00003E4C			2648+	DC	CL8'VMAH'	instruction name
00003DD4	00003E5C			2649+	DC	A(RE61+16)	address of v2 source
00003DD8	00003E6C			2650+	DC	A(RE61+32)	address of v3 source
00003DDC	00000010			2651+	DC	A(RE61+48)	address of v4 source
00003DE0	00003E3C			2652+	DC	A(16)	result length
00003DE8	00000000 00000000			2653+REA61	DC	A(RE61)	result address
00003DF0	00000000 00000000			2654+	DS	FD	gap
00003DF8	00000000 00000000			2655+V1061	DS	XL16	V1 output
00003E00	00000000 00000000						
				2656+	DS	FD	gap
				2657+*			
00003E08				2658+X61	DS	0F	
00003E08	E310 5010 0014		00000010	2659+	LGF	R1,V2ADDR	load v2 source
00003E0E	E761 0000 0806		00000000	2660+	VL	v22,0(R1)	use v22 to test decoder
00003E14	E310 5014 0014		00000014	2661+	LGF	R1,V3ADDR	load v3 source
00003E1A	E771 0000 0806		00000000	2662+	VL	v23,0(R1)	use v23 to test decoder
00003E20	E310 5018 0014		00000018	2663+	LGF	R1,V4ADDR	load v4 source
00003E26	E781 0000 0806		00000000	2664+	VL	v24,0(R1)	use v24 to test decoder
00003E2C	E766 7100 8FAB			2665+	VMAH	V22,V22,V23,V24,1	test instruction (dest is a source)
00003E32	E760 5030 080E		00003DF0	2666+	VST	V22,V1061	save v1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003E38	07FB			2667+	BR	R11	return
00003E3C				2668+RE61	DC	0F	xl16 expected result
00003E3C				2669+	DROP	R5	
00003E3C	FFFF0003 000A0015			2670	DC	XL16'FFFF0003000A0015 00240037FFB2FF97'	result
00003E44	00240037 FFB2FF97						
00003E4C	FF020304 05060708			2671	DC	XL16'FF02030405060708 090A0B0CF30EF110'	v2
00003E54	090A0B0C F30EF110						
00003E5C	00010102 02030304			2672	DC	XL16'0001010202030304 0405050606070708'	v3
00003E64	04050506 06070708						
00003E6C	FF020304 05060708			2673	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00003E74	090A0B0C 0D0E0F10						
				2674			
00003E80				2675	VRR_D	VMAH,1	
00003E80		00003E80		2676+	DS	0FD	
00003E80	00003EC8			2677+	USING	*,R5	base for test data and test routine
00003E84	003E			2678+T62	DC	A(X62)	address of test routine
00003E86	00			2679+	DC	H'62'	test number
00003E87	01			2680+	DC	X'00'	
00003E88	E5D4C1C8 40404040			2681+	DC	HL1'1'	m5
00003E90	00003F0C			2682+	DC	CL8'VMAH'	instruction name
00003E94	00003F1C			2683+	DC	A(RE62+16)	address of v2 source
00003E98	00003F2C			2684+	DC	A(RE62+32)	address of v3 source
00003E9C	00000010			2685+	DC	A(RE62+48)	address of v4 source
00003EA0	00003EFC			2686+	DC	A(16)	result length
00003EA8	00000000 00000000			2687+REA62	DC	A(RE62)	result address
00003EB0	00000000 00000000			2688+	DS	FD	gap
00003EB8	00000000 00000000			2689+V1062	DS	XL16	V1 output
00003EC0	00000000 00000000						
				2690+	DS	FD	gap
00003EC8				2691+*			
00003EC8	E310 5010 0014		00000010	2692+X62	DS	0F	
00003ECE	E761 0000 0806		00000000	2693+	LGF	R1,V2ADDR	load v2 source
00003ED4	E310 5014 0014		00000014	2694+	VL	v22,0(R1)	use v22 to test decoder
00003EDA	E771 0000 0806		00000000	2695+	LGF	R1,V3ADDR	load v3 source
00003EE0	E310 5018 0014		00000018	2696+	VL	v23,0(R1)	use v23 to test decoder
00003EE6	E781 0000 0806		00000000	2697+	LGF	R1,V4ADDR	load v4 source
00003EEC	E781 0000 0806		00000000	2698+	VL	v24,0(R1)	use v24 to test decoder
00003EEC	E766 7100 8FAB			2699+	VMAH	V22,V22,V23,V24,1	test instruction (dest is a source)
00003EF2	E760 5030 080E		00003EB0	2700+	VST	V22,V1062	save v1 output
00003EF8	07FB			2701+	BR	R11	return
00003EFC				2702+RE62	DC	0F	xl16 expected result
00003EFC				2703+	DROP	R5	
00003EFC	FFFF0000 00000000			2704	DC	XL16'FFFF000000000000 0009000BFFF3FFF1'	result
00003F04	0009000B FFF3FFF1						
00003F0C	FF020304 05060708			2705	DC	XL16'FF02030405060708 090A0B0CF30EF110'	v2
00003F14	090A0B0C F30EF110						
00003F1C	00000000 00000001			2706	DC	XL16'0000000000000001 0101010101010102'	v3
00003F24	01010101 01010102						
00003F2C	FF020304 05060708			2707	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00003F34	090A0B0C 0D0E0F10						
				2708			
				2709 * Word			
				2710	VRR_D	VMAH,2	
00003F40				2711+	DS	0FD	
00003F40		00003F40		2712+	USING	*,R5	base for test data and test routine
00003F40	00003F88			2713+T63	DC	A(X63)	address of test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00003F44	003F			2714+	DC	H'63'	test number
00003F46	00			2715+	DC	X'00'	
00003F47	02			2716+	DC	HL1'2'	m5
00003F48	E5D4C1C8 40404040			2717+	DC	CL8'VMAH'	instruction name
00003F50	00003FCC			2718+	DC	A(RE63+16)	address of v2 source
00003F54	00003FDC			2719+	DC	A(RE63+32)	address of v3 source
00003F58	00003FEC			2720+	DC	A(RE63+48)	address of v4 source
00003F5C	00000010			2721+	DC	A(16)	result length
00003F60	00003FBC			2722+REA63	DC	A(RE63)	result address
00003F68	00000000 00000000			2723+	DS	FD	gap
00003F70	00000000 00000000			2724+V1063	DS	XL16	V1 output
00003F78	00000000 00000000						
00003F80	00000000 00000000			2725+	DS	FD	gap
				2726+*			
00003F88				2727+X63	DS	0F	
00003F88	E310 5010 0014		00000010	2728+	LGF	R1,V2ADDR	load v2 source
00003F8E	E761 0000 0806		00000000	2729+	VL	v22,0(R1)	use v22 to test decoder
00003F94	E310 5014 0014		00000014	2730+	LGF	R1,V3ADDR	load v3 source
00003F9A	E771 0000 0806		00000000	2731+	VL	v23,0(R1)	use v23 to test decoder
00003FA0	E310 5018 0014		00000018	2732+	LGF	R1,V4ADDR	load v4 source
00003FA6	E781 0000 0806		00000000	2733+	VL	v24,0(R1)	use v24 to test decoder
00003FAC	E766 7200 8FAB			2734+	VMAH	V22,V22,V23,V24,2	test instruction (dest is a source)
00003FB2	E760 5030 080E		00003F70	2735+	VST	V22,V1063	save v1 output
00003FB8	07FB			2736+	BR	R11	return
00003FBC				2737+RE63	DC	0F	xl16 expected result
00003FBC				2738+	DROP	R5	
00003FBC	00010000 00000000			2739	DC	XL16'0001000000000000 0000000000000000'	result
00003FC4	00000000 00000000						
00003FCC	FF000000 00000019			2740	DC	XL16'FF00000000000019 00000038000000FA'	v2
00003FD4	00000038 000000FA						
00003FDC	FF000000 00000019			2741	DC	XL16'FF00000000000019 00000038000000FA'	v3
00003FE4	00000038 000000FA						
00003FEC	00000000 00000000			2742	DC	XL16'0000000000000000 0000000000000000'	v4
00003FF4	00000000 00000000						
				2743			
00004000				2744	VRR_D	VMAH,2	
00004000		00004000		2745+	DS	0FD	
00004000	00004048			2746+	USING	*,R5	base for test data and test routine
00004004	0040			2747+T64	DC	A(X64)	address of test routine
00004006	00			2748+	DC	H'64'	test number
00004007	02			2749+	DC	X'00'	
00004008	E5D4C1C8 40404040			2750+	DC	HL1'2'	m5
00004010	0000408C			2751+	DC	CL8'VMAH'	instruction name
00004014	0000409C			2752+	DC	A(RE64+16)	address of v2 source
00004018	000040AC			2753+	DC	A(RE64+32)	address of v3 source
0000401C	00000010			2754+	DC	A(RE64+48)	address of v4 source
00004020	0000407C			2755+	DC	A(16)	result length
00004028	00000000 00000000			2756+REA64	DC	A(RE64)	result address
00004030	00000000 00000000			2757+	DS	FD	gap
00004038	00000000 00000000			2758+V1064	DS	XL16	V1 output
00004040	00000000 00000000			2759+	DS	FD	gap
				2760+*			
00004048				2761+X64	DS	0F	
00004048	E310 5010 0014		00000010	2762+	LGF	R1,V2ADDR	load v2 source
0000404E	E761 0000 0806		00000000	2763+	VL	v22,0(R1)	use v22 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004054	E310 5014 0014		00000014	2764+	LGF	R1,V3ADDR	load v3 source
0000405A	E771 0000 0806		00000000	2765+	VL	v23,0(R1)	use v23 to test decoder
00004060	E310 5018 0014		00000018	2766+	LGF	R1,V4ADDR	load v4 source
00004066	E781 0000 0806		00000000	2767+	VL	v24,0(R1)	use v24 to test decoder
0000406C	E766 7200 8FAB			2768+	VMAH	V22,V22,V23,V24,2	test instruction (dest is a source)
00004072	E760 5030 080E		00004030	2769+	VST	V22,V1064	save v1 output
00004078	07FB			2770+	BR	R11	return
0000407C				2771+RE64	DC	0F	xl16 expected result
0000407C				2772+	DROP	R5	
0000407C	0000FFFF 00000000			2773	DC	XL16'0000FFFF00000000 0000000000000000'	result
00004084	00000000 00000000						
0000408C	FF0000FF 00000029			2774	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00004094	00000038 000000FA						
0000409C	FF000001 00000029			2775	DC	XL16'FF00000100000029 00000038000000FA'	v3
000040A4	00000038 000000FA						
000040AC	00000001 0000002F			2776	DC	XL16'000000010000002F 0000003000000002'	v4
000040B4	00000003 00000002						
000040C0				2777			
000040C0				2778	VRR_D	VMAH,2	
000040C0		000040C0		2779+	DS	0FD	
000040C0	00004108			2780+	USING	*,R5	base for test data and test routine
000040C4	0041			2781+T65	DC	A(X65)	address of test routine
000040C6	00			2782+	DC	H'65'	test number
000040C7	02			2783+	DC	X'00'	
000040C8	E5D4C1C8 40404040			2784+	DC	HL1'2'	m5
000040D0	0000414C			2785+	DC	CL8'VMAH'	instruction name
000040D4	0000415C			2786+	DC	A(RE65+16)	address of v2 source
000040D8	0000416C			2787+	DC	A(RE65+32)	address of v3 source
000040DC	00000010			2788+	DC	A(RE65+48)	address of v4 source
000040E0	0000413C			2789+	DC	A(16)	result length
000040E8	00000000 00000000			2790+REA65	DC	A(RE65)	result address
000040F0	00000000 00000000			2791+	DS	FD	gap
000040F8	00000000 00000000			2792+V1065	DS	XL16	V1 output
00004100	00000000 00000000						
00004108				2793+	DS	FD	gap
00004108				2794+*			
00004108	E310 5010 0014		00000010	2795+X65	DS	0F	
0000410E	E761 0000 0806		00000000	2796+	LGF	R1,V2ADDR	load v2 source
00004114	E310 5014 0014		00000014	2797+	VL	v22,0(R1)	use v22 to test decoder
0000411A	E771 0000 0806		00000000	2798+	LGF	R1,V3ADDR	load v3 source
0000411A	E771 0000 0806		00000000	2799+	VL	v23,0(R1)	use v23 to test decoder
00004120	E310 5018 0014		00000018	2800+	LGF	R1,V4ADDR	load v4 source
00004126	E781 0000 0806		00000000	2801+	VL	v24,0(R1)	use v24 to test decoder
0000412C	E766 7200 8FAB			2802+	VMAH	V22,V22,V23,V24,2	test instruction (dest is a source)
00004132	E760 5030 080E		000040F0	2803+	VST	V22,V1065	save v1 output
00004138	07FB			2804+	BR	R11	return
0000413C				2805+RE65	DC	0F	xl16 expected result
0000413C				2806+	DROP	R5	
0000413C	FFFF0004 00193C6A			2807	DC	XL16'FFFF000400193C6A 0051B52BFF5700C5'	result
00004144	0051B52B FF5700C5						
0000414C	FF020304 05060708			2808	DC	XL16'FF02030405060708 090A0B0CF30E0F10'	v2
00004154	090A0B0C F30E0F10						
0000415C	01020304 05060708			2809	DC	XL16'0102030405060708 090A0B0C0D0E0F10'	v3
00004164	090A0B0C 0D0E0F10						
0000416C	FF020304 05060708			2810	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00004174	090A0B0C 0D0E0F10						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				2811		
				2812	VRR_D	VMAH, 2
00004180				2813+	DS	0FD
00004180		00004180		2814+	USING	*, R5
00004180	000041C8			2815+T66	DC	A(X66)
00004184	0042			2816+	DC	H'66'
00004186	00			2817+	DC	X'00'
00004187	02			2818+	DC	HL1'2'
00004188	E5D4C1C8 40404040			2819+	DC	CL8'VMAH'
00004190	0000420C			2820+	DC	A(RE66+16)
00004194	0000421C			2821+	DC	A(RE66+32)
00004198	0000422C			2822+	DC	A(RE66+48)
0000419C	00000010			2823+	DC	A(16)
000041A0	000041FC			2824+REA66	DC	A(RE66)
000041A8	00000000 00000000			2825+	DS	FD
000041B0	00000000 00000000			2826+V1066	DS	XL16
000041B8	00000000 00000000					
000041C0	00000000 00000000			2827+	DS	FD
				2828+*		gap
000041C8				2829+X66	DS	0F
000041C8	E310 5010 0014		00000010	2830+	LGF	R1, V2ADDR
000041CE	E761 0000 0806		00000000	2831+	VL	v22, 0(R1)
000041D4	E310 5014 0014		00000014	2832+	LGF	R1, V3ADDR
000041DA	E771 0000 0806		00000000	2833+	VL	v23, 0(R1)
000041E0	E310 5018 0014		00000018	2834+	LGF	R1, V4ADDR
000041E6	E781 0000 0806		00000000	2835+	VL	v24, 0(R1)
000041EC	E766 7200 8FAB			2836+	VMAH	V22, V22, V23, V24, 2
000041F2	E760 5030 080E		000041B0	2837+	VST	V22, V1066
000041F8	07FB			2838+	BR	R11
000041FC				2839+RE66	DC	0F
000041FC				2840+	DROP	R5
000041FC	FFFFFF01 000A1B2F			2841	DC	XL16'FFFFFF01000A1B2F 0024558BFFB1F961'
00004204	0024558B FFB1F961					result
0000420C	FF020304 05060708			2842	DC	XL16'FF02030405060708 090A0B0CF30E0F10'
00004214	090A0B0C F30E0F10					v2
0000421C	00010102 02030304			2843	DC	XL16'0001010202030304 0405050606070708'
00004224	04050506 06070708					v3
0000422C	FF020304 05060708			2844	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'
00004234	090A0B0C 0D0E0F10					v4
				2845		
				2846	VRR_D	VMAH, 2
00004240				2847+	DS	0FD
00004240		00004240		2848+	USING	*, R5
00004240	00004288			2849+T67	DC	A(X67)
00004244	0043			2850+	DC	H'67'
00004246	00			2851+	DC	X'00'
00004247	02			2852+	DC	HL1'2'
00004248	E5D4C1C8 40404040			2853+	DC	CL8'VMAH'
00004250	000042CC			2854+	DC	A(RE67+16)
00004254	000042DC			2855+	DC	A(RE67+32)
00004258	000042EC			2856+	DC	A(RE67+48)
0000425C	00000010			2857+	DC	A(16)
00004260	000042BC			2858+REA67	DC	A(RE67)
00004268	00000000 00000000			2859+	DS	FD
00004270	00000000 00000000			2860+V1067	DS	XL16
00004278	00000000 00000000					V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004280	00000000 00000000			2861+ 2862+*	DS	FD	gap
00004288				2863+X67	DS	0F	
00004288	E310 5010 0014		00000010	2864+	LGF	R1,V2ADDR	load v2 source
0000428E	E761 0000 0806		00000000	2865+	VL	v22,0(R1)	use v22 to test decoder
00004294	E310 5014 0014		00000014	2866+	LGF	R1,V3ADDR	load v3 source
0000429A	E771 0000 0806		00000000	2867+	VL	v23,0(R1)	use v23 to test decoder
000042A0	E310 5018 0014		00000018	2868+	LGF	R1,V4ADDR	load v4 source
000042A6	E781 0000 0806		00000000	2869+	VL	v24,0(R1)	use v24 to test decoder
000042AC	E766 7200 8FAB			2870+	VMAH	V22,V22,V23,V24,2	test instruction (dest is a source)
000042B2	E760 5030 080E		00004270	2871+	VST	V22,V1067	save v1 output
000042B8	07FB			2872+	BR	R11	return
000042BC				2873+RE67	DC	0F	xl16 expected result
000042BC				2874+	DROP	R5	
000042BC	FFFFFFFF 00000000			2875	DC	XL16'FFFFFFFF00000000 0009131EFFF30110'	result
000042C4	0009131E FFF30110						
000042CC	FF020304 05060708			2876	DC	XL16'FF02030405060708 090A0B0CF30E0F10'	v2
000042D4	090A0B0C F30E0F10						
000042DC	00000000 00000001			2877	DC	XL16'0000000000000001 0101010101010102'	v3
000042E4	01010101 01010102						
000042EC	FF020304 05060708			2878	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000042F4	090A0B0C 0D0E0F10						
				2879			
				2880 * Doubleword			
00004300				2881	VRR_D	VMAH,3	
00004300		00004300		2882+	DS	0FD	
00004300	00004348			2883+	USING	*,R5	base for test data and test routine
00004304	0044			2884+T68	DC	A(X68)	address of test routine
00004306	00			2885+	DC	H'68'	test number
00004307	03			2886+	DC	X'00'	
00004308	E5D4C1C8 40404040			2887+	DC	HL1'3'	m5
00004310	0000438C			2888+	DC	CL8'VMAH'	instruction name
00004314	0000439C			2889+	DC	A(RE68+16)	address of v2 source
00004318	000043AC			2890+	DC	A(RE68+32)	address of v3 source
0000431C	00000010			2891+	DC	A(RE68+48)	address of v4 source
00004320	0000437C			2892+	DC	A(16)	result length
00004328	00000000 00000000			2893+REA68	DC	A(RE68)	result address
00004330	00000000 00000000			2894+	DS	FD	gap
00004338	00000000 00000000			2895+V1068	DS	XL16	V1 output
00004340	00000000 00000000						
				2896+	DS	FD	gap
				2897+*			
00004348				2898+X68	DS	0F	
00004348	E310 5010 0014		00000010	2899+	LGF	R1,V2ADDR	load v2 source
0000434E	E761 0000 0806		00000000	2900+	VL	v22,0(R1)	use v22 to test decoder
00004354	E310 5014 0014		00000014	2901+	LGF	R1,V3ADDR	load v3 source
0000435A	E771 0000 0806		00000000	2902+	VL	v23,0(R1)	use v23 to test decoder
00004360	E310 5018 0014		00000018	2903+	LGF	R1,V4ADDR	load v4 source
00004366	E781 0000 0806		00000000	2904+	VL	v24,0(R1)	use v24 to test decoder
0000436C	E766 7300 8FAB			2905+	VMAH	V22,V22,V23,V24,3	test instruction (dest is a source)
00004372	E760 5030 080E		00004330	2906+	VST	V22,V1068	save v1 output
00004378	07FB			2907+	BR	R11	return
0000437C				2908+RE68	DC	0F	xl16 expected result
0000437C				2909+	DROP	R5	
0000437C	00000000 00000000			2910	DC	XL16'0000000000000000 0000000000000C77'	result
00004384	00000000 00000C77						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000438C	FFFFFFFF	00019000		2911	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00004394	00000038	EEEEEEFA					
0000439C	FFFFFFFF	00019000		2912	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
000043A4	00000038	0EEEEEEFA					
000043AC	00000000	00000000		2913	DC	XL16'0000000000000000 0000000000000000'	v4
000043B4	00000000	00000000					
				2914			
				2915	VRR_D	VMAH, 3	
000043C0				2916+	DS	0FD	
000043C0		000043C0		2917+	USING	*, R5	base for test data and test routine
000043C0	00004408			2918+T69	DC	A(X69)	address of test routine
000043C4	0045			2919+	DC	H'69'	test number
000043C6	00			2920+	DC	X'00'	
000043C7	03			2921+	DC	HL1'3'	m5
000043C8	E5D4C1C8	40404040		2922+	DC	CL8'VMAH'	instruction name
000043D0	0000444C			2923+	DC	A(RE69+16)	address of v2 source
000043D4	0000445C			2924+	DC	A(RE69+32)	address of v3 source
000043D8	0000446C			2925+	DC	A(RE69+48)	address of v4 source
000043DC	00000010			2926+	DC	A(16)	result length
000043E0	0000443C			2927+REA69	DC	A(RE69)	result address
000043E8	00000000	00000000		2928+	DS	FD	gap
000043F0	00000000	00000000		2929+V1069	DS	XL16	V1 output
000043F8	00000000	00000000					
00004400	00000000	00000000		2930+	DS	FD	gap
				2931+*			
				2932+X69	DS	0F	
00004408				2933+	LGF	R1, V2ADDR	load v2 source
00004408	E310 5010 0014	00000010		2934+	VL	v22, 0(R1)	use v22 to test decoder
0000440E	E761 0000 0806	00000000		2935+	LGF	R1, V3ADDR	load v3 source
00004414	E310 5014 0014	00000014		2936+	VL	v23, 0(R1)	use v23 to test decoder
0000441A	E771 0000 0806	00000000		2937+	LGF	R1, V4ADDR	load v4 source
00004420	E310 5018 0014	00000018		2938+	VL	v24, 0(R1)	use v24 to test decoder
00004426	E781 0000 0806	00000000		2939+	VMAH	V22, V22, V23, V24, 3	test instruction (dest is a source)
0000442C	E766 7300 8FAB			2940+	VST	V22, V1069	save v1 output
00004432	E760 5030 080E	000043F0		2941+	BR	R11	return
00004438	07FB			2942+RE69	DC	0F	xl16 expected result
0000443C				2943+	DROP	R5	
0000443C	FFFF0004	0C192C46		2944	DC	XL16'FFFF00040C192C46 0051B52F8692B4F6'	result
00004444	0051B52F	8692B4F6					
0000444C	FF020304	05060750		2945	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00004454	090A0B0C	0D0E0F7F					
0000445C	01020304	05060750		2946	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3
00004464	090A0B78	0D0E0F7F					
0000446C	00000000	00000001		2947	DC	XL16'0000000000000001 0000000000000001'	v4
00004474	00000000	00000001					
				2948			
				2949	VRR_D	VMAH, 3	
00004480				2950+	DS	0FD	
00004480		00004480		2951+	USING	*, R5	base for test data and test routine
00004480	000044C8			2952+T70	DC	A(X70)	address of test routine
00004484	0046			2953+	DC	H'70'	test number
00004486	00			2954+	DC	X'00'	
00004487	03			2955+	DC	HL1'3'	m5
00004488	E5D4C1C8	40404040		2956+	DC	CL8'VMAH'	instruction name
00004490	0000450C			2957+	DC	A(RE70+16)	address of v2 source
00004494	0000451C			2958+	DC	A(RE70+32)	address of v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004498	0000452C			2959+	DC	A(RE70+48)	address of v4 source
0000449C	00000010			2960+	DC	A(16)	result length
000044A0	000044FC			2961+REA70	DC	A(RE70)	result address
000044A8	00000000 00000000			2962+	DS	FD	gap
000044B0	00000000 00000000			2963+V1070	DS	XL16	V1 output
000044B8	00000000 00000000						
000044C0	00000000 00000000			2964+	DS	FD	gap
				2965+*			
000044C8				2966+X70	DS	0F	
000044C8	E310 5010 0014		00000010	2967+	LGF	R1,V2ADDR	load v2 source
000044CE	E761 0000 0806		00000000	2968+	VL	v22,0(R1)	use v22 to test decoder
000044D4	E310 5014 0014		00000014	2969+	LGF	R1,V3ADDR	load v3 source
000044DA	E771 0000 0806		00000000	2970+	VL	v23,0(R1)	use v23 to test decoder
000044E0	E310 5018 0014		00000018	2971+	LGF	R1,V4ADDR	load v4 source
000044E6	E781 0000 0806		00000000	2972+	VL	v24,0(R1)	use v24 to test decoder
000044EC	E766 7300 8FAB			2973+	VMAH	V22,V22,V23,V24,3	test instruction (dest is a source)
000044F2	E760 5030 080E		000044B0	2974+	VST	V22,V1070	save v1 output
000044F8	07FB			2975+	BR	R11	return
000044FC				2976+RE70	DC	0F	xl16 expected result
000044FC				2977+	DROP	R5	
000044FC	FFFFFF01 0309101C			2978	DC	XL16'FFFFFF010309101C 0024558DB838C862'	result
00004504	0024558D B838C862						
0000450C	FF020304 05060750			2979	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00004514	090A0B0C 0D0E0F7F						
0000451C	00010102 02030328			2980	DC	XL16'0001010202030328 0405053C0607073F'	v3
00004524	0405053C 0607073F						
0000452C	20000000 00000000			2981	DC	XL16'2000000000000000 FFFFFFFFFFFFFFFFFF'	v4
00004534	FFFFFFFF FFFFFFFF						
				2982			
00004540				2983	VRR_D	VMAH,3	
00004540		00004540		2984+	DS	0FD	
00004540	00004588			2985+	USING	*,R5	base for test data and test routine
00004544	0047			2986+T71	DC	A(X71)	address of test routine
00004546	00			2987+	DC	H'71'	test number
00004547	03			2988+	DC	X'00'	
00004548	E5D4C1C8 40404040			2989+	DC	HL1'3'	m5
00004550	000045CC			2990+	DC	CL8'VMAH'	instruction name
00004554	000045DC			2991+	DC	A(RE71+16)	address of v2 source
00004558	000045EC			2992+	DC	A(RE71+32)	address of v3 source
0000455C	00000010			2993+	DC	A(RE71+48)	address of v4 source
00004560	00000010			2994+	DC	A(16)	result length
00004560	000045BC			2995+REA71	DC	A(RE71)	result address
00004568	00000000 00000000			2996+	DS	FD	gap
00004570	00000000 00000000			2997+V1071	DS	XL16	V1 output
00004578	00000000 00000000						
00004580	00000000 00000000			2998+	DS	FD	gap
				2999+*			
00004588				3000+X71	DS	0F	
00004588	E310 5010 0014		00000010	3001+	LGF	R1,V2ADDR	load v2 source
0000458E	E761 0000 0806		00000000	3002+	VL	v22,0(R1)	use v22 to test decoder
00004594	E310 5014 0014		00000014	3003+	LGF	R1,V3ADDR	load v3 source
0000459A	E771 0000 0806		00000000	3004+	VL	v23,0(R1)	use v23 to test decoder
000045A0	E310 5018 0014		00000018	3005+	LGF	R1,V4ADDR	load v4 source
000045A6	E781 0000 0806		00000000	3006+	VL	v24,0(R1)	use v24 to test decoder
000045AC	E766 7300 8FAB			3007+	VMAH	V22,V22,V23,V24,3	test instruction (dest is a source)
000045B2	E760 5030 080E		00004570	3008+	VST	V22,V1071	save v1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
000045B8	07FB			3009+	BR	R11	return		
000045BC				3010+RE71	DC	0F	xl16 expected result		
000045BC				3011+	DROP	R5			
000045BC	FFFFFFFF	FFFFFFFF		3012	DC	XL16'	FFFFFFFFFFFFFFFF 0009131EA8C3DFFE'	result	
000045C4	0009131E	A8C3DFFE							
000045CC	FF020304	05060750		3013	DC	XL16'	FF02030405060750 090A0B0C0D0E0F7F'	v2	
000045D4	090A0B0C	0D0E0F7F							
000045DC	00000000	0000000A		3014	DC	XL16'	000000000000000A 0101010F0101010F'	v3	
000045E4	0101010F	0101010F							
000045EC	00001000	00000001		3015	DC	XL16'	0000100000000001 1000000000000001'	v4	
000045F4	10000000	00000001							
				3016					
				3017 * Quadword					
				3018	VRR_D	VMAH, 4			
00004600				3019+	DS	0FD			
00004600		00004600		3020+	USING	*, R5	base for test data and test routine		
00004600	00004648			3021+T72	DC	A(X72)	address of test routine		
00004604	0048			3022+	DC	H'72'	test number		
00004606	00			3023+	DC	X'00'			
00004607	04			3024+	DC	HL1'4'	m5		
00004608	E5D4C1C8	40404040		3025+	DC	CL8'VMAH'	instruction name		
00004610	0000468C			3026+	DC	A(RE72+16)	address of v2 source		
00004614	0000469C			3027+	DC	A(RE72+32)	address of v3 source		
00004618	000046AC			3028+	DC	A(RE72+48)	address of v4 source		
0000461C	00000010			3029+	DC	A(16)	result length		
00004620	0000467C			3030+REA72	DC	A(RE72)	result address		
00004628	00000000	00000000		3031+	DS	FD	gap		
00004630	00000000	00000000		3032+V1072	DS	XL16	V1 output		
00004638	00000000	00000000							
00004640	00000000	00000000		3033+	DS	FD	gap		
				3034+*					
00004648				3035+X72	DS	0F			
00004648	E310 5010 0014		00000010	3036+	LGF	R1, V2ADDR	load v2 source		
0000464E	E761 0000 0806		00000000	3037+	VL	v22, 0(R1)	use v22 to test decoder		
00004654	E310 5014 0014		00000014	3038+	LGF	R1, V3ADDR	load v3 source		
0000465A	E771 0000 0806		00000000	3039+	VL	v23, 0(R1)	use v23 to test decoder		
00004660	E310 5018 0014		00000018	3040+	LGF	R1, V4ADDR	load v4 source		
00004666	E781 0000 0806		00000000	3041+	VL	v24, 0(R1)	use v24 to test decoder		
0000466C	E766 7400 8FAB			3042+	VMAH	V22, V22, V23, V24, 4	test instruction (dest is a source)		
00004672	E760 5030 080E		00004630	3043+	VST	V22, V1072	save v1 output		
00004678	07FB			3044+	BR	R11	return		
0000467C				3045+RE72	DC	0F	xl16 expected result		
0000467C				3046+	DROP	R5			
0000467C	31000000	00000000		3047	DC	XL16'	31000000 00000000 00000000 00000000'	result	
00004684	00000000	00000000							
0000468C	70000000	00000000		3048	DC	XL16'	70000000 00000000 00000000 00000000'	v2	
00004694	00000000	00000000							
0000469C	70000000	00000000		3049	DC	XL16'	70000000 00000000 00000000 00000000'	v3	
000046A4	00000000	00000000							
000046AC	7F000000	00000000		3050	DC	XL16'	7F000000 00000000 20000000 00000000'	v4	
000046B4	20000000	00000000							
				3051					
				3052	VRR_D	VMAH, 4			
000046C0				3053+	DS	0FD			
000046C0		000046C0		3054+	USING	*, R5	base for test data and test routine		
000046C0	00004708			3055+T73	DC	A(X73)	address of test routine		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
000046C4	0049			3056+	DC	H'73'		test number	
000046C6	00			3057+	DC	X'00'			
000046C7	04			3058+	DC	HL1'4'		m5	
000046C8	E5D4C1C8	40404040		3059+	DC	CL8'VMAH'		instruction name	
000046D0	0000474C			3060+	DC	A(RE73+16)		address of v2 source	
000046D4	0000475C			3061+	DC	A(RE73+32)		address of v3 source	
000046D8	0000476C			3062+	DC	A(RE73+48)		address of v4 source	
000046DC	00000010			3063+	DC	A(16)		result length	
000046E0	0000473C			3064+REA73	DC	A(RE73)		result address	
000046E8	00000000	00000000		3065+	DS	FD		gap	
000046F0	00000000	00000000		3066+V1073	DS	XL16		V1 output	
000046F8	00000000	00000000							
00004700	00000000	00000000		3067+	DS	FD		gap	
				3068+*					
00004708				3069+X73	DS	0F			
00004708	E310	5010	0014	3070+	LGF	R1,V2ADDR		load v2 source	
0000470E	E761	0000	0806	3071+	VL	v22,0(R1)		use v22 to test decoder	
00004714	E310	5014	0014	3072+	LGF	R1,V3ADDR		load v3 source	
0000471A	E771	0000	0806	3073+	VL	v23,0(R1)		use v23 to test decoder	
00004720	E310	5018	0014	3074+	LGF	R1,V4ADDR		load v4 source	
00004726	E781	0000	0806	3075+	VL	v24,0(R1)		use v24 to test decoder	
0000472C	E766	7400	8FAB	3076+	VMAH	V22,V22,V23,V24,4		test instruction (dest is a source)	
00004732	E760	5030	080E	3077+	VST	V22,V1073		save v1 output	
00004738	07FB			3078+	BR	R11		return	
0000473C				3079+RE73	DC	0F		xl16 expected result	
0000473C				3080+	DROP	R5			
0000473C	3100D200	E0FFFFFF		3081	DC	XL16'3100D200	E0FFFFFF FFFFFFFF FFFFFFFF	result	
00004744	FFFFFFFF	FFFFFFFF							
0000474C	7000F000	00000000		3082	DC	XL16'7000F000	00000000 00000000 00000000	v2	
00004754	00000000	00000000							
0000475C	7000F000	00000000		3083	DC	XL16'7000F000	00000000 00000000 00000000	v3	
00004764	00000000	00000000							
0000476C	FFFFFFFF	FFFFFFFF		3084	DC	XL16'FFFFFFFF	FFFFFFFF FFFFFFFF FFFFFFFF	v4	
00004774	FFFFFFFF	FFFFFFFF							
				3085					
00004780				3086	VRR_D	VMAH,4			
00004780			00004780	3087+	DS	0FD			
00004780	000047C8			3088+	USING	*,R5		base for test data and test routine	
00004784	004A			3089+T74	DC	A(X74)		address of test routine	
00004786	00			3090+	DC	H'74'		test number	
00004787	04			3091+	DC	X'00'			
00004788	E5D4C1C8	40404040		3092+	DC	HL1'4'		m5	
00004790	0000480C			3093+	DC	CL8'VMAH'		instruction name	
00004794	0000481C			3094+	DC	A(RE74+16)		address of v2 source	
00004798	0000482C			3095+	DC	A(RE74+32)		address of v3 source	
0000479C	00000010			3096+	DC	A(RE74+48)		address of v4 source	
000047A0	000047FC			3097+	DC	A(16)		result length	
000047A8	00000000	00000000		3098+REA74	DC	A(RE74)		result address	
000047B0	00000000	00000000		3099+	DS	FD		gap	
000047B8	00000000	00000000		3100+V1074	DS	XL16		V1 output	
000047C0	00000000	00000000		3101+	DS	FD		gap	
				3102+*					
000047C8				3103+X74	DS	0F			
000047C8	E310	5010	0014	3104+	LGF	R1,V2ADDR		load v2 source	
000047CE	E761	0000	0806	3105+	VL	v22,0(R1)		use v22 to test decoder	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
000047D4	E310 5014 0014		00000014	3106+	LGF	R1,V3ADDR	load v3 source		
000047DA	E771 0000 0806		00000000	3107+	VL	v23,0(R1)	use v23 to test decoder		
000047E0	E310 5018 0014		00000018	3108+	LGF	R1,V4ADDR	load v4 source		
000047E6	E781 0000 0806		00000000	3109+	VL	v24,0(R1)	use v24 to test decoder		
000047EC	E766 7400 8FAB			3110+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)		
000047F2	E760 5030 080E		000047B0	3111+	VST	V22,V1074	save v1 output		
000047F8	07FB			3112+	BR	R11	return		
000047FC				3113+RE74	DC	0F	xl16 expected result		
000047FC				3114+	DROP	R5			
000047FC	3100D201 4A00E100			3115	DC	XL16'3100D201 4A00E100 00000000 00000000'	result		
00004804	00000000 00000000								
0000480C	7000F000 F0000000			3116	DC	XL16'7000F000 F0000000 00000000 00000000'	v2		
00004814	00000000 00000000								
0000481C	7000F000 00000000			3117	DC	XL16'7000F000 00000000 00000000 00000000'	v3		
00004824	00000000 00000000								
0000482C	7F000000 00000000			3118	DC	XL16'7F000000 00000000 20000000 00000000'	v4		
00004834	20000000 00000000								
00004840				3119					
00004840				3120	VRR_D	VMAH,4			
00004840		00004840		3121+	DS	0FD			
00004840	00004888			3122+	USING	*,R5	base for test data and test routine		
00004844	004B			3123+T75	DC	A(X75)	address of test routine		
00004846	00			3124+	DC	H'75'	test number		
00004847	04			3125+	DC	X'00'			
00004848	E5D4C1C8 40404040			3126+	DC	HL1'4'	m5		
00004850	000048CC			3127+	DC	CL8'VMAH'	instruction name		
00004854	000048DC			3128+	DC	A(RE75+16)	address of v2 source		
00004858	000048EC			3129+	DC	A(RE75+32)	address of v3 source		
0000485C	00000010			3130+	DC	A(RE75+48)	address of v4 source		
00004860	000048BC			3131+	DC	A(16)	result length		
00004868	00000000 00000000			3132+REA75	DC	A(RE75)	result address		
00004870	00000000 00000000			3133+	DS	FD	gap		
00004878	00000000 00000000			3134+V1075	DS	XL16	V1 output		
00004880	00000000 00000000								
00004888				3135+	DS	FD	gap		
00004888				3136+*					
00004888	E310 5010 0014		00000010	3137+X75	DS	0F			
0000488E	E761 0000 0806		00000000	3138+	LGF	R1,V2ADDR	load v2 source		
00004894	E310 5014 0014		00000014	3139+	VL	v22,0(R1)	use v22 to test decoder		
0000489A	E771 0000 0806		00000000	3140+	LGF	R1,V3ADDR	load v3 source		
0000489A	E771 0000 0806		00000000	3141+	VL	v23,0(R1)	use v23 to test decoder		
000048A0	E310 5018 0014		00000018	3142+	LGF	R1,V4ADDR	load v4 source		
000048A6	E781 0000 0806		00000000	3143+	VL	v24,0(R1)	use v24 to test decoder		
000048AC	E766 7400 8FAB			3144+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)		
000048B2	E760 5030 080E		00004870	3145+	VST	V22,V1075	save v1 output		
000048B8	07FB			3146+	BR	R11	return		
000048BC				3147+RE75	DC	0F	xl16 expected result		
000048BC				3148+	DROP	R5			
000048BC	3100D201 4A00E100			3149	DC	XL16'3100D201 4A00E100 6900E100 E1000000'	result		
000048C4	6900E100 E1000000								
000048CC	7000F000 F0000000			3150	DC	XL16'7000F000 F0000000 00000000 00000000'	v2		
000048D4	00000000 00000000								
000048DC	7000F000 00000000			3151	DC	XL16'7000F000 00000000 F0000000 00000000'	v3		
000048E4	F0000000 00000000								
000048EC	7F000000 00000000			3152	DC	XL16'7F000000 00000000 20000000 00000000'	v4		
000048F4	20000000 00000000								

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
				3153		
				3154	VRR_D	VMAH, 4
00004900				3155+	DS	0FD
00004900		00004900		3156+	USING	*, R5
00004900	00004948			3157+T76	DC	A(X76)
00004904	004C			3158+	DC	H'76'
00004906	00			3159+	DC	X'00'
00004907	04			3160+	DC	HL1'4'
00004908	E5D4C1C8 40404040			3161+	DC	CL8'VMAH'
00004910	0000498C			3162+	DC	A(RE76+16)
00004914	0000499C			3163+	DC	A(RE76+32)
00004918	000049AC			3164+	DC	A(RE76+48)
0000491C	00000010			3165+	DC	A(16)
00004920	0000497C			3166+REA76	DC	A(RE76)
00004928	00000000 00000000			3167+	DS	FD
00004930	00000000 00000000			3168+V1076	DS	XL16
00004938	00000000 00000000					
00004940	00000000 00000000			3169+	DS	FD
				3170+*		gap
00004948				3171+X76	DS	0F
00004948	E310 5010 0014		00000010	3172+	LGF	R1, V2ADDR
0000494E	E761 0000 0806		00000000	3173+	VL	v22, 0(R1)
00004954	E310 5014 0014		00000014	3174+	LGF	R1, V3ADDR
0000495A	E771 0000 0806		00000000	3175+	VL	v23, 0(R1)
00004960	E310 5018 0014		00000018	3176+	LGF	R1, V4ADDR
00004966	E781 0000 0806		00000000	3177+	VL	v24, 0(R1)
0000496C	E766 7400 8FAB			3178+	VMAH	V22, V22, V23, V24, 4
00004972	E760 5030 080E		00004930	3179+	VST	V22, V1076
00004978	07FB			3180+	BR	R11
0000497C				3181+RE76	DC	0F
0000497C				3182+	DROP	R5
0000497C	00000000 80000000			3183	DC	XL16'00000000 80000000 80000000 80000000'
00004984	80000000 80000000					result
0000498C	7FFFFFFF FFFFFFFF			3184	DC	XL16'7FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'
00004994	FFFFFFF FFFFFFFF					v2
0000499C	00000001 00000001			3185	DC	XL16'00000001 00000001 00000001 00000001'
000049A4	00000001 00000001					v3
000049AC	FFFFFFF FFFFFFFF			3186	DC	XL16'FFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'
000049B4	FFFFFFF FFFFFFFF					v4
				3187		
				3188	VRR_D	VMAH, 4
000049C0				3189+	DS	0FD
000049C0		000049C0		3190+	USING	*, R5
000049C0	00004A08			3191+T77	DC	A(X77)
000049C4	004D			3192+	DC	H'77'
000049C6	00			3193+	DC	X'00'
000049C7	04			3194+	DC	HL1'4'
000049C8	E5D4C1C8 40404040			3195+	DC	CL8'VMAH'
000049D0	00004A4C			3196+	DC	A(RE77+16)
000049D4	00004A5C			3197+	DC	A(RE77+32)
000049D8	00004A6C			3198+	DC	A(RE77+48)
000049DC	00000010			3199+	DC	A(16)
000049E0	00004A3C			3200+REA77	DC	A(RE77)
000049E8	00000000 00000000			3201+	DS	FD
000049F0	00000000 00000000			3202+V1077	DS	XL16
000049F8	00000000 00000000					V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00004A00	00000000 00000000			3203+ 3204+*	DS	FD	gap		
00004A08				3205+X77	DS	0F			
00004A08	E310 5010 0014		00000010	3206+	LGF	R1,V2ADDR	load v2 source		
00004A0E	E761 0000 0806		00000000	3207+	VL	v22,0(R1)	use v22 to test decoder		
00004A14	E310 5014 0014		00000014	3208+	LGF	R1,V3ADDR	load v3 source		
00004A1A	E771 0000 0806		00000000	3209+	VL	v23,0(R1)	use v23 to test decoder		
00004A20	E310 5018 0014		00000018	3210+	LGF	R1,V4ADDR	load v4 source		
00004A26	E781 0000 0806		00000000	3211+	VL	v24,0(R1)	use v24 to test decoder		
00004A2C	E766 7400 8FAB			3212+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00004A32	E760 5030 080E		000049F0	3213+	VST	V22,V1077	save v1 output		
00004A38	07FB			3214+	BR	R11	return		
00004A3C				3215+RE77	DC	0F	xl16 expected result		
00004A3C				3216+	DROP	R5			
00004A3C	3FFFFFFF 00019000			3217	DC	XL16'3FFFFFFF 00019000 FFFCE03A EFEEEE89'	result		
00004A44	FFFCE03A EFEEEE89								
00004A4C	7FFFFFFF 00019000			3218	DC	XL16'7FFFFFFF 00019000 00000038 EEEEEFEFA'	v2		
00004A54	00000038 EEEEEFEFA								
00004A5C	7FFFFFFF 00019000			3219	DC	XL16'7FFFFFFF 00019000 00000038 0EEEEFEFA'	v3		
00004A64	00000038 0EEEEFEFA								
00004A6C	FF000000 00000000			3220	DC	XL16'FF000000 00000000 20000000 00000000'	v4		
00004A74	20000000 00000000								
				3221					
00004A80				3222	VRR_D	VMAH,4			
00004A80		00004A80		3223+	DS	0FD			
00004A80	00004AC8			3224+	USING	*,R5	base for test data and test routine		
00004A84	004E			3225+T78	DC	A(X78)	address of test routine		
00004A86	00			3226+	DC	H'78'	test number		
00004A87	04			3227+	DC	X'00'			
00004A88	E5D4C1C8 40404040			3228+	DC	HL1'4'	m5		
00004A90	00004B0C			3229+	DC	CL8'VMAH'	instruction name		
00004A94	00004B1C			3230+	DC	A(RE78+16)	address of v2 source		
00004A98	00004B2C			3231+	DC	A(RE78+32)	address of v3 source		
00004A9C	00000010			3232+	DC	A(RE78+48)	address of v4 source		
00004AA0	00004AFC			3233+	DC	A(16)	result length		
00004AA8	00000000 00000000			3234+REA78	DC	A(RE78)	result address		
00004AB0	00000000 00000000			3235+	DS	FD	gap		
00004AB8	00000000 00000000			3236+V1078	DS	XL16	V1 output		
00004AC0	00000000 00000000			3237+ 3238+*	DS	FD	gap		
00004AC8				3239+X78	DS	0F			
00004AC8	E310 5010 0014		00000010	3240+	LGF	R1,V2ADDR	load v2 source		
00004ACE	E761 0000 0806		00000000	3241+	VL	v22,0(R1)	use v22 to test decoder		
00004AD4	E310 5014 0014		00000014	3242+	LGF	R1,V3ADDR	load v3 source		
00004ADA	E771 0000 0806		00000000	3243+	VL	v23,0(R1)	use v23 to test decoder		
00004AE0	E310 5018 0014		00000018	3244+	LGF	R1,V4ADDR	load v4 source		
00004AE6	E781 0000 0806		00000000	3245+	VL	v24,0(R1)	use v24 to test decoder		
00004AEC	E766 7400 8FAB			3246+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)		
00004AF2	E760 5030 080E		00004AB0	3247+	VST	V22,V1078	save v1 output		
00004AF8	07FB			3248+	BR	R11	return		
00004AFC				3249+RE78	DC	0F	xl16 expected result		
00004AFC				3250+	DROP	R5			
00004AFC	00000000 00000000			3251	DC	XL16'00000000 00000000 FFFCE002 70FFFF8F'	result		
00004B04	FFFCE002 70FFFF8F								
00004B0C	FFFFFFFF 00019000			3252	DC	XL16'FFFFFFFF 00019000 00000038 EEEEEFEFA'	v2		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00004B14	00000038	EEEEEEFA							
00004B1C	FFFFFFFF	00019000		3253	DC	XL16'	FFFFFFFF 00019000 00000038 0EEEEEEFA'	v3	
00004B24	00000038	0EEEEEEFA							
00004B2C	00000000	00000000		3254	DC	XL16'	00000000 00000000 00000000 00000000'	v4	
00004B34	00000000	00000000							
				3255					
00004B40				3256	VRR_D	VMAH, 4			
00004B40			00004B40	3257+	DS	0FD			
00004B40	00004B88			3258+	USING	*, R5		base for test data and test routine	
00004B44	004F			3259+T79	DC	A(X79)		address of test routine	
00004B46	00			3260+	DC	H'79'		test number	
00004B47	04			3261+	DC	X'00'			
00004B48	E5D4C1C8	40404040		3262+	DC	HL1'4'		m5	
00004B50	00004BCC			3263+	DC	CL8'VMAH'		instruction name	
00004B54	00004BDC			3264+	DC	A(RE79+16)		address of v2 source	
00004B58	00004BEC			3265+	DC	A(RE79+32)		address of v3 source	
00004B5C	00000010			3266+	DC	A(RE79+48)		address of v4 source	
00004B60	00004BBC			3267+	DC	A(16)		result length	
00004B68	00000000	00000000		3268+REA79	DC	A(RE79)		result address	
00004B70	00000000	00000000		3269+	DS	FD		gap	
00004B78	00000000	00000000		3270+V1079	DS	XL16		V1 output	
00004B80	00000000	00000000							
				3271+	DS	FD		gap	
00004B88				3272+*					
00004B88	E310 5010 0014		00000010	3273+X79	DS	0F			
00004B8E	E761 0000 0806		00000000	3274+	LGF	R1, V2ADDR		load v2 source	
00004B94	E310 5014 0014		00000014	3275+	VL	v22, 0(R1)		use v22 to test decoder	
00004B9A	E771 0000 0806		00000000	3276+	LGF	R1, V3ADDR		load v3 source	
00004BA0	E310 5018 0014		00000018	3277+	VL	v23, 0(R1)		use v23 to test decoder	
00004BA6	E781 0000 0806		00000000	3278+	LGF	R1, V4ADDR		load v4 source	
00004BAC	E766 7400 8FAB			3279+	VL	v24, 0(R1)		use v24 to test decoder	
00004BB2	E760 5030 080E		00004B70	3280+	VMAH	V22, V22, V23, V24, 4		test instruction (dest is a source)	
00004BB8	07FB			3281+	VST	V22, V1079		save v1 output	
00004BBC				3282+	BR	R11		return	
00004BBC				3283+RE79	DC	0F		xl16 expected result	
00004BBC	FFFF0004	0C192C46		3284+	DROP	R5			
00004BC4	69B57B4B	BDEC6504		3285	DC	XL16'	FFFF0004 0C192C46 69B57B4B BDEC6504'	result	
00004BCC	FF020304	05060750		3286	DC	XL16'	FF020304 05060750 090A0B0C 0D0E0F7F'	v2	
00004BD4	090A0B0C	0D0E0F7F							
00004BDC	01020304	05060750		3287	DC	XL16'	01020304 05060750 090A0B78 0D0E0F7F'	v3	
00004BE4	090A0B78	0D0E0F7F							
00004BEC	10000000	00000001		3288	DC	XL16'	10000000 00000001 10000000 00000001'	v4	
00004BF4	10000000	00000001							
				3289					
00004C00				3290	VRR_D	VMAH, 4			
00004C00			00004C00	3291+	DS	0FD			
00004C00	00004C48			3292+	USING	*, R5		base for test data and test routine	
00004C04	0050			3293+T80	DC	A(X80)		address of test routine	
00004C06	00			3294+	DC	H'80'		test number	
00004C07	04			3295+	DC	X'00'			
00004C08	E5D4C1C8	40404040		3296+	DC	HL1'4'		m5	
00004C10	00004C8C			3297+	DC	CL8'VMAH'		instruction name	
00004C14	00004C9C			3298+	DC	A(RE80+16)		address of v2 source	
00004C18	00004CAC			3299+	DC	A(RE80+32)		address of v3 source	
				3300+	DC	A(RE80+48)		address of v4 source	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004C1C	00000010			3301+	DC	A(16)	result length
00004C20	00004C7C			3302+REA80	DC	A(RE80)	result address
00004C28	00000000 00000000			3303+	DS	FD	gap
00004C30	00000000 00000000			3304+V1080	DS	XL16	V1 output
00004C38	00000000 00000000						
00004C40	00000000 00000000			3305+	DS	FD	gap
				3306+*			
00004C48				3307+X80	DS	0F	
00004C48	E310 5010 0014		00000010	3308+	LGF	R1,V2ADDR	load v2 source
00004C4E	E761 0000 0806		00000000	3309+	VL	v22,0(R1)	use v22 to test decoder
00004C54	E310 5014 0014		00000014	3310+	LGF	R1,V3ADDR	load v3 source
00004C5A	E771 0000 0806		00000000	3311+	VL	v23,0(R1)	use v23 to test decoder
00004C60	E310 5018 0014		00000018	3312+	LGF	R1,V4ADDR	load v4 source
00004C66	E781 0000 0806		00000000	3313+	VL	v24,0(R1)	use v24 to test decoder
00004C6C	E766 7400 8FAB			3314+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)
00004C72	E760 5030 080E		00004C30	3315+	VST	V22,V1080	save v1 output
00004C78	07FB			3316+	BR	R11	return
00004C7C				3317+RE80	DC	0F	xl16 expected result
00004C7C				3318+	DROP	R5	
00004C7C	FFFFFF01 0309101C			3319	DC	XL16'FFFFFF01 0309101C 06CF0A94 A5DE7262'	result
00004C84	06CF0A94 A5DE7262						
00004C8C	FF020304 05060750			3320	DC	XL16'FF020304 05060750 090A0B0C 0D0E0F7F'	v2
00004C94	090A0B0C 0D0E0F7F						
00004C9C	00010102 02030328			3321	DC	XL16'00010102 02030328 0405053C 0607073F'	v3
00004CA4	0405053C 0607073F						
00004CAC	FFFFFFFF FFFFFFFF			3322	DC	XL16'FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF'	v4
00004CB4	FFFFFFFF FFFFFFFF						
				3323			
				3324	VRR_D	VMAH,4	
00004CC0				3325+	DS	0FD	
00004CC0		00004CC0		3326+	USING	*,R5	base for test data and test routine
00004CC0	00004D08			3327+T81	DC	A(X81)	address of test routine
00004CC4	0051			3328+	DC	H'81'	test number
00004CC6	00			3329+	DC	X'00'	
00004CC7	04			3330+	DC	HL1'4'	m5
00004CC8	E5D4C1C8 40404040			3331+	DC	CL8'VMAH'	instruction name
00004CD0	00004D4C			3332+	DC	A(RE81+16)	address of v2 source
00004CD4	00004D5C			3333+	DC	A(RE81+32)	address of v3 source
00004CD8	00004D6C			3334+	DC	A(RE81+48)	address of v4 source
00004CDC	00000010			3335+	DC	A(16)	result length
00004CE0	00004D3C			3336+REA81	DC	A(RE81)	result address
00004CE8	00000000 00000000			3337+	DS	FD	gap
00004CF0	00000000 00000000			3338+V1081	DS	XL16	V1 output
00004CF8	00000000 00000000						
00004D00	00000000 00000000			3339+	DS	FD	gap
				3340+*			
00004D08				3341+X81	DS	0F	
00004D08	E310 5010 0014		00000010	3342+	LGF	R1,V2ADDR	load v2 source
00004D0E	E761 0000 0806		00000000	3343+	VL	v22,0(R1)	use v22 to test decoder
00004D14	E310 5014 0014		00000014	3344+	LGF	R1,V3ADDR	load v3 source
00004D1A	E771 0000 0806		00000000	3345+	VL	v23,0(R1)	use v23 to test decoder
00004D20	E310 5018 0014		00000018	3346+	LGF	R1,V4ADDR	load v4 source
00004D26	E781 0000 0806		00000000	3347+	VL	v24,0(R1)	use v24 to test decoder
00004D2C	E766 7400 8FAB			3348+	VMAH	V22,V22,V23,V24,4	test instruction (dest is a source)
00004D32	E760 5030 080E		00004CF0	3349+	VST	V22,V1081	save v1 output
00004D38	07FB			3350+	BR	R11	return

LOC	OBJECT CODE	ADDR1	ADDR2	STMT					
00004D3C				3351+RE81	DC	0F	xl16 expected result		
00004D3C				3352+	DROP	R5			
00004D3C	FFFFFFFF	FFFFFFFF		3353	DC	XL16'	FFFFFFFF	FFFFFFFF	F6131F2C 2C658673' result
00004D44	F6131F2C	2C658673							
00004D4C	FF020304	05060750		3354	DC	XL16'	FF020304	05060750	090A0B0C 0D0E0F7F' v2
00004D54	090A0B0C	0D0E0F7F							
00004D5C	00000000	0000000A		3355	DC	XL16'	00000000	0000000A	0101010F 0101010F' v3
00004D64	0101010F	0101010F							
00004D6C	7FFFFFFFF	FFFFFFFF		3356	DC	XL16'	7FFFFFFFF	FFFFFFFF	FFFFFFFF FFFFFFFF' v4
00004D74	FFFFFFFF	FFFFFFFF							
				3357					
00004D80				3358	VRR_D	VMAH,4			
00004D80			00004D80	3359+	DS	0FD			
00004D80	00004DC8			3360+	USING	*,R5			base for test data and test routine
00004D84	0052			3361+T82	DC	A(X82)			address of test routine
00004D86	00			3362+	DC	H'82'			test number
00004D87	04			3363+	DC	X'00'			
00004D88	E5D4C1C8	40404040		3364+	DC	HL1'4'			m5
00004D90	00004E0C			3365+	DC	CL8'VMAH'			instruction name
00004D94	00004E1C			3366+	DC	A(RE82+16)			address of v2 source
00004D98	00004E2C			3367+	DC	A(RE82+32)			address of v3 source
00004D9C	00000010			3368+	DC	A(RE82+48)			address of v4 source
00004DA0	00004DFC			3369+	DC	A(16)			result length
00004DA8	00000000	00000000		3370+REA82	DC	A(RE82)			result address
00004DB0	00000000	00000000		3371+	DS	FD			gap
00004DB8	00000000	00000000		3372+V1082	DS	XL16			V1 output
00004DC0	00000000	00000000		3373+	DS	FD			gap
				3374+*					
00004DC8				3375+X82	DS	0F			
00004DC8	E310 5010 0014		00000010	3376+	LGF	R1,V2ADDR			load v2 source
00004DCE	E761 0000 0806		00000000	3377+	VL	v22,0(R1)			use v22 to test decoder
00004DD4	E310 5014 0014		00000014	3378+	LGF	R1,V3ADDR			load v3 source
00004DDA	E771 0000 0806		00000000	3379+	VL	v23,0(R1)			use v23 to test decoder
00004DE0	E310 5018 0014		00000018	3380+	LGF	R1,V4ADDR			load v4 source
00004DE6	E781 0000 0806		00000000	3381+	VL	v24,0(R1)			use v24 to test decoder
00004DEC	E766 7400 8FAB			3382+	VMAH	V22,V22,V23,V24,4			test instruction (dest is a source)
00004DF2	E760 5030 080E		00004DB0	3383+	VST	V22,V1082			save v1 output
00004DF8	07FB			3384+	BR	R11			return
00004DFC				3385+RE82	DC	0F			xl16 expected result
00004DFC				3386+	DROP	R5			
00004DFC	F6146364	177E9A6B		3387	DC	XL16'	F6146364	177E9A6B	F4342CDF 4C755AFE' result
00004E04	F4342CDF	4C755AFE							
00004E0C	50515253	54555657		3388	DC	XL16'	50515253	54555657	58595A5B 5C5D5E5F' v2
00004E14	58595A5B	5C5D5E5F							
00004E1C	E0616263	64656667		3389	DC	XL16'	E0616263	64656667	68696A6B 6C6D6E6F' v3
00004E24	68696A6B	6C6D6E6F							
00004E2C	FFFFFFFF	FFFFFFFF		3390	DC	XL16'	FFFFFFFF	FFFFFFFF	7FFFFFFFF FFFFFFFF' v4
00004E34	7FFFFFFFF	FFFFFFFF							
				3391	*-----				
				3392	* VMALE - Vector Multiply and Add Logical Even				
				3393	*-----				
				3394	* Byte				
				3395	VRR_D	VMALE,0			
00004E40				3396+	DS	0FD			
00004E40			00004E40	3397+	USING	*,R5			base for test data and test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004E40	00004E88			3398+T83	DC	A(X83)	address of test routine
00004E44	0053			3399+	DC	H'83'	test number
00004E46	00			3400+	DC	X'00'	
00004E47	00			3401+	DC	HL1'0'	m5
00004E48	E5D4C1D3 C5404040			3402+	DC	CL8'VMALE'	instruction name
00004E50	00004ECC			3403+	DC	A(RE83+16)	address of v2 source
00004E54	00004EDC			3404+	DC	A(RE83+32)	address of v3 source
00004E58	00004EEC			3405+	DC	A(RE83+48)	address of v4 source
00004E5C	00000010			3406+	DC	A(16)	result length
00004E60	00004EBC			3407+REA83	DC	A(RE83)	result address
00004E68	00000000 00000000			3408+	DS	FD	gap
00004E70	00000000 00000000			3409+V1083	DS	XL16	V1 output
00004E78	00000000 00000000						
00004E80	00000000 00000000			3410+	DS	FD	gap
				3411+*			
00004E88				3412+X83	DS	0F	
00004E88	E310 5010 0014		00000010	3413+	LGF	R1,V2ADDR	load v2 source
00004E8E	E761 0000 0806		00000000	3414+	VL	v22,0(R1)	use v22 to test decoder
00004E94	E310 5014 0014		00000014	3415+	LGF	R1,V3ADDR	load v3 source
00004E9A	E771 0000 0806		00000000	3416+	VL	v23,0(R1)	use v23 to test decoder
00004EA0	E310 5018 0014		00000018	3417+	LGF	R1,V4ADDR	load v4 source
00004EA6	E781 0000 0806		00000000	3418+	VL	v24,0(R1)	use v24 to test decoder
00004EAC	E766 7000 8FAC			3419+	VMALE	V22,V22,V23,V24,0	test instruction (dest is a source)
00004EB2	E760 5030 080E		00004E70	3420+	VST	V22,V1083	save v1 output
00004EB8	07FB			3421+	BR	R11	return
00004EBC				3422+RE83	DC	0F	xl16 expected result
00004EBC				3423+	DROP	R5	
00004EBC	FE010000 00000000			3424	DC	XL16'FE01000000000000 0000000000000000'	result
00004EC4	00000000 00000000						
00004ECC	FF000000 00000019			3425	DC	XL16'FF00000000000019 00000038000000FA'	v2
00004ED4	00000038 000000FA						
00004EDC	FF000000 00000019			3426	DC	XL16'FF00000000000019 00000038000000FA'	v3
00004EE4	00000038 000000FA						
00004EEC	00000000 00000000			3427	DC	XL16'0000000000000000 0000000000000000'	v4
00004EF4	00000000 00000000						
				3428			
				3429	VRR_D	VMALE,0	
00004F00				3430+	DS	0FD	
00004F00		00004F00		3431+	USING	*,R5	base for test data and test routine
00004F00	00004F48			3432+T84	DC	A(X84)	address of test routine
00004F04	0054			3433+	DC	H'84'	test number
00004F06	00			3434+	DC	X'00'	
00004F07	00			3435+	DC	HL1'0'	m5
00004F08	E5D4C1D3 C5404040			3436+	DC	CL8'VMALE'	instruction name
00004F10	00004F8C			3437+	DC	A(RE84+16)	address of v2 source
00004F14	00004F9C			3438+	DC	A(RE84+32)	address of v3 source
00004F18	00004FAC			3439+	DC	A(RE84+48)	address of v4 source
00004F1C	00000010			3440+	DC	A(16)	result length
00004F20	00004F7C			3441+REA84	DC	A(RE84)	result address
00004F28	00000000 00000000			3442+	DS	FD	gap
00004F30	00000000 00000000			3443+V1084	DS	XL16	V1 output
00004F38	00000000 00000000						
00004F40	00000000 00000000			3444+	DS	FD	gap
				3445+*			
00004F48				3446+X84	DS	0F	
00004F48	E310 5010 0014		00000010	3447+	LGF	R1,V2ADDR	load v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00004F4E	E761 0000 0806		00000000	3448+	VL	v22,0(R1)	use v22 to test decoder
00004F54	E310 5014 0014		00000014	3449+	LGF	R1,V3ADDR	load v3 source
00004F5A	E771 0000 0806		00000000	3450+	VL	v23,0(R1)	use v23 to test decoder
00004F60	E310 5018 0014		00000018	3451+	LGF	R1,V4ADDR	load v4 source
00004F66	E781 0000 0806		00000000	3452+	VL	v24,0(R1)	use v24 to test decoder
00004F6C	E766 7000 8FAC			3453+	VMALE	V22,V22,V23,V24,0	test instruction (dest is a source)
00004F72	E760 5030 080E		00004F30	3454+	VST	V22,V1084	save v1 output
00004F78	07FB			3455+	BR	R11	return
00004F7C				3456+RE84	DC	0F	xl16 expected result
00004F7C				3457+	DROP	R5	
00004F7C	FE030001 0000002F			3458	DC	XL16'FE0300010000002F 0000000300000002'	result
00004F84	00000003 00000002						
00004F8C	FF0000FF 00000029			3459	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00004F94	00000038 000000FA						
00004F9C	FF000001 00000029			3460	DC	XL16'FF00000100000029 00000038000000FA'	v3
00004FA4	00000038 000000FA						
00004FAC	00020001 0000002F			3461	DC	XL16'000200010000002F 0000000300000002'	v4
00004FB4	00000003 00000002						
				3462			
				3463	VRR_D	VMALE,0	
00004FC0				3464+	DS	0FD	
00004FC0		00004FC0		3465+	USING	*,R5	base for test data and test routine
00004FC0	00005008			3466+T85	DC	A(X85)	address of test routine
00004FC4	0055			3467+	DC	H'85'	test number
00004FC6	00			3468+	DC	X'00'	
00004FC7	00			3469+	DC	HL1'0'	m5
00004FC8	E5D4C1D3 C5404040			3470+	DC	CL8'VMALE'	instruction name
00004FD0	0000504C			3471+	DC	A(RE85+16)	address of v2 source
00004FD4	0000505C			3472+	DC	A(RE85+32)	address of v3 source
00004FD8	0000506C			3473+	DC	A(RE85+48)	address of v4 source
00004FDC	00000010			3474+	DC	A(16)	result length
00004FE0	0000503C			3475+REA85	DC	A(RE85)	result address
00004FE8	00000000 00000000			3476+	DS	FD	gap
00004FF0	00000000 00000000			3477+V1085	DS	XL16	V1 output
00004FF8	00000000 00000000						
00005000	00000000 00000000			3478+	DS	FD	gap
				3479+*			
00005008				3480+X85	DS	0F	
00005008	E310 5010 0014		00000010	3481+	LGF	R1,V2ADDR	load v2 source
0000500E	E761 0000 0806		00000000	3482+	VL	v22,0(R1)	use v22 to test decoder
00005014	E310 5014 0014		00000014	3483+	LGF	R1,V3ADDR	load v3 source
0000501A	E771 0000 0806		00000000	3484+	VL	v23,0(R1)	use v23 to test decoder
00005020	E310 5018 0014		00000018	3485+	LGF	R1,V4ADDR	load v4 source
00005026	E781 0000 0806		00000000	3486+	VL	v24,0(R1)	use v24 to test decoder
0000502C	E766 7000 8FAC			3487+	VMALE	V22,V22,V23,V24,0	test instruction (dest is a source)
00005032	E760 5030 080E		00004FF0	3488+	VST	V22,V1085	save v1 output
00005038	07FB			3489+	BR	R11	return
0000503C				3490+RE85	DC	0F	xl16 expected result
0000503C				3491+	DROP	R5	
0000503C	FD03030D 051F0739			3492	DC	XL16'FD03030D051F0739 095B0B850DB70FF1'	result
00005044	095B0B85 0DB70FF1						
0000504C	FF020304 05060708			3493	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005054	090A0B0C 0D0E0F10						
0000505C	FF020304 05060708			3494	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00005064	090A0B0C 0D0E0F10						
0000506C	FF020304 05060708			3495	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005074	090A0B0C 0D0E0F10			3496			
				3497	VRR_D	VMALE, 0	
00005080				3498+	DS	0FD	
00005080		00005080		3499+	USING	*, R5	base for test data and test routine
00005080	000050C8			3500+T86	DC	A(X86)	address of test routine
00005084	0056			3501+	DC	H'86'	test number
00005086	00			3502+	DC	X'00'	
00005087	00			3503+	DC	HL1'0'	m5
00005088	E5D4C1D3 C5404040			3504+	DC	CL8'VMALE'	instruction name
00005090	0000510C			3505+	DC	A(RE86+16)	address of v2 source
00005094	0000511C			3506+	DC	A(RE86+32)	address of v3 source
00005098	0000512C			3507+	DC	A(RE86+48)	address of v4 source
0000509C	00000010			3508+	DC	A(16)	result length
000050A0	000050FC			3509+REA86	DC	A(RE86)	result address
000050A8	00000000 00000000			3510+	DS	FD	gap
000050B0	00000000 00000000			3511+V1086	DS	XL16	V1 output
000050B8	00000000 00000000						
000050C0	00000000 00000000			3512+	DS	FD	gap
				3513+*			
000050C8				3514+X86	DS	0F	
000050C8	E310 5010 0014		00000010	3515+	LGF	R1, V2ADDR	load v2 source
000050CE	E761 0000 0806		00000000	3516+	VL	v22, 0(R1)	use v22 to test decoder
000050D4	E310 5014 0014		00000014	3517+	LGF	R1, V3ADDR	load v3 source
000050DA	E771 0000 0806		00000000	3518+	VL	v23, 0(R1)	use v23 to test decoder
000050E0	E310 5018 0014		00000018	3519+	LGF	R1, V4ADDR	load v4 source
000050E6	E781 0000 0806		00000000	3520+	VL	v24, 0(R1)	use v24 to test decoder
000050EC	E766 7000 8FAC			3521+	VMALE	V22, V22, V23, V24, 0	test instruction (dest is a source)
000050F2	E760 5030 080E		000050B0	3522+	VST	V22, V1086	save v1 output
000050F8	07FB			3523+	BR	R11	return
000050FC				3524+RE86	DC	0F	xl16 expected result
000050FC				3525+	DROP	R5	
000050FC	FD030307 0510071D			3526	DC	XL16'FD0303070510071D 092E0B430D5C0F79'	result
00005104	092E0B43 0D5C0F79						
0000510C	FF020304 05060708			3527	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005114	090A0B0C 0D0E0F10						
0000511C	FF010102 02030304			3528	DC	XL16'FF01010202030304 0405050606070708'	v3
00005124	04050506 06070708						
0000512C	FF020304 05060708			3529	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00005134	090A0B0C 0D0E0F10						
				3530			
				3531	VRR_D	VMALE, 0	
00005140				3532+	DS	0FD	
00005140		00005140		3533+	USING	*, R5	base for test data and test routine
00005140	00005188			3534+T87	DC	A(X87)	address of test routine
00005144	0057			3535+	DC	H'87'	test number
00005146	00			3536+	DC	X'00'	
00005147	00			3537+	DC	HL1'0'	m5
00005148	E5D4C1D3 C5404040			3538+	DC	CL8'VMALE'	instruction name
00005150	000051CC			3539+	DC	A(RE87+16)	address of v2 source
00005154	000051DC			3540+	DC	A(RE87+32)	address of v3 source
00005158	000051EC			3541+	DC	A(RE87+48)	address of v4 source
0000515C	00000010			3542+	DC	A(16)	result length
00005160	000051BC			3543+REA87	DC	A(RE87)	result address
00005168	00000000 00000000			3544+	DS	FD	gap
00005170	00000000 00000000			3545+V1087	DS	XL16	V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005178	00000000 00000000						
00005180	00000000 00000000			3546+	DS	FD	gap
				3547+*			
00005188				3548+X87	DS	0F	
00005188	E310 5010 0014		00000010	3549+	LGF	R1,V2ADDR	load v2 source
0000518E	E761 0000 0806		00000000	3550+	VL	v22,0(R1)	use v22 to test decoder
00005194	E310 5014 0014		00000014	3551+	LGF	R1,V3ADDR	load v3 source
0000519A	E771 0000 0806		00000000	3552+	VL	v23,0(R1)	use v23 to test decoder
000051A0	E310 5018 0014		00000018	3553+	LGF	R1,V4ADDR	load v4 source
000051A6	E781 0000 0806		00000000	3554+	VL	v24,0(R1)	use v24 to test decoder
000051AC	E766 7000 8FAC			3555+	VMALE	V22,V22,V23,V24,0	test instruction (dest is a source)
000051B2	E760 5030 080E		00005170	3556+	VST	V22,V1087	save v1 output
000051B8	07FB			3557+	BR	R11	return
000051BC				3558+RE87	DC	0F	xl16 expected result
000051BC				3559+	DROP	R5	
000051BC	FD030304 05060708			3560	DC	XL16'FD03030405060708 09130B170D1B0F1F'	result
000051C4	09130B17 0D1B0F1F						
000051CC	FF020304 05060708			3561	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000051D4	090A0B0C 0D0E0F10						
000051DC	FF000000 00000001			3562	DC	XL16'FF00000000000001 0101010101010102'	v3
000051E4	01010101 01010102						
000051EC	FF020304 05060708			3563	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000051F4	090A0B0C 0D0E0F10						
				3564			
				3565 * Halfword			
				3566	VRR_D	VMALE,1	
00005200				3567+	DS	0FD	
00005200		00005200		3568+	USING	*,R5	base for test data and test routine
00005200	00005248			3569+T88	DC	A(X88)	address of test routine
00005204	0058			3570+	DC	H'88'	test number
00005206	00			3571+	DC	X'00'	
00005207	01			3572+	DC	HL1'1'	m5
00005208	E5D4C1D3 C5404040			3573+	DC	CL8'VMALE'	instruction name
00005210	0000528C			3574+	DC	A(RE88+16)	address of v2 source
00005214	0000529C			3575+	DC	A(RE88+32)	address of v3 source
00005218	000052AC			3576+	DC	A(RE88+48)	address of v4 source
0000521C	00000010			3577+	DC	A(16)	result length
00005220	0000527C			3578+REA88	DC	A(RE88)	result address
00005228	00000000 00000000			3579+	DS	FD	gap
00005230	00000000 00000000			3580+V1088	DS	XL16	V1 output
00005238	00000000 00000000						
00005240	00000000 00000000			3581+	DS	FD	gap
				3582+*			
00005248				3583+X88	DS	0F	
00005248	E310 5010 0014		00000010	3584+	LGF	R1,V2ADDR	load v2 source
0000524E	E761 0000 0806		00000000	3585+	VL	v22,0(R1)	use v22 to test decoder
00005254	E310 5014 0014		00000014	3586+	LGF	R1,V3ADDR	load v3 source
0000525A	E771 0000 0806		00000000	3587+	VL	v23,0(R1)	use v23 to test decoder
00005260	E310 5018 0014		00000018	3588+	LGF	R1,V4ADDR	load v4 source
00005266	E781 0000 0806		00000000	3589+	VL	v24,0(R1)	use v24 to test decoder
0000526C	E766 7100 8FAC			3590+	VMALE	V22,V22,V23,V24,1	test instruction (dest is a source)
00005272	E760 5030 080E		00005230	3591+	VST	V22,V1088	save v1 output
00005278	07FB			3592+	BR	R11	return
0000527C				3593+RE88	DC	0F	xl16 expected result
0000527C				3594+	DROP	R5	
0000527C	FE010000 00000000			3595	DC	XL16'FE01000000000000 0000000000000000'	result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005284	00000000 00000000						
0000528C	FF000000 00000019			3596	DC	XL16'FF00000000000019 00000038000000FA'	v2
00005294	00000038 000000FA						
0000529C	FF000000 00000019			3597	DC	XL16'FF00000000000019 00000038000000FA'	v3
000052A4	00000038 000000FA						
000052AC	00000000 00000000			3598	DC	XL16'0000000000000000 0000000000000000'	v4
000052B4	00000000 00000000						
				3599			
000052C0				3600	VRR_D	VMALE,1	
000052C0		000052C0		3601+	DS	0FD	
000052C0	00005308			3602+	USING	*,R5	base for test data and test routine
000052C4	0059			3603+T89	DC	A(X89)	address of test routine
000052C6	00			3604+	DC	H'89'	test number
000052C7	01			3605+	DC	X'00'	
000052C8	E5D4C1D3 C5404040			3606+	DC	HL1'1'	m5
000052D0	0000534C			3607+	DC	CL8'VMALE'	instruction name
000052D4	0000535C			3608+	DC	A(RE89+16)	address of v2 source
000052D8	0000536C			3609+	DC	A(RE89+32)	address of v3 source
000052DC	00000010			3610+	DC	A(RE89+48)	address of v4 source
000052E0	0000533C			3611+	DC	A(16)	result length
000052E8	00000000 00000000			3612+REA89	DC	A(RE89)	result address
000052F0	00000000 00000000			3613+	DS	FD	gap
000052F8	00000000 00000000			3614+V1089	DS	XL16	V1 output
00005300	00000000 00000000						
				3615+	DS	FD	gap
				3616+*			
00005308				3617+X89	DS	0F	
00005308	E310 5010 0014		00000010	3618+	LGF	R1,V2ADDR	load v2 source
0000530E	E761 0000 0806		00000000	3619+	VL	v22,0(R1)	use v22 to test decoder
00005314	E310 5014 0014		00000014	3620+	LGF	R1,V3ADDR	load v3 source
0000531A	E771 0000 0806		00000000	3621+	VL	v23,0(R1)	use v23 to test decoder
00005320	E310 5018 0014		00000018	3622+	LGF	R1,V4ADDR	load v4 source
00005326	E781 0000 0806		00000000	3623+	VL	v24,0(R1)	use v24 to test decoder
0000532C	E766 7100 8FAC			3624+	VMALE	V22,V22,V23,V24,1	test instruction (dest is a source)
00005332	E760 5030 080E		000052F0	3625+	VST	V22,V1089	save v1 output
00005338	07FB			3626+	BR	R11	return
0000533C				3627+RE89	DC	0F	xl16 expected result
0000533C				3628+	DROP	R5	
0000533C	FE030001 0000002F			3629	DC	XL16'FE0300010000002F 0000000300000002'	result
00005344	00000003 00000002						
0000534C	FF0000FF 00000029			3630	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00005354	00000038 000000FA						
0000535C	FF000001 00000029			3631	DC	XL16'FF00000100000029 00000038000000FA'	v3
00005364	00000038 000000FA						
0000536C	00020001 0000002F			3632	DC	XL16'000200010000002F 0000000300000002'	v4
00005374	00000003 00000002						
				3633			
00005380				3634	VRR_D	VMALE,1	
00005380		00005380		3635+	DS	0FD	
00005380	000053C8			3636+	USING	*,R5	base for test data and test routine
00005384	005A			3637+T90	DC	A(X90)	address of test routine
00005386	00			3638+	DC	H'90'	test number
00005387	01			3639+	DC	X'00'	
00005388	E5D4C1D3 C5404040			3640+	DC	HL1'1'	m5
00005390	0000540C			3641+	DC	CL8'VMALE'	instruction name
				3642+	DC	A(RE90+16)	address of v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005394	0000541C			3643+	DC	A(RE90+32)	address of v3 source
00005398	0000542C			3644+	DC	A(RE90+48)	address of v4 source
0000539C	00000010			3645+	DC	A(16)	result length
000053A0	000053FC			3646+REA90	DC	A(RE90)	result address
000053A8	00000000 00000000			3647+	DS	FD	gap
000053B0	00000000 00000000			3648+V1090	DS	XL16	V1 output
000053B8	00000000 00000000						
000053C0	00000000 00000000			3649+	DS	FD	gap
				3650+*			
000053C8				3651+X90	DS	0F	
000053C8	E310 5010 0014		00000010	3652+	LGF	R1,V2ADDR	load v2 source
000053CE	E761 0000 0806		00000000	3653+	VL	v22,0(R1)	use v22 to test decoder
000053D4	E310 5014 0014		00000014	3654+	LGF	R1,V3ADDR	load v3 source
000053DA	E771 0000 0806		00000000	3655+	VL	v23,0(R1)	use v23 to test decoder
000053E0	E310 5018 0014		00000018	3656+	LGF	R1,V4ADDR	load v4 source
000053E6	E781 0000 0806		00000000	3657+	VL	v24,0(R1)	use v24 to test decoder
000053EC	E766 7100 8FAC			3658+	VMALE	V22,V22,V23,V24,1	test instruction (dest is a source)
000053F2	E760 5030 080E		000053B0	3659+	VST	V22,V1090	save v1 output
000053F8	07FB			3660+	BR	R11	return
000053FC				3661+RE90	DC	0F	xl16 expected result
000053FC				3662+	DROP	R5	
000053FC	FD06FF08 051F432C			3663	DC	XL16'FD06FF08051F432C 095BBF700DB87BD4'	result
00005404	095BBF70 0DB87BD4						
0000540C	FF020304 05060708			3664	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005414	090A0B0C 0D0E0F10						
0000541C	FF020304 05060708			3665	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00005424	090A0B0C 0D0E0F10						
0000542C	FF020304 05060708			3666	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00005434	090A0B0C 0D0E0F10						
				3667			
00005440				3668	VRR_D	VMALE,1	
00005440		00005440		3669+	DS	0FD	
00005440	00005488			3670+	USING	*,R5	base for test data and test routine
00005444	005B			3671+T91	DC	A(X91)	address of test routine
00005446	00			3672+	DC	H'91'	test number
00005447	01			3673+	DC	X'00'	
00005448	E5D4C1D3 C5404040			3674+	DC	HL1'1'	m5
00005450	000054CC			3675+	DC	CL8'VMALE'	instruction name
00005454	000054DC			3676+	DC	A(RE91+16)	address of v2 source
00005458	000054EC			3677+	DC	A(RE91+32)	address of v3 source
0000545C	00000010			3678+	DC	A(RE91+48)	address of v4 source
00005460	000054BC			3679+	DC	A(16)	result length
00005468	00000000 00000000			3680+REA91	DC	A(RE91)	result address
00005470	00000000 00000000			3681+	DS	FD	gap
00005478	00000000 00000000			3682+V1091	DS	XL16	V1 output
00005480	00000000 00000000						
				3683+	DS	FD	gap
				3684+*			
00005488				3685+X91	DS	0F	
00005488	E310 5010 0014		00000010	3686+	LGF	R1,V2ADDR	load v2 source
0000548E	E761 0000 0806		00000000	3687+	VL	v22,0(R1)	use v22 to test decoder
00005494	E310 5014 0014		00000014	3688+	LGF	R1,V3ADDR	load v3 source
0000549A	E771 0000 0806		00000000	3689+	VL	v23,0(R1)	use v23 to test decoder
000054A0	E310 5018 0014		00000018	3690+	LGF	R1,V4ADDR	load v4 source
000054A6	E781 0000 0806		00000000	3691+	VL	v24,0(R1)	use v24 to test decoder
000054AC	E766 7100 8FAC			3692+	VMALE	V22,V22,V23,V24,1	test instruction (dest is a source)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000054B2	E760 5030 080E		00005470	3693+	VST	V22,V1091	save v1 output
000054B8	07FB			3694+	BR	R11	return
000054BC				3695+RE91	DC	0F	xl16 expected result
000054BC				3696+	DROP	R5	
000054BC	FD060006 0510221A			3697	DC	XL16'FD0600060510221A 092E603E0D5CBE72'	result
000054C4	092E603E 0D5CBE72						
000054CC	FF020304 05060708			3698	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000054D4	090A0B0C 0D0E0F10						
000054DC	FF010102 02030304			3699	DC	XL16'FF01010202030304 0405050606070708'	v3
000054E4	04050506 06070708						
000054EC	FF020304 05060708			3700	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000054F4	090A0B0C 0D0E0F10						
				3701			
				3702	VRR_D	VMALE,1	
00005500				3703+	DS	0FD	
00005500		00005500		3704+	USING	*,R5	base for test data and test routine
00005500	00005548			3705+T92	DC	A(X92)	address of test routine
00005504	005C			3706+	DC	H'92'	test number
00005506	00			3707+	DC	X'00'	
00005507	01			3708+	DC	HL1'1'	m5
00005508	E5D4C1D3 C5404040			3709+	DC	CL8'VMALE'	instruction name
00005510	0000558C			3710+	DC	A(RE92+16)	address of v2 source
00005514	0000559C			3711+	DC	A(RE92+32)	address of v3 source
00005518	000055AC			3712+	DC	A(RE92+48)	address of v4 source
0000551C	00000010			3713+	DC	A(16)	result length
00005520	0000557C			3714+REA92	DC	A(RE92)	result address
00005528	00000000 00000000			3715+	DS	FD	gap
00005530	00000000 00000000			3716+V1092	DS	XL16	V1 output
00005538	00000000 00000000						
00005540	00000000 00000000			3717+	DS	FD	gap
				3718+*			
00005548				3719+X92	DS	0F	
00005548	E310 5010 0014		00000010	3720+	LGF	R1,V2ADDR	load v2 source
0000554E	E761 0000 0806		00000000	3721+	VL	v22,0(R1)	use v22 to test decoder
00005554	E310 5014 0014		00000014	3722+	LGF	R1,V3ADDR	load v3 source
0000555A	E771 0000 0806		00000000	3723+	VL	v23,0(R1)	use v23 to test decoder
00005560	E310 5018 0014		00000018	3724+	LGF	R1,V4ADDR	load v4 source
00005566	E781 0000 0806		00000000	3725+	VL	v24,0(R1)	use v24 to test decoder
0000556C	E766 7100 8FAC			3726+	VMALE	V22,V22,V23,V24,1	test instruction (dest is a source)
00005572	E760 5030 080E		00005530	3727+	VST	V22,V1092	save v1 output
00005578	07FB			3728+	BR	R11	return
0000557C				3729+RE92	DC	0F	xl16 expected result
0000557C				3730+	DROP	R5	
0000557C	FD050104 05060708			3731	DC	XL16'FD05010405060708 09131E160D1B2A1E'	result
00005584	09131E16 0D1B2A1E						
0000558C	FF020304 05060708			3732	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005594	090A0B0C 0D0E0F10						
0000559C	FF000000 00000001			3733	DC	XL16'FF00000000000001 0101010101010102'	v3
000055A4	01010101 01010102						
000055AC	FF020304 05060708			3734	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000055B4	090A0B0C 0D0E0F10						
				3735			
				3736 * Word			
				3737	VRR_D	VMALE,2	
000055C0				3738+	DS	0FD	
000055C0		000055C0		3739+	USING	*,R5	base for test data and test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000055C0	00005608			3740+T93	DC	A(X93)	address of test routine
000055C4	005D			3741+	DC	H'93'	test number
000055C6	00			3742+	DC	X'00'	
000055C7	02			3743+	DC	HL1'2'	m5
000055C8	E5D4C1D3 C5404040			3744+	DC	CL8'VMALE'	instruction name
000055D0	0000564C			3745+	DC	A(RE93+16)	address of v2 source
000055D4	0000565C			3746+	DC	A(RE93+32)	address of v3 source
000055D8	0000566C			3747+	DC	A(RE93+48)	address of v4 source
000055DC	00000010			3748+	DC	A(16)	result length
000055E0	0000563C			3749+REA93	DC	A(RE93)	result address
000055E8	00000000 00000000			3750+	DS	FD	gap
000055F0	00000000 00000000			3751+V1093	DS	XL16	V1 output
000055F8	00000000 00000000						
00005600	00000000 00000000			3752+	DS	FD	gap
				3753+*			
00005608				3754+X93	DS	0F	
00005608	E310 5010 0014		00000010	3755+	LGF	R1,V2ADDR	load v2 source
0000560E	E761 0000 0806		00000000	3756+	VL	v22,0(R1)	use v22 to test decoder
00005614	E310 5014 0014		00000014	3757+	LGF	R1,V3ADDR	load v3 source
0000561A	E771 0000 0806		00000000	3758+	VL	v23,0(R1)	use v23 to test decoder
00005620	E310 5018 0014		00000018	3759+	LGF	R1,V4ADDR	load v4 source
00005626	E781 0000 0806		00000000	3760+	VL	v24,0(R1)	use v24 to test decoder
0000562C	E766 7200 8FAC			3761+	VMALE	V22,V22,V23,V24,2	test instruction (dest is a source)
00005632	E760 5030 080E		000055F0	3762+	VST	V22,V1093	save v1 output
00005638	07FB			3763+	BR	R11	return
0000563C				3764+RE93	DC	0F	xl16 expected result
0000563C				3765+	DROP	R5	
0000563C	FE010000 00000000			3766	DC	XL16'FE01000000000000 00000000000000C40'	result
00005644	00000000 00000C40						
0000564C	FF000000 00000019			3767	DC	XL16'FF00000000000019 00000038000000FA'	v2
00005654	00000038 000000FA						
0000565C	FF000000 00000019			3768	DC	XL16'FF00000000000019 00000038000000FA'	v3
00005664	00000038 000000FA						
0000566C	00000000 00000000			3769	DC	XL16'0000000000000000 0000000000000000'	v4
00005674	00000000 00000000						
				3770			
00005680				3771	VRR_D	VMALE,2	
00005680		00005680		3772+	DS	0FD	
00005680	000056C8			3773+	USING	*,R5	base for test data and test routine
00005684	005E			3774+T94	DC	A(X94)	address of test routine
00005686	00			3775+	DC	H'94'	test number
00005687	02			3776+	DC	X'00'	
00005688	E5D4C1D3 C5404040			3777+	DC	HL1'2'	m5
00005690	0000570C			3778+	DC	CL8'VMALE'	instruction name
00005694	0000571C			3779+	DC	A(RE94+16)	address of v2 source
00005698	0000572C			3780+	DC	A(RE94+32)	address of v3 source
0000569C	00000010			3781+	DC	A(RE94+48)	address of v4 source
000056A0	000056FC			3782+	DC	A(16)	result length
000056A8	00000000 00000000			3783+REA94	DC	A(RE94)	result address
000056B0	00000000 00000000			3784+	DS	FD	gap
000056B8	00000000 00000000			3785+V1094	DS	XL16	V1 output
000056C0	00000000 00000000						
				3786+	DS	FD	gap
				3787+*			
000056C8				3788+X94	DS	0F	
000056C8	E310 5010 0014		00000010	3789+	LGF	R1,V2ADDR	load v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000056CE	E761 0000 0806		00000000	3790+	VL	v22,0(R1)	use v22 to test decoder
000056D4	E310 5014 0014		00000014	3791+	LGF	R1,V3ADDR	load v3 source
000056DA	E771 0000 0806		00000000	3792+	VL	v23,0(R1)	use v23 to test decoder
000056E0	E310 5018 0014		00000018	3793+	LGF	R1,V4ADDR	load v4 source
000056E6	E781 0000 0806		00000000	3794+	VL	v24,0(R1)	use v24 to test decoder
000056EC	E766 7200 8FAC			3795+	VMALE	V22,V22,V23,V24,2	test instruction (dest is a source)
000056F2	E760 5030 080E		000056B0	3796+	VST	V22,V1094	save v1 output
000056F8	07FB			3797+	BR	R11	return
000056FC				3798+RE94	DC	0F	xl16 expected result
000056FC				3799+	DROP	R5	
000056FC	FE030100 0000012E			3800	DC	XL16'FE0301000000012E 00000003000000C42'	result
00005704	00000003 00000C42						
0000570C	FF0000FF 00000029			3801	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00005714	00000038 000000FA						
0000571C	FF000001 00000029			3802	DC	XL16'FF00000100000029 00000038000000FA'	v3
00005724	00000038 000000FA						
0000572C	00020001 0000002F			3803	DC	XL16'000200010000002F 0000000300000002'	v4
00005734	00000003 00000002						
				3804			
				3805	VRR_D	VMALE,2	
00005740				3806+	DS	0FD	
00005740		00005740		3807+	USING	*,R5	base for test data and test routine
00005740	00005788			3808+T95	DC	A(X95)	address of test routine
00005744	005F			3809+	DC	H'95'	test number
00005746	00			3810+	DC	X'00'	
00005747	02			3811+	DC	HL1'2'	m5
00005748	E5D4C1D3 C5404040			3812+	DC	CL8'VMALE'	instruction name
00005750	000057CC			3813+	DC	A(RE95+16)	address of v2 source
00005754	000057DC			3814+	DC	A(RE95+32)	address of v3 source
00005758	000057EC			3815+	DC	A(RE95+48)	address of v4 source
0000575C	00000010			3816+	DC	A(16)	result length
00005760	000057BC			3817+REA95	DC	A(RE95)	result address
00005768	00000000 00000000			3818+	DS	FD	gap
00005770	00000000 00000000			3819+V1095	DS	XL16	V1 output
00005778	00000000 00000000						
00005780	00000000 00000000			3820+	DS	FD	gap
				3821+*			
00005788				3822+X95	DS	0F	
00005788	E310 5010 0014		00000010	3823+	LGF	R1,V2ADDR	load v2 source
0000578E	E761 0000 0806		00000000	3824+	VL	v22,0(R1)	use v22 to test decoder
00005794	E310 5014 0014		00000014	3825+	LGF	R1,V3ADDR	load v3 source
0000579A	E771 0000 0806		00000000	3826+	VL	v23,0(R1)	use v23 to test decoder
000057A0	E310 5018 0014		00000018	3827+	LGF	R1,V4ADDR	load v4 source
000057A6	E781 0000 0806		00000000	3828+	VL	v24,0(R1)	use v24 to test decoder
000057AC	E766 7200 8FAC			3829+	VMALE	V22,V22,V23,V24,2	test instruction (dest is a source)
000057B2	E760 5030 080E		00005770	3830+	VST	V22,V1095	save v1 output
000057B8	07FB			3831+	BR	R11	return
000057BC				3832+RE95	DC	0F	xl16 expected result
000057BC				3833+	DROP	R5	
000057BC	FD07050A 091F1F18			3834	DC	XL16'FD07050A091F1F18 095BC037C27817A0'	result
000057C4	095BC037 C27817A0						
000057CC	FF020304 05060708			3835	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000057D4	090A0B0C 0D0E0F10						
000057DC	FF020304 05060708			3836	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000057E4	090A0B0C 0D0E0F10						
000057EC	FF020304 05060708			3837	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000057F4	090A0B0C 0D0E0F10			3838			
				3839	VRR_D	VMALE, 2	
00005800				3840+	DS	0FD	
00005800		00005800		3841+	USING	*, R5	base for test data and test routine
00005800	00005848			3842+T96	DC	A(X96)	address of test routine
00005804	0060			3843+	DC	H'96'	test number
00005806	00			3844+	DC	X'00'	
00005807	02			3845+	DC	HL1'2'	m5
00005808	E5D4C1D3 C5404040			3846+	DC	CL8'VMALE'	instruction name
00005810	0000588C			3847+	DC	A(RE96+16)	address of v2 source
00005814	0000589C			3848+	DC	A(RE96+32)	address of v3 source
00005818	000058AC			3849+	DC	A(RE96+48)	address of v4 source
0000581C	00000010			3850+	DC	A(16)	result length
00005820	0000587C			3851+REA96	DC	A(RE96)	result address
00005828	00000000 00000000			3852+	DS	FD	gap
00005830	00000000 00000000			3853+V1096	DS	XL16	V1 output
00005838	00000000 00000000						
00005840	00000000 00000000			3854+	DS	FD	gap
				3855+*			
00005848				3856+X96	DS	0F	
00005848	E310 5010 0014		00000010	3857+	LGF	R1, V2ADDR	load v2 source
0000584E	E761 0000 0806		00000000	3858+	VL	v22, 0(R1)	use v22 to test decoder
00005854	E310 5014 0014		00000014	3859+	LGF	R1, V3ADDR	load v3 source
0000585A	E771 0000 0806		00000000	3860+	VL	v23, 0(R1)	use v23 to test decoder
00005860	E310 5018 0014		00000018	3861+	LGF	R1, V4ADDR	load v4 source
00005866	E781 0000 0806		00000000	3862+	VL	v24, 0(R1)	use v24 to test decoder
0000586C	E766 7200 8FAC			3863+	VMALE	V22, V22, V23, V24, 2	test instruction (dest is a source)
00005872	E760 5030 080E		00005830	3864+	VST	V22, V1096	save v1 output
00005878	07FB			3865+	BR	R11	return
0000587C				3866+RE96	DC	0F	xl16 expected result
0000587C				3867+	DROP	R5	
0000587C	FD060408 04111110			3868	DC	XL16'FD06040804111110 092E6097DCBD8D58'	result
00005884	092E6097 DCBD8D58						
0000588C	FF020304 05060708			3869	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005894	090A0B0C 0D0E0F10						
0000589C	FF010102 02030304			3870	DC	XL16'FF01010202030304 0405050606070708'	v3
000058A4	04050506 06070708						
000058AC	FF020304 05060708			3871	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000058B4	090A0B0C 0D0E0F10						
				3872			
				3873	VRR_D	VMALE, 2	
000058C0				3874+	DS	0FD	
000058C0		000058C0		3875+	USING	*, R5	base for test data and test routine
000058C0	00005908			3876+T97	DC	A(X97)	address of test routine
000058C4	0061			3877+	DC	H'97'	test number
000058C6	00			3878+	DC	X'00'	
000058C7	02			3879+	DC	HL1'2'	m5
000058C8	E5D4C1D3 C5404040			3880+	DC	CL8'VMALE'	instruction name
000058D0	0000594C			3881+	DC	A(RE97+16)	address of v2 source
000058D4	0000595C			3882+	DC	A(RE97+32)	address of v3 source
000058D8	0000596C			3883+	DC	A(RE97+48)	address of v4 source
000058DC	00000010			3884+	DC	A(16)	result length
000058E0	0000593C			3885+REA97	DC	A(RE97)	result address
000058E8	00000000 00000000			3886+	DS	FD	gap
000058F0	00000000 00000000			3887+V1097	DS	XL16	V1 output

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000058F8	00000000 00000000						
00005900	00000000 00000000			3888+	DS	FD	gap
				3889+*			
00005908				3890+X97	DS	0F	
00005908	E310 5010 0014		00000010	3891+	LGF	R1,V2ADDR	load v2 source
0000590E	E761 0000 0806		00000000	3892+	VL	v22,0(R1)	use v22 to test decoder
00005914	E310 5014 0014		00000014	3893+	LGF	R1,V3ADDR	load v3 source
0000591A	E771 0000 0806		00000000	3894+	VL	v23,0(R1)	use v23 to test decoder
00005920	E310 5018 0014		00000018	3895+	LGF	R1,V4ADDR	load v4 source
00005926	E781 0000 0806		00000000	3896+	VL	v24,0(R1)	use v24 to test decoder
0000592C	E766 7200 8FAC			3897+	VMALE	V22,V22,V23,V24,2	test instruction (dest is a source)
00005932	E760 5030 080E		000058F0	3898+	VST	V22,V1097	save v1 output
00005938	07FB			3899+	BR	R11	return
0000593C				3900+RE97	DC	0F	xl16 expected result
0000593C				3901+	DROP	R5	
0000593C	FD050405 01060708			3902	DC	XL16'FD05040501060708 09131E2A372F261C'	result
00005944	09131E2A 372F261C						
0000594C	FF020304 05060708			3903	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00005954	090A0B0C 0D0E0F10						
0000595C	FF000000 00000001			3904	DC	XL16'FF00000000000001 0101010101010102'	v3
00005964	01010101 01010102						
0000596C	FF020304 05060708			3905	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00005974	090A0B0C 0D0E0F10						
				3906			
				3907 * Doubleword			
				3908	VRR_D	VMALE,3	
00005980				3909+	DS	0FD	
00005980		00005980		3910+	USING	*,R5	base for test data and test routine
00005980	000059C8			3911+T98	DC	A(X98)	address of test routine
00005984	0062			3912+	DC	H'98'	test number
00005986	00			3913+	DC	X'00'	
00005987	03			3914+	DC	HL1'3'	m5
00005988	E5D4C1D3 C5404040			3915+	DC	CL8'VMALE'	instruction name
00005990	00005A0C			3916+	DC	A(RE98+16)	address of v2 source
00005994	00005A1C			3917+	DC	A(RE98+32)	address of v3 source
00005998	00005A2C			3918+	DC	A(RE98+48)	address of v4 source
0000599C	00000010			3919+	DC	A(16)	result length
000059A0	000059FC			3920+REA98	DC	A(RE98)	result address
000059A8	00000000 00000000			3921+	DS	FD	gap
000059B0	00000000 00000000			3922+V1098	DS	XL16	V1 output
000059B8	00000000 00000000						
000059C0	00000000 00000000			3923+	DS	FD	gap
				3924+*			
000059C8				3925+X98	DS	0F	
000059C8	E310 5010 0014		00000010	3926+	LGF	R1,V2ADDR	load v2 source
000059CE	E761 0000 0806		00000000	3927+	VL	v22,0(R1)	use v22 to test decoder
000059D4	E310 5014 0014		00000014	3928+	LGF	R1,V3ADDR	load v3 source
000059DA	E771 0000 0806		00000000	3929+	VL	v23,0(R1)	use v23 to test decoder
000059E0	E310 5018 0014		00000018	3930+	LGF	R1,V4ADDR	load v4 source
000059E6	E781 0000 0806		00000000	3931+	VL	v24,0(R1)	use v24 to test decoder
000059EC	E766 7300 8FAC			3932+	VMALE	V22,V22,V23,V24,3	test instruction (dest is a source)
000059F2	E760 5030 080E		000059B0	3933+	VST	V22,V1098	save v1 output
000059F8	07FB			3934+	BR	R11	return
000059FC				3935+RE98	DC	0F	xl16 expected result
000059FC				3936+	DROP	R5	
000059FC	FFFFFFFFE 00032000			3937	DC	XL16'FFFFFFFFE00032000 FFFCE00271000000'	result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005A04	FFFCE002 71000000						
00005A0C	FFFFFFFF 00019000			3938	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00005A14	00000038 EEEEEEEFA						
00005A1C	FFFFFFFF 00019000			3939	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
00005A24	00000038 0EEEEEEFA						
00005A2C	00000000 00000000			3940	DC	XL16'0000000000000000 0000000000000000'	v4
00005A34	00000000 00000000						
				3941			
				3942	VRR_D	VMALE, 3	
00005A40				3943+	DS	0FD	
00005A40		00005A40		3944+	USING	*, R5	base for test data and test routine
00005A40	00005A88			3945+T99	DC	A(X99)	address of test routine
00005A44	0063			3946+	DC	H'99'	test number
00005A46	00			3947+	DC	X'00'	
00005A47	03			3948+	DC	HL1'3'	m5
00005A48	E5D4C1D3 C5404040			3949+	DC	CL8'VMALE'	instruction name
00005A50	00005ACC			3950+	DC	A(RE99+16)	address of v2 source
00005A54	00005ADC			3951+	DC	A(RE99+32)	address of v3 source
00005A58	00005AEC			3952+	DC	A(RE99+48)	address of v4 source
00005A5C	00000010			3953+	DC	A(16)	result length
00005A60	00005ABC			3954+REA99	DC	A(RE99)	result address
00005A68	00000000 00000000			3955+	DS	FD	gap
00005A70	00000000 00000000			3956+V1099	DS	XL16	V1 output
00005A78	00000000 00000000						
00005A80	00000000 00000000			3957+	DS	FD	gap
				3958+*			
00005A88				3959+X99	DS	0F	
00005A88	E310 5010 0014		00000010	3960+	LGF	R1, V2ADDR	load v2 source
00005A8E	E761 0000 0806		00000000	3961+	VL	v22, 0(R1)	use v22 to test decoder
00005A94	E310 5014 0014		00000014	3962+	LGF	R1, V3ADDR	load v3 source
00005A9A	E771 0000 0806		00000000	3963+	VL	v23, 0(R1)	use v23 to test decoder
00005AA0	E310 5018 0014		00000018	3964+	LGF	R1, V4ADDR	load v4 source
00005AA6	E781 0000 0806		00000000	3965+	VL	v24, 0(R1)	use v24 to test decoder
00005AAC	E766 7300 8FAC			3966+	VMALE	V22, V22, V23, V24, 3	test instruction (dest is a source)
00005AB2	E760 5030 080E		00005A70	3967+	VST	V22, V1099	save v1 output
00005AB8	07FB			3968+	BR	R11	return
00005ABC				3969+RE99	DC	0F	xl16 expected result
00005ABC				3970+	DROP	R5	
00005ABC	11010308 111F3397			3971	DC	XL16'11010308111F3397 79B556ED77F57901'	result
00005AC4	79B556ED 77F57901						
00005ACC	FF020304 05060750			3972	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00005AD4	090A0B0C 0D0E0F7F						
00005ADC	01020304 05060750			3973	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3
00005AE4	090A0B78 0D0E0F7F						
00005AEC	10000000 00000001			3974	DC	XL16'1000000000000001 1000000000000001'	v4
00005AF4	10000000 00000001						
				3975			
				3976	VRR_D	VMALE, 3	
00005B00				3977+	DS	0FD	
00005B00		00005B00		3978+	USING	*, R5	base for test data and test routine
00005B00	00005B48			3979+T100	DC	A(X100)	address of test routine
00005B04	0064			3980+	DC	H'100'	test number
00005B06	00			3981+	DC	X'00'	
00005B07	03			3982+	DC	HL1'3'	m5
00005B08	E5D4C1D3 C5404040			3983+	DC	CL8'VMALE'	instruction name
00005B10	00005B8C			3984+	DC	A(RE100+16)	address of v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005B14	00005B9C			3985+	DC	A(RE100+32)	address of v3 source
00005B18	00005BAC			3986+	DC	A(RE100+48)	address of v4 source
00005B1C	00000010			3987+	DC	A(16)	result length
00005B20	00005B7C			3988+REA100	DC	A(RE100)	result address
00005B28	00000000 00000000			3989+	DS	FD	gap
00005B30	00000000 00000000			3990+V10100	DS	XL16	V1 output
00005B38	00000000 00000000						
00005B40	00000000 00000000			3991+	DS	FD	gap
				3992+*			
00005B48				3993+X100	DS	0F	
00005B48	E310 5010 0014		00000010	3994+	LGF	R1,V2ADDR	load v2 source
00005B4E	E761 0000 0806		00000000	3995+	VL	v22,0(R1)	use v22 to test decoder
00005B54	E310 5014 0014		00000014	3996+	LGF	R1,V3ADDR	load v3 source
00005B5A	E771 0000 0806		00000000	3997+	VL	v23,0(R1)	use v23 to test decoder
00005B60	E310 5018 0014		00000018	3998+	LGF	R1,V4ADDR	load v4 source
00005B66	E781 0000 0806		00000000	3999+	VL	v24,0(R1)	use v24 to test decoder
00005B6C	E766 7300 8FAC			4000+	VMALE	V22,V22,V23,V24,3	test instruction (dest is a source)
00005B72	E760 5030 080E		00005B30	4001+	VST	V22,V10100	save v1 output
00005B78	07FB			4002+	BR	R11	return
00005B7C				4003+RE100	DC	0F	xl16 expected result
00005B7C				4004+	DROP	R5	
00005B7C	EE010003 050C1352			4005	DC	XL16'EE010003050C1352 E6D2FE7090F7148E'	result
00005B84	E6D2FE70 90F7148E						
00005B8C	FF020304 05060750			4006	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00005B94	090A0B0C 0D0E0F7F						
00005B9C	00010102 02030328			4007	DC	XL16'0001010202030328 0405053C0607073F'	v3
00005BA4	0405053C 0607073F						
00005BAC	EE000000 0000000E			4008	DC	XL16'EE0000000000000E E00000000000000E'	v4
00005BB4	E0000000 0000000E						
				4009			
00005BC0				4010	VRR_D	VMALE,3	
00005BC0		00005BC0		4011+	DS	0FD	
00005BC0	00005C08			4012+	USING	*,R5	base for test data and test routine
00005BC4	0065			4013+T101	DC	A(X101)	address of test routine
00005BC6	00			4014+	DC	H'101'	test number
00005BC7	03			4015+	DC	X'00'	
00005BC8	E5D4C1D3 C5404040			4016+	DC	HL1'3'	m5
00005BD0	00005C4C			4017+	DC	CL8'VMALE'	instruction name
00005BD4	00005C5C			4018+	DC	A(RE101+16)	address of v2 source
00005BD8	00005C6C			4019+	DC	A(RE101+32)	address of v3 source
00005BDC	00000010			4020+	DC	A(RE101+48)	address of v4 source
00005BE0	00005C3C			4021+	DC	A(16)	result length
00005BE8	00000000 00000000			4022+REA101	DC	A(RE101)	result address
00005BF0	00000000 00000000			4023+	DS	FD	gap
00005BF8	00000000 00000000			4024+V10101	DS	XL16	V1 output
00005C00	00000000 00000000						
				4025+	DS	FD	gap
				4026+*			
00005C08				4027+X101	DS	0F	
00005C08	E310 5010 0014		00000010	4028+	LGF	R1,V2ADDR	load v2 source
00005C0E	E761 0000 0806		00000000	4029+	VL	v22,0(R1)	use v22 to test decoder
00005C14	E310 5014 0014		00000014	4030+	LGF	R1,V3ADDR	load v3 source
00005C1A	E771 0000 0806		00000000	4031+	VL	v23,0(R1)	use v23 to test decoder
00005C20	E310 5018 0014		00000018	4032+	LGF	R1,V4ADDR	load v4 source
00005C26	E781 0000 0806		00000000	4033+	VL	v24,0(R1)	use v24 to test decoder
00005C2C	E766 7300 8FAC			4034+	VMALE	V22,V22,V23,V24,3	test instruction (dest is a source)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00005C32	E760 5030 080E		00005BF0	4035+	VST	V22,V10101	save v1 output	
00005C38	07FB			4036+	BR	R11	return	
00005C3C				4037+RE101	DC	0F	xl16 expected result	
00005C3C				4038+	DROP	R5		
00005C3C	FFF00000 00000009			4039	DC	XL16'FFF0000000000009	F6141E28323C492F'	result
00005C44	F6141E28 323C492F							
00005C4C	FF020304 05060750			4040	DC	XL16'FF02030405060750	090A0B0C0D0E0F7F'	v2
00005C54	090A0B0C 0D0E0F7F							
00005C5C	00000000 0000000A			4041	DC	XL16'000000000000000A	0101010F0101010F'	v3
00005C64	0101010F 0101010F							
00005C6C	FFF00000 00000000			4042	DC	XL16'FFF0000000000000	000000000000000F'	v4
00005C74	00000000 0000000F							
				4043				
				4044	VRR_D	VMALE, 3		
00005C80				4045+	DS	0FD		
00005C80		00005C80		4046+	USING	*,R5	base for test data and test routine	
00005C80	00005CC8			4047+T102	DC	A(X102)	address of test routine	
00005C84	0066			4048+	DC	H'102'	test number	
00005C86	00			4049+	DC	X'00'		
00005C87	03			4050+	DC	HL1'3'	m5	
00005C88	E5D4C1D3 C5404040			4051+	DC	CL8'VMALE'	instruction name	
00005C90	00005D0C			4052+	DC	A(RE102+16)	address of v2 source	
00005C94	00005D1C			4053+	DC	A(RE102+32)	address of v3 source	
00005C98	00005D2C			4054+	DC	A(RE102+48)	address of v4 source	
00005C9C	00000010			4055+	DC	A(16)	result length	
00005CA0	00005CFC			4056+REA102	DC	A(RE102)	result address	
00005CA8	00000000 00000000			4057+	DS	FD	gap	
00005CB0	00000000 00000000			4058+V10102	DS	XL16	V1 output	
00005CB8	00000000 00000000							
00005CC0	00000000 00000000			4059+	DS	FD	gap	
				4060+*				
00005CC8				4061+X102	DS	0F		
00005CC8	E310 5010 0014		00000010	4062+	LGF	R1,V2ADDR	load v2 source	
00005CCE	E761 0000 0806		00000000	4063+	VL	v22,0(R1)	use v22 to test decoder	
00005CD4	E310 5014 0014		00000014	4064+	LGF	R1,V3ADDR	load v3 source	
00005CDA	E771 0000 0806		00000000	4065+	VL	v23,0(R1)	use v23 to test decoder	
00005CE0	E310 5018 0014		00000018	4066+	LGF	R1,V4ADDR	load v4 source	
00005CE6	E781 0000 0806		00000000	4067+	VL	v24,0(R1)	use v24 to test decoder	
00005CEC	E766 7300 8FAC			4068+	VMALE	V22,V22,V23,V24,3	test instruction (dest is a source)	
00005CF2	E760 5030 080E		00005CB0	4069+	VST	V22,V10102	save v1 output	
00005CF8	07FB			4070+	BR	R11	return	
00005CFC				4071+RE102	DC	0F	xl16 expected result	
00005CFC				4072+	DROP	R5		
00005CFC	1009131E A8C3DFFF			4073	DC	XL16'1009131EA8C3DFFF	191C345060616772'	result
00005D04	191C3450 60616772							
00005D0C	090A0B0C 0D0E0F7F			4074	DC	XL16'090A0B0C0D0E0F7F	FF02030405060750'	v2
00005D14	FF020304 05060750							
00005D1C	0101010F 0101010F			4075	DC	XL16'0101010F0101010F	000000000000000A'	v3
00005D24	00000000 0000000A							
00005D2C	10000000 00000001			4076	DC	XL16'1000000000000001	1000000000000001'	v4
00005D34	10000000 00000001							
				4077				
				4078	*	-----		
				4079	*	VMALO - Vector Multiply and Add Logical Odd		
				4080	*	-----		
				4081	*	Byte		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				4082	VRR_D	VMALO,0	
00005D40				4083+	DS	0FD	
00005D40		00005D40		4084+	USING	*,R5	base for test data and test routine
00005D40	00005D88			4085+T103	DC	A(X103)	address of test routine
00005D44	0067			4086+	DC	H'103'	test number
00005D46	00			4087+	DC	X'00'	
00005D47	00			4088+	DC	HL1'0'	m5
00005D48	E5D4C1D3 D6404040			4089+	DC	CL8'VMALO'	instruction name
00005D50	00005DCC			4090+	DC	A(RE103+16)	address of v2 source
00005D54	00005DDC			4091+	DC	A(RE103+32)	address of v3 source
00005D58	00005DEC			4092+	DC	A(RE103+48)	address of v4 source
00005D5C	00000010			4093+	DC	A(16)	result length
00005D60	00005DBC			4094+REA103	DC	A(RE103)	result address
00005D68	00000000 00000000			4095+	DS	FD	gap
00005D70	00000000 00000000			4096+V10103	DS	XL16	V1 output
00005D78	00000000 00000000						
00005D80	00000000 00000000			4097+	DS	FD	gap
				4098+*			
00005D88				4099+X103	DS	0F	
00005D88	E310 5010 0014		00000010	4100+	LGF	R1,V2ADDR	load v2 source
00005D8E	E761 0000 0806		00000000	4101+	VL	v22,0(R1)	use v22 to test decoder
00005D94	E310 5014 0014		00000014	4102+	LGF	R1,V3ADDR	load v3 source
00005D9A	E771 0000 0806		00000000	4103+	VL	v23,0(R1)	use v23 to test decoder
00005DA0	E310 5018 0014		00000018	4104+	LGF	R1,V4ADDR	load v4 source
00005DA6	E781 0000 0806		00000000	4105+	VL	v24,0(R1)	use v24 to test decoder
00005DAC	E766 7000 8FAD			4106+	VMALO	V22,V22,V23,V24,0	test instruction (dest is a source)
00005DB2	E760 5030 080E		00005D70	4107+	VST	V22,V10103	save v1 output
00005DB8	07FB			4108+	BR	R11	return
00005DBC				4109+RE103	DC	0F	xl16 expected result
00005DBC				4110+	DROP	R5	
00005DBC	00000000 00000271			4111	DC	XL16'00000000000000271 00000C400000F424'	result
00005DC4	00000C40 0000F424						
00005DCC	FF000000 00000019			4112	DC	XL16'FF00000000000019 00000038000000FA'	v2
00005DD4	00000038 000000FA						
00005DDC	FF000000 00000019			4113	DC	XL16'FF00000000000019 00000038000000FA'	v3
00005DE4	00000038 000000FA						
00005DEC	00000000 00000000			4114	DC	XL16'0000000000000000 0000000000000000'	v4
00005DF4	00000000 00000000						
				4115			
				4116	VRR_D	VMALO,0	
00005E00				4117+	DS	0FD	
00005E00		00005E00		4118+	USING	*,R5	base for test data and test routine
00005E00	00005E48			4119+T104	DC	A(X104)	address of test routine
00005E04	0068			4120+	DC	H'104'	test number
00005E06	00			4121+	DC	X'00'	
00005E07	00			4122+	DC	HL1'0'	m5
00005E08	E5D4C1D3 D6404040			4123+	DC	CL8'VMALO'	instruction name
00005E10	00005E8C			4124+	DC	A(RE104+16)	address of v2 source
00005E14	00005E9C			4125+	DC	A(RE104+32)	address of v3 source
00005E18	00005EAC			4126+	DC	A(RE104+48)	address of v4 source
00005E1C	00000010			4127+	DC	A(16)	result length
00005E20	00005E7C			4128+REA104	DC	A(RE104)	result address
00005E28	00000000 00000000			4129+	DS	FD	gap
00005E30	00000000 00000000			4130+V10104	DS	XL16	V1 output
00005E38	00000000 00000000						
00005E40	00000000 00000000			4131+	DS	FD	gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
				4132+*				
00005E48				4133+X104	DS	0F		
00005E48	E310 5010 0014		00000010	4134+	LGF	R1,V2ADDR	load v2 source	
00005E4E	E761 0000 0806		00000000	4135+	VL	v22,0(R1)	use v22 to test decoder	
00005E54	E310 5014 0014		00000014	4136+	LGF	R1,V3ADDR	load v3 source	
00005E5A	E771 0000 0806		00000000	4137+	VL	v23,0(R1)	use v23 to test decoder	
00005E60	E310 5018 0014		00000018	4138+	LGF	R1,V4ADDR	load v4 source	
00005E66	E781 0000 0806		00000000	4139+	VL	v24,0(R1)	use v24 to test decoder	
00005E6C	E766 7000 8FAD			4140+	VMALO	V22,V22,V23,V24,0	test instruction (dest is a source)	
00005E72	E760 5030 080E		00005E30	4141+	VST	V22,V10104	save v1 output	
00005E78	07FB			4142+	BR	R11	return	
00005E7C				4143+RE104	DC	0F	xl16 expected result	
00005E7C				4144+	DROP	R5		
00005E7C	00020100 000006C0			4145	DC	XL16'000201000000006C0 00000C430000F426'	result	
00005E84	00000C43 0000F426							
00005E8C	FF0000FF 00000029			4146	DC	XL16'FF0000FF00000029 00000038000000FA'	v2	
00005E94	00000038 000000FA							
00005E9C	FF000001 00000029			4147	DC	XL16'FF00000100000029 00000038000000FA'	v3	
00005EA4	00000038 000000FA							
00005EAC	00020001 0000002F			4148	DC	XL16'000200010000002F 0000000300000002'	v4	
00005EB4	00000003 00000002							
				4149				
				4150	VRR_D	VMALO,0		
00005EC0				4151+	DS	0FD		
00005EC0		00005EC0		4152+	USING	*,R5	base for test data and test routine	
00005EC0	00005F08			4153+T105	DC	A(X105)	address of test routine	
00005EC4	0069			4154+	DC	H'105'	test number	
00005EC6	00			4155+	DC	X'00'		
00005EC7	00			4156+	DC	HL1'0'	m5	
00005EC8	E5D4C1D3 D6404040			4157+	DC	CL8'VMALO'	instruction name	
00005ED0	00005F4C			4158+	DC	A(RE105+16)	address of v2 source	
00005ED4	00005F5C			4159+	DC	A(RE105+32)	address of v3 source	
00005ED8	00005F6C			4160+	DC	A(RE105+48)	address of v4 source	
00005EDC	00000010			4161+	DC	A(16)	result length	
00005EE0	00005F3C			4162+REA105	DC	A(RE105)	result address	
00005EE8	00000000 00000000			4163+	DS	FD	gap	
00005EF0	00000000 00000000			4164+V10105	DS	XL16	V1 output	
00005EF8	00000000 00000000							
00005F00	00000000 00000000			4165+	DS	FD	gap	
				4166+*				
00005F08				4167+X105	DS	0F		
00005F08	E310 5010 0014		00000010	4168+	LGF	R1,V2ADDR	load v2 source	
00005F0E	E761 0000 0806		00000000	4169+	VL	v22,0(R1)	use v22 to test decoder	
00005F14	E310 5014 0014		00000014	4170+	LGF	R1,V3ADDR	load v3 source	
00005F1A	E771 0000 0806		00000000	4171+	VL	v23,0(R1)	use v23 to test decoder	
00005F20	E310 5018 0014		00000018	4172+	LGF	R1,V4ADDR	load v4 source	
00005F26	E781 0000 0806		00000000	4173+	VL	v24,0(R1)	use v24 to test decoder	
00005F2C	E766 7000 8FAD			4174+	VMALO	V22,V22,V23,V24,0	test instruction (dest is a source)	
00005F32	E760 5030 080E		00005EF0	4175+	VST	V22,V10105	save v1 output	
00005F38	07FB			4176+	BR	R11	return	
00005F3C				4177+RE105	DC	0F	xl16 expected result	
00005F3C				4178+	DROP	R5		
00005F3C	FF060314 052A0748			4179	DC	XL16'FF060314052A0748 096E0B9C0DD21010'	result	
00005F44	096E0B9C 0DD21010							
00005F4C	FF020304 05060708			4180	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2	
00005F54	090A0B0C 0D0E0F10							

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00005F5C	FF020304 05060708			4181	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00005F64	090A0B0C 0D0E0F10						
00005F6C	FF020304 05060708			4182	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00005F74	090A0B0C 0D0E0F10						
				4183			
00005F80				4184	VRR_D	VMALO,0	
00005F80		00005F80		4185+	DS	0FD	
00005F80	00005FC8			4186+	USING	*,R5	base for test data and test routine
00005F84	006A			4187+T106	DC	A(X106)	address of test routine
00005F86	00			4188+	DC	H'106'	test number
00005F87	00			4189+	DC	X'00'	
00005F88	E5D4C1D3 D6404040			4190+	DC	HL1'0'	m5
00005F90	0000600C			4191+	DC	CL8'VMALO'	instruction name
00005F94	0000601C			4192+	DC	A(RE106+16)	address of v2 source
00005F98	0000602C			4193+	DC	A(RE106+32)	address of v3 source
00005F9C	00000010			4194+	DC	A(RE106+48)	address of v4 source
00005FA0	00005FFC			4195+	DC	A(16)	result length
00005FA8	00000000 00000000			4196+REA106	DC	A(RE106)	result address
00005FB0	00000000 00000000			4197+	DS	FD	gap
00005FB8	00000000 00000000			4198+V10106	DS	XL16	V1 output
00005FC0	00000000 00000000			4199+	DS	FD	gap
				4200+*			
00005FC8				4201+X106	DS	0F	
00005FC8	E310 5010 0014		00000010	4202+	LGF	R1,V2ADDR	load v2 source
00005FCE	E761 0000 0806		00000000	4203+	VL	v22,0(R1)	use v22 to test decoder
00005FD4	E310 5014 0014		00000014	4204+	LGF	R1,V3ADDR	load v3 source
00005FDA	E771 0000 0806		00000000	4205+	VL	v23,0(R1)	use v23 to test decoder
00005FE0	E310 5018 0014		00000018	4206+	LGF	R1,V4ADDR	load v4 source
00005FE6	E781 0000 0806		00000000	4207+	VL	v24,0(R1)	use v24 to test decoder
00005FEC	E766 7000 8FAD			4208+	VMALO	V22,V22,V23,V24,0	test instruction (dest is a source)
00005FF2	E760 5030 080E		00005FB0	4209+	VST	V22,V10106	save v1 output
00005FF8	07FB			4210+	BR	R11	return
00005FFC				4211+RE106	DC	0F	xl16 expected result
00005FFC				4212+	DROP	R5	
00005FFC	FF04030C 05180728			4213	DC	XL16'FF04030C05180728 093C0B540D700F90'	result
00006004	093C0B54 0D700F90						
0000600C	FF020304 05060708			4214	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006014	090A0B0C 0D0E0F10						
0000601C	FF010102 02030304			4215	DC	XL16'FF01010202030304 0405050606070708'	v3
00006024	04050506 06070708						
0000602C	FF020304 05060708			4216	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006034	090A0B0C 0D0E0F10						
				4217			
00006040				4218	VRR_D	VMALO,0	
00006040		00006040		4219+	DS	0FD	
00006040	00006088			4220+	USING	*,R5	base for test data and test routine
00006044	006B			4221+T107	DC	A(X107)	address of test routine
00006046	00			4222+	DC	H'107'	test number
00006047	00			4223+	DC	X'00'	
00006048	E5D4C1D3 D6404040			4224+	DC	HL1'0'	m5
00006050	000060CC			4225+	DC	CL8'VMALO'	instruction name
00006054	000060DC			4226+	DC	A(RE107+16)	address of v2 source
00006058	000060EC			4227+	DC	A(RE107+32)	address of v3 source
0000605C	00000010			4228+	DC	A(RE107+48)	address of v4 source
				4229+	DC	A(16)	result length

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006060	000060BC			4230+REA107	DC	A(RE107)	result address
00006068	00000000 00000000			4231+	DS	FD	gap
00006070	00000000 00000000			4232+V10107	DS	XL16	V1 output
00006078	00000000 00000000						
00006080	00000000 00000000			4233+	DS	FD	gap
				4234+*			
00006088				4235+X107	DS	0F	
00006088	E310 5010 0014		00000010	4236+	LGF	R1,V2ADDR	load v2 source
0000608E	E761 0000 0806		00000000	4237+	VL	v22,0(R1)	use v22 to test decoder
00006094	E310 5014 0014		00000014	4238+	LGF	R1,V3ADDR	load v3 source
0000609A	E771 0000 0806		00000000	4239+	VL	v23,0(R1)	use v23 to test decoder
000060A0	E310 5018 0014		00000018	4240+	LGF	R1,V4ADDR	load v4 source
000060A6	E781 0000 0806		00000000	4241+	VL	v24,0(R1)	use v24 to test decoder
000060AC	E766 7000 8FAD			4242+	VMALO	V22,V22,V23,V24,0	test instruction (dest is a source)
000060B2	E760 5030 080E		00006070	4243+	VST	V22,V10107	save v1 output
000060B8	07FB			4244+	BR	R11	return
000060BC				4245+RE107	DC	0F	xl16 expected result
000060BC				4246+	DROP	R5	
000060BC	FF020304 05060710			4247	DC	XL16'FF02030405060710 09140B180D1C0F30'	result
000060C4	09140B18 0D1C0F30						
000060CC	FF020304 05060708			4248	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000060D4	090A0B0C 0D0E0F10						
000060DC	FF000000 00000001			4249	DC	XL16'FF00000000000001 0101010101010102'	v3
000060E4	01010101 01010102						
000060EC	FF020304 05060708			4250	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000060F4	090A0B0C 0D0E0F10						
				4251			
				4252 * Halfword			
				4253	VRR_D	VMALO,1	
00006100				4254+	DS	0FD	
00006100		00006100		4255+	USING	*,R5	base for test data and test routine
00006100	00006148			4256+T108	DC	A(X108)	address of test routine
00006104	006C			4257+	DC	H'108'	test number
00006106	00			4258+	DC	X'00'	
00006107	01			4259+	DC	HL1'1'	m5
00006108	E5D4C1D3 D6404040			4260+	DC	CL8'VMALO'	instruction name
00006110	0000618C			4261+	DC	A(RE108+16)	address of v2 source
00006114	0000619C			4262+	DC	A(RE108+32)	address of v3 source
00006118	000061AC			4263+	DC	A(RE108+48)	address of v4 source
0000611C	00000010			4264+	DC	A(16)	result length
00006120	0000617C			4265+REA108	DC	A(RE108)	result address
00006128	00000000 00000000			4266+	DS	FD	gap
00006130	00000000 00000000			4267+V10108	DS	XL16	V1 output
00006138	00000000 00000000						
00006140	00000000 00000000			4268+	DS	FD	gap
				4269+*			
00006148				4270+X108	DS	0F	
00006148	E310 5010 0014		00000010	4271+	LGF	R1,V2ADDR	load v2 source
0000614E	E761 0000 0806		00000000	4272+	VL	v22,0(R1)	use v22 to test decoder
00006154	E310 5014 0014		00000014	4273+	LGF	R1,V3ADDR	load v3 source
0000615A	E771 0000 0806		00000000	4274+	VL	v23,0(R1)	use v23 to test decoder
00006160	E310 5018 0014		00000018	4275+	LGF	R1,V4ADDR	load v4 source
00006166	E781 0000 0806		00000000	4276+	VL	v24,0(R1)	use v24 to test decoder
0000616C	E766 7100 8FAD			4277+	VMALO	V22,V22,V23,V24,1	test instruction (dest is a source)
00006172	E760 5030 080E		00006130	4278+	VST	V22,V10108	save v1 output
00006178	07FB			4279+	BR	R11	return

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000617C				4280+RE108	DC	0F	xl16 expected result
0000617C				4281+	DROP	R5	
0000617C	00000000 00000271			4282	DC	XL16'0000000000000271 00000C400000F424'	result
00006184	00000C40 0000F424						
0000618C	FF000000 00000019			4283	DC	XL16'FF00000000000019 00000038000000FA'	v2
00006194	00000038 000000FA						
0000619C	FF000000 00000019			4284	DC	XL16'FF00000000000019 00000038000000FA'	v3
000061A4	00000038 000000FA						
000061AC	00000000 00000000			4285	DC	XL16'0000000000000000 0000000000000000'	v4
000061B4	00000000 00000000						
				4286			
000061C0				4287	VRR_D	VMAL0,1	
000061C0		000061C0		4288+	DS	0FD	
000061C0	00006208			4289+	USING	*,R5	base for test data and test routine
000061C4	006D			4290+T109	DC	A(X109)	address of test routine
000061C6	00			4291+	DC	H'109'	test number
000061C7	01			4292+	DC	X'00'	
000061C8	E5D4C1D3 D6404040			4293+	DC	HL1'1'	m5
000061D0	0000624C			4294+	DC	CL8'VMAL0'	instruction name
000061D4	0000625C			4295+	DC	A(RE109+16)	address of v2 source
000061D8	0000626C			4296+	DC	A(RE109+32)	address of v3 source
000061DC	00000010			4297+	DC	A(RE109+48)	address of v4 source
000061E0	0000623C			4298+	DC	A(16)	result length
000061E8	00000000 00000000			4299+REA109	DC	A(RE109)	result address
000061F0	00000000 00000000			4300+	DS	FD	gap
000061F8	00000000 00000000			4301+V10109	DS	XL16	V1 output
00006200	00000000 00000000						
				4302+	DS	FD	gap
				4303+*			
00006208				4304+X109	DS	0F	
00006208	E310 5010 0014		00000010	4305+	LGF	R1,V2ADDR	load v2 source
0000620E	E761 0000 0806		00000000	4306+	VL	v22,0(R1)	use v22 to test decoder
00006214	E310 5014 0014		00000014	4307+	LGF	R1,V3ADDR	load v3 source
0000621A	E771 0000 0806		00000000	4308+	VL	v23,0(R1)	use v23 to test decoder
00006220	E310 5018 0014		00000018	4309+	LGF	R1,V4ADDR	load v4 source
00006226	E781 0000 0806		00000000	4310+	VL	v24,0(R1)	use v24 to test decoder
0000622C	E766 7100 8FAD			4311+	VMAL0	V22,V22,V23,V24,1	test instruction (dest is a source)
00006232	E760 5030 080E		000061F0	4312+	VST	V22,V10109	save v1 output
00006238	07FB			4313+	BR	R11	return
0000623C				4314+RE109	DC	0F	xl16 expected result
0000623C				4315+	DROP	R5	
0000623C	00020100 000006C0			4316	DC	XL16'00020100000006C0 00000C430000F426'	result
00006244	00000C43 0000F426						
0000624C	FF0000FF 00000029			4317	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00006254	00000038 000000FA						
0000625C	FF000001 00000029			4318	DC	XL16'FF00000100000029 00000038000000FA'	v3
00006264	00000038 000000FA						
0000626C	00020001 0000002F			4319	DC	XL16'000200010000002F 0000000300000002'	v4
00006274	00000003 00000002						
				4320			
00006280				4321	VRR_D	VMAL0,1	
00006280		00006280		4322+	DS	0FD	
00006280	000062C8			4323+	USING	*,R5	base for test data and test routine
00006284	006E			4324+T110	DC	A(X110)	address of test routine
00006286	00			4325+	DC	H'110'	test number
				4326+	DC	X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006287	01			4327+	DC	HL1'1'	m5
00006288	E5D4C1D3 D6404040			4328+	DC	CL8'VMALO'	instruction name
00006290	0000630C			4329+	DC	A(RE110+16)	address of v2 source
00006294	0000631C			4330+	DC	A(RE110+32)	address of v3 source
00006298	0000632C			4331+	DC	A(RE110+48)	address of v4 source
0000629C	00000010			4332+	DC	A(16)	result length
000062A0	000062FC			4333+REA110	DC	A(RE110)	result address
000062A8	00000000 00000000			4334+	DS	FD	gap
000062B0	00000000 00000000			4335+V10110	DS	XL16	V1 output
000062B8	00000000 00000000						
000062C0	00000000 00000000			4336+	DS	FD	gap
				4337+*			
000062C8				4338+X110	DS	0F	
000062C8	E310 5010 0014		00000010	4339+	LGF	R1,V2ADDR	load v2 source
000062CE	E761 0000 0806		00000000	4340+	VL	v22,0(R1)	use v22 to test decoder
000062D4	E310 5014 0014		00000014	4341+	LGF	R1,V3ADDR	load v3 source
000062DA	E771 0000 0806		00000000	4342+	VL	v23,0(R1)	use v23 to test decoder
000062E0	E310 5018 0014		00000018	4343+	LGF	R1,V4ADDR	load v4 source
000062E6	E781 0000 0806		00000000	4344+	VL	v24,0(R1)	use v24 to test decoder
000062EC	E766 7100 8FAD			4345+	VMALO	V22,V22,V23,V24,1	test instruction (dest is a source)
000062F2	E760 5030 080E		000062B0	4346+	VST	V22,V10110	save v1 output
000062F8	07FB			4347+	BR	R11	return
000062FC				4348+RE110	DC	0F	xl16 expected result
000062FC				4349+	DROP	R5	
000062FC	FF0B1B14 05377748			4350	DC	XL16'FF0B1B1405377748 0984139C0DF0F010'	result
00006304	0984139C 0DF0F010						
0000630C	FF020304 05060708			4351	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006314	090A0B0C 0D0E0F10						
0000631C	FF020304 05060708			4352	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00006324	090A0B0C 0D0E0F10						
0000632C	FF020304 05060708			4353	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006334	090A0B0C 0D0E0F10						
				4354			
				4355	VRR_D	VMALO,1	
00006340				4356+	DS	0FD	
00006340		00006340		4357+	USING	*,R5	base for test data and test routine
00006340	00006388			4358+T111	DC	A(X111)	address of test routine
00006344	006F			4359+	DC	H'111'	test number
00006346	00			4360+	DC	X'00'	
00006347	01			4361+	DC	HL1'1'	m5
00006348	E5D4C1D3 D6404040			4362+	DC	CL8'VMALO'	instruction name
00006350	000063CC			4363+	DC	A(RE111+16)	address of v2 source
00006354	000063DC			4364+	DC	A(RE111+32)	address of v3 source
00006358	000063EC			4365+	DC	A(RE111+48)	address of v4 source
0000635C	00000010			4366+	DC	A(16)	result length
00006360	000063BC			4367+REA111	DC	A(RE111)	result address
00006368	00000000 00000000			4368+	DS	FD	gap
00006370	00000000 00000000			4369+V10111	DS	XL16	V1 output
00006378	00000000 00000000						
00006380	00000000 00000000			4370+	DS	FD	gap
				4371+*			
00006388				4372+X111	DS	0F	
00006388	E310 5010 0014		00000010	4373+	LGF	R1,V2ADDR	load v2 source
0000638E	E761 0000 0806		00000000	4374+	VL	v22,0(R1)	use v22 to test decoder
00006394	E310 5014 0014		00000014	4375+	LGF	R1,V3ADDR	load v3 source
0000639A	E771 0000 0806		00000000	4376+	VL	v23,0(R1)	use v23 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000063A0	E310 5018 0014		00000018	4377+	LGF	R1,V4ADDR	load v4 source
000063A6	E781 0000 0806		00000000	4378+	VL	v24,0(R1)	use v24 to test decoder
000063AC	E766 7100 8FAD			4379+	VMAL0	V22,V22,V23,V24,1	test instruction (dest is a source)
000063B2	E760 5030 080E		00006370	4380+	VST	V22,V10111	save v1 output
000063B8	07FB			4381+	BR	R11	return
000063BC				4382+RE111	DC	0F	xl16 expected result
000063BC				4383+	DROP	R5	
000063BC	FF050D0C 051B3B28			4384	DC	XL16'FF050D0C051B3B28 094189540D77F790'	result
000063C4	09418954 0D77F790						
000063CC	FF020304 05060708			4385	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000063D4	090A0B0C 0D0E0F10						
000063DC	FF010102 02030304			4386	DC	XL16'FF01010202030304 0405050606070708'	v3
000063E4	04050506 06070708						
000063EC	FF020304 05060708			4387	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000063F4	090A0B0C 0D0E0F10						
				4388			
				4389	VRR_D	VMAL0,1	
00006400				4390+	DS	0FD	
00006400		00006400		4391+	USING	*,R5	base for test data and test routine
00006400	00006448			4392+T112	DC	A(X112)	address of test routine
00006404	0070			4393+	DC	H'112'	test number
00006406	00			4394+	DC	X'00'	
00006407	01			4395+	DC	HL1'1'	m5
00006408	E5D4C1D3 D6404040			4396+	DC	CL8'VMAL0'	instruction name
00006410	0000648C			4397+	DC	A(RE112+16)	address of v2 source
00006414	0000649C			4398+	DC	A(RE112+32)	address of v3 source
00006418	000064AC			4399+	DC	A(RE112+48)	address of v4 source
0000641C	00000010			4400+	DC	A(16)	result length
00006420	0000647C			4401+REA112	DC	A(RE112)	result address
00006428	00000000 00000000			4402+	DS	FD	gap
00006430	00000000 00000000			4403+V10112	DS	XL16	V1 output
00006438	00000000 00000000						
00006440	00000000 00000000			4404+	DS	FD	gap
				4405+*			
00006448				4406+X112	DS	0F	
00006448	E310 5010 0014		00000010	4407+	LGF	R1,V2ADDR	load v2 source
0000644E	E761 0000 0806		00000000	4408+	VL	v22,0(R1)	use v22 to test decoder
00006454	E310 5014 0014		00000014	4409+	LGF	R1,V3ADDR	load v3 source
0000645A	E771 0000 0806		00000000	4410+	VL	v23,0(R1)	use v23 to test decoder
00006460	E310 5018 0014		00000018	4411+	LGF	R1,V4ADDR	load v4 source
00006466	E781 0000 0806		00000000	4412+	VL	v24,0(R1)	use v24 to test decoder
0000646C	E766 7100 8FAD			4413+	VMAL0	V22,V22,V23,V24,1	test instruction (dest is a source)
00006472	E760 5030 080E		00006430	4414+	VST	V22,V10112	save v1 output
00006478	07FB			4415+	BR	R11	return
0000647C				4416+RE112	DC	0F	xl16 expected result
0000647C				4417+	DROP	R5	
0000647C	FF020304 05060E10			4418	DC	XL16'FF02030405060E10 091522180D1D3D30'	result
00006484	09152218 0D1D3D30						
0000648C	FF020304 05060708			4419	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006494	090A0B0C 0D0E0F10						
0000649C	FF000000 00000001			4420	DC	XL16'FF0000000000000001 0101010101010102'	v3
000064A4	01010101 01010102						
000064AC	FF020304 05060708			4421	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000064B4	090A0B0C 0D0E0F10						
				4422			
				4423	*	Word	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				4424	VRR_D	VMALO, 2	
000064C0				4425+	DS	0FD	
000064C0		000064C0		4426+	USING	*, R5	base for test data and test routine
000064C0	00006508			4427+T113	DC	A(X113)	address of test routine
000064C4	0071			4428+	DC	H'113'	test number
000064C6	00			4429+	DC	X'00'	
000064C7	02			4430+	DC	HL1'2'	m5
000064C8	E5D4C1D3 D6404040			4431+	DC	CL8'VMALO'	instruction name
000064D0	0000654C			4432+	DC	A(RE113+16)	address of v2 source
000064D4	0000655C			4433+	DC	A(RE113+32)	address of v3 source
000064D8	0000656C			4434+	DC	A(RE113+48)	address of v4 source
000064DC	00000010			4435+	DC	A(16)	result length
000064E0	0000653C			4436+REA113	DC	A(RE113)	result address
000064E8	00000000 00000000			4437+	DS	FD	gap
000064F0	00000000 00000000			4438+V10113	DS	XL16	V1 output
000064F8	00000000 00000000						
00006500	00000000 00000000			4439+	DS	FD	gap
				4440+*			
00006508				4441+X113	DS	0F	
00006508	E310 5010 0014		00000010	4442+	LGF	R1, V2ADDR	load v2 source
0000650E	E761 0000 0806		00000000	4443+	VL	v22, 0(R1)	use v22 to test decoder
00006514	E310 5014 0014		00000014	4444+	LGF	R1, V3ADDR	load v3 source
0000651A	E771 0000 0806		00000000	4445+	VL	v23, 0(R1)	use v23 to test decoder
00006520	E310 5018 0014		00000018	4446+	LGF	R1, V4ADDR	load v4 source
00006526	E781 0000 0806		00000000	4447+	VL	v24, 0(R1)	use v24 to test decoder
0000652C	E766 7200 8FAD			4448+	VMALO	V22, V22, V23, V24, 2	test instruction (dest is a source)
00006532	E760 5030 080E		000064F0	4449+	VST	V22, V10113	save v1 output
00006538	07FB			4450+	BR	R11	return
0000653C				4451+RE113	DC	0F	xl16 expected result
0000653C				4452+	DROP	R5	
0000653C	00000000 00000271			4453	DC	XL16'00000000000000271 0000000000000F424'	result
00006544	00000000 0000F424						
0000654C	FF000000 00000019			4454	DC	XL16'FF00000000000019 00000038000000FA'	v2
00006554	00000038 000000FA						
0000655C	FF000000 00000019			4455	DC	XL16'FF00000000000019 00000038000000FA'	v3
00006564	00000038 000000FA						
0000656C	00000000 00000000			4456	DC	XL16'0000000000000000 0000000000000000'	v4
00006574	00000000 00000000						
				4457			
00006580				4458	VRR_D	VMALO, 2	
00006580		00006580		4459+	DS	0FD	
00006580	000065C8			4460+	USING	*, R5	base for test data and test routine
00006584	0072			4461+T114	DC	A(X114)	address of test routine
00006586	00			4462+	DC	H'114'	test number
00006587	02			4463+	DC	X'00'	
00006588	E5D4C1D3 D6404040			4464+	DC	HL1'2'	m5
00006590	0000660C			4465+	DC	CL8'VMALO'	instruction name
00006594	0000661C			4466+	DC	A(RE114+16)	address of v2 source
00006598	0000662C			4467+	DC	A(RE114+32)	address of v3 source
0000659C	00000010			4468+	DC	A(RE114+48)	address of v4 source
000065A0	000065FC			4469+	DC	A(16)	result length
000065A8	00000000 00000000			4470+REA114	DC	A(RE114)	result address
000065B0	00000000 00000000			4471+	DS	FD	gap
000065B8	00000000 00000000			4472+V10114	DS	XL16	V1 output
000065C0	00000000 00000000			4473+	DS	FD	gap

LOC	OBJECT	CODE	ADDR1	ADDR2	STMT			
					4474+*			
000065C8					4475+X114	DS	0F	
000065C8	E310	5010 0014		00000010	4476+	LGF	R1,V2ADDR	load v2 source
000065CE	E761	0000 0806		00000000	4477+	VL	v22,0(R1)	use v22 to test decoder
000065D4	E310	5014 0014		00000014	4478+	LGF	R1,V3ADDR	load v3 source
000065DA	E771	0000 0806		00000000	4479+	VL	v23,0(R1)	use v23 to test decoder
000065E0	E310	5018 0014		00000018	4480+	LGF	R1,V4ADDR	load v4 source
000065E6	E781	0000 0806		00000000	4481+	VL	v24,0(R1)	use v24 to test decoder
000065EC	E766	7200 8FAD			4482+	VMALO	V22,V22,V23,V24,2	test instruction (dest is a source)
000065F2	E760	5030 080E		000065B0	4483+	VST	V22,V10114	save v1 output
000065F8	07FB				4484+	BR	R11	return
000065FC					4485+RE114	DC	0F	xl16 expected result
000065FC					4486+	DROP	R5	
000065FC	00020001	0000006C0			4487	DC	XL16'000200010000006C0 000000030000F426'	result
00006604	00000003	0000F426						
0000660C	FF0000FF	00000029			4488	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00006614	00000038	000000FA						
0000661C	FF000001	00000029			4489	DC	XL16'FF00000100000029 00000038000000FA'	v3
00006624	00000038	000000FA						
0000662C	00020001	0000002F			4490	DC	XL16'000200010000002F 0000000300000002'	v4
00006634	00000003	00000002						
					4491			
					4492	VRR_D	VMALO,2	
00006640					4493+	DS	0FD	
00006640			00006640		4494+	USING	*,R5	base for test data and test routine
00006640	00006688				4495+T115	DC	A(X115)	address of test routine
00006644	0073				4496+	DC	H'115'	test number
00006646	00				4497+	DC	X'00'	
00006647	02				4498+	DC	HL1'2'	m5
00006648	E5D4C1D3	D6404040			4499+	DC	CL8'VMALO'	instruction name
00006650	000066CC				4500+	DC	A(RE115+16)	address of v2 source
00006654	000066DC				4501+	DC	A(RE115+32)	address of v3 source
00006658	000066EC				4502+	DC	A(RE115+48)	address of v4 source
0000665C	00000010				4503+	DC	A(16)	result length
00006660	000066BC				4504+REA115	DC	A(RE115)	result address
00006668	00000000	00000000			4505+	DS	FD	gap
00006670	00000000	00000000			4506+V10115	DS	XL16	V1 output
00006678	00000000	00000000						
00006680	00000000	00000000			4507+	DS	FD	gap
					4508+*			
00006688					4509+X115	DS	0F	
00006688	E310	5010 0014		00000010	4510+	LGF	R1,V2ADDR	load v2 source
0000668E	E761	0000 0806		00000000	4511+	VL	v22,0(R1)	use v22 to test decoder
00006694	E310	5014 0014		00000014	4512+	LGF	R1,V3ADDR	load v3 source
0000669A	E771	0000 0806		00000000	4513+	VL	v23,0(R1)	use v23 to test decoder
000066A0	E310	5018 0014		00000018	4514+	LGF	R1,V4ADDR	load v4 source
000066A6	E781	0000 0806		00000000	4515+	VL	v24,0(R1)	use v24 to test decoder
000066AC	E766	7200 8FAD			4516+	VMALO	V22,V22,V23,V24,2	test instruction (dest is a source)
000066B2	E760	5030 080E		00006670	4517+	VST	V22,V10115	save v1 output
000066B8	07FB				4518+	BR	R11	return
000066BC					4519+RE115	DC	0F	xl16 expected result
000066BC					4520+	DROP	R5	
000066BC	FF1B3F6E	A9977748			4521	DC	XL16'FF1B3F6EA9977748 09B4795953B0F010'	result
000066C4	09B47959	53B0F010						
000066CC	FF020304	05060708			4522	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000066D4	090A0B0C	0D0E0F10						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000066DC	FF020304 05060708			4523	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000066E4	090A0B0C 0D0E0F10						
000066EC	FF020304 05060708			4524	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000066F4	090A0B0C 0D0E0F10						
				4525			
00006700				4526	VRR_D	VMALO,2	
00006700		00006700		4527+	DS	0FD	
00006700	00006748			4528+	USING	*,R5	base for test data and test routine
00006704	0074			4529+T116	DC	A(X116)	address of test routine
00006706	00			4530+	DC	H'116'	test number
00006707	02			4531+	DC	X'00'	
00006708	E5D4C1D3 D6404040			4532+	DC	HL1'2'	m5
00006710	0000678C			4533+	DC	CL8'VMALO'	instruction name
00006714	0000679C			4534+	DC	A(RE116+16)	address of v2 source
00006718	000067AC			4535+	DC	A(RE116+32)	address of v3 source
0000671C	00000010			4536+	DC	A(RE116+48)	address of v4 source
00006720	0000677C			4537+	DC	A(16)	result length
00006728	00000000 00000000			4538+REA116	DC	A(RE116)	result address
00006730	00000000 00000000			4539+	DS	FD	gap
00006738	00000000 00000000			4540+V10116	DS	XL16	V1 output
00006740	00000000 00000000						
				4541+	DS	FD	gap
				4542+*			
00006748				4543+X116	DS	0F	
00006748	E310 5010 0014		00000010	4544+	LGF	R1,V2ADDR	load v2 source
0000674E	E761 0000 0806		00000000	4545+	VL	v22,0(R1)	use v22 to test decoder
00006754	E310 5014 0014		00000014	4546+	LGF	R1,V3ADDR	load v3 source
0000675A	E771 0000 0806		00000000	4547+	VL	v23,0(R1)	use v23 to test decoder
00006760	E310 5018 0014		00000018	4548+	LGF	R1,V4ADDR	load v4 source
00006766	E781 0000 0806		00000000	4549+	VL	v24,0(R1)	use v24 to test decoder
0000676C	E766 7200 8FAD			4550+	VMALO	V22,V22,V23,V24,2	test instruction (dest is a source)
00006772	E760 5030 080E		00006730	4551+	VST	V22,V10116	save v1 output
00006778	07FB			4552+	BR	R11	return
0000677C				4553+RE116	DC	0F	xl16 expected result
0000677C				4554+	DROP	R5	
0000677C	FF0C1E33 504B3B28			4555	DC	XL16'FF0C1E33504B3B28 0958BB24A157F790'	result
00006784	0958BB24 A157F790						
0000678C	FF020304 05060708			4556	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006794	090A0B0C 0D0E0F10						
0000679C	FF010102 02030304			4557	DC	XL16'FF01010202030304 0405050606070708'	v3
000067A4	04050506 06070708						
000067AC	FF020304 05060708			4558	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000067B4	090A0B0C 0D0E0F10						
				4559			
000067C0				4560	VRR_D	VMALO,2	
000067C0		000067C0		4561+	DS	0FD	
000067C0	00006808			4562+	USING	*,R5	base for test data and test routine
000067C4	0075			4563+T117	DC	A(X117)	address of test routine
000067C6	00			4564+	DC	H'117'	test number
000067C7	02			4565+	DC	X'00'	
000067C8	E5D4C1D3 D6404040			4566+	DC	HL1'2'	m5
000067D0	0000684C			4567+	DC	CL8'VMALO'	instruction name
000067D4	0000685C			4568+	DC	A(RE117+16)	address of v2 source
000067D8	0000686C			4569+	DC	A(RE117+32)	address of v3 source
000067DC	00000010			4570+	DC	A(RE117+48)	address of v4 source
				4571+	DC	A(16)	result length

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000067E0	0000683C			4572+REA117	DC	A(RE117)	result address
000067E8	00000000 00000000			4573+	DS	FD	gap
000067F0	00000000 00000000			4574+V10117	DS	XL16	V1 output
000067F8	00000000 00000000						
00006800	00000000 00000000			4575+	DS	FD	gap
				4576+*			
00006808				4577+X117	DS	0F	
00006808	E310 5010 0014		00000010	4578+	LGF	R1,V2ADDR	load v2 source
0000680E	E761 0000 0806		00000000	4579+	VL	v22,0(R1)	use v22 to test decoder
00006814	E310 5014 0014		00000014	4580+	LGF	R1,V3ADDR	load v3 source
0000681A	E771 0000 0806		00000000	4581+	VL	v23,0(R1)	use v23 to test decoder
00006820	E310 5018 0014		00000018	4582+	LGF	R1,V4ADDR	load v4 source
00006826	E781 0000 0806		00000000	4583+	VL	v24,0(R1)	use v24 to test decoder
0000682C	E766 7200 8FAD			4584+	VMALO	V22,V22,V23,V24,2	test instruction (dest is a source)
00006832	E760 5030 080E		000067F0	4585+	VST	V22,V10117	save v1 output
00006838	07FB			4586+	BR	R11	return
0000683C				4587+RE117	DC	0F	xl16 expected result
0000683C				4588+	DROP	R5	
0000683C	FF020304 0A0C0E10			4589	DC	XL16'FF0203040A0C0E10 0917263654493D30'	result
00006844	09172636 54493D30						
0000684C	FF020304 05060708			4590	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006854	090A0B0C 0D0E0F10						
0000685C	FF000000 00000001			4591	DC	XL16'FF0000000000000001 0101010101010102'	v3
00006864	01010101 01010102						
0000686C	FF020304 05060708			4592	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006874	090A0B0C 0D0E0F10						
				4593			
				4594 * Doubleword			
				4595	VRR_D	VMALO,3	
00006880				4596+	DS	0FD	
00006880		00006880		4597+	USING	*,R5	base for test data and test routine
00006880	000068C8			4598+T118	DC	A(X118)	address of test routine
00006884	0076			4599+	DC	H'118'	test number
00006886	00			4600+	DC	X'00'	
00006887	03			4601+	DC	HL1'3'	m5
00006888	E5D4C1D3 D6404040			4602+	DC	CL8'VMALO'	instruction name
00006890	0000690C			4603+	DC	A(RE118+16)	address of v2 source
00006894	0000691C			4604+	DC	A(RE118+32)	address of v3 source
00006898	0000692C			4605+	DC	A(RE118+48)	address of v4 source
0000689C	00000010			4606+	DC	A(16)	result length
000068A0	000068FC			4607+REA118	DC	A(RE118)	result address
000068A8	00000000 00000000			4608+	DS	FD	gap
000068B0	00000000 00000000			4609+V10118	DS	XL16	V1 output
000068B8	00000000 00000000						
000068C0	00000000 00000000			4610+	DS	FD	gap
				4611+*			
000068C8				4612+X118	DS	0F	
000068C8	E310 5010 0014		00000010	4613+	LGF	R1,V2ADDR	load v2 source
000068CE	E761 0000 0806		00000000	4614+	VL	v22,0(R1)	use v22 to test decoder
000068D4	E310 5014 0014		00000014	4615+	LGF	R1,V3ADDR	load v3 source
000068DA	E771 0000 0806		00000000	4616+	VL	v23,0(R1)	use v23 to test decoder
000068E0	E310 5018 0014		00000018	4617+	LGF	R1,V4ADDR	load v4 source
000068E6	E781 0000 0806		00000000	4618+	VL	v24,0(R1)	use v24 to test decoder
000068EC	E766 7300 8FAD			4619+	VMALO	V22,V22,V23,V24,3	test instruction (dest is a source)
000068F2	E760 5030 080E		000068B0	4620+	VST	V22,V10118	save v1 output
000068F8	07FB			4621+	BR	R11	return

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000068FC				4622+RE118	DC	0F	xl16 expected result
000068FC				4623+	DROP	R5	
000068FC	00000000 00000C77			4624	DC	XL16'00000000000000C77	result
00006904	96789F9F 4FEDCC24						
0000690C	FFFFFFFF 00019000			4625	DC	XL16'FFFFFFFF00019000	v2
00006914	00000038 EEEEEFEFA						
0000691C	FFFFFFFF 00019000			4626	DC	XL16'FFFFFFFF00019000	v3
00006924	00000038 0EEEEFEFA						
0000692C	00000000 00000000			4627	DC	XL16'0000000000000000	v4
00006934	00000000 00000000						
				4628			
00006940				4629	VRR_D	VMALO, 3	
00006940		00006940		4630+	DS	0FD	
00006940	00006988			4631+	USING	*, R5	base for test data and test routine
00006944	0077			4632+T119	DC	A(X119)	address of test routine
00006946	00			4633+	DC	H'119'	test number
00006947	03			4634+	DC	X'00'	
00006948	E5D4C1D3 D6404040			4635+	DC	HL1'3'	m5
00006950	000069CC			4636+	DC	CL8'VMALO'	instruction name
00006954	000069DC			4637+	DC	A(RE119+16)	address of v2 source
00006958	000069EC			4638+	DC	A(RE119+32)	address of v3 source
0000695C	00000010			4639+	DC	A(RE119+48)	address of v4 source
00006960	000069BC			4640+	DC	A(16)	result length
00006968	00000000 00000000			4641+REA119	DC	A(RE119)	result address
00006970	00000000 00000000			4642+	DS	FD	gap
00006978	00000000 00000000			4643+V10119	DS	XL16	V1 output
00006980	00000000 00000000						
				4644+	DS	FD	gap
				4645+*			
00006988				4646+X119	DS	0F	
00006988	E310 5010 0014		00000010	4647+	LGF	R1, V2ADDR	load v2 source
0000698E	E761 0000 0806		00000000	4648+	VL	v22, 0(R1)	use v22 to test decoder
00006994	E310 5014 0014		00000014	4649+	LGF	R1, V3ADDR	load v3 source
0000699A	E771 0000 0806		00000000	4650+	VL	v23, 0(R1)	use v23 to test decoder
000069A0	E310 5018 0014		00000018	4651+	LGF	R1, V4ADDR	load v4 source
000069A6	E781 0000 0806		00000000	4652+	VL	v24, 0(R1)	use v24 to test decoder
000069AC	E766 7300 8FAD			4653+	VMALO	V22, V22, V23, V24, 3	test instruction (dest is a source)
000069B2	E760 5030 080E		00006970	4654+	VST	V22, V10119	save v1 output
000069B8	07FB			4655+	BR	R11	return
000069BC				4656+RE119	DC	0F	xl16 expected result
000069BC				4657+	DROP	R5	
000069BC	1051B52F 8692B4F7			4658	DC	XL16'1051B52F8692B4F7	result
000069C4	252B55D4 98D42102						
000069CC	FF020304 05060750			4659	DC	XL16'FF02030405060750	v2
000069D4	090A0B0C 0D0E0F7F						
000069DC	01020304 05060750			4660	DC	XL16'0102030405060750	v3
000069E4	090A0B78 0D0E0F7F						
000069EC	10000000 00000001			4661	DC	XL16'1000000000000001	v4
000069F4	10000000 00000001						
				4662			
00006A00				4663	VRR_D	VMALO, 3	
00006A00		00006A00		4664+	DS	0FD	
00006A00	00006A48			4665+	USING	*, R5	base for test data and test routine
00006A04	0078			4666+T120	DC	A(X120)	address of test routine
00006A06	00			4667+	DC	H'120'	test number
				4668+	DC	X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006A07	03			4669+	DC	HL1'3'	m5
00006A08	E5D4C1D3 D6404040			4670+	DC	CL8'VMALO'	instruction name
00006A10	00006A8C			4671+	DC	A(RE120+16)	address of v2 source
00006A14	00006A9C			4672+	DC	A(RE120+32)	address of v3 source
00006A18	00006AAC			4673+	DC	A(RE120+48)	address of v4 source
00006A1C	00000010			4674+	DC	A(16)	result length
00006A20	00006A7C			4675+REA120	DC	A(RE120)	result address
00006A28	00000000 00000000			4676+	DS	FD	gap
00006A30	00000000 00000000			4677+V10120	DS	XL16	V1 output
00006A38	00000000 00000000						
00006A40	00000000 00000000			4678+	DS	FD	gap
				4679+*			
00006A48				4680+X120	DS	0F	
00006A48	E310 5010 0014		00000010	4681+	LGF	R1,V2ADDR	load v2 source
00006A4E	E761 0000 0806		00000000	4682+	VL	v22,0(R1)	use v22 to test decoder
00006A54	E310 5014 0014		00000014	4683+	LGF	R1,V3ADDR	load v3 source
00006A5A	E771 0000 0806		00000000	4684+	VL	v23,0(R1)	use v23 to test decoder
00006A60	E310 5018 0014		00000018	4685+	LGF	R1,V4ADDR	load v4 source
00006A66	E781 0000 0806		00000000	4686+	VL	v24,0(R1)	use v24 to test decoder
00006A6C	E766 7300 8FAD			4687+	VMALO	V22,V22,V23,V24,3	test instruction (dest is a source)
00006A72	E760 5030 080E		00006A30	4688+	VST	V22,V10120	save v1 output
00006A78	07FB			4689+	BR	R11	return
00006A7C				4690+RE120	DC	0F	xl16 expected result
00006A7C				4691+	DROP	R5	
00006A7C	F024558D B838C872			4692	DC	XL16'F024558DB838C872 A47CD8D5FF5B4950'	result
00006A84	A47CD8D5 FF5B4950						
00006A8C	FF020304 05060750			4693	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00006A94	090A0B0C 0D0E0F7F						
00006A9C	00010102 02030328			4694	DC	XL16'0001010202030328 0405053C0607073F'	v3
00006AA4	0405053C 0607073F						
00006AAC	F0000000 0000000F			4695	DC	XL16'F000000000000000F F000000000000000F'	v4
00006AB4	F0000000 0000000F						
				4696			
				4697	VRR_D	VMALO,3	
00006AC0				4698+	DS	0FD	
00006AC0		00006AC0		4699+	USING	*,R5	base for test data and test routine
00006AC0	00006B08			4700+T121	DC	A(X121)	address of test routine
00006AC4	0079			4701+	DC	H'121'	test number
00006AC6	00			4702+	DC	X'00'	
00006AC7	03			4703+	DC	HL1'3'	m5
00006AC8	E5D4C1D3 D6404040			4704+	DC	CL8'VMALO'	instruction name
00006AD0	00006B4C			4705+	DC	A(RE121+16)	address of v2 source
00006AD4	00006B5C			4706+	DC	A(RE121+32)	address of v3 source
00006AD8	00006B6C			4707+	DC	A(RE121+48)	address of v4 source
00006ADC	00000010			4708+	DC	A(16)	result length
00006AE0	00006B3C			4709+REA121	DC	A(RE121)	result address
00006AE8	00000000 00000000			4710+	DS	FD	gap
00006AF0	00000000 00000000			4711+V10121	DS	XL16	V1 output
00006AF8	00000000 00000000						
00006B00	00000000 00000000			4712+	DS	FD	gap
				4713+*			
00006B08				4714+X121	DS	0F	
00006B08	E310 5010 0014		00000010	4715+	LGF	R1,V2ADDR	load v2 source
00006B0E	E761 0000 0806		00000000	4716+	VL	v22,0(R1)	use v22 to test decoder
00006B14	E310 5014 0014		00000014	4717+	LGF	R1,V3ADDR	load v3 source
00006B1A	E771 0000 0806		00000000	4718+	VL	v23,0(R1)	use v23 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00006B20	E310 5018 0014		00000018	4719+	LGF	R1,V4ADDR	load v4 source	
00006B26	E781 0000 0806		00000000	4720+	VL	v24,0(R1)	use v24 to test decoder	
00006B2C	E766 7300 8FAD			4721+	VMALO	V22,V22,V23,V24,3	test instruction (dest is a source)	
00006B32	E760 5030 080E		00006AF0	4722+	VST	V22,V10121	save v1 output	
00006B38	07FB			4723+	BR	R11	return	
00006B3C				4724+RE121	DC	0F	xl16 expected result	
00006B3C				4725+	DROP	R5		
00006B3C	DD09131E A8C3DFFE			4726	DC	XL16'DD09131EA8C3DFFE D91C345D6D616771'	result	
00006B44	D91C345D 6D616771							
00006B4C	FF020304 05060750			4727	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2	
00006B54	090A0B0C 0D0E0F7F							
00006B5C	00000000 0000000A			4728	DC	XL16'000000000000000A 0101010F0101010F'	v3	
00006B64	0101010F 0101010F							
00006B6C	DD000000 00000000			4729	DC	XL16'DD00000000000000 D000000D0D000000'	v4	
00006B74	D000000D 0D000000							
				4730				
				4731	VRR_D	VMALO,3		
00006B80				4732+	DS	0FD		
00006B80		00006B80		4733+	USING	*,R5	base for test data and test routine	
00006B80	00006BC8			4734+T122	DC	A(X122)	address of test routine	
00006B84	007A			4735+	DC	H'122'	test number	
00006B86	00			4736+	DC	X'00'		
00006B87	03			4737+	DC	HL1'3'	m5	
00006B88	E5D4C1D3 D6404040			4738+	DC	CL8'VMALO'	instruction name	
00006B90	00006C0C			4739+	DC	A(RE122+16)	address of v2 source	
00006B94	00006C1C			4740+	DC	A(RE122+32)	address of v3 source	
00006B98	00006C2C			4741+	DC	A(RE122+48)	address of v4 source	
00006B9C	00000010			4742+	DC	A(16)	result length	
00006BA0	00006BFC			4743+REA122	DC	A(RE122)	result address	
00006BA8	00000000 00000000			4744+	DS	FD	gap	
00006BB0	00000000 00000000			4745+V10122	DS	XL16	V1 output	
00006BB8	00000000 00000000							
00006BC0	00000000 00000000			4746+	DS	FD	gap	
				4747+*				
00006BC8				4748+X122	DS	0F		
00006BC8	E310 5010 0014		00000010	4749+	LGF	R1,V2ADDR	load v2 source	
00006BCE	E761 0000 0806		00000000	4750+	VL	v22,0(R1)	use v22 to test decoder	
00006BD4	E310 5014 0014		00000014	4751+	LGF	R1,V3ADDR	load v3 source	
00006BDA	E771 0000 0806		00000000	4752+	VL	v23,0(R1)	use v23 to test decoder	
00006BE0	E310 5018 0014		00000018	4753+	LGF	R1,V4ADDR	load v4 source	
00006BE6	E781 0000 0806		00000000	4754+	VL	v24,0(R1)	use v24 to test decoder	
00006BEC	E766 7300 8FAD			4755+	VMALO	V22,V22,V23,V24,3	test instruction (dest is a source)	
00006BF2	E760 5030 080E		00006BB0	4756+	VST	V22,V10122	save v1 output	
00006BF8	07FB			4757+	BR	R11	return	
00006BFC				4758+RE122	DC	0F	xl16 expected result	
00006BFC				4759+	DROP	R5		
00006BFC	FED00000 00000009			4760	DC	XL16'FED0000000000009 F6141E28323C492C'	result	
00006C04	F6141E28 323C492C							
00006C0C	090A0B0C 0D0E0F7F			4761	DC	XL16'090A0B0C0D0E0F7F FF02030405060750'	v2	
00006C14	FF020304 05060750							
00006C1C	0101010F 0101010F			4762	DC	XL16'0101010F0101010F 000000000000000A'	v3	
00006C24	00000000 0000000A							
00006C2C	FED00000 00000000			4763	DC	XL16'FED0000000000000 000000000000000C'	v4	
00006C34	00000000 0000000C							
				4764				
				4765	*	-----		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				4766 * VMAE	- Vector Multiply and Add Even		
				4767 *	-----		
				4768 * Byte			
00006C40				4769	VRR_D VMAE,0		
00006C40		00006C40		4770+	DS 0FD		
00006C40	00006C88			4771+	USING *,R5	base for test data and test routine	
00006C44	007B			4772+T123	DC A(X123)	address of test routine	
00006C46	00			4773+	DC H'123'	test number	
00006C47	00			4774+	DC X'00'		
00006C48	E5D4C1C5 40404040			4775+	DC HL1'0'	m5	
00006C50	00006CCC			4776+	DC CL8'VMAE'	instruction name	
00006C54	00006CDC			4777+	DC A(RE123+16)	address of v2 source	
00006C58	00006CEC			4778+	DC A(RE123+32)	address of v3 source	
00006C5C	00000010			4779+	DC A(RE123+48)	address of v4 source	
00006C60	00006CBC			4780+	DC A(16)	result length	
00006C68	00000000 00000000			4781+REA123	DC A(RE123)	result address	
00006C70	00000000 00000000			4782+	DS FD	gap	
00006C78	00000000 00000000			4783+V10123	DS XL16	V1 output	
00006C80	00000000 00000000			4784+	DS FD	gap	
				4785+*			
00006C88				4786+X123	DS 0F		
00006C88	E310 5010 0014		00000010	4787+	LGF R1,V2ADDR	load v2 source	
00006C8E	E761 0000 0806		00000000	4788+	VL v22,0(R1)	use v22 to test decoder	
00006C94	E310 5014 0014		00000014	4789+	LGF R1,V3ADDR	load v3 source	
00006C9A	E771 0000 0806		00000000	4790+	VL v23,0(R1)	use v23 to test decoder	
00006CA0	E310 5018 0014		00000018	4791+	LGF R1,V4ADDR	load v4 source	
00006CA6	E781 0000 0806		00000000	4792+	VL v24,0(R1)	use v24 to test decoder	
00006CAC	E766 7000 8FAE			4793+	VMAE V22,V22,V23,V24,0	test instruction (dest is a source)	
00006CB2	E760 5030 080E		00006C70	4794+	VST V22,V10123	save v1 output	
00006CB8	07FB			4795+	BR R11	return	
00006CBC				4796+RE123	DC 0F	xl16 expected result	
00006CBC				4797+	DROP R5		
00006CBC	00010000 00000000			4798	DC XL16'0001000000000000 0000000000000000'	result	
00006CC4	00000000 00000000						
00006CCC	FF000000 00000019			4799	DC XL16'FF00000000000019 00000038000000FA'	v2	
00006CD4	00000038 000000FA						
00006CDC	FF000000 00000019			4800	DC XL16'FF00000000000019 00000038000000FA'	v3	
00006CE4	00000038 000000FA						
00006CEC	00000000 00000000			4801	DC XL16'0000000000000000 0000000000000000'	v4	
00006CF4	00000000 00000000						
				4802			
00006D00				4803	VRR_D VMAE,0		
00006D00		00006D00		4804+	DS 0FD		
00006D00	00006D48			4805+	USING *,R5	base for test data and test routine	
00006D04	007C			4806+T124	DC A(X124)	address of test routine	
00006D06	00			4807+	DC H'124'	test number	
00006D07	00			4808+	DC X'00'		
00006D08	E5D4C1C5 40404040			4809+	DC HL1'0'	m5	
00006D10	00006D8C			4810+	DC CL8'VMAE'	instruction name	
00006D14	00006D9C			4811+	DC A(RE124+16)	address of v2 source	
00006D18	00006DAC			4812+	DC A(RE124+32)	address of v3 source	
00006D1C	00000010			4813+	DC A(RE124+48)	address of v4 source	
00006D20	00006D7C			4814+	DC A(16)	result length	
00006D28	00000000 00000000			4815+REA124	DC A(RE124)	result address	
				4816+	DS FD	gap	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006D30	00000000 00000000			4817+V10124	DS	XL16	V1 output
00006D38	00000000 00000000						
00006D40	00000000 00000000			4818+ 4819+*	DS	FD	gap
00006D48				4820+X124	DS	0F	
00006D48	E310 5010 0014		00000010	4821+	LGF	R1,V2ADDR	load v2 source
00006D4E	E761 0000 0806		00000000	4822+	VL	v22,0(R1)	use v22 to test decoder
00006D54	E310 5014 0014		00000014	4823+	LGF	R1,V3ADDR	load v3 source
00006D5A	E771 0000 0806		00000000	4824+	VL	v23,0(R1)	use v23 to test decoder
00006D60	E310 5018 0014		00000018	4825+	LGF	R1,V4ADDR	load v4 source
00006D66	E781 0000 0806		00000000	4826+	VL	v24,0(R1)	use v24 to test decoder
00006D6C	E766 7000 8FAE			4827+	VMAE	V22,V22,V23,V24,0	test instruction (dest is a source)
00006D72	E760 5030 080E		00006D30	4828+	VST	V22,V10124	save v1 output
00006D78	07FB			4829+	BR	R11	return
00006D7C				4830+RE124	DC	0F	xl16 expected result
00006D7C				4831+	DROP	R5	
00006D7C	00030001 0000002F			4832	DC	XL16'000300010000002F 0000000300000002'	result
00006D84	00000003 00000002						
00006D8C	FF0000FF 00000029			4833	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00006D94	00000038 000000FA						
00006D9C	FF000001 00000029			4834	DC	XL16'FF00000100000029 00000038000000FA'	v3
00006DA4	00000038 000000FA						
00006DAC	00020001 0000002F			4835	DC	XL16'000200010000002F 0000000300000002'	v4
00006DB4	00000003 00000002						
				4836			
				4837	VRR_D	VMAE,0	
00006DC0				4838+	DS	0FD	
00006DC0		00006DC0		4839+	USING	*,R5	base for test data and test routine
00006DC0	00006E08			4840+T125	DC	A(X125)	address of test routine
00006DC4	007D			4841+	DC	H'125'	test number
00006DC6	00			4842+	DC	X'00'	
00006DC7	00			4843+	DC	HL1'0'	m5
00006DC8	E5D4C1C5 40404040			4844+	DC	CL8'VMAE'	instruction name
00006DD0	00006E4C			4845+	DC	A(RE125+16)	address of v2 source
00006DD4	00006E5C			4846+	DC	A(RE125+32)	address of v3 source
00006DD8	00006E6C			4847+	DC	A(RE125+48)	address of v4 source
00006DDC	00000010			4848+	DC	A(16)	result length
00006DE0	00006E3C			4849+REA125	DC	A(RE125)	result address
00006DE8	00000000 00000000			4850+	DS	FD	gap
00006DF0	00000000 00000000			4851+V10125	DS	XL16	V1 output
00006DF8	00000000 00000000						
00006E00	00000000 00000000			4852+ 4853+*	DS	FD	gap
				4854+X125	DS	0F	
00006E08				4855+	LGF	R1,V2ADDR	load v2 source
00006E08	E310 5010 0014		00000010	4856+	LGF	R1,V2ADDR	load v2 source
00006E0E	E761 0000 0806		00000000	4856+	VL	v22,0(R1)	use v22 to test decoder
00006E14	E310 5014 0014		00000014	4857+	LGF	R1,V3ADDR	load v3 source
00006E1A	E771 0000 0806		00000000	4858+	VL	v23,0(R1)	use v23 to test decoder
00006E20	E310 5018 0014		00000018	4859+	LGF	R1,V4ADDR	load v4 source
00006E26	E781 0000 0806		00000000	4860+	VL	v24,0(R1)	use v24 to test decoder
00006E2C	E766 7000 8FAE			4861+	VMAE	V22,V22,V23,V24,0	test instruction (dest is a source)
00006E32	E760 5030 080E		00006DF0	4862+	VST	V22,V10125	save v1 output
00006E38	07FB			4863+	BR	R11	return
00006E3C				4864+RE125	DC	0F	xl16 expected result
00006E3C				4865+	DROP	R5	
00006E3C	FF03030D 051F0739			4866	DC	XL16'FF03030D051F0739 095B0B850DB70FF1'	result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006E44	095B0B85 0DB70FF1						
00006E4C	FF020304 05060708			4867	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006E54	090A0B0C 0D0E0F10						
00006E5C	FF020304 05060708			4868	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00006E64	090A0B0C 0D0E0F10						
00006E6C	FF020304 05060708			4869	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006E74	090A0B0C 0D0E0F10						
				4870			
00006E80				4871	VRR_D	VMAE,0	
00006E80		00006E80		4872+	DS	0FD	
00006E80	00006EC8			4873+	USING	*,R5	base for test data and test routine
00006E84	007E			4874+T126	DC	A(X126)	address of test routine
00006E86	00			4875+	DC	H'126'	test number
00006E87	00			4876+	DC	X'00'	
00006E88	E5D4C1C5 40404040			4877+	DC	HL1'0'	m5
00006E90	00006F0C			4878+	DC	CL8'VMAE'	instruction name
00006E94	00006F1C			4879+	DC	A(RE126+16)	address of v2 source
00006E98	00006F2C			4880+	DC	A(RE126+32)	address of v3 source
00006E9C	00000010			4881+	DC	A(RE126+48)	address of v4 source
00006EA0	00006EFC			4882+	DC	A(16)	result length
00006EA8	00000000 00000000			4883+REA126	DC	A(RE126)	result address
00006EB0	00000000 00000000			4884+	DS	FD	gap
00006EB8	00000000 00000000			4885+V10126	DS	XL16	V1 output
00006EC0	00000000 00000000						
				4886+	DS	FD	gap
				4887+*			
00006EC8				4888+X126	DS	0F	
00006EC8	E310 5010 0014		00000010	4889+	LGF	R1,V2ADDR	load v2 source
00006ECE	E761 0000 0806		00000000	4890+	VL	v22,0(R1)	use v22 to test decoder
00006ED4	E310 5014 0014		00000014	4891+	LGF	R1,V3ADDR	load v3 source
00006EDA	E771 0000 0806		00000000	4892+	VL	v23,0(R1)	use v23 to test decoder
00006EE0	E310 5018 0014		00000018	4893+	LGF	R1,V4ADDR	load v4 source
00006EE6	E781 0000 0806		00000000	4894+	VL	v24,0(R1)	use v24 to test decoder
00006EEC	E766 7000 8FAE			4895+	VMAE	V22,V22,V23,V24,0	test instruction (dest is a source)
00006EF2	E760 5030 080E		00006EB0	4896+	VST	V22,V10126	save v1 output
00006EF8	07FB			4897+	BR	R11	return
00006EFC				4898+RE126	DC	0F	xl16 expected result
00006EFC				4899+	DROP	R5	
00006EFC	FF030307 0510071D			4900	DC	XL16'FF0303070510071D 092E0B430D5C0F79'	result
00006F04	092E0B43 0D5C0F79						
00006F0C	FF020304 05060708			4901	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006F14	090A0B0C 0D0E0F10						
00006F1C	FF010102 02030304			4902	DC	XL16'FF01010202030304 0405050606070708'	v3
00006F24	04050506 06070708						
00006F2C	FF020304 05060708			4903	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006F34	090A0B0C 0D0E0F10						
				4904			
00006F40				4905	VRR_D	VMAE,0	
00006F40		00006F40		4906+	DS	0FD	
00006F40	00006F88			4907+	USING	*,R5	base for test data and test routine
00006F44	007F			4908+T127	DC	A(X127)	address of test routine
00006F46	00			4909+	DC	H'127'	test number
00006F47	00			4910+	DC	X'00'	
00006F48	E5D4C1C5 40404040			4911+	DC	HL1'0'	m5
00006F50	00006FCC			4912+	DC	CL8'VMAE'	instruction name
				4913+	DC	A(RE127+16)	address of v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00006F54	00006FDC			4914+	DC	A(RE127+32)	address of v3 source
00006F58	00006FEC			4915+	DC	A(RE127+48)	address of v4 source
00006F5C	00000010			4916+	DC	A(16)	result length
00006F60	00006FBC			4917+REA127	DC	A(RE127)	result address
00006F68	00000000 00000000			4918+	DS	FD	gap
00006F70	00000000 00000000			4919+V10127	DS	XL16	V1 output
00006F78	00000000 00000000						
00006F80	00000000 00000000			4920+	DS	FD	gap
				4921+*			
00006F88				4922+X127	DS	0F	
00006F88	E310 5010 0014		00000010	4923+	LGF	R1,V2ADDR	load v2 source
00006F8E	E761 0000 0806		00000000	4924+	VL	v22,0(R1)	use v22 to test decoder
00006F94	E310 5014 0014		00000014	4925+	LGF	R1,V3ADDR	load v3 source
00006F9A	E771 0000 0806		00000000	4926+	VL	v23,0(R1)	use v23 to test decoder
00006FA0	E310 5018 0014		00000018	4927+	LGF	R1,V4ADDR	load v4 source
00006FA6	E781 0000 0806		00000000	4928+	VL	v24,0(R1)	use v24 to test decoder
00006FAC	E766 7000 8FAE			4929+	VMAE	V22,V22,V23,V24,0	test instruction (dest is a source)
00006FB2	E760 5030 080E		00006F70	4930+	VST	V22,V10127	save v1 output
00006FB8	07FB			4931+	BR	R11	return
00006FBC				4932+RE127	DC	0F	xl16 expected result
00006FBC				4933+	DROP	R5	
00006FBC	FF030304 05060708			4934	DC	XL16'FF03030405060708 09130B170D1B0F1F'	result
00006FC4	09130B17 0D1B0F1F						
00006FCC	FF020304 05060708			4935	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00006FD4	090A0B0C 0D0E0F10						
00006FDC	FF000000 00000001			4936	DC	XL16'FF0000000000000001 0101010101010102'	v3
00006FE4	01010101 01010102						
00006FEC	FF020304 05060708			4937	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00006FF4	090A0B0C 0D0E0F10						
				4938			
				4939 * Halfword			
				4940	VRR_D	VMAE,1	
00007000				4941+	DS	0FD	
00007000		00007000		4942+	USING	*,R5	base for test data and test routine
00007000	00007048			4943+T128	DC	A(X128)	address of test routine
00007004	0080			4944+	DC	H'128'	test number
00007006	00			4945+	DC	X'00'	
00007007	01			4946+	DC	HL1'1'	m5
00007008	E5D4C1C5 40404040			4947+	DC	CL8'VMAE'	instruction name
00007010	0000708C			4948+	DC	A(RE128+16)	address of v2 source
00007014	0000709C			4949+	DC	A(RE128+32)	address of v3 source
00007018	000070AC			4950+	DC	A(RE128+48)	address of v4 source
0000701C	00000010			4951+	DC	A(16)	result length
00007020	0000707C			4952+REA128	DC	A(RE128)	result address
00007028	00000000 00000000			4953+	DS	FD	gap
00007030	00000000 00000000			4954+V10128	DS	XL16	V1 output
00007038	00000000 00000000						
00007040	00000000 00000000			4955+	DS	FD	gap
				4956+*			
00007048				4957+X128	DS	0F	
00007048	E310 5010 0014		00000010	4958+	LGF	R1,V2ADDR	load v2 source
0000704E	E761 0000 0806		00000000	4959+	VL	v22,0(R1)	use v22 to test decoder
00007054	E310 5014 0014		00000014	4960+	LGF	R1,V3ADDR	load v3 source
0000705A	E771 0000 0806		00000000	4961+	VL	v23,0(R1)	use v23 to test decoder
00007060	E310 5018 0014		00000018	4962+	LGF	R1,V4ADDR	load v4 source
00007066	E781 0000 0806		00000000	4963+	VL	v24,0(R1)	use v24 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000706C	E766 7100 8FAE			4964+	VMAE	V22,V22,V23,V24,1	test instruction (dest is a source)
00007072	E760 5030 080E		00007030	4965+	VST	V22,V10128	save v1 output
00007078	07FB			4966+	BR	R11	return
0000707C				4967+RE128	DC	0F	xl16 expected result
0000707C				4968+	DROP	R5	
0000707C	00010000 00000000			4969	DC	XL16'0001000000000000 0000000000000000'	result
00007084	00000000 00000000						
0000708C	FF000000 00000019			4970	DC	XL16'FF00000000000019 00000038000000FA'	v2
00007094	00000038 000000FA						
0000709C	FF000000 00000019			4971	DC	XL16'FF00000000000019 00000038000000FA'	v3
000070A4	00000038 000000FA						
000070AC	00000000 00000000			4972	DC	XL16'0000000000000000 0000000000000000'	v4
000070B4	00000000 00000000						
				4973			
000070C0				4974	VRR_D	VMAE,1	
000070C0		000070C0		4975+	DS	0FD	
000070C0	00007108			4976+	USING	*,R5	base for test data and test routine
000070C4	0081			4977+T129	DC	A(X129)	address of test routine
000070C6	00			4978+	DC	H'129'	test number
000070C7	01			4979+	DC	X'00'	
000070C8	E5D4C1C5 40404040			4980+	DC	HL1'1'	m5
000070D0	0000714C			4981+	DC	CL8'VMAE'	instruction name
000070D4	0000715C			4982+	DC	A(RE129+16)	address of v2 source
000070D8	0000716C			4983+	DC	A(RE129+32)	address of v3 source
000070DC	00000010			4984+	DC	A(RE129+48)	address of v4 source
000070E0	0000713C			4985+	DC	A(16)	result length
000070E8	00000000 00000000			4986+REA129	DC	A(RE129)	result address
000070F0	00000000 00000000			4987+	DS	FD	gap
000070F8	00000000 00000000			4988+V10129	DS	XL16	V1 output
00007100	00000000 00000000						
				4989+	DS	FD	gap
				4990+*			
00007108				4991+X129	DS	0F	
00007108	E310 5010 0014	00000010		4992+	LGF	R1,V2ADDR	load v2 source
0000710E	E761 0000 0806	00000000		4993+	VL	v22,0(R1)	use v22 to test decoder
00007114	E310 5014 0014	00000014		4994+	LGF	R1,V3ADDR	load v3 source
0000711A	E771 0000 0806	00000000		4995+	VL	v23,0(R1)	use v23 to test decoder
00007120	E310 5018 0014	00000018		4996+	LGF	R1,V4ADDR	load v4 source
00007126	E781 0000 0806	00000000		4997+	VL	v24,0(R1)	use v24 to test decoder
0000712C	E766 7100 8FAE			4998+	VMAE	V22,V22,V23,V24,1	test instruction (dest is a source)
00007132	E760 5030 080E	000070F0		4999+	VST	V22,V10129	save v1 output
00007138	07FB			5000+	BR	R11	return
0000713C				5001+RE129	DC	0F	xl16 expected result
0000713C				5002+	DROP	R5	
0000713C	00030001 0000002F			5003	DC	XL16'000300010000002F 0000000300000002'	result
00007144	00000003 00000002						
0000714C	FF0000FF 00000029			5004	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00007154	00000038 000000FA						
0000715C	FF000001 00000029			5005	DC	XL16'FF00000100000029 00000038000000FA'	v3
00007164	00000038 000000FA						
0000716C	00020001 0000002F			5006	DC	XL16'000200010000002F 0000000300000002'	v4
00007174	00000003 00000002						
				5007			
00007180				5008	VRR_D	VMAE,1	
00007180		00007180		5009+	DS	0FD	
				5010+	USING	*,R5	base for test data and test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007180	000071C8			5011+T130	DC	A(X130)	address of test routine
00007184	0082			5012+	DC	H'130'	test number
00007186	00			5013+	DC	X'00'	
00007187	01			5014+	DC	HL1'1'	m5
00007188	E5D4C1C5 40404040			5015+	DC	CL8'VMAE'	instruction name
00007190	0000720C			5016+	DC	A(RE130+16)	address of v2 source
00007194	0000721C			5017+	DC	A(RE130+32)	address of v3 source
00007198	0000722C			5018+	DC	A(RE130+48)	address of v4 source
0000719C	00000010			5019+	DC	A(16)	result length
000071A0	000071FC			5020+REA130	DC	A(RE130)	result address
000071A8	00000000 00000000			5021+	DS	FD	gap
000071B0	00000000 00000000			5022+V10130	DS	XL16	V1 output
000071B8	00000000 00000000						
000071C0	00000000 00000000			5023+	DS	FD	gap
				5024+*			
000071C8				5025+X130	DS	0F	
000071C8	E310 5010 0014		00000010	5026+	LGF	R1,V2ADDR	load v2 source
000071CE	E761 0000 0806		00000000	5027+	VL	v22,0(R1)	use v22 to test decoder
000071D4	E310 5014 0014		00000014	5028+	LGF	R1,V3ADDR	load v3 source
000071DA	E771 0000 0806		00000000	5029+	VL	v23,0(R1)	use v23 to test decoder
000071E0	E310 5018 0014		00000018	5030+	LGF	R1,V4ADDR	load v4 source
000071E6	E781 0000 0806		00000000	5031+	VL	v24,0(R1)	use v24 to test decoder
000071EC	E766 7100 8FAE			5032+	VMAE	V22,V22,V23,V24,1	test instruction (dest is a source)
000071F2	E760 5030 080E		000071B0	5033+	VST	V22,V10130	save v1 output
000071F8	07FB			5034+	BR	R11	return
000071FC				5035+RE130	DC	0F	xl16 expected result
000071FC				5036+	DROP	R5	
000071FC	FF02FF08 051F432C			5037	DC	XL16'FF02FF08051F432C 095BBF700DB87BD4'	result
00007204	095BBF70 0DB87BD4						
0000720C	FF020304 05060708			5038	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007214	090A0B0C 0D0E0F10						
0000721C	FF020304 05060708			5039	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00007224	090A0B0C 0D0E0F10						
0000722C	FF020304 05060708			5040	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00007234	090A0B0C 0D0E0F10						
				5041			
				5042	VRR_D	VMAE,1	
00007240				5043+	DS	0FD	
00007240		00007240		5044+	USING	*,R5	base for test data and test routine
00007240	00007288			5045+T131	DC	A(X131)	address of test routine
00007244	0083			5046+	DC	H'131'	test number
00007246	00			5047+	DC	X'00'	
00007247	01			5048+	DC	HL1'1'	m5
00007248	E5D4C1C5 40404040			5049+	DC	CL8'VMAE'	instruction name
00007250	000072CC			5050+	DC	A(RE131+16)	address of v2 source
00007254	000072DC			5051+	DC	A(RE131+32)	address of v3 source
00007258	000072EC			5052+	DC	A(RE131+48)	address of v4 source
0000725C	00000010			5053+	DC	A(16)	result length
00007260	000072BC			5054+REA131	DC	A(RE131)	result address
00007268	00000000 00000000			5055+	DS	FD	gap
00007270	00000000 00000000			5056+V10131	DS	XL16	V1 output
00007278	00000000 00000000						
00007280	00000000 00000000			5057+	DS	FD	gap
				5058+*			
00007288				5059+X131	DS	0F	
00007288	E310 5010 0014		00000010	5060+	LGF	R1,V2ADDR	load v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000728E	E761 0000 0806		00000000	5061+	VL	v22,0(R1)	use v22 to test decoder
00007294	E310 5014 0014		00000014	5062+	LGF	R1,V3ADDR	load v3 source
0000729A	E771 0000 0806		00000000	5063+	VL	v23,0(R1)	use v23 to test decoder
000072A0	E310 5018 0014		00000018	5064+	LGF	R1,V4ADDR	load v4 source
000072A6	E781 0000 0806		00000000	5065+	VL	v24,0(R1)	use v24 to test decoder
000072AC	E766 7100 8FAE			5066+	VMAE	V22,V22,V23,V24,1	test instruction (dest is a source)
000072B2	E760 5030 080E		00007270	5067+	VST	V22,V10131	save v1 output
000072B8	07FB			5068+	BR	R11	return
000072BC				5069+RE131	DC	0F	xl16 expected result
000072BC				5070+	DROP	R5	
000072BC	FF030006 0510221A			5071	DC	XL16'FF0300060510221A 092E603E0D5CBE72'	result
000072C4	092E603E 0D5CBE72						
000072CC	FF020304 05060708			5072	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000072D4	090A0B0C 0D0E0F10						
000072DC	FF010102 02030304			5073	DC	XL16'FF01010202030304 0405050606070708'	v3
000072E4	04050506 06070708						
000072EC	FF020304 05060708			5074	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000072F4	090A0B0C 0D0E0F10						
				5075			
				5076	VRR_D	VMAE,1	
00007300				5077+	DS	0FD	
00007300		00007300		5078+	USING	*,R5	base for test data and test routine
00007300	00007348			5079+T132	DC	A(X132)	address of test routine
00007304	0084			5080+	DC	H'132'	test number
00007306	00			5081+	DC	X'00'	
00007307	01			5082+	DC	HL1'1'	m5
00007308	E5D4C1C5 40404040			5083+	DC	CL8'VMAE'	instruction name
00007310	0000738C			5084+	DC	A(RE132+16)	address of v2 source
00007314	0000739C			5085+	DC	A(RE132+32)	address of v3 source
00007318	000073AC			5086+	DC	A(RE132+48)	address of v4 source
0000731C	00000010			5087+	DC	A(16)	result length
00007320	0000737C			5088+REA132	DC	A(RE132)	result address
00007328	00000000 00000000			5089+	DS	FD	gap
00007330	00000000 00000000			5090+V10132	DS	XL16	V1 output
00007338	00000000 00000000						
00007340	00000000 00000000			5091+	DS	FD	gap
				5092+*			
00007348				5093+X132	DS	0F	
00007348	E310 5010 0014		00000010	5094+	LGF	R1,V2ADDR	load v2 source
0000734E	E761 0000 0806		00000000	5095+	VL	v22,0(R1)	use v22 to test decoder
00007354	E310 5014 0014		00000014	5096+	LGF	R1,V3ADDR	load v3 source
0000735A	E771 0000 0806		00000000	5097+	VL	v23,0(R1)	use v23 to test decoder
00007360	E310 5018 0014		00000018	5098+	LGF	R1,V4ADDR	load v4 source
00007366	E781 0000 0806		00000000	5099+	VL	v24,0(R1)	use v24 to test decoder
0000736C	E766 7100 8FAE			5100+	VMAE	V22,V22,V23,V24,1	test instruction (dest is a source)
00007372	E760 5030 080E		00007330	5101+	VST	V22,V10132	save v1 output
00007378	07FB			5102+	BR	R11	return
0000737C				5103+RE132	DC	0F	xl16 expected result
0000737C				5104+	DROP	R5	
0000737C	FF030104 05060708			5105	DC	XL16'FF03010405060708 09131E160D1B2A1E'	result
00007384	09131E16 0D1B2A1E						
0000738C	FF020304 05060708			5106	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007394	090A0B0C 0D0E0F10						
0000739C	FF000000 00000001			5107	DC	XL16'FF0000000000000001 0101010101010102'	v3
000073A4	01010101 01010102						
000073AC	FF020304 05060708			5108	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000073B4	090A0B0C 0D0E0F10			5109			
				5110 * Word			
				5111	VRR_D	VMAE, 2	
000073C0				5112+	DS	0FD	
000073C0		000073C0		5113+	USING	*, R5	base for test data and test routine
000073C0	00007408			5114+T133	DC	A(X133)	address of test routine
000073C4	0085			5115+	DC	H'133'	test number
000073C6	00			5116+	DC	X'00'	
000073C7	02			5117+	DC	HL1'2'	m5
000073C8	E5D4C1C5 40404040			5118+	DC	CL8'VMAE'	instruction name
000073D0	0000744C			5119+	DC	A(RE133+16)	address of v2 source
000073D4	0000745C			5120+	DC	A(RE133+32)	address of v3 source
000073D8	0000746C			5121+	DC	A(RE133+48)	address of v4 source
000073DC	00000010			5122+	DC	A(16)	result length
000073E0	0000743C			5123+REA133	DC	A(RE133)	result address
000073E8	00000000 00000000			5124+	DS	FD	gap
000073F0	00000000 00000000			5125+V10133	DS	XL16	V1 output
000073F8	00000000 00000000						
00007400	00000000 00000000			5126+	DS	FD	gap
				5127+*			
00007408				5128+X133	DS	0F	
00007408	E310 5010 0014		00000010	5129+	LGF	R1, V2ADDR	load v2 source
0000740E	E761 0000 0806		00000000	5130+	VL	v22, 0(R1)	use v22 to test decoder
00007414	E310 5014 0014		00000014	5131+	LGF	R1, V3ADDR	load v3 source
0000741A	E771 0000 0806		00000000	5132+	VL	v23, 0(R1)	use v23 to test decoder
00007420	E310 5018 0014		00000018	5133+	LGF	R1, V4ADDR	load v4 source
00007426	E781 0000 0806		00000000	5134+	VL	v24, 0(R1)	use v24 to test decoder
0000742C	E766 7200 8FAE			5135+	VMAE	V22, V22, V23, V24, 2	test instruction (dest is a source)
00007432	E760 5030 080E		000073F0	5136+	VST	V22, V10133	save v1 output
00007438	07FB			5137+	BR	R11	return
0000743C				5138+RE133	DC	0F	xl16 expected result
0000743C				5139+	DROP	R5	
0000743C	00010000 00000000			5140	DC	XL16'0001000000000000 00000000000000C40'	result
00007444	00000000 00000C40						
0000744C	FF000000 00000019			5141	DC	XL16'FF00000000000019 00000038000000FA'	v2
00007454	00000038 000000FA						
0000745C	FF000000 00000019			5142	DC	XL16'FF00000000000019 00000038000000FA'	v3
00007464	00000038 000000FA						
0000746C	00000000 00000000			5143	DC	XL16'0000000000000000 0000000000000000'	v4
00007474	00000000 00000000						
				5144			
				5145	VRR_D	VMAE, 2	
00007480				5146+	DS	0FD	
00007480		00007480		5147+	USING	*, R5	base for test data and test routine
00007480	000074C8			5148+T134	DC	A(X134)	address of test routine
00007484	0086			5149+	DC	H'134'	test number
00007486	00			5150+	DC	X'00'	
00007487	02			5151+	DC	HL1'2'	m5
00007488	E5D4C1C5 40404040			5152+	DC	CL8'VMAE'	instruction name
00007490	0000750C			5153+	DC	A(RE134+16)	address of v2 source
00007494	0000751C			5154+	DC	A(RE134+32)	address of v3 source
00007498	0000752C			5155+	DC	A(RE134+48)	address of v4 source
0000749C	00000010			5156+	DC	A(16)	result length
000074A0	000074FC			5157+REA134	DC	A(RE134)	result address
000074A8	00000000 00000000			5158+	DS	FD	gap

LOC	OBJECT CODE			ADDR1	ADDR2	STMT			
000074B0	00000000	00000000				5159+V10134	DS	XL16	V1 output
000074B8	00000000	00000000							
000074C0	00000000	00000000				5160+	DS	FD	gap
						5161+*			
000074C8						5162+X134	DS	0F	
000074C8	E310	5010	0014		00000010	5163+	LGF	R1,V2ADDR	load v2 source
000074CE	E761	0000	0806		00000000	5164+	VL	v22,0(R1)	use v22 to test decoder
000074D4	E310	5014	0014		00000014	5165+	LGF	R1,V3ADDR	load v3 source
000074DA	E771	0000	0806		00000000	5166+	VL	v23,0(R1)	use v23 to test decoder
000074E0	E310	5018	0014		00000018	5167+	LGF	R1,V4ADDR	load v4 source
000074E6	E781	0000	0806		00000000	5168+	VL	v24,0(R1)	use v24 to test decoder
000074EC	E766	7200	8FAE			5169+	VMAE	V22,V22,V23,V24,2	test instruction (dest is a source)
000074F2	E760	5030	080E		000074B0	5170+	VST	V22,V10134	save v1 output
000074F8	07FB					5171+	BR	R11	return
000074FC						5172+RE134	DC	0F	xl16 expected result
000074FC						5173+	DROP	R5	
000074FC	00030000	0000012E				5174	DC	XL16'000300000000012E 00000003000000C42'	result
00007504	00000003	00000C42							
0000750C	FF0000FF	00000029				5175	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00007514	00000038	000000FA							
0000751C	FF000001	00000029				5176	DC	XL16'FF00000100000029 00000038000000FA'	v3
00007524	00000038	000000FA							
0000752C	00020001	0000002F				5177	DC	XL16'000200010000002F 0000000300000002'	v4
00007534	00000003	00000002							
						5178			
						5179	VRR_D	VMAE,2	
00007540						5180+	DS	0FD	
00007540				00007540		5181+	USING	*,R5	base for test data and test routine
00007540	00007588					5182+T135	DC	A(X135)	address of test routine
00007544	0087					5183+	DC	H'135'	test number
00007546	00					5184+	DC	X'00'	
00007547	02					5185+	DC	HL1'2'	m5
00007548	E5D4C1C5	40404040				5186+	DC	CL8'VMAE'	instruction name
00007550	000075CC					5187+	DC	A(RE135+16)	address of v2 source
00007554	000075DC					5188+	DC	A(RE135+32)	address of v3 source
00007558	000075EC					5189+	DC	A(RE135+48)	address of v4 source
0000755C	00000010					5190+	DC	A(16)	result length
00007560	000075BC					5191+REA135	DC	A(RE135)	result address
00007568	00000000	00000000				5192+	DS	FD	gap
00007570	00000000	00000000				5193+V10135	DS	XL16	V1 output
00007578	00000000	00000000							
00007580	00000000	00000000				5194+	DS	FD	gap
						5195+*			
00007588						5196+X135	DS	0F	
00007588	E310	5010	0014		00000010	5197+	LGF	R1,V2ADDR	load v2 source
0000758E	E761	0000	0806		00000000	5198+	VL	v22,0(R1)	use v22 to test decoder
00007594	E310	5014	0014		00000014	5199+	LGF	R1,V3ADDR	load v3 source
0000759A	E771	0000	0806		00000000	5200+	VL	v23,0(R1)	use v23 to test decoder
000075A0	E310	5018	0014		00000018	5201+	LGF	R1,V4ADDR	load v4 source
000075A6	E781	0000	0806		00000000	5202+	VL	v24,0(R1)	use v24 to test decoder
000075AC	E766	7200	8FAE			5203+	VMAE	V22,V22,V23,V24,2	test instruction (dest is a source)
000075B2	E760	5030	080E		00007570	5204+	VST	V22,V10135	save v1 output
000075B8	07FB					5205+	BR	R11	return
000075BC						5206+RE135	DC	0F	xl16 expected result
000075BC						5207+	DROP	R5	
000075BC	FF02FF02	091F1F18				5208	DC	XL16'FF02FF02091F1F18 095BC037C27817A0'	result

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000075C4	095BC037 C27817A0						
000075CC	FF020304 05060708			5209	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000075D4	090A0B0C 0D0E0F10						
000075DC	FF020304 05060708			5210	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000075E4	090A0B0C 0D0E0F10						
000075EC	FF020304 05060708			5211	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000075F4	090A0B0C 0D0E0F10						
				5212			
00007600				5213	VRR_D	VMAE, 2	
00007600		00007600		5214+	DS	0FD	
00007600	00007648			5215+	USING	*, R5	base for test data and test routine
00007604	0088			5216+T136	DC	A(X136)	address of test routine
00007606	00			5217+	DC	H'136'	test number
00007607	02			5218+	DC	X'00'	
00007608	E5D4C1C5 40404040			5219+	DC	HL1'2'	m5
00007610	0000768C			5220+	DC	CL8'VMAE'	instruction name
00007614	0000769C			5221+	DC	A(RE136+16)	address of v2 source
00007618	000076AC			5222+	DC	A(RE136+32)	address of v3 source
0000761C	00000010			5223+	DC	A(RE136+48)	address of v4 source
00007620	0000767C			5224+	DC	A(16)	result length
00007628	00000000 00000000			5225+REA136	DC	A(RE136)	result address
00007630	00000000 00000000			5226+	DS	FD	gap
00007638	00000000 00000000			5227+V10136	DS	XL16	V1 output
00007640	00000000 00000000						
				5228+	DS	FD	gap
				5229+*			
00007648				5230+X136	DS	0F	
00007648	E310 5010 0014		00000010	5231+	LGF	R1, V2ADDR	load v2 source
0000764E	E761 0000 0806		00000000	5232+	VL	v22, 0(R1)	use v22 to test decoder
00007654	E310 5014 0014		00000014	5233+	LGF	R1, V3ADDR	load v3 source
0000765A	E771 0000 0806		00000000	5234+	VL	v23, 0(R1)	use v23 to test decoder
00007660	E310 5018 0014		00000018	5235+	LGF	R1, V4ADDR	load v4 source
00007666	E781 0000 0806		00000000	5236+	VL	v24, 0(R1)	use v24 to test decoder
0000766C	E766 7200 8FAE			5237+	VMAE	V22, V22, V23, V24, 2	test instruction (dest is a source)
00007672	E760 5030 080E		00007630	5238+	VST	V22, V10136	save v1 output
00007678	07FB			5239+	BR	R11	return
0000767C				5240+RE136	DC	0F	xl16 expected result
0000767C				5241+	DROP	R5	
0000767C	FF030002 04111110			5242	DC	XL16'FF03000204111110 092E6097DCBD8D58'	result
00007684	092E6097 DCBD8D58						
0000768C	FF020304 05060708			5243	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007694	090A0B0C 0D0E0F10						
0000769C	FF010102 02030304			5244	DC	XL16'FF01010202030304 0405050606070708'	v3
000076A4	04050506 06070708						
000076AC	FF020304 05060708			5245	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000076B4	090A0B0C 0D0E0F10						
				5246			
000076C0				5247	VRR_D	VMAE, 2	
000076C0		000076C0		5248+	DS	0FD	
000076C0	00007708			5249+	USING	*, R5	base for test data and test routine
000076C4	0089			5250+T137	DC	A(X137)	address of test routine
000076C6	00			5251+	DC	H'137'	test number
000076C7	02			5252+	DC	X'00'	
000076C8	E5D4C1C5 40404040			5253+	DC	HL1'2'	m5
000076D0	0000774C			5254+	DC	CL8'VMAE'	instruction name
				5255+	DC	A(RE137+16)	address of v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000076D4	0000775C			5256+	DC	A(RE137+32)	address of v3 source
000076D8	0000776C			5257+	DC	A(RE137+48)	address of v4 source
000076DC	00000010			5258+	DC	A(16)	result length
000076E0	0000773C			5259+REA137	DC	A(RE137)	result address
000076E8	00000000 00000000			5260+	DS	FD	gap
000076F0	00000000 00000000			5261+V10137	DS	XL16	V1 output
000076F8	00000000 00000000						
00007700	00000000 00000000			5262+	DS	FD	gap
				5263+*			
00007708				5264+X137	DS	0F	
00007708	E310 5010 0014		00000010	5265+	LGF	R1,V2ADDR	load v2 source
0000770E	E761 0000 0806		00000000	5266+	VL	v22,0(R1)	use v22 to test decoder
00007714	E310 5014 0014		00000014	5267+	LGF	R1,V3ADDR	load v3 source
0000771A	E771 0000 0806		00000000	5268+	VL	v23,0(R1)	use v23 to test decoder
00007720	E310 5018 0014		00000018	5269+	LGF	R1,V4ADDR	load v4 source
00007726	E781 0000 0806		00000000	5270+	VL	v24,0(R1)	use v24 to test decoder
0000772C	E766 7200 8FAE			5271+	VMAE	V22,V22,V23,V24,2	test instruction (dest is a source)
00007732	E760 5030 080E		000076F0	5272+	VST	V22,V10137	save v1 output
00007738	07FB			5273+	BR	R11	return
0000773C				5274+RE137	DC	0F	xl16 expected result
0000773C				5275+	DROP	R5	
0000773C	FF030101 01060708			5276	DC	XL16'FF03010101060708 09131E2A372F261C'	result
00007744	09131E2A 372F261C						
0000774C	FF020304 05060708			5277	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007754	090A0B0C 0D0E0F10						
0000775C	FF000000 00000001			5278	DC	XL16'FF00000000000001 0101010101010102'	v3
00007764	01010101 01010102						
0000776C	FF020304 05060708			5279	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00007774	090A0B0C 0D0E0F10						
				5280			
				5281 * Doubleword			
				5282	VRR_D	VMAE,3	
00007780				5283+	DS	0FD	
00007780		00007780		5284+	USING	*,R5	base for test data and test routine
00007780	000077C8			5285+T138	DC	A(X138)	address of test routine
00007784	008A			5286+	DC	H'138'	test number
00007786	00			5287+	DC	X'00'	
00007787	03			5288+	DC	HL1'3'	m5
00007788	E5D4C1C5 40404040			5289+	DC	CL8'VMAE'	instruction name
00007790	0000780C			5290+	DC	A(RE138+16)	address of v2 source
00007794	0000781C			5291+	DC	A(RE138+32)	address of v3 source
00007798	0000782C			5292+	DC	A(RE138+48)	address of v4 source
0000779C	00000010			5293+	DC	A(16)	result length
000077A0	000077FC			5294+REA138	DC	A(RE138)	result address
000077A8	00000000 00000000			5295+	DS	FD	gap
000077B0	00000000 00000000			5296+V10138	DS	XL16	V1 output
000077B8	00000000 00000000						
000077C0	00000000 00000000			5297+	DS	FD	gap
				5298+*			
000077C8				5299+X138	DS	0F	
000077C8	E310 5010 0014		00000010	5300+	LGF	R1,V2ADDR	load v2 source
000077CE	E761 0000 0806		00000000	5301+	VL	v22,0(R1)	use v22 to test decoder
000077D4	E310 5014 0014		00000014	5302+	LGF	R1,V3ADDR	load v3 source
000077DA	E771 0000 0806		00000000	5303+	VL	v23,0(R1)	use v23 to test decoder
000077E0	E310 5018 0014		00000018	5304+	LGF	R1,V4ADDR	load v4 source
000077E6	E781 0000 0806		00000000	5305+	VL	v24,0(R1)	use v24 to test decoder

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000077EC	E766 7300 8FAE			5306+	VMAE	V22,V22,V23,V24,3	test instruction (dest is a source)
000077F2	E760 5030 080E		000077B0	5307+	VST	V22,V10138	save v1 output
000077F8	07FB			5308+	BR	R11	return
000077FC				5309+RE138	DC	0F	xl16 expected result
000077FC				5310+	DROP	R5	
000077FC	00000000 00000000			5311	DC	XL16'0000000000000000 FFFCE00271000000'	result
00007804	FFFCE002 71000000						
0000780C	FFFFFFFF 00019000			5312	DC	XL16'FFFFFFFF00019000 00000038EEEEEEFA'	v2
00007814	00000038 EEEEEEEFA						
0000781C	FFFFFFFF 00019000			5313	DC	XL16'FFFFFFFF00019000 000000380EEEEEEFA'	v3
00007824	00000038 0EEEEEEFA						
0000782C	00000000 00000000			5314	DC	XL16'0000000000000000 0000000000000000'	v4
00007834	00000000 00000000						
				5315			
00007840				5316	VRR_D	VMAE, 3	
00007840		00007840		5317+	DS	0FD	
00007840	00007888			5318+	USING	*,R5	base for test data and test routine
00007844	008B			5319+T139	DC	A(X139)	address of test routine
00007846	00			5320+	DC	H'139'	test number
00007847	03			5321+	DC	X'00'	
00007848	E5D4C1C5 40404040			5322+	DC	HL1'3'	m5
00007850	000078CC			5323+	DC	CL8'VMAE'	instruction name
00007854	000078DC			5324+	DC	A(RE139+16)	address of v2 source
00007858	000078EC			5325+	DC	A(RE139+32)	address of v3 source
0000785C	00000010			5326+	DC	A(RE139+48)	address of v4 source
00007860	000078BC			5327+	DC	A(16)	result length
00007868	00000000 00000000			5328+REA139	DC	A(RE139)	result address
00007870	00000000 00000000			5329+	DS	FD	gap
00007878	00000000 00000000			5330+V10139	DS	XL16	V1 output
00007880	00000000 00000000						
				5331+	DS	FD	gap
				5332+*			
00007888				5333+X139	DS	0F	
00007888	E310 5010 0014		00000010	5334+	LGF	R1,V2ADDR	load v2 source
0000788E	E761 0000 0806		00000000	5335+	VL	v22,0(R1)	use v22 to test decoder
00007894	E310 5014 0014		00000014	5336+	LGF	R1,V3ADDR	load v3 source
0000789A	E771 0000 0806		00000000	5337+	VL	v23,0(R1)	use v23 to test decoder
000078A0	E310 5018 0014		00000018	5338+	LGF	R1,V4ADDR	load v4 source
000078A6	E781 0000 0806		00000000	5339+	VL	v24,0(R1)	use v24 to test decoder
000078AC	E766 7300 8FAE			5340+	VMAE	V22,V22,V23,V24,3	test instruction (dest is a source)
000078B2	E760 5030 080E		00007870	5341+	VST	V22,V10139	save v1 output
000078B8	07FB			5342+	BR	R11	return
000078BC				5343+RE139	DC	0F	xl16 expected result
000078BC				5344+	DROP	R5	
000078BC	FFFF0004 0C192C46			5345	DC	XL16'FFFF00040C192C46 69B556ED77F578FF'	result
000078C4	69B556ED 77F578FF						
000078CC	FF020304 05060750			5346	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
000078D4	090A0B0C 0D0E0F7F						
000078DC	01020304 05060750			5347	DC	XL16'0102030405060750 090A0B780D0E0F7F'	v3
000078E4	090A0B78 0D0E0F7F						
000078EC	FFFFFFFF FFFFFFFF			5348	DC	XL16'FFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFF'	v4
000078F4	FFFFFFFF FFFFFFFF						
				5349			
				5350	VRR_D	VMAE, 3	
00007900				5351+	DS	0FD	
00007900		00007900		5352+	USING	*,R5	base for test data and test routine

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007900	00007948			5353+T140	DC	A(X140)	address of test routine
00007904	008C			5354+	DC	H'140'	test number
00007906	00			5355+	DC	X'00'	
00007907	03			5356+	DC	HL1'3'	m5
00007908	E5D4C1C5 40404040			5357+	DC	CL8'VMAE'	instruction name
00007910	0000798C			5358+	DC	A(RE140+16)	address of v2 source
00007914	0000799C			5359+	DC	A(RE140+32)	address of v3 source
00007918	000079AC			5360+	DC	A(RE140+48)	address of v4 source
0000791C	00000010			5361+	DC	A(16)	result length
00007920	0000797C			5362+REA140	DC	A(RE140)	result address
00007928	00000000 00000000			5363+	DS	FD	gap
00007930	00000000 00000000			5364+V10140	DS	XL16	V1 output
00007938	00000000 00000000						
00007940	00000000 00000000			5365+	DS	FD	gap
				5366+*			
00007948				5367+X140	DS	0F	
00007948	E310 5010 0014		00000010	5368+	LGF	R1,V2ADDR	load v2 source
0000794E	E761 0000 0806		00000000	5369+	VL	v22,0(R1)	use v22 to test decoder
00007954	E310 5014 0014		00000014	5370+	LGF	R1,V3ADDR	load v3 source
0000795A	E771 0000 0806		00000000	5371+	VL	v23,0(R1)	use v23 to test decoder
00007960	E310 5018 0014		00000018	5372+	LGF	R1,V4ADDR	load v4 source
00007966	E781 0000 0806		00000000	5373+	VL	v24,0(R1)	use v24 to test decoder
0000796C	E766 7300 8FAE			5374+	VMAE	V22,V22,V23,V24,3	test instruction (dest is a source)
00007972	E760 5030 080E		00007930	5375+	VST	V22,V10140	save v1 output
00007978	07FB			5376+	BR	R11	return
0000797C				5377+RE140	DC	0F	xl16 expected result
0000797C				5378+	DROP	R5	
0000797C	FFFFFF01 0309101D			5379	DC	XL16'FFFFFF010309101D 06D2FE7090F71480'	result
00007984	06D2FE70 90F71480						
0000798C	FF020304 05060750			5380	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00007994	090A0B0C 0D0E0F7F						
0000799C	00010102 02030328			5381	DC	XL16'0001010202030328 0405053C0607073F'	v3
000079A4	0405053C 0607073F						
000079AC	00000000 00000001			5382	DC	XL16'0000000000000001 0000000000000000'	v4
000079B4	00000000 00000000						
				5383			
000079C0				5384	VRR_D	VMAE,3	
000079C0		000079C0		5385+	DS	0FD	
000079C0	00007A08			5386+	USING	*,R5	base for test data and test routine
000079C4	008D			5387+T141	DC	A(X141)	address of test routine
000079C6	00			5388+	DC	H'141'	test number
000079C7	03			5389+	DC	X'00'	
000079C8	E5D4C1C5 40404040			5390+	DC	HL1'3'	m5
000079D0	00007A4C			5391+	DC	CL8'VMAE'	instruction name
000079D4	00007A5C			5392+	DC	A(RE141+16)	address of v2 source
000079D8	00007A6C			5393+	DC	A(RE141+32)	address of v3 source
000079DC	00000010			5394+	DC	A(RE141+48)	address of v4 source
000079E0	00007A3C			5395+	DC	A(16)	result length
000079E8	00000000 00000000			5396+REA141	DC	A(RE141)	result address
000079F0	00000000 00000000			5397+	DS	FD	gap
000079F8	00000000 00000000			5398+V10141	DS	XL16	V1 output
00007A00	00000000 00000000						
				5399+	DS	FD	gap
				5400+*			
00007A08				5401+X141	DS	0F	
00007A08	E310 5010 0014		00000010	5402+	LGF	R1,V2ADDR	load v2 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007A0E	E761 0000 0806		00000000	5403+	VL	v22,0(R1)	use v22 to test decoder
00007A14	E310 5014 0014		00000014	5404+	LGF	R1,V3ADDR	load v3 source
00007A1A	E771 0000 0806		00000000	5405+	VL	v23,0(R1)	use v23 to test decoder
00007A20	E310 5018 0014		00000018	5406+	LGF	R1,V4ADDR	load v4 source
00007A26	E781 0000 0806		00000000	5407+	VL	v24,0(R1)	use v24 to test decoder
00007A2C	E766 7300 8FAE			5408+	VMAE	V22,V22,V23,V24,3	test instruction (dest is a source)
00007A32	E760 5030 080E		000079F0	5409+	VST	V22,V10141	save v1 output
00007A38	07FB			5410+	BR	R11	return
00007A3C				5411+RE141	DC	0F	xl16 expected result
00007A3C				5412+	DROP	R5	
00007A3C	FFFFFFFF FFFFFFFF			5413	DC	XL16'FFFFFFFFFFFFFFFF F6141E28323C491C'	result
00007A44	F6141E28 323C491C						
00007A4C	FF020304 05060750			5414	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00007A54	090A0B0C 0D0E0F7F						
00007A5C	00000000 0000000A			5415	DC	XL16'000000000000000A 0101010F0101010F'	v3
00007A64	0101010F 0101010F						
00007A6C	FFFFFFFF FFFFFFFF			5416	DC	XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFC'	v4
00007A74	FFFFFFFF FFFFFFFC						
				5417			
				5418	VRR_D	VMAE,3	
00007A80				5419+	DS	0FD	
00007A80		00007A80		5420+	USING	*,R5	base for test data and test routine
00007A80	00007AC8			5421+T142	DC	A(X142)	address of test routine
00007A84	008E			5422+	DC	H'142'	test number
00007A86	00			5423+	DC	X'00'	
00007A87	03			5424+	DC	HL1'3'	m5
00007A88	E5D4C1C5 40404040			5425+	DC	CL8'VMAE'	instruction name
00007A90	00007B0C			5426+	DC	A(RE142+16)	address of v2 source
00007A94	00007B1C			5427+	DC	A(RE142+32)	address of v3 source
00007A98	00007B2C			5428+	DC	A(RE142+48)	address of v4 source
00007A9C	00000010			5429+	DC	A(16)	result length
00007AA0	00007AFC			5430+REA142	DC	A(RE142)	result address
00007AA8	00000000 00000000			5431+	DS	FD	gap
00007AB0	00000000 00000000			5432+V10142	DS	XL16	V1 output
00007AB8	00000000 00000000						
00007AC0	00000000 00000000			5433+	DS	FD	gap
				5434+*			
00007AC8				5435+X142	DS	0F	
00007AC8	E310 5010 0014		00000010	5436+	LGF	R1,V2ADDR	load v2 source
00007ACE	E761 0000 0806		00000000	5437+	VL	v22,0(R1)	use v22 to test decoder
00007AD4	E310 5014 0014		00000014	5438+	LGF	R1,V3ADDR	load v3 source
00007ADA	E771 0000 0806		00000000	5439+	VL	v23,0(R1)	use v23 to test decoder
00007AE0	E310 5018 0014		00000018	5440+	LGF	R1,V4ADDR	load v4 source
00007AE6	E781 0000 0806		00000000	5441+	VL	v24,0(R1)	use v24 to test decoder
00007AEC	E766 7300 8FAE			5442+	VMAE	V22,V22,V23,V24,3	test instruction (dest is a source)
00007AF2	E760 5030 080E		00007AB0	5443+	VST	V22,V10142	save v1 output
00007AF8	07FB			5444+	BR	R11	return
00007AFC				5445+RE142	DC	0F	xl16 expected result
00007AFC				5446+	DROP	R5	
00007AFC	7709131E A8C3DFFE			5447	DC	XL16'7709131EA8C3DFFE F91C345060616771'	result
00007B04	F91C3450 60616771						
00007B0C	090A0B0C 0D0E0F7F			5448	DC	XL16'090A0B0C0D0E0F7F FF02030405060750'	v2
00007B14	FF020304 05060750						
00007B1C	0101010F 0101010F			5449	DC	XL16'0101010F0101010F 000000000000000A'	v3
00007B24	00000000 0000000A						
00007B2C	77000000 00000000			5450	DC	XL16'7700000000000000 F000000000000000'	v4

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007B34	F0000000 00000000			5451			
				5452 *	-----		
				5453 * VMA0	- Vector Multiply and Add Odd		
				5454 *	-----		
				5455 * Byte			
00007B40				5456	VRR_D VMA0,0		
00007B40		00007B40		5457+	DS 0FD		
00007B40	00007B88			5458+	USING *,R5	base for test data and test routine	
00007B44	008F			5459+T143	DC A(X143)	address of test routine	
00007B46	00			5460+	DC H'143'	test number	
00007B47	00			5461+	DC X'00'		
00007B48	E5D4C1D6 40404040			5462+	DC HL1'0'	m5	
00007B50	00007BCC			5463+	DC CL8'VMA0'	instruction name	
00007B54	00007BDC			5464+	DC A(RE143+16)	address of v2 source	
00007B58	00007BEC			5465+	DC A(RE143+32)	address of v3 source	
00007B5C	00000010			5466+	DC A(RE143+48)	address of v4 source	
00007B60	00007BBC			5467+	DC A(16)	result length	
00007B68	00000000 00000000			5468+REA143	DC A(RE143)	result address	
00007B70	00000000 00000000			5469+	DS FD	gap	
00007B78	00000000 00000000			5470+V10143	DS XL16	V1 output	
00007B80	00000000 00000000			5471+	DS FD	gap	
				5472+*			
00007B88				5473+X143	DS 0F		
00007B88	E310 5010 0014		00000010	5474+	LGF R1,V2ADDR	load v2 source	
00007B8E	E761 0000 0806		00000000	5475+	VL v22,0(R1)	use v22 to test decoder	
00007B94	E310 5014 0014		00000014	5476+	LGF R1,V3ADDR	load v3 source	
00007B9A	E771 0000 0806		00000000	5477+	VL v23,0(R1)	use v23 to test decoder	
00007BA0	E310 5018 0014		00000018	5478+	LGF R1,V4ADDR	load v4 source	
00007BA6	E781 0000 0806		00000000	5479+	VL v24,0(R1)	use v24 to test decoder	
00007BAC	E766 7000 8FAF			5480+	VMA0 V22,V22,V23,V24,0	test instruction (dest is a source)	
00007BB2	E760 5030 080E		00007B70	5481+	VST V22,V10143	save v1 output	
00007BB8	07FB			5482+	BR R11	return	
00007BBC				5483+RE143	DC 0F	xl16 expected result	
00007BBC				5484+	DROP R5		
00007BBC	00000000 00000271			5485	DC XL16'0000000000000271 00000C4000000024'	result	
00007BC4	00000C40 00000024						
00007BCC	FF000000 00000019			5486	DC XL16'FF00000000000019 00000038000000FA'	v2	
00007BD4	00000038 000000FA						
00007BDC	FF000000 00000019			5487	DC XL16'FF00000000000019 00000038000000FA'	v3	
00007BE4	00000038 000000FA						
00007BEC	00000000 00000000			5488	DC XL16'0000000000000000 0000000000000000'	v4	
00007BF4	00000000 00000000						
				5489			
00007C00				5490	VRR_D VMA0,0		
00007C00		00007C00		5491+	DS 0FD		
00007C00	00007C48			5492+	USING *,R5	base for test data and test routine	
00007C04	0090			5493+T144	DC A(X144)	address of test routine	
00007C06	00			5494+	DC H'144'	test number	
00007C07	00			5495+	DC X'00'		
00007C08	E5D4C1D6 40404040			5496+	DC HL1'0'	m5	
00007C10	00007C8C			5497+	DC CL8'VMA0'	instruction name	
00007C14	00007C9C			5498+	DC A(RE144+16)	address of v2 source	
00007C18	00007CAC			5499+	DC A(RE144+32)	address of v3 source	
				5500+	DC A(RE144+48)	address of v4 source	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007C1C	00000010			5501+	DC	A(16)	result length
00007C20	00007C7C			5502+REA144	DC	A(RE144)	result address
00007C28	00000000 00000000			5503+	DS	FD	gap
00007C30	00000000 00000000			5504+V10144	DS	XL16	V1 output
00007C38	00000000 00000000						
00007C40	00000000 00000000			5505+	DS	FD	gap
				5506+*			
00007C48				5507+X144	DS	0F	
00007C48	E310 5010 0014		00000010	5508+	LGF	R1,V2ADDR	load v2 source
00007C4E	E761 0000 0806		00000000	5509+	VL	v22,0(R1)	use v22 to test decoder
00007C54	E310 5014 0014		00000014	5510+	LGF	R1,V3ADDR	load v3 source
00007C5A	E771 0000 0806		00000000	5511+	VL	v23,0(R1)	use v23 to test decoder
00007C60	E310 5018 0014		00000018	5512+	LGF	R1,V4ADDR	load v4 source
00007C66	E781 0000 0806		00000000	5513+	VL	v24,0(R1)	use v24 to test decoder
00007C6C	E766 7000 8FAF			5514+	VMA0	V22,V22,V23,V24,0	test instruction (dest is a source)
00007C72	E760 5030 080E		00007C30	5515+	VST	V22,V10144	save v1 output
00007C78	07FB			5516+	BR	R11	return
00007C7C				5517+RE144	DC	0F	xl16 expected result
00007C7C				5518+	DROP	R5	
00007C7C	00020000 000006C0			5519	DC	XL16'00020000000006C0 00000C4300000026'	result
00007C84	00000C43 00000026						
00007C8C	FF0000FF 00000029			5520	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00007C94	00000038 000000FA						
00007C9C	FF000001 00000029			5521	DC	XL16'FF00000100000029 00000038000000FA'	v3
00007CA4	00000038 000000FA						
00007CAC	00020001 0000002F			5522	DC	XL16'000200010000002F 0000000300000002'	v4
00007CB4	00000003 00000002						
				5523			
				5524	VRR_D	VMA0,0	
00007CC0				5525+	DS	0FD	
00007CC0		00007CC0		5526+	USING	*,R5	base for test data and test routine
00007CC0	00007D08			5527+T145	DC	A(X145)	address of test routine
00007CC4	0091			5528+	DC	H'145'	test number
00007CC6	00			5529+	DC	X'00'	
00007CC7	00			5530+	DC	HL1'0'	m5
00007CC8	E5D4C1D6 40404040			5531+	DC	CL8'VMA0'	instruction name
00007CD0	00007D4C			5532+	DC	A(RE145+16)	address of v2 source
00007CD4	00007D5C			5533+	DC	A(RE145+32)	address of v3 source
00007CD8	00007D6C			5534+	DC	A(RE145+48)	address of v4 source
00007CDC	00000010			5535+	DC	A(16)	result length
00007CE0	00007D3C			5536+REA145	DC	A(RE145)	result address
00007CE8	00000000 00000000			5537+	DS	FD	gap
00007CF0	00000000 00000000			5538+V10145	DS	XL16	V1 output
00007CF8	00000000 00000000						
00007D00	00000000 00000000			5539+	DS	FD	gap
				5540+*			
00007D08				5541+X145	DS	0F	
00007D08	E310 5010 0014		00000010	5542+	LGF	R1,V2ADDR	load v2 source
00007D0E	E761 0000 0806		00000000	5543+	VL	v22,0(R1)	use v22 to test decoder
00007D14	E310 5014 0014		00000014	5544+	LGF	R1,V3ADDR	load v3 source
00007D1A	E771 0000 0806		00000000	5545+	VL	v23,0(R1)	use v23 to test decoder
00007D20	E310 5018 0014		00000018	5546+	LGF	R1,V4ADDR	load v4 source
00007D26	E781 0000 0806		00000000	5547+	VL	v24,0(R1)	use v24 to test decoder
00007D2C	E766 7000 8FAF			5548+	VMA0	V22,V22,V23,V24,0	test instruction (dest is a source)
00007D32	E760 5030 080E		00007CF0	5549+	VST	V22,V10145	save v1 output
00007D38	07FB			5550+	BR	R11	return

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007D3C				5551+RE145	DC	0F	xl16 expected result
00007D3C				5552+	DROP	R5	
00007D3C	FF060314 052A0748			5553	DC	XL16'FF060314052A0748 096E0B9C0DD21010'	result
00007D44	096E0B9C 0DD21010						
00007D4C	FF020304 05060708			5554	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007D54	090A0B0C 0D0E0F10						
00007D5C	FF020304 05060708			5555	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00007D64	090A0B0C 0D0E0F10						
00007D6C	FF020304 05060708			5556	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00007D74	090A0B0C 0D0E0F10						
				5557			
00007D80				5558	VRR_D	VMA0,0	
00007D80		00007D80		5559+	DS	0FD	
00007D80	00007DC8			5560+	USING	*,R5	base for test data and test routine
00007D84	0092			5561+T146	DC	A(X146)	address of test routine
00007D86	00			5562+	DC	H'146'	test number
00007D87	00			5563+	DC	X'00'	
00007D88	E5D4C1D6 40404040			5564+	DC	HL1'0'	m5
00007D90	00007E0C			5565+	DC	CL8'VMA0'	instruction name
00007D94	00007E1C			5566+	DC	A(RE146+16)	address of v2 source
00007D98	00007E2C			5567+	DC	A(RE146+32)	address of v3 source
00007D9C	00000010			5568+	DC	A(RE146+48)	address of v4 source
00007DA0	00007DFC			5569+	DC	A(16)	result length
00007DA8	00000000 00000000			5570+REA146	DC	A(RE146)	result address
00007DB0	00000000 00000000			5571+	DS	FD	gap
00007DB8	00000000 00000000			5572+V10146	DS	XL16	V1 output
00007DC0	00000000 00000000						
				5573+	DS	FD	gap
				5574+*			
00007DC8				5575+X146	DS	0F	
00007DC8	E310 5010 0014		00000010	5576+	LGF	R1,V2ADDR	load v2 source
00007DCE	E761 0000 0806		00000000	5577+	VL	v22,0(R1)	use v22 to test decoder
00007DD4	E310 5014 0014		00000014	5578+	LGF	R1,V3ADDR	load v3 source
00007DDA	E771 0000 0806		00000000	5579+	VL	v23,0(R1)	use v23 to test decoder
00007DE0	E310 5018 0014		00000018	5580+	LGF	R1,V4ADDR	load v4 source
00007DE6	E781 0000 0806		00000000	5581+	VL	v24,0(R1)	use v24 to test decoder
00007DEC	E766 7000 8FAF			5582+	VMA0	V22,V22,V23,V24,0	test instruction (dest is a source)
00007DF2	E760 5030 080E		00007DB0	5583+	VST	V22,V10146	save v1 output
00007DF8	07FB			5584+	BR	R11	return
00007DFC				5585+RE146	DC	0F	xl16 expected result
00007DFC				5586+	DROP	R5	
00007DFC	FF04030C 05180728			5587	DC	XL16'FF04030C05180728 093C0B540D700F90'	result
00007E04	093C0B54 0D700F90						
00007E0C	FF020304 05060708			5588	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007E14	090A0B0C 0D0E0F10						
00007E1C	FF010102 02030304			5589	DC	XL16'FF01010202030304 0405050606070708'	v3
00007E24	04050506 06070708						
00007E2C	FF020304 05060708			5590	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00007E34	090A0B0C 0D0E0F10						
				5591			
00007E40				5592	VRR_D	VMA0,0	
00007E40		00007E40		5593+	DS	0FD	
00007E40	00007E88			5594+	USING	*,R5	base for test data and test routine
00007E44	0093			5595+T147	DC	A(X147)	address of test routine
00007E46	00			5596+	DC	H'147'	test number
				5597+	DC	X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00007E47	00			5598+	DC	HL1'0'	m5
00007E48	E5D4C1D6 40404040			5599+	DC	CL8'VMA0'	instruction name
00007E50	00007ECC			5600+	DC	A(RE147+16)	address of v2 source
00007E54	00007EDC			5601+	DC	A(RE147+32)	address of v3 source
00007E58	00007EEC			5602+	DC	A(RE147+48)	address of v4 source
00007E5C	00000010			5603+	DC	A(16)	result length
00007E60	00007EBC			5604+REA147	DC	A(RE147)	result address
00007E68	00000000 00000000			5605+	DS	FD	gap
00007E70	00000000 00000000			5606+V10147	DS	XL16	V1 output
00007E78	00000000 00000000						
00007E80	00000000 00000000			5607+	DS	FD	gap
				5608+*			
00007E88				5609+X147	DS	0F	
00007E88	E310 5010 0014		00000010	5610+	LGF	R1,V2ADDR	load v2 source
00007E8E	E761 0000 0806		00000000	5611+	VL	v22,0(R1)	use v22 to test decoder
00007E94	E310 5014 0014		00000014	5612+	LGF	R1,V3ADDR	load v3 source
00007E9A	E771 0000 0806		00000000	5613+	VL	v23,0(R1)	use v23 to test decoder
00007EA0	E310 5018 0014		00000018	5614+	LGF	R1,V4ADDR	load v4 source
00007EA6	E781 0000 0806		00000000	5615+	VL	v24,0(R1)	use v24 to test decoder
00007EAC	E766 7000 8FAF			5616+	VMA0	V22,V22,V23,V24,0	test instruction (dest is a source)
00007EB2	E760 5030 080E		00007E70	5617+	VST	V22,V10147	save v1 output
00007EB8	07FB			5618+	BR	R11	return
00007EBC				5619+RE147	DC	0F	xl16 expected result
00007EBC				5620+	DROP	R5	
00007EBC	FF020304 05060710			5621	DC	XL16'FF02030405060710 09140B180D1C0F30'	result
00007EC4	09140B18 0D1C0F30						
00007ECC	FF020304 05060708			5622	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00007ED4	090A0B0C 0D0E0F10						
00007EDC	FF000000 00000001			5623	DC	XL16'FF00000000000001 0101010101010102'	v3
00007EE4	01010101 01010102						
00007EEC	FF020304 05060708			5624	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00007EF4	090A0B0C 0D0E0F10						
				5625			
				5626 * Halfword			
				5627	VRR_D	VMA0,1	
00007F00				5628+	DS	0FD	
00007F00		00007F00		5629+	USING	*,R5	base for test data and test routine
00007F00	00007F48			5630+T148	DC	A(X148)	address of test routine
00007F04	0094			5631+	DC	H'148'	test number
00007F06	00			5632+	DC	X'00'	
00007F07	01			5633+	DC	HL1'1'	m5
00007F08	E5D4C1D6 40404040			5634+	DC	CL8'VMA0'	instruction name
00007F10	00007F8C			5635+	DC	A(RE148+16)	address of v2 source
00007F14	00007F9C			5636+	DC	A(RE148+32)	address of v3 source
00007F18	00007FAC			5637+	DC	A(RE148+48)	address of v4 source
00007F1C	00000010			5638+	DC	A(16)	result length
00007F20	00007F7C			5639+REA148	DC	A(RE148)	result address
00007F28	00000000 00000000			5640+	DS	FD	gap
00007F30	00000000 00000000			5641+V10148	DS	XL16	V1 output
00007F38	00000000 00000000						
00007F40	00000000 00000000			5642+	DS	FD	gap
				5643+*			
00007F48				5644+X148	DS	0F	
00007F48	E310 5010 0014		00000010	5645+	LGF	R1,V2ADDR	load v2 source
00007F4E	E761 0000 0806		00000000	5646+	VL	v22,0(R1)	use v22 to test decoder
00007F54	E310 5014 0014		00000014	5647+	LGF	R1,V3ADDR	load v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00007F5A	E771 0000 0806		00000000	5648+	VL	v23,0(R1)	use v23 to test decoder	
00007F60	E310 5018 0014		00000018	5649+	LGF	R1,V4ADDR	load v4 source	
00007F66	E781 0000 0806		00000000	5650+	VL	v24,0(R1)	use v24 to test decoder	
00007F6C	E766 7100 8FAF			5651+	VMA0	V22,V22,V23,V24,1	test instruction (dest is a source)	
00007F72	E760 5030 080E		00007F30	5652+	VST	V22,V10148	save v1 output	
00007F78	07FB			5653+	BR	R11	return	
00007F7C				5654+RE148	DC	0F	xl16 expected result	
00007F7C				5655+	DROP	R5		
00007F7C	00000000 00000271			5656	DC	XL16'0000000000000271 00000C400000F424'	result	
00007F84	00000C40 0000F424							
00007F8C	FF000000 00000019			5657	DC	XL16'FF00000000000019 00000038000000FA'	v2	
00007F94	00000038 000000FA							
00007F9C	FF000000 00000019			5658	DC	XL16'FF00000000000019 00000038000000FA'	v3	
00007FA4	00000038 000000FA							
00007FAC	00000000 00000000			5659	DC	XL16'0000000000000000 0000000000000000'	v4	
00007FB4	00000000 00000000							
				5660				
00007FC0				5661	VRR_D	VMA0,1		
00007FC0		00007FC0		5662+	DS	0FD		
00007FC0	00008008			5663+	USING	*,R5	base for test data and test routine	
00007FC4	0095			5664+T149	DC	A(X149)	address of test routine	
00007FC6	00			5665+	DC	H'149'	test number	
00007FC7	01			5666+	DC	X'00'		
00007FC8	E5D4C1D6 40404040			5667+	DC	HL1'1'	m5	
00007FD0	0000804C			5668+	DC	CL8'VMA0'	instruction name	
00007FD4	0000805C			5669+	DC	A(RE149+16)	address of v2 source	
00007FD8	0000806C			5670+	DC	A(RE149+32)	address of v3 source	
00007FDC	00000010			5671+	DC	A(RE149+48)	address of v4 source	
00007FE0	0000803C			5672+	DC	A(16)	result length	
00007FE8	00000000 00000000			5673+REA149	DC	A(RE149)	result address	
00007FF0	00000000 00000000			5674+	DS	FD	gap	
00007FF8	00000000 00000000			5675+V10149	DS	XL16	V1 output	
00008000	00000000 00000000							
				5676+	DS	FD	gap	
				5677+*				
00008008				5678+X149	DS	0F		
00008008	E310 5010 0014		00000010	5679+	LGF	R1,V2ADDR	load v2 source	
0000800E	E761 0000 0806		00000000	5680+	VL	v22,0(R1)	use v22 to test decoder	
00008014	E310 5014 0014		00000014	5681+	LGF	R1,V3ADDR	load v3 source	
0000801A	E771 0000 0806		00000000	5682+	VL	v23,0(R1)	use v23 to test decoder	
00008020	E310 5018 0014		00000018	5683+	LGF	R1,V4ADDR	load v4 source	
00008026	E781 0000 0806		00000000	5684+	VL	v24,0(R1)	use v24 to test decoder	
0000802C	E766 7100 8FAF			5685+	VMA0	V22,V22,V23,V24,1	test instruction (dest is a source)	
00008032	E760 5030 080E		00007FF0	5686+	VST	V22,V10149	save v1 output	
00008038	07FB			5687+	BR	R11	return	
0000803C				5688+RE149	DC	0F	xl16 expected result	
0000803C				5689+	DROP	R5		
0000803C	00020100 000006C0			5690	DC	XL16'00020100000006C0 00000C430000F426'	result	
00008044	00000C43 0000F426							
0000804C	FF0000FF 00000029			5691	DC	XL16'FF0000FF00000029 00000038000000FA'	v2	
00008054	00000038 000000FA							
0000805C	FF000001 00000029			5692	DC	XL16'FF00000100000029 00000038000000FA'	v3	
00008064	00000038 000000FA							
0000806C	00020001 0000002F			5693	DC	XL16'000200010000002F 0000000300000002'	v4	
00008074	00000003 00000002							
				5694				

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				5695	VRR_D	VMA0,1	
00008080				5696+	DS	0FD	
00008080		00008080		5697+	USING	*,R5	base for test data and test routine
00008080	000080C8			5698+T150	DC	A(X150)	address of test routine
00008084	0096			5699+	DC	H'150'	test number
00008086	00			5700+	DC	X'00'	
00008087	01			5701+	DC	HL1'1'	m5
00008088	E5D4C1D6 40404040			5702+	DC	CL8'VMA0'	instruction name
00008090	0000810C			5703+	DC	A(RE150+16)	address of v2 source
00008094	0000811C			5704+	DC	A(RE150+32)	address of v3 source
00008098	0000812C			5705+	DC	A(RE150+48)	address of v4 source
0000809C	00000010			5706+	DC	A(16)	result length
000080A0	000080FC			5707+REA150	DC	A(RE150)	result address
000080A8	00000000 00000000			5708+	DS	FD	gap
000080B0	00000000 00000000			5709+V10150	DS	XL16	V1 output
000080B8	00000000 00000000						
000080C0	00000000 00000000			5710+	DS	FD	gap
				5711+*			
000080C8				5712+X150	DS	0F	
000080C8	E310 5010 0014		00000010	5713+	LGF	R1,V2ADDR	load v2 source
000080CE	E761 0000 0806		00000000	5714+	VL	v22,0(R1)	use v22 to test decoder
000080D4	E310 5014 0014		00000014	5715+	LGF	R1,V3ADDR	load v3 source
000080DA	E771 0000 0806		00000000	5716+	VL	v23,0(R1)	use v23 to test decoder
000080E0	E310 5018 0014		00000018	5717+	LGF	R1,V4ADDR	load v4 source
000080E6	E781 0000 0806		00000000	5718+	VL	v24,0(R1)	use v24 to test decoder
000080EC	E766 7100 8FAF			5719+	VMA0	V22,V22,V23,V24,1	test instruction (dest is a source)
000080F2	E760 5030 080E		000080B0	5720+	VST	V22,V10150	save v1 output
000080F8	07FB			5721+	BR	R11	return
000080FC				5722+RE150	DC	0F	xl16 expected result
000080FC				5723+	DROP	R5	
000080FC	FF0B1B14 05377748			5724	DC	XL16'FF0B1B1405377748 0984139C0DF0F010'	result
00008104	0984139C 0DF0F010						
0000810C	FF020304 05060708			5725	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00008114	090A0B0C 0D0E0F10						
0000811C	FF020304 05060708			5726	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
00008124	090A0B0C 0D0E0F10						
0000812C	FF020304 05060708			5727	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00008134	090A0B0C 0D0E0F10						
				5728			
				5729	VRR_D	VMA0,1	
00008140				5730+	DS	0FD	
00008140		00008140		5731+	USING	*,R5	base for test data and test routine
00008140	00008188			5732+T151	DC	A(X151)	address of test routine
00008144	0097			5733+	DC	H'151'	test number
00008146	00			5734+	DC	X'00'	
00008147	01			5735+	DC	HL1'1'	m5
00008148	E5D4C1D6 40404040			5736+	DC	CL8'VMA0'	instruction name
00008150	000081CC			5737+	DC	A(RE151+16)	address of v2 source
00008154	000081DC			5738+	DC	A(RE151+32)	address of v3 source
00008158	000081EC			5739+	DC	A(RE151+48)	address of v4 source
0000815C	00000010			5740+	DC	A(16)	result length
00008160	000081BC			5741+REA151	DC	A(RE151)	result address
00008168	00000000 00000000			5742+	DS	FD	gap
00008170	00000000 00000000			5743+V10151	DS	XL16	V1 output
00008178	00000000 00000000						
00008180	00000000 00000000			5744+	DS	FD	gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				5745+*			
00008188				5746+X151	DS	0F	
00008188	E310 5010 0014		00000010	5747+	LGF	R1,V2ADDR	load v2 source
0000818E	E761 0000 0806		00000000	5748+	VL	v22,0(R1)	use v22 to test decoder
00008194	E310 5014 0014		00000014	5749+	LGF	R1,V3ADDR	load v3 source
0000819A	E771 0000 0806		00000000	5750+	VL	v23,0(R1)	use v23 to test decoder
000081A0	E310 5018 0014		00000018	5751+	LGF	R1,V4ADDR	load v4 source
000081A6	E781 0000 0806		00000000	5752+	VL	v24,0(R1)	use v24 to test decoder
000081AC	E766 7100 8FAF			5753+	VMA0	V22,V22,V23,V24,1	test instruction (dest is a source)
000081B2	E760 5030 080E		00008170	5754+	VST	V22,V10151	save v1 output
000081B8	07FB			5755+	BR	R11	return
000081BC				5756+RE151	DC	0F	xl16 expected result
000081BC				5757+	DROP	R5	
000081BC	FF050D0C 051B3B28			5758	DC	XL16'FF050D0C051B3B28 094189540D77F790'	result
000081C4	09418954 0D77F790						
000081CC	FF020304 05060708			5759	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000081D4	090A0B0C 0D0E0F10						
000081DC	FF010102 02030304			5760	DC	XL16'FF01010202030304 0405050606070708'	v3
000081E4	04050506 06070708						
000081EC	FF020304 05060708			5761	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000081F4	090A0B0C 0D0E0F10						
				5762			
				5763	VRR_D	VMA0,1	
00008200				5764+	DS	0FD	
00008200		00008200		5765+	USING	*,R5	base for test data and test routine
00008200	00008248			5766+T152	DC	A(X152)	address of test routine
00008204	0098			5767+	DC	H'152'	test number
00008206	00			5768+	DC	X'00'	
00008207	01			5769+	DC	HL1'1'	m5
00008208	E5D4C1D6 40404040			5770+	DC	CL8'VMA0'	instruction name
00008210	0000828C			5771+	DC	A(RE152+16)	address of v2 source
00008214	0000829C			5772+	DC	A(RE152+32)	address of v3 source
00008218	000082AC			5773+	DC	A(RE152+48)	address of v4 source
0000821C	00000010			5774+	DC	A(16)	result length
00008220	0000827C			5775+REA152	DC	A(RE152)	result address
00008228	00000000 00000000			5776+	DS	FD	gap
00008230	00000000 00000000			5777+V10152	DS	XL16	V1 output
00008238	00000000 00000000						
00008240	00000000 00000000			5778+	DS	FD	gap
				5779+*			
00008248				5780+X152	DS	0F	
00008248	E310 5010 0014		00000010	5781+	LGF	R1,V2ADDR	load v2 source
0000824E	E761 0000 0806		00000000	5782+	VL	v22,0(R1)	use v22 to test decoder
00008254	E310 5014 0014		00000014	5783+	LGF	R1,V3ADDR	load v3 source
0000825A	E771 0000 0806		00000000	5784+	VL	v23,0(R1)	use v23 to test decoder
00008260	E310 5018 0014		00000018	5785+	LGF	R1,V4ADDR	load v4 source
00008266	E781 0000 0806		00000000	5786+	VL	v24,0(R1)	use v24 to test decoder
0000826C	E766 7100 8FAF			5787+	VMA0	V22,V22,V23,V24,1	test instruction (dest is a source)
00008272	E760 5030 080E		00008230	5788+	VST	V22,V10152	save v1 output
00008278	07FB			5789+	BR	R11	return
0000827C				5790+RE152	DC	0F	xl16 expected result
0000827C				5791+	DROP	R5	
0000827C	FF020304 05060E10			5792	DC	XL16'FF02030405060E10 091522180D1D3D30'	result
00008284	09152218 0D1D3D30						
0000828C	FF020304 05060708			5793	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00008294	090A0B0C 0D0E0F10						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000829C	FF000000 00000001			5794	DC	XL16'FF00000000000001 0101010101010102'	v3
000082A4	01010101 01010102						
000082AC	FF020304 05060708			5795	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000082B4	090A0B0C 0D0E0F10						
				5796			
				5797 * Word			
				5798	VRR_D	VMA0,2	
000082C0				5799+	DS	0FD	
000082C0		000082C0		5800+	USING	*,R5	base for test data and test routine
000082C0	00008308			5801+T153	DC	A(X153)	address of test routine
000082C4	0099			5802+	DC	H'153'	test number
000082C6	00			5803+	DC	X'00'	
000082C7	02			5804+	DC	HL1'2'	m5
000082C8	E5D4C1D6 40404040			5805+	DC	CL8'VMA0'	instruction name
000082D0	0000834C			5806+	DC	A(RE153+16)	address of v2 source
000082D4	0000835C			5807+	DC	A(RE153+32)	address of v3 source
000082D8	0000836C			5808+	DC	A(RE153+48)	address of v4 source
000082DC	00000010			5809+	DC	A(16)	result length
000082E0	0000833C			5810+REA153	DC	A(RE153)	result address
000082E8	00000000 00000000			5811+	DS	FD	gap
000082F0	00000000 00000000			5812+V10153	DS	XL16	V1 output
000082F8	00000000 00000000						
00008300	00000000 00000000			5813+	DS	FD	gap
				5814+*			
00008308				5815+X153	DS	0F	
00008308	E310 5010 0014		00000010	5816+	LGF	R1,V2ADDR	load v2 source
0000830E	E761 0000 0806		00000000	5817+	VL	v22,0(R1)	use v22 to test decoder
00008314	E310 5014 0014		00000014	5818+	LGF	R1,V3ADDR	load v3 source
0000831A	E771 0000 0806		00000000	5819+	VL	v23,0(R1)	use v23 to test decoder
00008320	E310 5018 0014		00000018	5820+	LGF	R1,V4ADDR	load v4 source
00008326	E781 0000 0806		00000000	5821+	VL	v24,0(R1)	use v24 to test decoder
0000832C	E766 7200 8FAF			5822+	VMA0	V22,V22,V23,V24,2	test instruction (dest is a source)
00008332	E760 5030 080E		000082F0	5823+	VST	V22,V10153	save v1 output
00008338	07FB			5824+	BR	R11	return
0000833C				5825+RE153	DC	0F	xl16 expected result
0000833C				5826+	DROP	R5	
0000833C	00000000 00000271			5827	DC	XL16'0000000000000271 000000000000F424'	result
00008344	00000000 0000F424						
0000834C	FF000000 00000019			5828	DC	XL16'FF00000000000019 00000038000000FA'	v2
00008354	00000038 000000FA						
0000835C	FF000000 00000019			5829	DC	XL16'FF00000000000019 00000038000000FA'	v3
00008364	00000038 000000FA						
0000836C	00000000 00000000			5830	DC	XL16'0000000000000000 0000000000000000'	v4
00008374	00000000 00000000						
				5831			
				5832	VRR_D	VMA0,2	
00008380				5833+	DS	0FD	
00008380		00008380		5834+	USING	*,R5	base for test data and test routine
00008380	000083C8			5835+T154	DC	A(X154)	address of test routine
00008384	009A			5836+	DC	H'154'	test number
00008386	00			5837+	DC	X'00'	
00008387	02			5838+	DC	HL1'2'	m5
00008388	E5D4C1D6 40404040			5839+	DC	CL8'VMA0'	instruction name
00008390	0000840C			5840+	DC	A(RE154+16)	address of v2 source
00008394	0000841C			5841+	DC	A(RE154+32)	address of v3 source
00008398	0000842C			5842+	DC	A(RE154+48)	address of v4 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
0000839C	00000010			5843+	DC	A(16)	result length
000083A0	000083FC			5844+REA154	DC	A(RE154)	result address
000083A8	00000000 00000000			5845+	DS	FD	gap
000083B0	00000000 00000000			5846+V10154	DS	XL16	V1 output
000083B8	00000000 00000000						
000083C0	00000000 00000000			5847+	DS	FD	gap
				5848+*			
000083C8				5849+X154	DS	0F	
000083C8	E310 5010 0014		00000010	5850+	LGF	R1,V2ADDR	load v2 source
000083CE	E761 0000 0806		00000000	5851+	VL	v22,0(R1)	use v22 to test decoder
000083D4	E310 5014 0014		00000014	5852+	LGF	R1,V3ADDR	load v3 source
000083DA	E771 0000 0806		00000000	5853+	VL	v23,0(R1)	use v23 to test decoder
000083E0	E310 5018 0014		00000018	5854+	LGF	R1,V4ADDR	load v4 source
000083E6	E781 0000 0806		00000000	5855+	VL	v24,0(R1)	use v24 to test decoder
000083EC	E766 7200 8FAF			5856+	VMA0	V22,V22,V23,V24,2	test instruction (dest is a source)
000083F2	E760 5030 080E		000083B0	5857+	VST	V22,V10154	save v1 output
000083F8	07FB			5858+	BR	R11	return
000083FC				5859+RE154	DC	0F	xl16 expected result
000083FC				5860+	DROP	R5	
000083FC	00020001 000006C0			5861	DC	XL16'000200010000006C0 0000000300000F426'	result
00008404	00000003 0000F426						
0000840C	FF0000FF 00000029			5862	DC	XL16'FF0000FF00000029 00000038000000FA'	v2
00008414	00000038 000000FA						
0000841C	FF000001 00000029			5863	DC	XL16'FF00000100000029 00000038000000FA'	v3
00008424	00000038 000000FA						
0000842C	00020001 0000002F			5864	DC	XL16'000200010000002F 0000000300000002'	v4
00008434	00000003 00000002						
				5865			
				5866	VRR_D	VMA0,2	
00008440				5867+	DS	0FD	
00008440		00008440		5868+	USING	*,R5	base for test data and test routine
00008440	00008488			5869+T155	DC	A(X155)	address of test routine
00008444	009B			5870+	DC	H'155'	test number
00008446	00			5871+	DC	X'00'	
00008447	02			5872+	DC	HL1'2'	m5
00008448	E5D4C1D6 40404040			5873+	DC	CL8'VMA0'	instruction name
00008450	000084CC			5874+	DC	A(RE155+16)	address of v2 source
00008454	000084DC			5875+	DC	A(RE155+32)	address of v3 source
00008458	000084EC			5876+	DC	A(RE155+48)	address of v4 source
0000845C	00000010			5877+	DC	A(16)	result length
00008460	000084BC			5878+REA155	DC	A(RE155)	result address
00008468	00000000 00000000			5879+	DS	FD	gap
00008470	00000000 00000000			5880+V10155	DS	XL16	V1 output
00008478	00000000 00000000						
00008480	00000000 00000000			5881+	DS	FD	gap
				5882+*			
00008488				5883+X155	DS	0F	
00008488	E310 5010 0014		00000010	5884+	LGF	R1,V2ADDR	load v2 source
0000848E	E761 0000 0806		00000000	5885+	VL	v22,0(R1)	use v22 to test decoder
00008494	E310 5014 0014		00000014	5886+	LGF	R1,V3ADDR	load v3 source
0000849A	E771 0000 0806		00000000	5887+	VL	v23,0(R1)	use v23 to test decoder
000084A0	E310 5018 0014		00000018	5888+	LGF	R1,V4ADDR	load v4 source
000084A6	E781 0000 0806		00000000	5889+	VL	v24,0(R1)	use v24 to test decoder
000084AC	E766 7200 8FAF			5890+	VMA0	V22,V22,V23,V24,2	test instruction (dest is a source)
000084B2	E760 5030 080E		00008470	5891+	VST	V22,V10155	save v1 output
000084B8	07FB			5892+	BR	R11	return

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000084BC				5893+RE155	DC	0F	xl16 expected result
000084BC				5894+	DROP	R5	
000084BC	FF1B3F6E A9977748			5895	DC	XL16'FF1B3F6EA9977748 09B4795953B0F010'	result
000084C4	09B47959 53B0F010						
000084CC	FF020304 05060708			5896	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
000084D4	090A0B0C 0D0E0F10						
000084DC	FF020304 05060708			5897	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v3
000084E4	090A0B0C 0D0E0F10						
000084EC	FF020304 05060708			5898	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000084F4	090A0B0C 0D0E0F10						
				5899			
00008500				5900	VRR_D	VMA0,2	
00008500		00008500		5901+	DS	0FD	
00008500	00008548			5902+	USING	*,R5	base for test data and test routine
00008504	009C			5903+T156	DC	A(X156)	address of test routine
00008506	00			5904+	DC	H'156'	test number
00008507	02			5905+	DC	X'00'	
00008508	E5D4C1D6 40404040			5906+	DC	HL1'2'	m5
00008510	0000858C			5907+	DC	CL8'VMA0'	instruction name
00008514	0000859C			5908+	DC	A(RE156+16)	address of v2 source
00008518	000085AC			5909+	DC	A(RE156+32)	address of v3 source
0000851C	00000010			5910+	DC	A(RE156+48)	address of v4 source
00008520	0000857C			5911+	DC	A(16)	result length
00008528	00000000 00000000			5912+REA156	DC	A(RE156)	result address
00008530	00000000 00000000			5913+	DS	FD	gap
00008538	00000000 00000000			5914+V10156	DS	XL16	V1 output
00008540	00000000 00000000						
				5915+	DS	FD	gap
				5916+*			
00008548				5917+X156	DS	0F	
00008548	E310 5010 0014		00000010	5918+	LGF	R1,V2ADDR	load v2 source
0000854E	E761 0000 0806		00000000	5919+	VL	v22,0(R1)	use v22 to test decoder
00008554	E310 5014 0014		00000014	5920+	LGF	R1,V3ADDR	load v3 source
0000855A	E771 0000 0806		00000000	5921+	VL	v23,0(R1)	use v23 to test decoder
00008560	E310 5018 0014		00000018	5922+	LGF	R1,V4ADDR	load v4 source
00008566	E781 0000 0806		00000000	5923+	VL	v24,0(R1)	use v24 to test decoder
0000856C	E766 7200 8FAF			5924+	VMA0	V22,V22,V23,V24,2	test instruction (dest is a source)
00008572	E760 5030 080E		00008530	5925+	VST	V22,V10156	save v1 output
00008578	07FB			5926+	BR	R11	return
0000857C				5927+RE156	DC	0F	xl16 expected result
0000857C				5928+	DROP	R5	
0000857C	FF0C1E33 504B3B28			5929	DC	XL16'FF0C1E33504B3B28 0958BB24A157F790'	result
00008584	0958BB24 A157F790						
0000858C	FF020304 05060708			5930	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00008594	090A0B0C 0D0E0F10						
0000859C	FF010102 02030304			5931	DC	XL16'FF01010202030304 0405050606070708'	v3
000085A4	04050506 06070708						
000085AC	FF020304 05060708			5932	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
000085B4	090A0B0C 0D0E0F10						
				5933			
000085C0				5934	VRR_D	VMA0,2	
000085C0		000085C0		5935+	DS	0FD	
000085C0	00008608			5936+	USING	*,R5	base for test data and test routine
000085C4	009D			5937+T157	DC	A(X157)	address of test routine
000085C6	00			5938+	DC	H'157'	test number
				5939+	DC	X'00'	

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
000085C7	02			5940+	DC	HL1'2'	m5
000085C8	E5D4C1D6 40404040			5941+	DC	CL8'VMA0'	instruction name
000085D0	0000864C			5942+	DC	A(RE157+16)	address of v2 source
000085D4	0000865C			5943+	DC	A(RE157+32)	address of v3 source
000085D8	0000866C			5944+	DC	A(RE157+48)	address of v4 source
000085DC	00000010			5945+	DC	A(16)	result length
000085E0	0000863C			5946+REA157	DC	A(RE157)	result address
000085E8	00000000 00000000			5947+	DS	FD	gap
000085F0	00000000 00000000			5948+V10157	DS	XL16	V1 output
000085F8	00000000 00000000						
00008600	00000000 00000000			5949+	DS	FD	gap
				5950+*			
00008608				5951+X157	DS	0F	
00008608	E310 5010 0014		00000010	5952+	LGF	R1,V2ADDR	load v2 source
0000860E	E761 0000 0806		00000000	5953+	VL	v22,0(R1)	use v22 to test decoder
00008614	E310 5014 0014		00000014	5954+	LGF	R1,V3ADDR	load v3 source
0000861A	E771 0000 0806		00000000	5955+	VL	v23,0(R1)	use v23 to test decoder
00008620	E310 5018 0014		00000018	5956+	LGF	R1,V4ADDR	load v4 source
00008626	E781 0000 0806		00000000	5957+	VL	v24,0(R1)	use v24 to test decoder
0000862C	E766 7200 8FAF			5958+	VMA0	V22,V22,V23,V24,2	test instruction (dest is a source)
00008632	E760 5030 080E		000085F0	5959+	VST	V22,V10157	save v1 output
00008638	07FB			5960+	BR	R11	return
0000863C				5961+RE157	DC	0F	xl16 expected result
0000863C				5962+	DROP	R5	
0000863C	FF020304 0A0C0E10			5963	DC	XL16'FF0203040A0C0E10 0917263654493D30'	result
00008644	09172636 54493D30						
0000864C	FF020304 05060708			5964	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v2
00008654	090A0B0C 0D0E0F10						
0000865C	FF000000 00000001			5965	DC	XL16'FF0000000000000001 0101010101010102'	v3
00008664	01010101 01010102						
0000866C	FF020304 05060708			5966	DC	XL16'FF02030405060708 090A0B0C0D0E0F10'	v4
00008674	090A0B0C 0D0E0F10						
				5967			
				5968 * Doubleword			
				5969	VRR_D	VMA0,3	
00008680				5970+	DS	0FD	
00008680		00008680		5971+	USING	*,R5	base for test data and test routine
00008680	000086C8			5972+T158	DC	A(X158)	address of test routine
00008684	009E			5973+	DC	H'158'	test number
00008686	00			5974+	DC	X'00'	
00008687	03			5975+	DC	HL1'3'	m5
00008688	E5D4C1D6 40404040			5976+	DC	CL8'VMA0'	instruction name
00008690	0000870C			5977+	DC	A(RE158+16)	address of v2 source
00008694	0000871C			5978+	DC	A(RE158+32)	address of v3 source
00008698	0000872C			5979+	DC	A(RE158+48)	address of v4 source
0000869C	00000010			5980+	DC	A(16)	result length
000086A0	000086FC			5981+REA158	DC	A(RE158)	result address
000086A8	00000000 00000000			5982+	DS	FD	gap
000086B0	00000000 00000000			5983+V10158	DS	XL16	V1 output
000086B8	00000000 00000000						
000086C0	00000000 00000000			5984+	DS	FD	gap
				5985+*			
000086C8				5986+X158	DS	0F	
000086C8	E310 5010 0014		00000010	5987+	LGF	R1,V2ADDR	load v2 source
000086CE	E761 0000 0806		00000000	5988+	VL	v22,0(R1)	use v22 to test decoder
000086D4	E310 5014 0014		00000014	5989+	LGF	R1,V3ADDR	load v3 source

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
000086DA	E771 0000 0806		00000000	5990+	VL	v23,0(R1)	use v23 to test decoder	
000086E0	E310 5018 0014		00000018	5991+	LGF	R1,V4ADDR	load v4 source	
000086E6	E781 0000 0806		00000000	5992+	VL	v24,0(R1)	use v24 to test decoder	
000086EC	E766 7300 8FAF			5993+	VMA0	V22,V22,V23,V24,3	test instruction (dest is a source)	
000086F2	E760 5030 080E		000086B0	5994+	VST	V22,V10158	save v1 output	
000086F8	07FB			5995+	BR	R11	return	
000086FC				5996+RE158	DC	0F	xl16 expected result	
000086FC				5997+	DROP	R5		
000086FC	00000000 00000C77			5998	DC	XL16'00000000000000C77	96789F9F4FEDCC24'	result
00008704	96789F9F 4FEDCC24							
0000870C	FFFFFFFF 00019000			5999	DC	XL16'FFFFFFFF00019000	00000038EEEEEEFA'	v2
00008714	00000038 EEEEEEEFA							
0000871C	FFFFFFFF 00019000			6000	DC	XL16'FFFFFFFF00019000	000000380EEEEEEFA'	v3
00008724	00000038 0EEEEEEFA							
0000872C	00000000 00000000			6001	DC	XL16'0000000000000000	0000000000000000'	v4
00008734	00000000 00000000							
				6002				
00008740				6003	VRR_D	VMA0,3		
00008740		00008740		6004+	DS	0FD		
00008740	00008788			6005+	USING	*,R5	base for test data and test routine	
00008744	009F			6006+T159	DC	A(X159)	address of test routine	
00008746	00			6007+	DC	H'159'	test number	
00008747	03			6008+	DC	X'00'		
00008748	E5D4C1D6 40404040			6009+	DC	HL1'3'	m5	
00008750	000087CC			6010+	DC	CL8'VMA0'	instruction name	
00008754	000087DC			6011+	DC	A(RE159+16)	address of v2 source	
00008758	000087EC			6012+	DC	A(RE159+32)	address of v3 source	
0000875C	00000010			6013+	DC	A(RE159+48)	address of v4 source	
00008760	000087BC			6014+	DC	A(16)	result length	
00008760	000087BC			6015+REA159	DC	A(RE159)	result address	
00008768	00000000 00000000			6016+	DS	FD	gap	
00008770	00000000 00000000			6017+V10159	DS	XL16	V1 output	
00008778	00000000 00000000							
00008780	00000000 00000000			6018+	DS	FD	gap	
				6019+*				
00008788				6020+X159	DS	0F		
00008788	E310 5010 0014		00000010	6021+	LGF	R1,V2ADDR	load v2 source	
0000878E	E761 0000 0806		00000000	6022+	VL	v22,0(R1)	use v22 to test decoder	
00008794	E310 5014 0014		00000014	6023+	LGF	R1,V3ADDR	load v3 source	
0000879A	E771 0000 0806		00000000	6024+	VL	v23,0(R1)	use v23 to test decoder	
000087A0	E310 5018 0014		00000018	6025+	LGF	R1,V4ADDR	load v4 source	
000087A6	E781 0000 0806		00000000	6026+	VL	v24,0(R1)	use v24 to test decoder	
000087AC	E766 7300 8FAF			6027+	VMA0	V22,V22,V23,V24,3	test instruction (dest is a source)	
000087B2	E760 5030 080E		00008770	6028+	VST	V22,V10159	save v1 output	
000087B8	07FB			6029+	BR	R11	return	
000087BC				6030+RE159	DC	0F	xl16 expected result	
000087BC				6031+	DROP	R5		
000087BC	7051B52F 8692B4F6			6032	DC	XL16'7051B52F8692B4F6	152B55D498D421F1'	result
000087C4	152B55D4 98D421F1							
000087CC	FF020304 05060750			6033	DC	XL16'FF02030405060750	090A0B0C0D0E0F7F'	v2
000087D4	090A0B0C 0D0E0F7F							
000087DC	01020304 05060750			6034	DC	XL16'0102030405060750	090A0B780D0E0F7F'	v3
000087E4	090A0B78 0D0E0F7F							
000087EC	70000000 00000000			6035	DC	XL16'7000000000000000	00000000000000F0'	v4
000087F4	00000000 000000F0							
				6036				

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
00008800				6037	VRR_D	VMA0,3	
00008800				6038+	DS	0FD	
00008800		00008800		6039+	USING	*,R5	base for test data and test routine
00008800	00008848			6040+T160	DC	A(X160)	address of test routine
00008804	00A0			6041+	DC	H'160'	test number
00008806	00			6042+	DC	X'00'	
00008807	03			6043+	DC	HL1'3'	m5
00008808	E5D4C1D6	40404040		6044+	DC	CL8'VMA0'	instruction name
00008810	0000888C			6045+	DC	A(RE160+16)	address of v2 source
00008814	0000889C			6046+	DC	A(RE160+32)	address of v3 source
00008818	000088AC			6047+	DC	A(RE160+48)	address of v4 source
0000881C	00000010			6048+	DC	A(16)	result length
00008820	0000887C			6049+REA160	DC	A(RE160)	result address
00008828	00000000	00000000		6050+	DS	FD	gap
00008830	00000000	00000000		6051+V10160	DS	XL16	V1 output
00008838	00000000	00000000					
00008840	00000000	00000000		6052+	DS	FD	gap
				6053+*			
00008848				6054+X160	DS	0F	
00008848	E310 5010 0014		00000010	6055+	LGF	R1,V2ADDR	load v2 source
0000884E	E761 0000 0806		00000000	6056+	VL	v22,0(R1)	use v22 to test decoder
00008854	E310 5014 0014		00000014	6057+	LGF	R1,V3ADDR	load v3 source
0000885A	E771 0000 0806		00000000	6058+	VL	v23,0(R1)	use v23 to test decoder
00008860	E310 5018 0014		00000018	6059+	LGF	R1,V4ADDR	load v4 source
00008866	E781 0000 0806		00000000	6060+	VL	v24,0(R1)	use v24 to test decoder
0000886C	E766 7300 8FAF			6061+	VMA0	V22,V22,V23,V24,3	test instruction (dest is a source)
00008872	E760 5030 080E		00008830	6062+	VST	V22,V10160	save v1 output
00008878	07FB			6063+	BR	R11	return
0000887C				6064+RE160	DC	0F	xl16 expected result
0000887C				6065+	DROP	R5	
0000887C	0024558D B838C862			6066	DC	XL16'0024558DB838C862 B47CD8D5FF5B4940'	result
00008884	B47CD8D5 FF5B4940						
0000888C	FF020304 05060750			6067	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00008894	090A0B0C 0D0E0F7F						
0000889C	00010102 02030328			6068	DC	XL16'0001010202030328 0405053C0607073F'	v3
000088A4	0405053C 0607073F						
000088AC	FFFFFFFF FFFFFFFF			6069	DC	XL16'FFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFF'	v4
000088B4	FFFFFFFF FFFFFFFF						
				6070			
000088C0				6071	VRR_D	VMA0,3	
000088C0				6072+	DS	0FD	
000088C0		000088C0		6073+	USING	*,R5	base for test data and test routine
000088C0	00008908			6074+T161	DC	A(X161)	address of test routine
000088C4	00A1			6075+	DC	H'161'	test number
000088C6	00			6076+	DC	X'00'	
000088C7	03			6077+	DC	HL1'3'	m5
000088C8	E5D4C1D6	40404040		6078+	DC	CL8'VMA0'	instruction name
000088D0	0000894C			6079+	DC	A(RE161+16)	address of v2 source
000088D4	0000895C			6080+	DC	A(RE161+32)	address of v3 source
000088D8	0000896C			6081+	DC	A(RE161+48)	address of v4 source
000088DC	00000010			6082+	DC	A(16)	result length
000088E0	0000893C			6083+REA161	DC	A(RE161)	result address
000088E8	00000000	00000000		6084+	DS	FD	gap
000088F0	00000000	00000000		6085+V10161	DS	XL16	V1 output
000088F8	00000000	00000000					
00008900	00000000	00000000		6086+	DS	FD	gap

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				6087+*			
00008908				6088+X161	DS	0F	
00008908	E310 5010 0014		00000010	6089+	LGF	R1,V2ADDR	load v2 source
0000890E	E761 0000 0806		00000000	6090+	VL	v22,0(R1)	use v22 to test decoder
00008914	E310 5014 0014		00000014	6091+	LGF	R1,V3ADDR	load v3 source
0000891A	E771 0000 0806		00000000	6092+	VL	v23,0(R1)	use v23 to test decoder
00008920	E310 5018 0014		00000018	6093+	LGF	R1,V4ADDR	load v4 source
00008926	E781 0000 0806		00000000	6094+	VL	v24,0(R1)	use v24 to test decoder
0000892C	E766 7300 8FAF			6095+	VMA0	V22,V22,V23,V24,3	test instruction (dest is a source)
00008932	E760 5030 080E		000088F0	6096+	VST	V22,V10161	save v1 output
00008938	07FB			6097+	BR	R11	return
0000893C				6098+RE161	DC	0F	xl16 expected result
0000893C				6099+	DROP	R5	
0000893C	7009131E A8C3DFFE			6100	DC	XL16'7009131EA8C3DFFE F91C345060616771'	result
00008944	F91C3450 60616771						
0000894C	FF020304 05060750			6101	DC	XL16'FF02030405060750 090A0B0C0D0E0F7F'	v2
00008954	090A0B0C 0D0E0F7F						
0000895C	00000000 0000000A			6102	DC	XL16'000000000000000A 0101010F0101010F'	v3
00008964	0101010F 0101010F						
0000896C	70000000 00000000			6103	DC	XL16'7000000000000000 F000000000000000'	v4
00008974	F0000000 00000000						
				6104			
				6105	VRR_D	VMA0,3	
00008980				6106+	DS	0FD	
00008980		00008980		6107+	USING	*,R5	base for test data and test routine
00008980	000089C8			6108+T162	DC	A(X162)	address of test routine
00008984	00A2			6109+	DC	H'162'	test number
00008986	00			6110+	DC	X'00'	
00008987	03			6111+	DC	HL1'3'	m5
00008988	E5D4C1D6 40404040			6112+	DC	CL8'VMA0'	instruction name
00008990	00008A0C			6113+	DC	A(RE162+16)	address of v2 source
00008994	00008A1C			6114+	DC	A(RE162+32)	address of v3 source
00008998	00008A2C			6115+	DC	A(RE162+48)	address of v4 source
0000899C	00000010			6116+	DC	A(16)	result length
000089A0	000089FC			6117+REA162	DC	A(RE162)	result address
000089A8	00000000 00000000			6118+	DS	FD	gap
000089B0	00000000 00000000			6119+V10162	DS	XL16	V1 output
000089B8	00000000 00000000						
000089C0	00000000 00000000			6120+	DS	FD	gap
				6121+*			
000089C8				6122+X162	DS	0F	
000089C8	E310 5010 0014		00000010	6123+	LGF	R1,V2ADDR	load v2 source
000089CE	E761 0000 0806		00000000	6124+	VL	v22,0(R1)	use v22 to test decoder
000089D4	E310 5014 0014		00000014	6125+	LGF	R1,V3ADDR	load v3 source
000089DA	E771 0000 0806		00000000	6126+	VL	v23,0(R1)	use v23 to test decoder
000089E0	E310 5018 0014		00000018	6127+	LGF	R1,V4ADDR	load v4 source
000089E6	E781 0000 0806		00000000	6128+	VL	v24,0(R1)	use v24 to test decoder
000089EC	E766 7300 8FAF			6129+	VMA0	V22,V22,V23,V24,3	test instruction (dest is a source)
000089F2	E760 5030 080E		000089B0	6130+	VST	V22,V10162	save v1 output
000089F8	07FB			6131+	BR	R11	return
000089FC				6132+RE162	DC	0F	xl16 expected result
000089FC				6133+	DROP	R5	
000089FC	FFFFFFFF FFFFFFFF			6134	DC	XL16'FFFFFFFFFFFFFFFF F6141E28323C491C'	result
00008A04	F6141E28 323C491C						
00008A0C	090A0B0C 0D0E0F7F			6135	DC	XL16'090A0B0C0D0E0F7F FF02030405060750'	v2
00008A14	FF020304 05060750						

LOC	OBJECT CODE	ADDR1	ADDR2	STMT				
00008A1C	0101010F	0101010F		6136	DC	XL16'	0101010F0101010F 0000000000000000A'	v3
00008A24	00000000	0000000A						
00008A2C	FFFFFFFF	FFFFFFFF		6137	DC	XL16'	FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFC'	v4
00008A34	FFFFFFFF	FFFFFFFC						
				6138				
				6139				
00008A3C	00000000			6140	DC	F'0'	END OF TABLE	
00008A40	00000000			6141	DC	F'0'		
				6142	*			
				6143	*	table of pointers to individual load test		
				6144	*			
00008A44				6145	E7TESTS	DS	0F	
				6146		PTTABLE		
00008A44				6147+	TTABLE	DS	0F	
00008A44	000010C0			6148+	DC	A(T1)		
00008A48	00001180			6149+	DC	A(T2)		
00008A4C	00001240			6150+	DC	A(T3)		
00008A50	00001300			6151+	DC	A(T4)		
00008A54	000013C0			6152+	DC	A(T5)		
00008A58	00001480			6153+	DC	A(T6)		
00008A5C	00001540			6154+	DC	A(T7)		
00008A60	00001600			6155+	DC	A(T8)		
00008A64	000016C0			6156+	DC	A(T9)		
00008A68	00001780			6157+	DC	A(T10)		
00008A6C	00001840			6158+	DC	A(T11)		
00008A70	00001900			6159+	DC	A(T12)		
00008A74	000019C0			6160+	DC	A(T13)		
00008A78	00001A80			6161+	DC	A(T14)		
00008A7C	00001B40			6162+	DC	A(T15)		
00008A80	00001C00			6163+	DC	A(T16)		
00008A84	00001CC0			6164+	DC	A(T17)		
00008A88	00001D80			6165+	DC	A(T18)		
00008A8C	00001E40			6166+	DC	A(T19)		
00008A90	00001F00			6167+	DC	A(T20)		
00008A94	00001FC0			6168+	DC	A(T21)		
00008A98	00002080			6169+	DC	A(T22)		
00008A9C	00002140			6170+	DC	A(T23)		
00008AA0	00002200			6171+	DC	A(T24)		
00008AA4	000022C0			6172+	DC	A(T25)		
00008AA8	00002380			6173+	DC	A(T26)		
00008AAC	00002440			6174+	DC	A(T27)		
00008AB0	00002500			6175+	DC	A(T28)		
00008AB4	000025C0			6176+	DC	A(T29)		
00008AB8	00002680			6177+	DC	A(T30)		
00008ABC	00002740			6178+	DC	A(T31)		
00008AC0	00002800			6179+	DC	A(T32)		
00008AC4	000028C0			6180+	DC	A(T33)		
00008AC8	00002980			6181+	DC	A(T34)		
00008ACC	00002A40			6182+	DC	A(T35)		
00008AD0	00002B00			6183+	DC	A(T36)		
00008AD4	00002BC0			6184+	DC	A(T37)		
00008AD8	00002C80			6185+	DC	A(T38)		
00008ADC	00002D40			6186+	DC	A(T39)		
00008AE0	00002E00			6187+	DC	A(T40)		
00008AE4	00002EC0			6188+	DC	A(T41)		
00008AE8	00002F80			6189+	DC	A(T42)		

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00008AEC	00003040			6190+	DC	A(T43)
00008AF0	00003100			6191+	DC	A(T44)
00008AF4	000031C0			6192+	DC	A(T45)
00008AF8	00003280			6193+	DC	A(T46)
00008AFC	00003340			6194+	DC	A(T47)
00008B00	00003400			6195+	DC	A(T48)
00008B04	000034C0			6196+	DC	A(T49)
00008B08	00003580			6197+	DC	A(T50)
00008B0C	00003640			6198+	DC	A(T51)
00008B10	00003700			6199+	DC	A(T52)
00008B14	000037C0			6200+	DC	A(T53)
00008B18	00003880			6201+	DC	A(T54)
00008B1C	00003940			6202+	DC	A(T55)
00008B20	00003A00			6203+	DC	A(T56)
00008B24	00003AC0			6204+	DC	A(T57)
00008B28	00003B80			6205+	DC	A(T58)
00008B2C	00003C40			6206+	DC	A(T59)
00008B30	00003D00			6207+	DC	A(T60)
00008B34	00003DC0			6208+	DC	A(T61)
00008B38	00003E80			6209+	DC	A(T62)
00008B3C	00003F40			6210+	DC	A(T63)
00008B40	00004000			6211+	DC	A(T64)
00008B44	000040C0			6212+	DC	A(T65)
00008B48	00004180			6213+	DC	A(T66)
00008B4C	00004240			6214+	DC	A(T67)
00008B50	00004300			6215+	DC	A(T68)
00008B54	000043C0			6216+	DC	A(T69)
00008B58	00004480			6217+	DC	A(T70)
00008B5C	00004540			6218+	DC	A(T71)
00008B60	00004600			6219+	DC	A(T72)
00008B64	000046C0			6220+	DC	A(T73)
00008B68	00004780			6221+	DC	A(T74)
00008B6C	00004840			6222+	DC	A(T75)
00008B70	00004900			6223+	DC	A(T76)
00008B74	000049C0			6224+	DC	A(T77)
00008B78	00004A80			6225+	DC	A(T78)
00008B7C	00004B40			6226+	DC	A(T79)
00008B80	00004C00			6227+	DC	A(T80)
00008B84	00004CC0			6228+	DC	A(T81)
00008B88	00004D80			6229+	DC	A(T82)
00008B8C	00004E40			6230+	DC	A(T83)
00008B90	00004F00			6231+	DC	A(T84)
00008B94	00004FC0			6232+	DC	A(T85)
00008B98	00005080			6233+	DC	A(T86)
00008B9C	00005140			6234+	DC	A(T87)
00008BA0	00005200			6235+	DC	A(T88)
00008BA4	000052C0			6236+	DC	A(T89)
00008BA8	00005380			6237+	DC	A(T90)
00008BAC	00005440			6238+	DC	A(T91)
00008BB0	00005500			6239+	DC	A(T92)
00008BB4	000055C0			6240+	DC	A(T93)
00008BB8	00005680			6241+	DC	A(T94)
00008BBC	00005740			6242+	DC	A(T95)
00008BC0	00005800			6243+	DC	A(T96)
00008BC4	000058C0			6244+	DC	A(T97)
00008BC8	00005980			6245+	DC	A(T98)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT		
00008BCC	00005A40			6246+	DC	A(T99)
00008BD0	00005B00			6247+	DC	A(T100)
00008BD4	00005BC0			6248+	DC	A(T101)
00008BD8	00005C80			6249+	DC	A(T102)
00008BDC	00005D40			6250+	DC	A(T103)
00008BE0	00005E00			6251+	DC	A(T104)
00008BE4	00005EC0			6252+	DC	A(T105)
00008BE8	00005F80			6253+	DC	A(T106)
00008BEC	00006040			6254+	DC	A(T107)
00008BF0	00006100			6255+	DC	A(T108)
00008BF4	000061C0			6256+	DC	A(T109)
00008BF8	00006280			6257+	DC	A(T110)
00008BFC	00006340			6258+	DC	A(T111)
00008C00	00006400			6259+	DC	A(T112)
00008C04	000064C0			6260+	DC	A(T113)
00008C08	00006580			6261+	DC	A(T114)
00008C0C	00006640			6262+	DC	A(T115)
00008C10	00006700			6263+	DC	A(T116)
00008C14	000067C0			6264+	DC	A(T117)
00008C18	00006880			6265+	DC	A(T118)
00008C1C	00006940			6266+	DC	A(T119)
00008C20	00006A00			6267+	DC	A(T120)
00008C24	00006AC0			6268+	DC	A(T121)
00008C28	00006B80			6269+	DC	A(T122)
00008C2C	00006C40			6270+	DC	A(T123)
00008C30	00006D00			6271+	DC	A(T124)
00008C34	00006DC0			6272+	DC	A(T125)
00008C38	00006E80			6273+	DC	A(T126)
00008C3C	00006F40			6274+	DC	A(T127)
00008C40	00007000			6275+	DC	A(T128)
00008C44	000070C0			6276+	DC	A(T129)
00008C48	00007180			6277+	DC	A(T130)
00008C4C	00007240			6278+	DC	A(T131)
00008C50	00007300			6279+	DC	A(T132)
00008C54	000073C0			6280+	DC	A(T133)
00008C58	00007480			6281+	DC	A(T134)
00008C5C	00007540			6282+	DC	A(T135)
00008C60	00007600			6283+	DC	A(T136)
00008C64	000076C0			6284+	DC	A(T137)
00008C68	00007780			6285+	DC	A(T138)
00008C6C	00007840			6286+	DC	A(T139)
00008C70	00007900			6287+	DC	A(T140)
00008C74	000079C0			6288+	DC	A(T141)
00008C78	00007A80			6289+	DC	A(T142)
00008C7C	00007B40			6290+	DC	A(T143)
00008C80	00007C00			6291+	DC	A(T144)
00008C84	00007CC0			6292+	DC	A(T145)
00008C88	00007D80			6293+	DC	A(T146)
00008C8C	00007E40			6294+	DC	A(T147)
00008C90	00007F00			6295+	DC	A(T148)
00008C94	00007FC0			6296+	DC	A(T149)
00008C98	00008080			6297+	DC	A(T150)
00008C9C	00008140			6298+	DC	A(T151)
00008CA0	00008200			6299+	DC	A(T152)
00008CA4	000082C0			6300+	DC	A(T153)
00008CA8	00008380			6301+	DC	A(T154)

LOC	OBJECT CODE	ADDR1	ADDR2	STMT			
				6317	*****		
				6318	* Register equates		
				6319	*****		
		00000000	00000001	6321	R0	EQU	0
		00000001	00000001	6322	R1	EQU	1
		00000002	00000001	6323	R2	EQU	2
		00000003	00000001	6324	R3	EQU	3
		00000004	00000001	6325	R4	EQU	4
		00000005	00000001	6326	R5	EQU	5
		00000006	00000001	6327	R6	EQU	6
		00000007	00000001	6328	R7	EQU	7
		00000008	00000001	6329	R8	EQU	8
		00000009	00000001	6330	R9	EQU	9
		0000000A	00000001	6331	R10	EQU	10
		0000000B	00000001	6332	R11	EQU	11
		0000000C	00000001	6333	R12	EQU	12
		0000000D	00000001	6334	R13	EQU	13
		0000000E	00000001	6335	R14	EQU	14
		0000000F	00000001	6336	R15	EQU	15
				6338	*****		
				6339	* Register equates		
				6340	*****		
		00000000	00000001	6342	V0	EQU	0
		00000001	00000001	6343	V1	EQU	1
		00000002	00000001	6344	V2	EQU	2
		00000003	00000001	6345	V3	EQU	3
		00000004	00000001	6346	V4	EQU	4
		00000005	00000001	6347	V5	EQU	5
		00000006	00000001	6348	V6	EQU	6
		00000007	00000001	6349	V7	EQU	7
		00000008	00000001	6350	V8	EQU	8
		00000009	00000001	6351	V9	EQU	9
		0000000A	00000001	6352	V10	EQU	10
		0000000B	00000001	6353	V11	EQU	11
		0000000C	00000001	6354	V12	EQU	12
		0000000D	00000001	6355	V13	EQU	13
		0000000E	00000001	6356	V14	EQU	14
		0000000F	00000001	6357	V15	EQU	15
		00000010	00000001	6358	V16	EQU	16
		00000011	00000001	6359	V17	EQU	17
		00000012	00000001	6360	V18	EQU	18
		00000013	00000001	6361	V19	EQU	19
		00000014	00000001	6362	V20	EQU	20
		00000015	00000001	6363	V21	EQU	21

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES														
BEGIN	I	00000200	2	165	131	161	162	163											
CTLR0	F	0000053C	4	394	175	176	177	178											
DECNUM	C	00001073	16	446	308	310	316	318											
E7TEST	4	00000000	72	460	257														
E7TESTS	F	00008A44	4	6145	250														
EDIT	X	00001047	18	441	309	317													
ENDTEST	U	000003CE	1	294	255														
EOJ	I	00000520	4	384	210	243	297												
EOJPSW	D	00000510	8	382	384														
FAILCONT	U	000003BE	1	284															
FAILED	F	00001000	4	423	286	295													
FAILMSG	U	000003BA	1	278	268														
FAILPSW	D	00000528	8	386	388														
FAILTEST	I	00000538	4	388	298														
FB0001	F	00000280	8	194	198	199	201												
FB0002	F	00000330	8	227	231	232	234												
IMAGE	1	00000000	36060	0															
K	U	00000400	1	407	408	409	410												
K64	U	00010000	1	409															
M5	U	00000007	1	464	315														
MB	U	00100000	1	410															
MSG	I	00000458	4	344	209	242	327												
MSGCMD	C	000004A6	9	374	357	358													
MSGMSG	C	000004AF	95	375	351	372	349												
MSGMVC	I	000004A0	6	372	355														
MSGOK	I	0000046E	2	353	350														
MSGRET	I	0000048E	4	368	361	364													
MSGSAVE	F	00000494	4	371	347	368													
NEXTE7	U	00000384	1	252	271	289													
OPNAME	C	00000008	8	466	313														
PAGE	U	00001000	1	408															
PRT3	C	0000105D	18	444	309	310	311	317	318	319									
PRTLNE	C	00001008	16	429	436	326													
PRTLNG	U	0000003F	1	436	325														
PRTM5	C	00001044	2	434	319														
PRTNAME	C	00001033	8	432	313														
PRTNUM	C	00001018	3	430	311														
R0	U	00000000	1	6321	125	175	178	198	200	201	202	207	231	233	234	235	240		
R1	U	00000001	1	6322	259	260	285	286	324	325	328	344	347	349	351	353	368		
					208	241	266	267	295	296	326	358	372	602	603	604	605		
					606	607	636	637	638	639	640	641	670	671	672	673	674		
					675	704	705	706	707	708	709	738	739	740	741	742	743		
					773	774	775	776	777	778	807	808	809	810	811	812	841		
					842	843	844	845	846	875	876	877	878	879	880	909	910		
					911	912	913	914	944	945	946	947	948	949	978	979	980		
					981	982	983	1012	1013	1014	1015	1016	1017	1046	1047	1048	1049		
					1050	1051	1080	1081	1082	1083	1084	1085	1115	1116	1117	1118	1119		
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					1287	1288	1289	1290	1291	1320	1321	1322	1323	1324	1325	1354	1355		
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ASMA Ver. 0.7.0 zvector-e7-10-multiplyAdd										01 Apr 2026 11:14:03 Page 137							
SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
					1803	1804	1805	1806	1835	1836	1837	1838	1839	1840	1869	1870	1871
					1872	1873	1874	1903	1904	1905	1906	1907	1908	1938	1939	1940	1941
					1942	1943	1972	1973	1974	1975	1976	1977	2006	2007	2008	2009	2010
					2011	2040	2041	2042	2043	2044	2045	2074	2075	2076	2077	2078	2079
					2109	2110	2111	2112	2113	2114	2143	2144	2145	2146	2147	2148	2177
					2178	2179	2180	2181	2182	2211	2212	2213	2214	2215	2216	2246	2247
					2248	2249	2250	2251	2280	2281	2282	2283	2284	2285	2314	2315	2316
					2317	2318	2319	2348	2349	2350	2351	2352	2353	2386	2387	2388	2389
					2390	2391	2420	2421	2422	2423	2424	2425	2454	2455	2456	2457	2458
					2459	2488	2489	2490	2491	2492	2493	2522	2523	2524	2525	2526	2527
					2557	2558	2559	2560	2561	2562	2591	2592	2593	2594	2595	2596	2625
					2626	2627	2628	2629	2630	2659	2660	2661	2662	2663	2664	2693	2694
					2695	2696	2697	2698	2728	2729	2730	2731	2732	2733	2762	2763	2764
					2765	2766	2767	2796	2797	2798	2799	2800	2801	2830	2831	2832	2833
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					2904	2933	2934	2935	2936	2937	2938	2967	2968	2969	2970	2971	2972
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					3071	3072	3073	3074	3075	3104	3105	3106	3107	3108	3109	3138	3139
					3140	3141	3142	3143	3172	3173	3174	3175	3176	3177	3206	3207	3208
					3209	3210	3211	3240	3241	3242	3243	3244	3245	3274	3275	3276	3277
					3278	3279	3308	3309	3310	3311	3312	3313	3342	3343	3344	3345	3346
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					3447	3448	3449	3450	3451	3452	3481	3482	3483	3484	3485	3486	3515
					3516	3517	3518	3519	3520	3549	3550	3551	3552	3553	3554	3584	3585
					3586	3587	3588	3589	3618	3619	3620	3621	3622	3623	3652	3653	3654
					3655	3656	3657	3686	3687	3688	3689	3690	3691	3720	3721	3722	3723
					3724	3725	3755	3756	3757	3758	3759	3760	3789	3790	3791	3792	3793
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					3891	3892	3893	3894	3895	3896	3926	3927	3928	3929	3930	3931	3960
					3961	3962	3963	3964	3965	3994	3995	3996	3997	3998	3999	4028	4029
					4030	4031	4032	4033	4062	4063	4064	4065	4066	4067	4100	4101	4102
					4103	4104	4105	4134	4135	4136	4137	4138	4139	4168	4169	4170	4171
					4172	4173	4202	4203	4204	4205	4206	4207	4236	4237	4238	4239	4240
					4241	4271	4272	4273	4274	4275	4276	4305	4306	4307	4308	4309	4310
					4339	4340	4341	4342	4343	4344	4373	4374	4375	4376	4377	4378	4407
					4408	4409	4410	4411	4412	4442	4443	4444	4445	4446	4447	4476	4477
					4478	4479	4480	4481	4510	4511	4512	4513	4514	4515	4544	4545	4546
					4547	4548	4549	4578	4579	4580	4581	4582	4583	4613	4614	4615	4616
					4617	4618	4647	4648	4649	4650	4651	4652	4681	4682	4683	4684	4685
					4686	4715	4716	4717	4718	4719	4720	4749	4750	4751	4752	4753	4754
					4787	4788	4789	4790	4791	4792	4821	4822	4823	4824	4825	4826	4855
					4856	4857	4858	4859	4860	4889	4890	4891	4892	4893	4894	4923	4924
					4925	4926	4927	4928	4958	4959	4960	4961	4962	4963	4992	4993	4994
					4995	4996	4997	5026	5027	5028	5029	5030	5031	5060	5061	5062	5063
					5064	5065	5094	5095	5096	5097	5098	5099	5129	5130	5131	5132	5133
					5134	5163	5164	5165	5166	5167	5168	5197	5198	5199	5200	5201	5202
					5231	5232	5233	5234	5235	5236	5265	5266	5267	5268	5269	5270	5300
					5301	5302	5303	5304	5305	5334	5335	5336	5337	5338	5339	5368	5369
					5370	5371	5372	5373	5402	5403	5404	5405	5406	5407	5436	5437	5438
					5439	5440	5441	5474	5475	5476	5477	5478	5479	5508	5509	5510	5511
					5512	5513	5542	5543	5544	5545	5546	5547	5576	5577	5578	5579	5580
					5581	5610	5611	5612	5613	5614	5615	5645	5646	5647	5648	5649	5650
					5679	5680	5681	5682	5683	5684	5713	5714	5715	5716	5717	5718	5747
					5748	5749	5750	5751	5752	5781	5782	5783	5784	5785	5786	5816	5817
					5818	5819	5820	5821	5850	5851	5852	5853	5854	5855	5884	5885	5886
					5887	5888	5889	5918	5919	5920	5921	5922	5923	5952	5953	5954	5955

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
RE10	F	000017FC	4	918	899 900 901 903
RE100	F	00005B7C	4	4003	3984 3985 3986 3988
RE101	F	00005C3C	4	4037	4018 4019 4020 4022
RE102	F	00005CFC	4	4071	4052 4053 4054 4056
RE103	F	00005DBC	4	4109	4090 4091 4092 4094
RE104	F	00005E7C	4	4143	4124 4125 4126 4128
RE105	F	00005F3C	4	4177	4158 4159 4160 4162
RE106	F	00005FFC	4	4211	4192 4193 4194 4196
RE107	F	000060BC	4	4245	4226 4227 4228 4230
RE108	F	0000617C	4	4280	4261 4262 4263 4265
RE109	F	0000623C	4	4314	4295 4296 4297 4299
RE11	F	000018BC	4	953	934 935 936 938
RE110	F	000062FC	4	4348	4329 4330 4331 4333
RE111	F	000063BC	4	4382	4363 4364 4365 4367
RE112	F	0000647C	4	4416	4397 4398 4399 4401
RE113	F	0000653C	4	4451	4432 4433 4434 4436
RE114	F	000065FC	4	4485	4466 4467 4468 4470
RE115	F	000066BC	4	4519	4500 4501 4502 4504
RE116	F	0000677C	4	4553	4534 4535 4536 4538
RE117	F	0000683C	4	4587	4568 4569 4570 4572
RE118	F	000068FC	4	4622	4603 4604 4605 4607
RE119	F	000069BC	4	4656	4637 4638 4639 4641
RE12	F	0000197C	4	987	968 969 970 972
RE120	F	00006A7C	4	4690	4671 4672 4673 4675
RE121	F	00006B3C	4	4724	4705 4706 4707 4709
RE122	F	00006BFC	4	4758	4739 4740 4741 4743
RE123	F	00006CBC	4	4796	4777 4778 4779 4781
RE124	F	00006D7C	4	4830	4811 4812 4813 4815
RE125	F	00006E3C	4	4864	4845 4846 4847 4849
RE126	F	00006EFC	4	4898	4879 4880 4881 4883
RE127	F	00006FBC	4	4932	4913 4914 4915 4917
RE128	F	0000707C	4	4967	4948 4949 4950 4952
RE129	F	0000713C	4	5001	4982 4983 4984 4986
RE13	F	00001A3C	4	1021	1002 1003 1004 1006
RE130	F	000071FC	4	5035	5016 5017 5018 5020
RE131	F	000072BC	4	5069	5050 5051 5052 5054
RE132	F	0000737C	4	5103	5084 5085 5086 5088
RE133	F	0000743C	4	5138	5119 5120 5121 5123
RE134	F	000074FC	4	5172	5153 5154 5155 5157
RE135	F	000075BC	4	5206	5187 5188 5189 5191
RE136	F	0000767C	4	5240	5221 5222 5223 5225
RE137	F	0000773C	4	5274	5255 5256 5257 5259
RE138	F	000077FC	4	5309	5290 5291 5292 5294
RE139	F	000078BC	4	5343	5324 5325 5326 5328
RE14	F	00001AFC	4	1055	1036 1037 1038 1040
RE140	F	0000797C	4	5377	5358 5359 5360 5362
RE141	F	00007A3C	4	5411	5392 5393 5394 5396
RE142	F	00007AFC	4	5445	5426 5427 5428 5430
RE143	F	00007BBC	4	5483	5464 5465 5466 5468
RE144	F	00007C7C	4	5517	5498 5499 5500 5502
RE145	F	00007D3C	4	5551	5532 5533 5534 5536
RE146	F	00007DFC	4	5585	5566 5567 5568 5570
RE147	F	00007EBC	4	5619	5600 5601 5602 5604
RE148	F	00007F7C	4	5654	5635 5636 5637 5639
RE149	F	0000803C	4	5688	5669 5670 5671 5673
RE15	F	00001BBC	4	1089	1070 1071 1072 1074

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
RE150	F	000080FC	4	5722	5703	5704	5705	5707	
RE151	F	000081BC	4	5756	5737	5738	5739	5741	
RE152	F	0000827C	4	5790	5771	5772	5773	5775	
RE153	F	0000833C	4	5825	5806	5807	5808	5810	
RE154	F	000083FC	4	5859	5840	5841	5842	5844	
RE155	F	000084BC	4	5893	5874	5875	5876	5878	
RE156	F	0000857C	4	5927	5908	5909	5910	5912	
RE157	F	0000863C	4	5961	5942	5943	5944	5946	
RE158	F	000086FC	4	5996	5977	5978	5979	5981	
RE159	F	000087BC	4	6030	6011	6012	6013	6015	
RE16	F	00001C7C	4	1124	1105	1106	1107	1109	
RE160	F	0000887C	4	6064	6045	6046	6047	6049	
RE161	F	0000893C	4	6098	6079	6080	6081	6083	
RE162	F	000089FC	4	6132	6113	6114	6115	6117	
RE17	F	00001D3C	4	1158	1139	1140	1141	1143	
RE18	F	00001DFC	4	1192	1173	1174	1175	1177	
RE19	F	00001EBC	4	1226	1207	1208	1209	1211	
RE2	F	000011FC	4	645	626	627	628	630	
RE20	F	00001F7C	4	1261	1242	1243	1244	1246	
RE21	F	0000203C	4	1295	1276	1277	1278	1280	
RE22	F	000020FC	4	1329	1310	1311	1312	1314	
RE23	F	000021BC	4	1363	1344	1345	1346	1348	
RE24	F	0000227C	4	1397	1378	1379	1380	1382	
RE25	F	0000233C	4	1431	1412	1413	1414	1416	
RE26	F	000023FC	4	1465	1446	1447	1448	1450	
RE27	F	000024BC	4	1499	1480	1481	1482	1484	
RE28	F	0000257C	4	1533	1514	1515	1516	1518	
RE29	F	0000263C	4	1567	1548	1549	1550	1552	
RE3	F	000012BC	4	679	660	661	662	664	
RE30	F	000026FC	4	1605	1586	1587	1588	1590	
RE31	F	000027BC	4	1639	1620	1621	1622	1624	
RE32	F	0000287C	4	1673	1654	1655	1656	1658	
RE33	F	0000293C	4	1707	1688	1689	1690	1692	
RE34	F	000029FC	4	1741	1722	1723	1724	1726	
RE35	F	00002ABC	4	1776	1757	1758	1759	1761	
RE36	F	00002B7C	4	1810	1791	1792	1793	1795	
RE37	F	00002C3C	4	1844	1825	1826	1827	1829	
RE38	F	00002CFC	4	1878	1859	1860	1861	1863	
RE39	F	00002DBC	4	1912	1893	1894	1895	1897	
RE4	F	0000137C	4	713	694	695	696	698	
RE40	F	00002E7C	4	1947	1928	1929	1930	1932	
RE41	F	00002F3C	4	1981	1962	1963	1964	1966	
RE42	F	00002FFC	4	2015	1996	1997	1998	2000	
RE43	F	000030BC	4	2049	2030	2031	2032	2034	
RE44	F	0000317C	4	2083	2064	2065	2066	2068	
RE45	F	0000323C	4	2118	2099	2100	2101	2103	
RE46	F	000032FC	4	2152	2133	2134	2135	2137	
RE47	F	000033BC	4	2186	2167	2168	2169	2171	
RE48	F	0000347C	4	2220	2201	2202	2203	2205	
RE49	F	0000353C	4	2255	2236	2237	2238	2240	
RE5	F	0000143C	4	747	728	729	730	732	
RE50	F	000035FC	4	2289	2270	2271	2272	2274	
RE51	F	000036BC	4	2323	2304	2305	2306	2308	
RE52	F	0000377C	4	2357	2338	2339	2340	2342	
RE53	F	0000383C	4	2395	2376	2377	2378	2380	
RE54	F	000038FC	4	2429	2410	2411	2412	2414	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
RE55	F	000039BC	4	2463	2444 2445 2446 2448
RE56	F	00003A7C	4	2497	2478 2479 2480 2482
RE57	F	00003B3C	4	2531	2512 2513 2514 2516
RE58	F	00003BFC	4	2566	2547 2548 2549 2551
RE59	F	00003CBC	4	2600	2581 2582 2583 2585
RE6	F	000014FC	4	782	763 764 765 767
RE60	F	00003D7C	4	2634	2615 2616 2617 2619
RE61	F	00003E3C	4	2668	2649 2650 2651 2653
RE62	F	00003EFC	4	2702	2683 2684 2685 2687
RE63	F	00003FBC	4	2737	2718 2719 2720 2722
RE64	F	0000407C	4	2771	2752 2753 2754 2756
RE65	F	0000413C	4	2805	2786 2787 2788 2790
RE66	F	000041FC	4	2839	2820 2821 2822 2824
RE67	F	000042BC	4	2873	2854 2855 2856 2858
RE68	F	0000437C	4	2908	2889 2890 2891 2893
RE69	F	0000443C	4	2942	2923 2924 2925 2927
RE7	F	000015BC	4	816	797 798 799 801
RE70	F	000044FC	4	2976	2957 2958 2959 2961
RE71	F	000045BC	4	3010	2991 2992 2993 2995
RE72	F	0000467C	4	3045	3026 3027 3028 3030
RE73	F	0000473C	4	3079	3060 3061 3062 3064
RE74	F	000047FC	4	3113	3094 3095 3096 3098
RE75	F	000048BC	4	3147	3128 3129 3130 3132
RE76	F	0000497C	4	3181	3162 3163 3164 3166
RE77	F	00004A3C	4	3215	3196 3197 3198 3200
RE78	F	00004AFC	4	3249	3230 3231 3232 3234
RE79	F	00004BBC	4	3283	3264 3265 3266 3268
RE8	F	0000167C	4	850	831 832 833 835
RE80	F	00004C7C	4	3317	3298 3299 3300 3302
RE81	F	00004D3C	4	3351	3332 3333 3334 3336
RE82	F	00004DFC	4	3385	3366 3367 3368 3370
RE83	F	00004EBC	4	3422	3403 3404 3405 3407
RE84	F	00004F7C	4	3456	3437 3438 3439 3441
RE85	F	0000503C	4	3490	3471 3472 3473 3475
RE86	F	000050FC	4	3524	3505 3506 3507 3509
RE87	F	000051BC	4	3558	3539 3540 3541 3543
RE88	F	0000527C	4	3593	3574 3575 3576 3578
RE89	F	0000533C	4	3627	3608 3609 3610 3612
RE9	F	0000173C	4	884	865 866 867 869
RE90	F	000053FC	4	3661	3642 3643 3644 3646
RE91	F	000054BC	4	3695	3676 3677 3678 3680
RE92	F	0000557C	4	3729	3710 3711 3712 3714
RE93	F	0000563C	4	3764	3745 3746 3747 3749
RE94	F	000056FC	4	3798	3779 3780 3781 3783
RE95	F	000057BC	4	3832	3813 3814 3815 3817
RE96	F	0000587C	4	3866	3847 3848 3849 3851
RE97	F	0000593C	4	3900	3881 3882 3883 3885
RE98	F	000059FC	4	3935	3916 3917 3918 3920
RE99	F	00005ABC	4	3969	3950 3951 3952 3954
REA1	A	000010E0	4	596	
REA10	A	000017A0	4	903	
REA100	A	00005B20	4	3988	
REA101	A	00005BE0	4	4022	
REA102	A	00005CA0	4	4056	
REA103	A	00005D60	4	4094	
REA104	A	00005E20	4	4128	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
REA105	A	00005EE0	4	4162	
REA106	A	00005FA0	4	4196	
REA107	A	00006060	4	4230	
REA108	A	00006120	4	4265	
REA109	A	000061E0	4	4299	
REA11	A	00001860	4	938	
REA110	A	000062A0	4	4333	
REA111	A	00006360	4	4367	
REA112	A	00006420	4	4401	
REA113	A	000064E0	4	4436	
REA114	A	000065A0	4	4470	
REA115	A	00006660	4	4504	
REA116	A	00006720	4	4538	
REA117	A	000067E0	4	4572	
REA118	A	000068A0	4	4607	
REA119	A	00006960	4	4641	
REA12	A	00001920	4	972	
REA120	A	00006A20	4	4675	
REA121	A	00006AE0	4	4709	
REA122	A	00006BA0	4	4743	
REA123	A	00006C60	4	4781	
REA124	A	00006D20	4	4815	
REA125	A	00006DE0	4	4849	
REA126	A	00006EA0	4	4883	
REA127	A	00006F60	4	4917	
REA128	A	00007020	4	4952	
REA129	A	000070E0	4	4986	
REA13	A	000019E0	4	1006	
REA130	A	000071A0	4	5020	
REA131	A	00007260	4	5054	
REA132	A	00007320	4	5088	
REA133	A	000073E0	4	5123	
REA134	A	000074A0	4	5157	
REA135	A	00007560	4	5191	
REA136	A	00007620	4	5225	
REA137	A	000076E0	4	5259	
REA138	A	000077A0	4	5294	
REA139	A	00007860	4	5328	
REA14	A	00001AA0	4	1040	
REA140	A	00007920	4	5362	
REA141	A	000079E0	4	5396	
REA142	A	00007AA0	4	5430	
REA143	A	00007B60	4	5468	
REA144	A	00007C20	4	5502	
REA145	A	00007CE0	4	5536	
REA146	A	00007DA0	4	5570	
REA147	A	00007E60	4	5604	
REA148	A	00007F20	4	5639	
REA149	A	00007FE0	4	5673	
REA15	A	00001B60	4	1074	
REA150	A	000080A0	4	5707	
REA151	A	00008160	4	5741	
REA152	A	00008220	4	5775	
REA153	A	000082E0	4	5810	
REA154	A	000083A0	4	5844	
REA155	A	00008460	4	5878	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
REA156	A	00008520	4	5912	
REA157	A	000085E0	4	5946	
REA158	A	000086A0	4	5981	
REA159	A	00008760	4	6015	
REA16	A	00001C20	4	1109	
REA160	A	00008820	4	6049	
REA161	A	000088E0	4	6083	
REA162	A	000089A0	4	6117	
REA17	A	00001CE0	4	1143	
REA18	A	00001DA0	4	1177	
REA19	A	00001E60	4	1211	
REA2	A	000011A0	4	630	
REA20	A	00001F20	4	1246	
REA21	A	00001FE0	4	1280	
REA22	A	000020A0	4	1314	
REA23	A	00002160	4	1348	
REA24	A	00002220	4	1382	
REA25	A	000022E0	4	1416	
REA26	A	000023A0	4	1450	
REA27	A	00002460	4	1484	
REA28	A	00002520	4	1518	
REA29	A	000025E0	4	1552	
REA3	A	00001260	4	664	
REA30	A	000026A0	4	1590	
REA31	A	00002760	4	1624	
REA32	A	00002820	4	1658	
REA33	A	000028E0	4	1692	
REA34	A	000029A0	4	1726	
REA35	A	00002A60	4	1761	
REA36	A	00002B20	4	1795	
REA37	A	00002BE0	4	1829	
REA38	A	00002CA0	4	1863	
REA39	A	00002D60	4	1897	
REA4	A	00001320	4	698	
REA40	A	00002E20	4	1932	
REA41	A	00002EE0	4	1966	
REA42	A	00002FA0	4	2000	
REA43	A	00003060	4	2034	
REA44	A	00003120	4	2068	
REA45	A	000031E0	4	2103	
REA46	A	000032A0	4	2137	
REA47	A	00003360	4	2171	
REA48	A	00003420	4	2205	
REA49	A	000034E0	4	2240	
REA5	A	000013E0	4	732	
REA50	A	000035A0	4	2274	
REA51	A	00003660	4	2308	
REA52	A	00003720	4	2342	
REA53	A	000037E0	4	2380	
REA54	A	000038A0	4	2414	
REA55	A	00003960	4	2448	
REA56	A	00003A20	4	2482	
REA57	A	00003AE0	4	2516	
REA58	A	00003BA0	4	2551	
REA59	A	00003C60	4	2585	
REA6	A	000014A0	4	767	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
REA60	A	00003D20	4	2619		
REA61	A	00003DE0	4	2653		
REA62	A	00003EA0	4	2687		
REA63	A	00003F60	4	2722		
REA64	A	00004020	4	2756		
REA65	A	000040E0	4	2790		
REA66	A	000041A0	4	2824		
REA67	A	00004260	4	2858		
REA68	A	00004320	4	2893		
REA69	A	000043E0	4	2927		
REA7	A	00001560	4	801		
REA70	A	000044A0	4	2961		
REA71	A	00004560	4	2995		
REA72	A	00004620	4	3030		
REA73	A	000046E0	4	3064		
REA74	A	000047A0	4	3098		
REA75	A	00004860	4	3132		
REA76	A	00004920	4	3166		
REA77	A	000049E0	4	3200		
REA78	A	00004AA0	4	3234		
REA79	A	00004B60	4	3268		
REA8	A	00001620	4	835		
REA80	A	00004C20	4	3302		
REA81	A	00004CE0	4	3336		
REA82	A	00004DA0	4	3370		
REA83	A	00004E60	4	3407		
REA84	A	00004F20	4	3441		
REA85	A	00004FE0	4	3475		
REA86	A	000050A0	4	3509		
REA87	A	00005160	4	3543		
REA88	A	00005220	4	3578		
REA89	A	000052E0	4	3612		
REA9	A	000016E0	4	869		
REA90	A	000053A0	4	3646		
REA91	A	00005460	4	3680		
REA92	A	00005520	4	3714		
REA93	A	000055E0	4	3749		
REA94	A	000056A0	4	3783		
REA95	A	00005760	4	3817		
REA96	A	00005820	4	3851		
REA97	A	000058E0	4	3885		
REA98	A	000059A0	4	3920		
REA99	A	00005A60	4	3954		
READDR	A	00000020	4	471	266	
REG2LOW	U	000000DD	1	413		
REG2PATT	U	AABBCCDD	1	412		
RELEN	A	0000001C	4	470		
RPTDWSAV	D	00000448	8	337	324	328
RPTERROR	I	000003DC	4	304	279	
RPTSAVE	F	00000440	4	334	304	331
RPTSVR5	F	00000444	4	335	305	330
SKL0001	U	0000004E	1	191	207	
SKL0002	U	0000004E	1	224	240	
SKT0001	C	0000022A	20	188	191	208
SKT0002	C	000002D4	20	221	224	241
SVOLDPSW	U	00000140	0	127		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T1	A	000010C0	4	587	6148
T10	A	00001780	4	894	6157
T100	A	00005B00	4	3979	6247
T101	A	00005BC0	4	4013	6248
T102	A	00005C80	4	4047	6249
T103	A	00005D40	4	4085	6250
T104	A	00005E00	4	4119	6251
T105	A	00005EC0	4	4153	6252
T106	A	00005F80	4	4187	6253
T107	A	00006040	4	4221	6254
T108	A	00006100	4	4256	6255
T109	A	000061C0	4	4290	6256
T11	A	00001840	4	929	6158
T110	A	00006280	4	4324	6257
T111	A	00006340	4	4358	6258
T112	A	00006400	4	4392	6259
T113	A	000064C0	4	4427	6260
T114	A	00006580	4	4461	6261
T115	A	00006640	4	4495	6262
T116	A	00006700	4	4529	6263
T117	A	000067C0	4	4563	6264
T118	A	00006880	4	4598	6265
T119	A	00006940	4	4632	6266
T12	A	00001900	4	963	6159
T120	A	00006A00	4	4666	6267
T121	A	00006AC0	4	4700	6268
T122	A	00006B80	4	4734	6269
T123	A	00006C40	4	4772	6270
T124	A	00006D00	4	4806	6271
T125	A	00006DC0	4	4840	6272
T126	A	00006E80	4	4874	6273
T127	A	00006F40	4	4908	6274
T128	A	00007000	4	4943	6275
T129	A	000070C0	4	4977	6276
T13	A	000019C0	4	997	6160
T130	A	00007180	4	5011	6277
T131	A	00007240	4	5045	6278
T132	A	00007300	4	5079	6279
T133	A	000073C0	4	5114	6280
T134	A	00007480	4	5148	6281
T135	A	00007540	4	5182	6282
T136	A	00007600	4	5216	6283
T137	A	000076C0	4	5250	6284
T138	A	00007780	4	5285	6285
T139	A	00007840	4	5319	6286
T14	A	00001A80	4	1031	6161
T140	A	00007900	4	5353	6287
T141	A	000079C0	4	5387	6288
T142	A	00007A80	4	5421	6289
T143	A	00007B40	4	5459	6290
T144	A	00007C00	4	5493	6291
T145	A	00007CC0	4	5527	6292
T146	A	00007D80	4	5561	6293
T147	A	00007E40	4	5595	6294
T148	A	00007F00	4	5630	6295
T149	A	00007FC0	4	5664	6296

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T15	A	00001B40	4	1065	6162
T150	A	00008080	4	5698	6297
T151	A	00008140	4	5732	6298
T152	A	00008200	4	5766	6299
T153	A	000082C0	4	5801	6300
T154	A	00008380	4	5835	6301
T155	A	00008440	4	5869	6302
T156	A	00008500	4	5903	6303
T157	A	000085C0	4	5937	6304
T158	A	00008680	4	5972	6305
T159	A	00008740	4	6006	6306
T16	A	00001C00	4	1100	6163
T160	A	00008800	4	6040	6307
T161	A	000088C0	4	6074	6308
T162	A	00008980	4	6108	6309
T17	A	00001CC0	4	1134	6164
T18	A	00001D80	4	1168	6165
T19	A	00001E40	4	1202	6166
T2	A	00001180	4	621	6149
T20	A	00001F00	4	1237	6167
T21	A	00001FC0	4	1271	6168
T22	A	00002080	4	1305	6169
T23	A	00002140	4	1339	6170
T24	A	00002200	4	1373	6171
T25	A	000022C0	4	1407	6172
T26	A	00002380	4	1441	6173
T27	A	00002440	4	1475	6174
T28	A	00002500	4	1509	6175
T29	A	000025C0	4	1543	6176
T3	A	00001240	4	655	6150
T30	A	00002680	4	1581	6177
T31	A	00002740	4	1615	6178
T32	A	00002800	4	1649	6179
T33	A	000028C0	4	1683	6180
T34	A	00002980	4	1717	6181
T35	A	00002A40	4	1752	6182
T36	A	00002B00	4	1786	6183
T37	A	00002BC0	4	1820	6184
T38	A	00002C80	4	1854	6185
T39	A	00002D40	4	1888	6186
T4	A	00001300	4	689	6151
T40	A	00002E00	4	1923	6187
T41	A	00002EC0	4	1957	6188
T42	A	00002F80	4	1991	6189
T43	A	00003040	4	2025	6190
T44	A	00003100	4	2059	6191
T45	A	000031C0	4	2094	6192
T46	A	00003280	4	2128	6193
T47	A	00003340	4	2162	6194
T48	A	00003400	4	2196	6195
T49	A	000034C0	4	2231	6196
T5	A	000013C0	4	723	6152
T50	A	00003580	4	2265	6197
T51	A	00003640	4	2299	6198
T52	A	00003700	4	2333	6199
T53	A	000037C0	4	2371	6200

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T54	A	00003880	4	2405	6201
T55	A	00003940	4	2439	6202
T56	A	00003A00	4	2473	6203
T57	A	00003AC0	4	2507	6204
T58	A	00003B80	4	2542	6205
T59	A	00003C40	4	2576	6206
T6	A	00001480	4	758	6153
T60	A	00003D00	4	2610	6207
T61	A	00003DC0	4	2644	6208
T62	A	00003E80	4	2678	6209
T63	A	00003F40	4	2713	6210
T64	A	00004000	4	2747	6211
T65	A	000040C0	4	2781	6212
T66	A	00004180	4	2815	6213
T67	A	00004240	4	2849	6214
T68	A	00004300	4	2884	6215
T69	A	000043C0	4	2918	6216
T7	A	00001540	4	792	6154
T70	A	00004480	4	2952	6217
T71	A	00004540	4	2986	6218
T72	A	00004600	4	3021	6219
T73	A	000046C0	4	3055	6220
T74	A	00004780	4	3089	6221
T75	A	00004840	4	3123	6222
T76	A	00004900	4	3157	6223
T77	A	000049C0	4	3191	6224
T78	A	00004A80	4	3225	6225
T79	A	00004B40	4	3259	6226
T8	A	00001600	4	826	6155
T80	A	00004C00	4	3293	6227
T81	A	00004CC0	4	3327	6228
T82	A	00004D80	4	3361	6229
T83	A	00004E40	4	3398	6230
T84	A	00004F00	4	3432	6231
T85	A	00004FC0	4	3466	6232
T86	A	00005080	4	3500	6233
T87	A	00005140	4	3534	6234
T88	A	00005200	4	3569	6235
T89	A	000052C0	4	3603	6236
T9	A	000016C0	4	860	6156
T90	A	00005380	4	3637	6237
T91	A	00005440	4	3671	6238
T92	A	00005500	4	3705	6239
T93	A	000055C0	4	3740	6240
T94	A	00005680	4	3774	6241
T95	A	00005740	4	3808	6242
T96	A	00005800	4	3842	6243
T97	A	000058C0	4	3876	6244
T98	A	00005980	4	3911	6245
T99	A	00005A40	4	3945	6246
TESTING	F	00001004	4	424	260
TNUM	H	00000004	2	462	259
TSUB	A	00000000	4	461	263
TTABLE	F	00008A44	4	6147	
V0	U	00000000	1	6342	
V1	U	00000001	1	6343	262

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V10	U	0000000A	1	6352	
V11	U	0000000B	1	6353	
V12	U	0000000C	1	6354	
V13	U	0000000D	1	6355	
V14	U	0000000E	1	6356	
V15	U	0000000F	1	6357	
V16	U	00000010	1	6358	
V17	U	00000011	1	6359	
V18	U	00000012	1	6360	
V19	U	00000013	1	6361	
V1FUDGE	X	00001094	16	453	262
V101	X	000010F0	16	598	609
V1010	X	000017B0	16	905	916
V10100	X	00005B30	16	3990	4001
V10101	X	00005BF0	16	4024	4035
V10102	X	00005CB0	16	4058	4069
V10103	X	00005D70	16	4096	4107
V10104	X	00005E30	16	4130	4141
V10105	X	00005EF0	16	4164	4175
V10106	X	00005FB0	16	4198	4209
V10107	X	00006070	16	4232	4243
V10108	X	00006130	16	4267	4278
V10109	X	000061F0	16	4301	4312
V1011	X	00001870	16	940	951
V10110	X	000062B0	16	4335	4346
V10111	X	00006370	16	4369	4380
V10112	X	00006430	16	4403	4414
V10113	X	000064F0	16	4438	4449
V10114	X	000065B0	16	4472	4483
V10115	X	00006670	16	4506	4517
V10116	X	00006730	16	4540	4551
V10117	X	000067F0	16	4574	4585
V10118	X	000068B0	16	4609	4620
V10119	X	00006970	16	4643	4654
V1012	X	00001930	16	974	985
V10120	X	00006A30	16	4677	4688
V10121	X	00006AF0	16	4711	4722
V10122	X	00006BB0	16	4745	4756
V10123	X	00006C70	16	4783	4794
V10124	X	00006D30	16	4817	4828
V10125	X	00006DF0	16	4851	4862
V10126	X	00006EB0	16	4885	4896
V10127	X	00006F70	16	4919	4930
V10128	X	00007030	16	4954	4965
V10129	X	000070F0	16	4988	4999
V1013	X	000019F0	16	1008	1019
V10130	X	000071B0	16	5022	5033
V10131	X	00007270	16	5056	5067
V10132	X	00007330	16	5090	5101
V10133	X	000073F0	16	5125	5136
V10134	X	000074B0	16	5159	5170
V10135	X	00007570	16	5193	5204
V10136	X	00007630	16	5227	5238
V10137	X	000076F0	16	5261	5272
V10138	X	000077B0	16	5296	5307
V10139	X	00007870	16	5330	5341

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1014	X	00001AB0	16	1042	1053
V10140	X	00007930	16	5364	5375
V10141	X	000079F0	16	5398	5409
V10142	X	00007AB0	16	5432	5443
V10143	X	00007B70	16	5470	5481
V10144	X	00007C30	16	5504	5515
V10145	X	00007CF0	16	5538	5549
V10146	X	00007DB0	16	5572	5583
V10147	X	00007E70	16	5606	5617
V10148	X	00007F30	16	5641	5652
V10149	X	00007FF0	16	5675	5686
V1015	X	00001B70	16	1076	1087
V10150	X	000080B0	16	5709	5720
V10151	X	00008170	16	5743	5754
V10152	X	00008230	16	5777	5788
V10153	X	000082F0	16	5812	5823
V10154	X	000083B0	16	5846	5857
V10155	X	00008470	16	5880	5891
V10156	X	00008530	16	5914	5925
V10157	X	000085F0	16	5948	5959
V10158	X	000086B0	16	5983	5994
V10159	X	00008770	16	6017	6028
V1016	X	00001C30	16	1111	1122
V10160	X	00008830	16	6051	6062
V10161	X	000088F0	16	6085	6096
V10162	X	000089B0	16	6119	6130
V1017	X	00001CF0	16	1145	1156
V1018	X	00001DB0	16	1179	1190
V1019	X	00001E70	16	1213	1224
V102	X	000011B0	16	632	643
V1020	X	00001F30	16	1248	1259
V1021	X	00001FF0	16	1282	1293
V1022	X	000020B0	16	1316	1327
V1023	X	00002170	16	1350	1361
V1024	X	00002230	16	1384	1395
V1025	X	000022F0	16	1418	1429
V1026	X	000023B0	16	1452	1463
V1027	X	00002470	16	1486	1497
V1028	X	00002530	16	1520	1531
V1029	X	000025F0	16	1554	1565
V103	X	00001270	16	666	677
V1030	X	000026B0	16	1592	1603
V1031	X	00002770	16	1626	1637
V1032	X	00002830	16	1660	1671
V1033	X	000028F0	16	1694	1705
V1034	X	000029B0	16	1728	1739
V1035	X	00002A70	16	1763	1774
V1036	X	00002B30	16	1797	1808
V1037	X	00002BF0	16	1831	1842
V1038	X	00002CB0	16	1865	1876
V1039	X	00002D70	16	1899	1910
V104	X	00001330	16	700	711
V1040	X	00002E30	16	1934	1945
V1041	X	00002EF0	16	1968	1979
V1042	X	00002FB0	16	2002	2013
V1043	X	00003070	16	2036	2047

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1044	X	00003130	16	2070	2081
V1045	X	000031F0	16	2105	2116
V1046	X	000032B0	16	2139	2150
V1047	X	00003370	16	2173	2184
V1048	X	00003430	16	2207	2218
V1049	X	000034F0	16	2242	2253
V105	X	000013F0	16	734	745
V1050	X	000035B0	16	2276	2287
V1051	X	00003670	16	2310	2321
V1052	X	00003730	16	2344	2355
V1053	X	000037F0	16	2382	2393
V1054	X	000038B0	16	2416	2427
V1055	X	00003970	16	2450	2461
V1056	X	00003A30	16	2484	2495
V1057	X	00003AF0	16	2518	2529
V1058	X	00003BB0	16	2553	2564
V1059	X	00003C70	16	2587	2598
V106	X	000014B0	16	769	780
V1060	X	00003D30	16	2621	2632
V1061	X	00003DF0	16	2655	2666
V1062	X	00003EB0	16	2689	2700
V1063	X	00003F70	16	2724	2735
V1064	X	00004030	16	2758	2769
V1065	X	000040F0	16	2792	2803
V1066	X	000041B0	16	2826	2837
V1067	X	00004270	16	2860	2871
V1068	X	00004330	16	2895	2906
V1069	X	000043F0	16	2929	2940
V107	X	00001570	16	803	814
V1070	X	000044B0	16	2963	2974
V1071	X	00004570	16	2997	3008
V1072	X	00004630	16	3032	3043
V1073	X	000046F0	16	3066	3077
V1074	X	000047B0	16	3100	3111
V1075	X	00004870	16	3134	3145
V1076	X	00004930	16	3168	3179
V1077	X	000049F0	16	3202	3213
V1078	X	00004AB0	16	3236	3247
V1079	X	00004B70	16	3270	3281
V108	X	00001630	16	837	848
V1080	X	00004C30	16	3304	3315
V1081	X	00004CF0	16	3338	3349
V1082	X	00004DB0	16	3372	3383
V1083	X	00004E70	16	3409	3420
V1084	X	00004F30	16	3443	3454
V1085	X	00004FF0	16	3477	3488
V1086	X	000050B0	16	3511	3522
V1087	X	00005170	16	3545	3556
V1088	X	00005230	16	3580	3591
V1089	X	000052F0	16	3614	3625
V109	X	000016F0	16	871	882
V1090	X	000053B0	16	3648	3659
V1091	X	00005470	16	3682	3693
V1092	X	00005530	16	3716	3727
V1093	X	000055F0	16	3751	3762
V1094	X	000056B0	16	3785	3796

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
V1095	X	00005770	16	3819	3830												
V1096	X	00005830	16	3853	3864												
V1097	X	000058F0	16	3887	3898												
V1098	X	000059B0	16	3922	3933												
V1099	X	00005A70	16	3956	3967												
V10OUTPUT	X	00000030	16	473	267												
V2	U	00000002	1	6344													
V20	U	00000014	1	6362													
V21	U	00000015	1	6363													
V22	U	00000016	1	6364	603	608	609	637	642	643	671	676	677	705	710	711	739
					744	745	774	779	780	808	813	814	842	847	848	876	881
					882	910	915	916	945	950	951	979	984	985	1013	1018	1019
					1047	1052	1053	1081	1086	1087	1116	1121	1122	1150	1155	1156	1184
					1189	1190	1218	1223	1224	1253	1258	1259	1287	1292	1293	1321	1326
					1327	1355	1360	1361	1389	1394	1395	1423	1428	1429	1457	1462	1463
					1491	1496	1497	1525	1530	1531	1559	1564	1565	1597	1602	1603	1631
					1636	1637	1665	1670	1671	1699	1704	1705	1733	1738	1739	1768	1773
					1774	1802	1807	1808	1836	1841	1842	1870	1875	1876	1904	1909	1910
					1939	1944	1945	1973	1978	1979	2007	2012	2013	2041	2046	2047	2075
					2080	2081	2110	2115	2116	2144	2149	2150	2178	2183	2184	2212	2217
					2218	2247	2252	2253	2281	2286	2287	2315	2320	2321	2349	2354	2355
					2387	2392	2393	2421	2426	2427	2455	2460	2461	2489	2494	2495	2523
					2528	2529	2558	2563	2564	2592	2597	2598	2626	2631	2632	2660	2665
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					2831	2836	2837	2865	2870	2871	2900	2905	2906	2934	2939	2940	2968
					2973	2974	3002	3007	3008	3037	3042	3043	3071	3076	3077	3105	3110
					3111	3139	3144	3145	3173	3178	3179	3207	3212	3213	3241	3246	3247
					3275	3280	3281	3309	3314	3315	3343	3348	3349	3377	3382	3383	3414
					3419	3420	3448	3453	3454	3482	3487	3488	3516	3521	3522	3550	3555
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					3721	3726	3727	3756	3761	3762	3790	3795	3796	3824	3829	3830	3858
					3863	3864	3892	3897	3898	3927	3932	3933	3961	3966	3967	3995	4000
					4001	4029	4034	4035	4063	4068	4069	4101	4106	4107	4135	4140	4141
					4169	4174	4175	4203	4208	4209	4237	4242	4243	4272	4277	4278	4306
					4311	4312	4340	4345	4346	4374	4379	4380	4408	4413	4414	4443	4448
					4449	4477	4482	4483	4511	4516	4517	4545	4550	4551	4579	4584	4585
					4614	4619	4620	4648	4653	4654	4682	4687	4688	4716	4721	4722	4750
					4755	4756	4788	4793	4794	4822	4827	4828	4856	4861	4862	4890	4895
					4896	4924	4929	4930	4959	4964	4965	4993	4998	4999	5027	5032	5033
					5061	5066	5067	5095	5100	5101	5130	5135	5136	5164	5169	5170	5198
					5203	5204	5232	5237	5238	5266	5271	5272	5301	5306	5307	5335	5340
					5341	5369	5374	5375	5403	5408	5409	5437	5442	5443	5475	5480	5481
					5509	5514	5515	5543	5548	5549	5577	5582	5583	5611	5616	5617	5646
					5651	5652	5680	5685	5686	5714	5719	5720	5748	5753	5754	5782	5787
					5788	5817	5822	5823	5851	5856	5857	5885	5890	5891	5919	5924	5925
					5953	5958	5959	5988	5993	5994	6022	6027	6028	6056	6061	6062	6090
					6095	6096	6124	6129	6130								
V23	U	00000017	1	6365	605	608	639	642	673	676	707	710	741	744	776	779	810
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					1049	1052	1083	1086	1118	1121	1152	1155	1186	1189	1220	1223	1255
					1258	1289	1292	1323	1326	1357	1360	1391	1394	1425	1428	1459	1462
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					1704	1735	1738	1770	1773	1804	1807	1838	1841	1872	1875	1906	1909
					1941	1944	1975	1978	2009	2012	2043	2046	2077	2080	2112	2115	2146
					2149	2180	2183	2214	2217	2249	2252	2283	2286	2317	2320	2351	2354
					2389	2392	2423	2426	2457	2460	2491	2494	2525	2528	2560	2563	2594

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
V24	U	00000018	1	6366	2597	2628	2631	2662	2665	2696	2699	2731	2734	2765	2768	2799	2802
					2833	2836	2867	2870	2902	2905	2936	2939	2970	2973	3004	3007	3039
					3042	3073	3076	3107	3110	3141	3144	3175	3178	3209	3212	3243	3246
					3277	3280	3311	3314	3345	3348	3379	3382	3416	3419	3450	3453	3484
					3487	3518	3521	3552	3555	3587	3590	3621	3624	3655	3658	3689	3692
					3723	3726	3758	3761	3792	3795	3826	3829	3860	3863	3894	3897	3929
					3932	3963	3966	3997	4000	4031	4034	4065	4068	4103	4106	4137	4140
					4171	4174	4205	4208	4239	4242	4274	4277	4308	4311	4342	4345	4376
					4379	4410	4413	4445	4448	4479	4482	4513	4516	4547	4550	4581	4584
					4616	4619	4650	4653	4684	4687	4718	4721	4752	4755	4790	4793	4824
					4827	4858	4861	4892	4895	4926	4929	4961	4964	4995	4998	5029	5032
					5063	5066	5097	5100	5132	5135	5166	5169	5200	5203	5234	5237	5268
					5271	5303	5306	5337	5340	5371	5374	5405	5408	5439	5442	5477	5480
					5511	5514	5545	5548	5579	5582	5613	5616	5648	5651	5682	5685	5716
					5719	5750	5753	5784	5787	5819	5822	5853	5856	5887	5890	5921	5924
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					813	846	847	880	881	914	915	949	950	983	984	1017	1018
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					1495	1496	1529	1530	1563	1564	1601	1602	1635	1636	1669	1670	1703
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					2149	2182	2183	2216	2217	2251	2252	2285	2286	2319	2320	2353	2354
					2391	2392	2425	2426	2459	2460	2493	2494	2527	2528	2562	2563	2596
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					2835	2836	2869	2870	2904	2905	2938	2939	2972	2973	3006	3007	3041
					3042	3075	3076	3109	3110	3143	3144	3177	3178	3211	3212	3245	3246
					3279	3280	3313	3314	3347	3348	3381	3382	3418	3419	3452	3453	3486
					3487	3520	3521	3554	3555	3589	3590	3623	3624	3657	3658	3691	3692
					3725	3726	3760	3761	3794	3795	3828	3829	3862	3863	3896	3897	3931
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					4173	4174	4207	4208	4241	4242	4276	4277	4310	4311	4344	4345	4378
					4379	4412	4413	4447	4448	4481	4482	4515	4516	4549	4550	4583	4584
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					5719	5752	5753	5786	5787	5821	5822	5855	5856	5889	5890	5923	5924
					5957	5958	5992	5993	6026	6027	6060	6061	6094	6095	6128	6129	
V25	U	00000019	1	6367													
V26	U	0000001A	1	6368													
V27	U	0000001B	1	6369													
V28	U	0000001C	1	6370													
V29	U	0000001D	1	6371													
V2ADDR	A	00000010	4	467	602	636	670	704	738	773	807	841	875	909	944	978	1012
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					1490	1524	1558	1596	1630	1664	1698	1732	1767	1801	1835	1869	1903
					1938	1972	2006	2040	2074	2109	2143	2177	2211	2246	2280	2314	2348
					2386	2420	2454	2488	2522	2557	2591	2625	2659	2693	2728	2762	2796
					2830	2864	2899	2933	2967	3001	3036	3070	3104	3138	3172	3206	3240
					3274	3308	3342	3376	3413	3447	3481	3515	3549	3584	3618	3652	3686
					3720	3755	3789	3823	3857	3891	3926	3960	3994	4028	4062	4100	4134
					4168	4202	4236	4271	4305	4339	4373	4407	4442	4476	4510	4544	4578
					4613	4647	4681	4715	4749	4787	4821	4855	4889	4923	4958	4992	5026

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X113	F	00006508	4	4441	4427
X114	F	000065C8	4	4475	4461
X115	F	00006688	4	4509	4495
X116	F	00006748	4	4543	4529
X117	F	00006808	4	4577	4563
X118	F	000068C8	4	4612	4598
X119	F	00006988	4	4646	4632
X12	F	00001948	4	977	963
X120	F	00006A48	4	4680	4666
X121	F	00006B08	4	4714	4700
X122	F	00006BC8	4	4748	4734
X123	F	00006C88	4	4786	4772
X124	F	00006D48	4	4820	4806
X125	F	00006E08	4	4854	4840
X126	F	00006EC8	4	4888	4874
X127	F	00006F88	4	4922	4908
X128	F	00007048	4	4957	4943
X129	F	00007108	4	4991	4977
X13	F	00001A08	4	1011	997
X130	F	000071C8	4	5025	5011
X131	F	00007288	4	5059	5045
X132	F	00007348	4	5093	5079
X133	F	00007408	4	5128	5114
X134	F	000074C8	4	5162	5148
X135	F	00007588	4	5196	5182
X136	F	00007648	4	5230	5216
X137	F	00007708	4	5264	5250
X138	F	000077C8	4	5299	5285
X139	F	00007888	4	5333	5319
X14	F	00001AC8	4	1045	1031
X140	F	00007948	4	5367	5353
X141	F	00007A08	4	5401	5387
X142	F	00007AC8	4	5435	5421
X143	F	00007B88	4	5473	5459
X144	F	00007C48	4	5507	5493
X145	F	00007D08	4	5541	5527
X146	F	00007DC8	4	5575	5561
X147	F	00007E88	4	5609	5595
X148	F	00007F48	4	5644	5630
X149	F	00008008	4	5678	5664
X15	F	00001B88	4	1079	1065
X150	F	000080C8	4	5712	5698
X151	F	00008188	4	5746	5732
X152	F	00008248	4	5780	5766
X153	F	00008308	4	5815	5801
X154	F	000083C8	4	5849	5835
X155	F	00008488	4	5883	5869
X156	F	00008548	4	5917	5903
X157	F	00008608	4	5951	5937
X158	F	000086C8	4	5986	5972
X159	F	00008788	4	6020	6006
X16	F	00001C48	4	1114	1100
X160	F	00008848	4	6054	6040
X161	F	00008908	4	6088	6074
X162	F	000089C8	4	6122	6108
X17	F	00001D08	4	1148	1134

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X18	F	00001DC8	4	1182	1168
X19	F	00001E88	4	1216	1202
X2	F	000011C8	4	635	621
X20	F	00001F48	4	1251	1237
X21	F	00002008	4	1285	1271
X22	F	000020C8	4	1319	1305
X23	F	00002188	4	1353	1339
X24	F	00002248	4	1387	1373
X25	F	00002308	4	1421	1407
X26	F	000023C8	4	1455	1441
X27	F	00002488	4	1489	1475
X28	F	00002548	4	1523	1509
X29	F	00002608	4	1557	1543
X3	F	00001288	4	669	655
X30	F	000026C8	4	1595	1581
X31	F	00002788	4	1629	1615
X32	F	00002848	4	1663	1649
X33	F	00002908	4	1697	1683
X34	F	000029C8	4	1731	1717
X35	F	00002A88	4	1766	1752
X36	F	00002B48	4	1800	1786
X37	F	00002C08	4	1834	1820
X38	F	00002CC8	4	1868	1854
X39	F	00002D88	4	1902	1888
X4	F	00001348	4	703	689
X40	F	00002E48	4	1937	1923
X41	F	00002F08	4	1971	1957
X42	F	00002FC8	4	2005	1991
X43	F	00003088	4	2039	2025
X44	F	00003148	4	2073	2059
X45	F	00003208	4	2108	2094
X46	F	000032C8	4	2142	2128
X47	F	00003388	4	2176	2162
X48	F	00003448	4	2210	2196
X49	F	00003508	4	2245	2231
X5	F	00001408	4	737	723
X50	F	000035C8	4	2279	2265
X51	F	00003688	4	2313	2299
X52	F	00003748	4	2347	2333
X53	F	00003808	4	2385	2371
X54	F	000038C8	4	2419	2405
X55	F	00003988	4	2453	2439
X56	F	00003A48	4	2487	2473
X57	F	00003B08	4	2521	2507
X58	F	00003BC8	4	2556	2542
X59	F	00003C88	4	2590	2576
X6	F	000014C8	4	772	758
X60	F	00003D48	4	2624	2610
X61	F	00003E08	4	2658	2644
X62	F	00003EC8	4	2692	2678
X63	F	00003F88	4	2727	2713
X64	F	00004048	4	2761	2747
X65	F	00004108	4	2795	2781
X66	F	000041C8	4	2829	2815
X67	F	00004288	4	2863	2849
X68	F	00004348	4	2898	2884

DESC	SYMBOL	SIZE	POS	ADDR
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Entry: 0

Image	IMAGE	36060	0000-8CDB	0000-8CDB
Region		36060	0000-8CDB	0000-8CDB
CSECT	ZVE7TST	36060	0000-8CDB	0000-8CDB

STMT

FILE NAME

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1 /devstor/sharedvfp/tests/zvector-e7-10-multiplyAdd.asm
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**** NO ERRORS FOUND ****